

By Micro.IT.Corporation
Cell# 03344500853
Abaid Ur Rehman

Date: 08-11-2010

Question # 1 of 10

Information hiding can be achieved through_____.

- 1.Encapsulation, Inheritance
- 2.Encapsulation, Polymorphism
- 3.Encapsulation, Abstraction**
- 4.Overloading

Question # 2 of 10 (Start time: 01:11:21 AM) Total Marks: 1

A good model is related to a real life problem.

Select correct option:

- 1.Loosely
- 2.Openly
- 3.Closely**

Question # 3 of 10 (Start time: 01:12:33 AM) Total Marks: 1

Which of the following features of OOP is used to derive a class from another?

Select correct option:

- 1.Encapsulation
- 2.Polymorphism
- 3.Data hiding
- 4. Inheritance**

Question # 4 of 10 (Start time: 01:13:51 AM) Total Marks: 1

Which of the following is a weak relationship between two objects?

Select correct option:

1. Inheritance
- 2.Composition

3.Agregation

4.None of given

Question # 5 of 10 (Start time: 01:14:56 AM) Total Marks: 1

Data items in a class must be private.

Select correct option:

1.True

2.False

Question # 6 of 10 (Start time: 01:15:52 AM) Total Marks: 1

Which one is a class association

Select correct option:

1.Simple Association

2. Inheritance

3.Composition

4.Aggregation

Question # 7 of 10 (Start time: 01:16:55 AM) Total Marks: 1

Suppose there is an object of type Person, which of the following can be considered as one of its attributes

Select correct option:

1.Name

2.Age

3.Work()

4.Both Name and Age

Question # 8 of 10 (Start time: 01:17:52 AM) Total Marks: 1

Which one is not an object association?

Select correct option:

1.Simple association

2. Inheritance

3.Aggregation

4.Association

Question # 9 of 10 (Start time: 01:18:50 AM) Total Marks: 1

Using encapsulation we can achieve

Select correct option:

1. Information hiding

2.Least interdependencies among modules

3. Implementation independence

4.All of given options

Question # 10 of 10 (Start time: 01:19:43 AM) Total Marks: 1

In constant member function the type of this pointer is:

Select correct option:

1.Constant pointer

2.Constant pointer to object

3.Constant pointer to class

4.Constant pointer to constant object

Question # 1 of 10

Which of the following is the way to extract common behavior and attributes from the given classes and make a separate class of those common behaviors and attributes?

1.Generalization

- 2.Sub-typing
- 3.Specialization
- 4.Extension

Question # 2 of 10

The ability to derive a class from more than one class is called

- 1.Single inheritance
- 2.Encapsulation

3.Multiple inheritance

- 4.Polymorphism

Question # 3 of 10:

If MyClass has a destructor what is the destructor named?

- 1.MyClass
- 2.~MyClass**
- 3.My~Class
- 4.MyClass~

Question # 4 of 10:

Class abc{ ----- }; Is a valid class declaration?

- 1.yes**
- 2.no

Question # 5 of 10:

Without using Deep copy constructor, A _____ problem can occur

- 1.System crash
- 2.Memory Leakage
- 3.Dangling pointer

4.All of the given

Question # 6 of 10:

If only one behaviour of a derived class is incompatible with base class, then it is:

- 1.Generalization
- 2.Specialization**
- 3.Extension
- 4. Inheritance

Question # 7 of 10:

Which of the following may not be an integral part of an object?

- 1. state
- 2.behavior
- 3.Protected data members

4.All of given

Question # 8 of 10:

Only tangible things can be chosen as an object.

1.True

2.False

1.

2. Question # 1 of 10

Information hiding can be achieved through_____.

Encapsulation, Inheritance

Encapsulation, Polymorphism

Encapsulation, Abstraction

Overloading

Question # 2 of 10 (Start time: 01:11:21 AM) Total Marks: 1

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Data hiding

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Composition

Aggregation

None of given

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Data items in a class must be private.

Select correct option:

True

False

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Aggregation

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Age

Work()

Both Name and Age

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Inheritance

Aggregation

Association

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Select correct option:

Information hiding

Least interdependencies among modules

Implementation independence

All of given options

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Constant pointer

Constant pointer to object

Constant pointer to class

Constant pointer to constant object

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Sub-typing

Specialization

Extension

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The ability to derive a class from more than one class is called
Single inheritance

Encapsulation

Multiple inheritance

Polymorphism

Question # 3 of 10:

If MyClass has a destructor what is the destructor named?

MyClass

~MyClass

My~Class

MyClass~

Question # 4 of 10:

Class abc{ ----- }; Is a valid class declaration?

yes

no

Question # 5 of 10:

Without using Deep copy constructor, A _____ problem can occur

System crash

Memory Leakage

Dangling pointer

All of the given

Question # 6 of 10:

If only one behaviour of a derived class is incompatible with base class, then it is:

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Specialization

Extension

Inheritance

Question # 7 of 10:

Which of the following may not be an integral part of an object?

state

behavior

Protected data members

All of given

Question # 8 of 10:

Only tangible things can be chosen as an object.

True

False

Class is not a mechanism to create objects and define user data types.

1. true

2. false

Memory is allocated to non static members only, when:

1. Class is created
2. Object is defined
3. Object is initialized
4. Object is created

The sub-object's life is not dependent on the life of master class in _____.

1. Composition
2. Aggregation
3. Separation
4. non of the given

Unary operators and assignment operator are right associative.

1. true
2. false

The >= operator can't be overloaded.

1. true
2. false

_____ is creating objects of one class inside another class.

1. Association
2. Composition
3. Aggregation
4. Inheritance

If we are create array of objects through new operator, then

1. We can call overloaded constructor through new
2. We can't call overloaded constructor through new
3. We can call default constructor through new
4. None of the given

Object can be declared constant with the use of Constant keyword.

1. true
2. false

_____ Operator will take only one operand.

1. New
2. int
3. object
4. none of the given

Which of the following operator(s) take(s) one or no argument if overloaded?

1. ++
2. *
3. %
4. All of the given choices

this pointer does not pass implicitly to _____ functions.

1. Static Member
2. Non-Static Member
3. Instance Number
4. None of the given

Operator overloading is

1. making C++ operators work with objects.
2. giving C++ operators more than they can handle.
3. giving new meanings to existing Class members.
4. making new C++ operators

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 1 of 10 (Start time: 09:57:41 AM) Total Marks: 1

Consider the code below, class class1{ public: void func1(); }; class class2 :
private
class1 { }; Function func1 of class1 is
_____ in class2,

Select correct option:

- public
- protected
- private
- none of the given options

[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 2 of 10 (Start time: 09:59:01 AM) Total Marks: 1

User can make virtual table explicitly.

Select correct option:

- True
- False

[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 3 of 10 (Start time: 10:00:15 AM) Total Marks: 1

In private inheritance derived class pointer can be assigned to base class pointer in,

Select correct option:

Main function

In derived class member and friend functions

In base class member and friend functions

None of the given options

[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 4 of 10 (Start time: 10:01:15 AM) Total Marks: 1

In C++, we declare a function virtual by preceding the function header with keyword

“Inline”

Select correct option:

True

False

[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 5 of 10 (Start time: 10:02:45 AM) Total Marks: 1

Outside world can access only _____ members of a class using its object.

Select correct option:

Public

Private

Protected

No member is accessible.

[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 6 of 10 (Start time: 10:03:10 AM) Total Marks: 1

Friend Functions of a class are _____ members of that class.

Select correct option:

Public

Private

Protected

None of the given options.

[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Time Left

Quiz Start Time: 09:57 AM

Question # 7 of 10 (Start time: 10:03:54 AM) Total Marks: 1

Consider the following two lines of code written for a class Student, 1. Student subj1,subj2; 2. subj2 = subj1; In line No.2 what constructor of Student class will be called,

Select correct option:

Default constructor of Student class.

Copy constructor of student class

Both default and copy constructor of Student class

No constructor will be called.

[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left Class is not a mechanism to create objects and define user data types.

1. true

2. false

Memory is allocated to non static members only, when:

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2. Object is defined

3. Object is initialized

4. Object is created

The sub-object's life is not dependent on the life of master class in _____.

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2. Aggregation

3. Separation

4. non of the given

Unary operators and assignment operator are right associative.

1. true

2. false

The >= operator can't be overloaded.

1. true

2. false

_____ is creating objects of one class inside another class.

1. Association

2. Composition

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4. Inheritance

If we are create array of objects through new operator, then

1. We can call overloaded constructor through new

2. We can't call overloaded constructor through new

3. We can call default constructor through new

4. None of the given

Object can be declared constant with the use of Constant keyword.

1. true

2. false

_____ Operator will take only one operand.

1. New

2. int

3. object

4. none of the given

Which of the following operator(s) take(s) one or no argument if overloaded?

1. ++

2. *

3. %

4. All of the given choices

this pointer does not pass implicitly to _____ functions.

1. Static Member

2. Non-Static Member

3. Instance Number

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Operator overloading is

1. making C++ operators work with objects.

2. giving C++ operators more than they can handle.

3. giving new meanings to existing Class members.

4. making new C++ operators

BC080400849 : Nimra Qamar

Quiz Start Time: 10:39 PM

Time Left 88

sec(s)

Question # 1 of 8 (Start time: 10:39:47 PM) Total Marks: 1

Which of the following operator(s) take(s) one or no argument if overloaded?

Select correct option:

++

*

%

All of the given choices

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 10:39 PM

Time Left 89

sec(s)

Question # 2 of 8 (Start time: 10:40:38 PM) Total Marks: 1

Object can be declared constant with the use of Constant keyword.

Select correct option:

True

False

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 10:39 PM

Time Left 89

sec(s)

Question # 3 of 8 (Start time: 10:41:41 PM) Total Marks: 1

Static data members are called _____ variable

Select correct option:

Class

Object

Structure

none of the given

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 10:39 PM

Time Left 89

sec(s)

Question # 4 of 8 (Start time: 10:42:35 PM) Total Marks: 1

Associatively can be change in operator overloading.

Select correct option:

True

False

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 10:39 PM

Time Left 89

sec(s)

Question # 5 of 8 (Start time: 10:43:56 PM) Total Marks: 1

_____ and _____ methods may not be declared abstract.

Select correct option:

Private,static

private,public

static,public

none of the given

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 10:39 PM

Time Left 89

sec(s)

Question # 6 of 8 (Start time: 10:45:17 PM) Total Marks: 1

Let Suppose a class Student with objects std1, std2, and std3. For the statement
std3 =

std1 - std2 to work correctly, if the overloaded - operator must

Select correct option:

take two arguments.

None of the given choices

take single argument

take three arguments

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 10:39 PM

Time Left 88

sec(s)

Question # 7 of 8 (Start time: 10:46:48 PM) Total Marks: 1

To initialize an array of objects, only _____ will be called

Select correct option:

Default Constructor

Overloaded Constructor

Default Object

None of the above

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 10:39 PM

Time Left 87

sec(s)

Question # 8 of 8 (Start time: 10:47:49 PM) Total Marks: 1

_____ provide the facility to access the data member.

Select correct option:

accessor function

private function

inline function

None of the given

[Click here to Save Answer & Move to Next Question](#)

Question # 8 of 10 (Start time: 10:04:41 AM) Total Marks: 1

Consider the following two lines of code written for a class Student, 1. Student subj1; 2.

Student subj2 = subj1; In line No.1

what constructor of student class will be called,

Select correct option:

Default constructor of Student class.

Copy constructor of student class

Both default and copy constructor of student class

None the given options

[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 9 of 10 (Start time: 10:05:09 AM) Total Marks: 1

Consider the code below, class class1{ protected: void func1(); }; class class2 : protected

class1 { }; Function func1 of class1 is _____ in class2,

Select correct option:

public

protected

private

none of the given options

[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 10 of 10 (Start time: 10:05:50 AM) Total Marks: 1

Virtual functions allow you to

Select correct option:

create an array of type pointer-to-base class that can hold pointers to derived classes.

create functions that can never be accessed.

group objects of different classes so they can all be accessed by the same function code.

use the same function call to execute member functions of objects from different classes.

[Click here to Save Answer & Move to Next Question](#)

Question # 1 of 10

Information hiding can be achieved through_____.

1. Encapsulation, Inheritance
2. Encapsulation, Polymorphism
- 3. Encapsulation, Abstraction**
4. Overloading

Question # 2 of 10 (Start time: 01:11:21 AM) Total Marks: 1

A good model is related to a real life problem.

Select correct option:

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2. Openly
- 3. Closely**

Question # 3 of 10 (Start time: 01:12:33 AM) Total Marks: 1

Which of the following features of OOP is used to derive a class from another?

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2. Polymorphism
3. Data hiding

4. Inheritance

Question # 4 of 10 (Start time: 01:13:51 AM) Total Marks: 1

Which of the following is a weak relationship between two objects?

Select correct option:

1. Inheritance
2. Composition

3. Aggregation

4. None of given

Question # 5 of 10 (Start time: 01:14:56 AM) Total Marks: 1

Data items in a class must be private.

Select correct option:

1. True
- 2. False**

Question # 6 of 10 (Start time: 01:15:52 AM) Total Marks: 1

Which one is a class association

Select correct option:

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Question # 7 of 10 (Start time: 01:16:55 AM) Total Marks: 1

Suppose there is an object of type Person, which of the following can be considered as

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Select correct option:

1. Name

2. Age

3. Work()

4. Both Name and Age

Question # 8 of 10 (Start time: 01:17:52 AM) Total Marks: 1

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Question # 9 of 10 (Start time: 01:18:50 AM) Total Marks: 1

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In constant member function the type of this pointer is:

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2. Constant pointer to object

3. Constant pointer to class

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Which of the following is the way to extract common behavior and attributes from the

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The ability to derive a class from more than one class is called

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If MyClass has a destructor what is the destructor named?

- 1. MyClass

2. ~MyClass

- 3. My~Class
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Class abc{ ----- }; Is a valid class declaration?

1. yes

- 2. no

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Without using Deep copy constructor, A _____ problem can occur

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- 2. behavior
- 3. Protected data members

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Only tangible things can be chosen as an object.

1. True

2. False

BC080400849 : Nimra Qamar

Quiz Start Time: 08:14 PM

Time Left 79

sec(s)

Question # 1 of 10 (Start time: 08:14:14 PM) Total Marks: 1

When we create objects, then space is allocated to:

Select correct option:

Member functions

Access specifier

Data members

None of the given

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 08:14 PM

Time Left 87

sec(s)

Question # 2 of 10 (Start time: 08:15:18 PM) Total Marks: 1

Constructor and destructor can be declared constant

Select correct option:

True

False

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 08:14 PM

Time Left 88

sec(s)

Question # 3 of 10 (Start time: 08:16:04 PM) Total Marks: 1

Information hiding can be achieved through_____.

Select correct option:

Encapsulation, Inheritance

Encapsulation, Polymorphism

Encapsulation, Abstraction

Encapsulation, Overloading

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 08:14 PM

Time Left 88

sec(s)

Question # 4 of 10 (Start time: 08:16:47 PM) Total Marks: 1

A real world object can be transformed into programming entity by defining its respective

Select correct option:

Class

Function

Only states

Only behaviour

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 08:14 PM

Time Left 88

sec(s)

Question # 5 of 10 (Start time: 08:17:50 PM) Total Marks: 1

Which of the following is a weak relationship between two objects?

Select correct option:

Inheritance

Composition

Aggregation

None of given

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 08:14 PM

Time Left 88

sec(s)

Question # 6 of 10 (Start time: 08:18:45 PM) Total Marks: 1

Which of the following is a necessary ingredient in an object model?

Select correct option:

Class

Objects

Association

All of given

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 08:14 PM

Time Left 88

sec(s)

Question # 7 of 10 (Start time: 08:19:48 PM) Total Marks: 1

If a class A inherits from class B, then class A is called.

Select correct option:

Child class

Derived class

Parent class

Child and derived class

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 08:14 PM

Time Left 88

sec(s)

Question # 8 of 10 (Start time: 08:20:54 PM) Total Marks: 1

Which of the following is the way to extract common behaviour and attributes from the

given classes and make a separate class of those common behaviours and attributes?

Select correct option:

Generalization

Sub-typing

Specialization

Extension

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 08:14 PM

Time Left 87

sec(s)

Question # 9 of 10 (Start time: 08:21:27 PM) Total Marks: 1

The _____ keyword tells the compiler to substitute the code within the function definition for every instance of a function call

Select correct option:

virtual

inline

instance

none of the given

[Click here to Save Answer & Move to Next Question](#)

BC080400849 : Nimra Qamar

Quiz Start Time: 08:14 PM

Time Left 88

sec(s)

Question # 10 of 10 (Start time: 08:22:16 PM) Total Marks: 1

The process of hiding unwanted details from users is called _____.

Select correct option:

Protection

Encapsulation

Argumentation

Abstraction

[Click here to Save Answer & Move to Next Question](#)

BC090401541 : Azmat Ur Rehman

Quiz Start Time: 12:40 PM

Time Left

Question # 1 of 10 (Start time: 12:40:20 PM) Total Marks: 1

Which of the following is the way to extract common behaviour and attributes from the given classes and make a separate class of those common behaviours and attributes?

Select correct option:

Generalization

Sub-typing

Specialization

Extension

[Click here to Save Answer & Move to Next Question](#)

BC090401541 : Azmat Ur Rehman

Quiz Start Time: 12:40 PM

Time Left

Question # 2 of 10 (Start time: 12:41:52 PM) Total Marks: 1

“A fan has wings”. Which type of relation exists between fan and wings in this sentence?

Select correct option:

Aggregation

Association

Generalization

Composition

[Click here to Save Answer & Move to Next Question](#)

BC090401541 : Azmat Ur Rehman

Quiz Start Time: 12:40 PM

Time Left

Question # 3 of 10 (Start time: 12:42:46 PM) Total Marks: 1

A good model is related to a real life problem.

Select correct option:

Loosely

Openly

Closely

Not

[Click here to Save Answer & Move to Next Question](#)

BC090401541 : Azmat Ur Rehman

Quiz Start Time: 12:40 PM

Time Left

Question # 5 of 10 (Start time: 12:44:45 PM) Total Marks: 1

When we create objects, then space is allocated to:

Select correct option:

Member functions

Access specifier

Data members

None of the given

[Click here to Save Answer & Move to Next Question](#)

BC090401541 : Azmat Ur Rehman

Quiz Start Time: 12:40 PM

Time Left

Question # 6 of 10 (Start time: 12:45:21 PM) Total Marks: 1

There is only one form of copy constructor.

Select correct option:

True

False

[Click here to Save Answer & Move to Next Question](#)

BC090401541 : Azmat Ur Rehman

Quiz Start Time: 12:40 PM

Time Left 22

sec(s)

Question # 7 of 10 (Start time: 12:45:38 PM) Total Marks: 1

Which of the following features of OOP is used to deal with only relevant details?

Select correct option:

Abstraction

Information hiding

Object

Inheritance

[Click here to Save Answer & Move to Next Question](#)

BC090401541 : Azmat Ur Rehman

Quiz Start Time: 12:40 PM

Time Left 59

sec(s)

Question # 8 of 10 (Start time: 12:48:26 PM) Total Marks: 1

Suppose there is an object of type Person, which of the following can be considered as

one of its attributes

Select correct option:

Age

Work()

Both Name and Age

[Click here to Save Answer & Move to Next Question](#)

BC090401541 : Azmat Ur Rehman

Quiz Start Time: 12:40 PM

Time Left

Question # 9 of 10 (Start time: 12:56:04 PM) Total Marks: 1

Through interface we access object_____.

Select correct option:

States

Data members

Behaviour

None of the given

[Click here to Save Answer & Move to Next Question](#)

BC090401541 : Azmat Ur Rehman

Quiz Start Time: 12:40 PM

Time Left

Question # 10 of 10 (Start time: 12:57:00 PM) Total Marks: 1

If a class A inherits from class B, then class A is called.

Select correct option:

Child class

Derived class

Parent class

Child and derived class

[Click here to Save Answer & Move to Next Question](#)

MC090405816 : Sohail Aslam

Quiz Start Time: 12:59 PM

Time Left

Question # 1 of 10 (Start time: 12:59:51 PM) Total Marks: 1

If some of objects exhibit identical characteristics, then they belong to:

Select correct option:

Different classes

Multiple classes

Same class

None of the given

[Click here to Save Answer & Move to Next Question](#)

MC090405816 : Sohail Aslam

Quiz Start Time: 12:59 PM

Time Left 82

sec(s)

Question # 2 of 10 (Start time: 01:00:41 PM) Total Marks: 1

_____ is automatically called when the object is created.

Select correct option:

member function

object

constructor

None of the given

[Click here to Save Answer & Move to Next Question](#)

MC090405816 : Sohail Aslam

Quiz Start Time: 12:59 PM

Time Left

Question # 3 of 10 (Start time: 01:03:09 PM) Total Marks: 1

Which is true about sub-typing in case of inheritance?

Select correct option:

In sub-typing a new class is derived from existing w
extended behavior of its parent.

In sub-typing a new class is derived from existing w

In sub-typing a class is derived from existing one wh

None of the given.

[Click here to Save Answer & Move to Next Question](#)

MC090405816 : Sohail Aslam

Quiz Start Time: 12:59 PM

Time Left

Question # 4 of 10 (Start time: 01:04:28 PM) Total Marks: 1

If a class involves dynamic memory allocation, then:

Select correct option:

Default copy constructor, shallow copy is implement
User defined copy constructor, shallow copy is impl
Default copy constructor, deep copy is implemented
User defined copy constructor, deep copy is implem
Click here to Save Answer & Move to Next Question

MC090405816 : Sohail Aslam

Quiz Start Time: 12:59 PM

Time Left

Question # 5 of 10 (Start time: 01:05:37 PM) Total Marks: 1

Which one is a class association

Select correct option:

Simple Association

Inheritance

Composition

Aggregation

Click here to Save Answer & Move to Next Question

MC090405816 : Sohail Aslam

Quiz Start Time: 12:59 PM

Time Left

Question # 6 of 10 (Start time: 01:06:50 PM) Total Marks: 1

Data items in a class must be private.

Select correct option:

True

False

Click here to Save Answer & Move to Next Question

MC090405816 : Sohail Aslam

Quiz Start Time: 12:59 PM

Time Left

Question # 7 of 10 (Start time: 01:07:16 PM) Total Marks: 1

Three main characteristics of "Object Oriented programming" are,

Select correct option:

Encapsulation,dynamic binding,polymerphism

polymorphism, overloading, overriding

encapsulation, inheritance, dynamic binding

encapsulation, inheritance, polymorphism

Click here to Save Answer & Move to Next Question

MC090405816 : Sohail Aslam

Quiz Start Time: 12:59 PM
Time Left

Question # 8 of 10 (Start time: 01:08:14 PM) Total Marks: 1

Which of the following is the way to extract common behaviour and attributes from the given classes and make a separate class of those common behaviours and attributes?

Select correct option:

Generalization

Sub-typing

Specialization

Extension

[Click here to Save Answer & Move to Next Question](#)

MC090405816 : Sohail Aslam

Quiz Start Time: 12:59 PM

Time Left 69

sec(s)

Question # 9 of 10 (Start time: 01:09:04 PM) Total Marks: 1

The sentence “Object Oriented Programming book in bookshelf” is an example of:

Select correct option:

Association

Multiple association

Aggregation

[Click here to Save Answer & Move to Next Question](#)

MC090405816 : Sohail Aslam

Quiz Start Time: 12:59 PM

Time Left

Question # 10 of 10 (Start time: 01:16:05 PM) Total Marks: 1

Data members are the attributes of objects.

Select correct option:

True

False

[Click here to Save Answer & Move to Next Question](#)

MC090406317 : Aamer Abbas

Quiz Start Time: 01:18 PM

Time Left

Question # 1 of 10 (Start time: 01:18:48 PM) Total Marks: 1

Constructor have same name as the class name.

Select correct option:

True

False

[Click here to Save Answer & Move to Next Question](#)

MC090406317 : Aamer Abbas

Quiz Start Time: 01:18 PM

Time Left

Question # 2 of 10 (Start time: 01:19:03 PM) Total Marks: 1

Which of the following features of OOP is used to derive a class from another?

Select correct option:

Encapsulation

Polymorphism

Data hiding

Inheritance

[Click here to Save Answer & Move to Next Question](#)

MC090406317 : Aamer Abbas

Quiz Start Time: 01:18 PM

Time Left 81

sec(s)

Question # 3 of 10 (Start time: 01:19:29 PM) Total Marks: 1

Class abc{ ----- }; Is a valid class declaration?

Select correct option:

Yes

No

[Click here to Save Answer & Move to Next Question](#)

MC090406317 : Aamer Abbas

Quiz Start Time: 01:18 PM

Time Left

Question # 6 of 10 (Start time: 01:22:47 PM) Total Marks: 1

Which of the following is a weak relationship between two objects?

Select correct option:

Inheritance

Composition

Aggregation

None of given

[Click here to Save Answer & Move to Next Question](#)

MC090406317 : Aamer Abbas

Quiz Start Time: 01:18 PM
Time Left

Question # 4 of 10 (Start time: 01:20:47 PM) Total Marks: 1

Without using Deep copy constructor, A _____ problem can occur

Select correct option:

System crash

Memory Leakage

Dangling pointer

All of the given

[Click here to Save Answer & Move to Next Question](#)

MC090406317 : Aamer Abbas

Quiz Start Time: 01:18 PM

Time Left

Question # 5 of 10 (Start time: 01:21:20 PM) Total Marks: 1

An abstract class shows _____ behaviour.

Select correct option:

Overriding

Specific

General

None of the given

[Click here to Save Answer & Move to Next Question](#)

MC090406317 : Aamer Abbas

Quiz Start Time: 01:18 PM

Time Left

Question # 7 of 10 (Start time: 01:22:59 PM) Total Marks: 1

Which of the following are benefits of encapsulation?

Select correct option:

All variables can be manipulated as Objects instead

by making all variables protected they are protected

The implementation of a class can be changed witho

Making all methods protected prevents accidental co

[Click here to Save Answer & Move to Next Question](#)

MC090406317 : Aamer Abbas

Quiz Start Time: 01:18 PM

Time Left

Question # 8 of 10 (Start time: 01:24:19 PM) Total Marks: 1

If a class A inherits from class B, then class A is called.

Select correct option:

Child class

Derived class

Parent class

Child and derived class

[Click here to Save Answer & Move to Next Question](#)

MC090406317 : Aamer Abbas

Quiz Start Time: 01:18 PM

Time Left

Question # 9 of 10 (Start time: 01:24:44 PM) Total Marks: 1

Consider the statement “room has chair” Which of the following type of association

exists between room and chair?

Select correct option:

Inheritance

Composition

There is no association

Aggregation

[Click here to Save Answer & Move to Next Question](#)

MC090406317 : Aamer Abbas

Quiz Start Time: 01:18 PM

Time Left

Question # 10 of 10 (Start time: 01:25:05 PM) Total Marks: 1

The dot operator (or class member access operator) connects the following two entities

(reading from left to right):

Select correct option:

A class member and a class object

A class object and a class

A class and a member of that class

A class object and a member of that class

[Click here to Save Answer & Move to Next Question](#)

—

—

FINAL TERM EXAMINATION

Spring 2010

CS304- Object Oriented Programming (Session - 4)

Question No: 1 (Marks: 1) - Please choose one

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be

concrete, while classes like Sphere and Cube would normally be abstract.

☐ True

☐ False

Question No: 2 (Marks: 1) - Please choose one

Virtual functions allow you to

☐ create an array of type pointer-to-base class that can hold pointers to derived classes.

☐ create functions that can never be accessed.

☐ group objects of different classes so they can all be accessed by the same function code.

☐ use the same function call to execute member functions of objects from different classes

Question No: 3 (Marks: 1) - Please choose one

A pointer to a base class can point to objects of a derived class.

☐ True

☐ False

Question No: 4 (Marks: 1) - Please choose one

A copy constructor is invoked when

☐ a function do not returns by value.

☐ an argument is passed by value.

☐

☐

☐ a function returns by reference.

☐ an argument is passed by reference.

Question No: 5 (Marks: 1) - Please choose one

Each try block can have _____ no. of catch blocks.

☐ 1

☐ 2

☐ 3

☐ As many as necessary.

Question No: 6 (Marks: 1) - Please choose one

Non Template Friend functions of a class are friends of _____instance/s of that class.

- ☐ All
- ☐ One specific
- ☐ All instances of one data type
- ☐ None of the given options

Question No: 7 (Marks: 1) - Please choose one

Template functions use _____ than ordinary functions.

- ☐ Greater Memory
- ☐ Lesser Memory
- ☐ Equal Memory
- ☐ None of the given options

Question No: 8 (Marks: 1) - Please choose one

The find() algorithm

- ☐ finds matching sequences of elements in two containers.
- ☐ finds a container that matches a specified container.
- ☐ takes iterators as its first two arguments.
- ☐ takes container elements as its first two arguments.

Question No: 9 (Marks: 1) - Please choose one

The copy() algorithm returns an iterator to

—

—

- ☐ the last element copied from.
- ☐ the last element copied to.
- ☐ the element one past the last element copied from.
- ☐ the element one past the last element copied to.

Question No: 10 (Marks: 1) - Please choose one

If you define a vector v with the default constructor, and define another vector w with a

one-argument constructor to a size of 11, and insert 3 elements into each of these vectors

with push_back(), then the size() member function will return _____ for v and _____ for

w.

- ☐ 11 for v and 3 for w.
- ☐ 0 for v and 0 for w.
- ☐ 0 for v and 3 for w.
- ☐ 3 for v and 11 for w.

Question No: 11 (Marks: 1) - Please choose one
Which is not the Advantage of inheritance?

- _ providing class growth through natural selection.
- _ facilitating class libraries.
- _ avoiding the rewriting of code.
- _ providing a useful conceptual framework.

Question No: 12 (Marks: 1) - Please choose one
class DocElement

```
{
public:
virtual void Print() { cout << "Generic element"; }
};
class Heading : public DocElement
{
public:
void Print() { cout << "Heading element"; }
};
```

—

```
—
class Paragraph : public DocElement
{
public:
void Print() { cout << "Paragraph element"; }
};
void main()
{
DocElement * p = new Paragraph();
p->Print();
}
```

When you run this program, it will print out a single line to the console output.
What will be in that line?

Select one correct answer from the following list:

- _ Generic element
- _ Heading element
- _ Paragraph element
- _ Nothing will be printed.

Question No: 13 (Marks: 1) - Please choose one

Which type of inheritance is being represented by the following statement,

class X : public A, public B { };

- _ Single inheritance
- _ Multiple inheritance
- _ Double inheritance
- _ None of the given options

Question No: 14 (Marks: 1) - Please choose one

When we write a class template the first line must be:

—

—

- _ template < class class_name>
- _ template < class data_type>
- _ template < class T >

Here T can be replaced with any name but it is preferable.

- _ class class-name()
- class template<class_name>

Question No: 15 (Marks: 1) - Please choose one

Function templates should be used where code and behavior must be identical.

- _ True
- _ False

Question No: 16 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- _ Reusability
- _ Writability
- _ Maintainability
- _ All of given

Question No: 17 (Marks: 1) - Please choose one

The specialization pattern <T*> after the name says that this specialization is to be used

for every,

- _ data type
- _ meta type
- _ virtual type

—

—

- _ pointer type

Question No: 18 (Marks: 1) - Please choose one

A range is often supplied to an algorithm by two _____ values.

- ☐ italic
- ☐ iteration
- ☐ iterator
- ☐ None of given

Question No: 19 (Marks: 1) - Please choose one

Which of the following is an integral part of an object?

- ☐ State
- ☐ Behavior
- ☐ Unique identity
- ☐ All of the given

Question No: 20 (Marks: 1) - Please choose one

Consider the following statement

Cupboard has books

What is the relationship between Cupboard and books?

- ☐ Composition
- ☐ Aggregation
- ☐ Inheritance
- ☐ None of the given options

Question No: 21 (Marks: 1) - Please choose one

Which sentence clearly defines an object?

—

—

- ☐ one instance of a class.
- ☐ another word for a class.
- ☐ a class with static methods.
- ☐ a method that accesses class attributes.

Question No: 22 (Marks: 1) - Please choose one

_____, which means if A declares B as its friend it does NOT mean that A can

access private data of B. It only means that B can access all data of A.

- ☐ Friendship is one way only
- ☐ Friendship is two way only
- ☐ NO Friendship between classes
- ☐ Any kind of friendship

Question No: 23 (Marks: 1) - Please choose one

The statement `objA=objB;` will cause a compiler error if the objects are of different classes.

☐ True

☐ False

Question No: 24 (Marks: 1) - Please choose one

Consider the call given below of an overloaded operator "+",

Rational_number_1 + Rational_number_2

Where Rational_number_1 and Rational_number_2 are the two objects of Rational_number class (a user defined class). Identify which of the above two objects

will be passed as an argument to the overloaded operator function?

—

—

☐ Rational_number_1

☐ Rational_number_2

☐ Both Rational_number_1 & Rational_number_2

☐ any of the two objects, randomly

Question No: 25 (Marks: 1) - Please choose one

If a class D has been derived using protected inheritance from class B (If B is a protected

base and D is derived class) then public and protected members of B -----
accessed by

member functions and friends of class D and classes derived from D

☐ can be

☐ cannot be

☐ does restrict to be

☐ not given

Question No: 26 (Marks: 1) - Please choose one

In Private ----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

☐ specialization

☐ inheritance

☐ abstraction

☐ composition

Question No: 27 (Marks: 2)

Give two uses of a destructor.

Question No: 28 (Marks: 2)

Describe the way to declare a template class as a friend class of any other class.

Question No: 29 (Marks: 2)

Give the name of two basic types of containers collectively called First class containers?

Question No: 30 (Marks: 2)

State any conflict that may rise due to multiple inheritance?

Question No: 31 (Marks: 3)

What will be the output after executing the following code?

```
class c1{
public:
virtual void function(){
cout<<"I am in c1"<<endl;
```

—

—
}

};

```
class c2: public c1{
```

```
public:
```

```
void function(){
```

```
cout<<"I am in c2"<<endl;
```

```
}
```

```
};
```

```
class c3: public c1 {
```

```
public:
```

```
void function(){
```

```
cout<<"I am in c3"<<endl;
```

```
}
```

```
};
```

```
int main(){
```

```
c1 * test1 = new c2();
```

```
c1 * test2 = new c3();
```

```
test1->function();
```

```
test2->function();
```

```
system("PAUSE");
```

```
return 0;
```

```
}
```

Question No: 32 (Marks: 3)

If we declare a function as friend of a template class will it be a friend for a particular

data type or for all data types of that class.

Question No: 33 (Marks: 3)

Tell the logical error/s in the code given below with reference to resource management;

also describe how we can correct that error/s.

```
class Test{
public:
int function1(){
try{
```

—

```
—
FILE *fileptr = fopen("filename.txt","w");
throw exception();
fclose(fileptr);
return 0;
}
catch(Exception e){
...
}
}
};
```

Question No: 34 (Marks: 5)

What is the output produced by the following program?

```
#include<iostream.h>
void sample_function(double test) throw (int);
int main()
{
try
{
cout <<"Trying.\n";
sample_function(98.6);
cout << "Trying after call.\n";
}
catch(int)
{
cout << "Catching.\n";
}
cout << "End program.\n";
return 0;
```

```

}
void sample_function(double test) throw (int)
{
cout << "Starting sample_function.\n";
if(test < 100)
throw 42;
}

```

Question No: 35 (Marks: 5)

The code given below has one template function as a friend of a template class,

1. You have to identify any error/s in this code and describe the reason for error/s.
2. Give the correct code after removing the error/s.

—

—

```

template<typename U>
void Test(U);
template< class T >
class B {
int data;
public:
friend void Test<>( T );
};
template<typename U>
void Test(U u){
B < int> b1;
b1.data = 7;
}
int main(int argc, char *argv[])
{
char i;
Test(i);
system("PAUSE");
return 0;
}

```

Question No: 36 (Marks: 5)

Consider the following class,

```

class Base
{
char * p;

```

```

public:
Base() { p = new char[10]; }
~Base() { delete [] p; }
};
class Derived : public Base
{
char * q;
public:
Derived() { q = new char[20]; }
~Derived() { delete [] q; }
};
void foo()
{
Base* p = new Derived();

```

—

```

—
delete p;
}

```

With this program, every time function foo is called, some memory will leak. Explain why memory will leak. Also, explain how to fix this problem.

FINAL TERM EXAMINATION

Fall 2009

CS304- Object Oriented Programming (Session - 1)

Time: 120 min

Marks: 75

Question No: 1 (Marks: 1) - Please choose one

Which one of the following terms must relate to polymorphism?

- _ Static allocation
- _ Static typing
- _ [Dynamic binding](#)
- _ Dynamic allocation

Question No: 2 (Marks: 1) - Please choose one

Multiple inheritance can be of type

- _ Public
- _ Private

—

- ☐ Protected
- ☒ All of the given

Question No: 3 (Marks: 1) - Please choose one

When a subclass specifies an alternative definition for an attribute or method of its superclass, it is _____ the definition in the superclass.

- ☐ overload
- ☒ overriding
- ☐ copy riding
- ☐ none of given

Question No: 4 (Marks: 1) - Please choose one

Like template functions, a class template may not handle all the types successfully.

- ☒ True
- ☐ False

Question No: 5 (Marks: 1) - Please choose one

It is sometimes useful to specify a class from which no objects will ever be created.

- ☐ True
- ☒ False

Question No: 6 (Marks: 1) - Please choose one

Assume a class Derv that is privately derived from class Base. An object of class Derv

located in main() can access

- ☐ public members of Derv.
- ☐ protected members of Derv.

—

—

- ☐ private members of Derv.
- ☒ protected members of Base.

Question No: 7 (Marks: 1) - Please choose one

A pointer to a base class can point to objects of a derived class.

- ☒ True
- ☐ False

Question No: 8 (Marks: 1) - Please choose one

A copy constructor is invoked when

- ☐ a function do not returns by value.
- ☐ an argument is passed by value.
- ☒ a function returns by reference.
- ☐ an argument is passed by reference.

Question No: 9 (Marks: 1) - Please choose one

A function call is resolved at run-time in _____

- ☐ non-virtual member function.
- ☒ virtual member function.
- ☐ Both non-virtual member and virtual member function.
- ☐ None of given

Question No: 10 (Marks: 1) - Please choose one

When the base class and the derived class have a member function with the same name,

you must be more specific which function you want to call (using _____).

—

—

- ☒ scope resolution operator
- ☐ dot operator
- ☐ null operator
- ☐ Operator overloading

Question No: 11 (Marks: 1) - Please choose one

Each try block can have _____ no. of catch blocks.

- ☐ 1
- ☐ 2
- ☐ 3
- ☒ As many as necessary.

Question No: 12 (Marks: 1) - Please choose one

Two important STL associative containers are _____ and _____.

- ☐ set,map
- ☒ sequence,mapping
- ☐ setmet,multipule
- ☐ sit,mat

Question No: 13 (Marks: 1) - Please choose one

The mechanism of selecting function at run time according to the nature of calling object

is called,

- ☐ late binding
- ☐ static binding
- ☐ virtual binding
- ☒ None of the given options

Question No: 14 (Marks: 1) - Please choose one

An abstract class is useful when,

—

—

- We do not derive any class from it.
- There are multiple paths from one derived class to another.
- [We do not want to instantiate its object.](#)
- You want to defer the declaration of the class.

Question No: 15 (Marks: 1) - Please choose one

Which of the following is incorrect line regarding function template?

- `template<class T>`
- `template <typename U>`
- [Class<template T>](#)
- `template < class T, class U>`

Question No: 16 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- Reusability
- Writability
- Maintainability
- [All of given](#)

Question No: 17 (Marks: 1) - Please choose one

By default the vector data items are initialized to _____

- [0](#)
- 0.0
- 1
- null

—

—

Question No: 18 (Marks: 1) - Please choose one

Which one of the following functions returns the total number of elements in a vector.

- `length();`
- [size\(\);](#)
- `ele();`
- `veclen();`

Question No: 19 (Marks: 1) - Please choose one

Suppose you create an uninitialized vector as follows:

`vector<int> evec;`

After adding the statment,

`evvec.push_back(21);`

what will happen?

_ The following statement will add an element to the start (the back) of `evvec` and will initialize it with the value 21.

_ The following statement will add an element to the center of `evvec` and will reinitialize it with the value 21.

_ The following statement will delete an element to the end (the back) of `evvec` and will reinitialize it with the value 21.

_ The following statement will add an element to the end (the back) of `evvec` and initialize it with the value 21.

Question No: 20 (Marks: 1) - Please choose one

An STL container can not be used to,

_ hold objects of class `employee`.

_ store elements in a way that makes them quickly accessible.

_ compile c++ programs.

_ organize the way objects are stored in memory

Question No: 21 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

_ True

_ False

Question No: 22 (Marks: 1) - Please choose one

The main function of scope resolution operator (`::`) is,

_

_

_ To define an object

_ To define a data member

_ To link the definition of an identifier to its declaration

_ To make a class private

Question No: 23 (Marks: 1) - Please choose one

When is a constructor called?

_ Each time the constructor identifier is used in a program statement

_ During the instantiation of a new object

_ During the construction of a new class

_ At the beginning of any program execution

Question No: 24 (Marks: 1) - Please choose one

Consider the code below,

`class Fred {`

```

public:
Fred();
...
};
int main()
{
Fred a[10];
Fred* p = new Fred[10];
...
}

```

Select the best option,

- ☐ Fred a[10]; calls the default constructor 09 times
- ☐ Fred* p = new Fred[10]; calls the default constructor 10 times
- ☒ Produce an error
- ☐ Fred a[10]; calls the default constructor 11 times
- ☐ Fred* p = new Fred[10]; calls the default constructor 11 times
- ☐ Fred a[10]; calls the default constructor 10 times
- ☐ Fred* p = new Fred[10]; calls the default constructor 10 times

—

—

Question No: 25 (Marks: 1) - Please choose one

Associativity can be changed in operator overloading.

- ☐ True
- ☒ False

Question No: 26 (Marks: 1) - Please choose one

A normal C++ operator that acts in special ways on newly defined data types is said to be

- ☐ glorified.
- ☐ encapsulated.
- ☒ classified.
- ☐ overloaded.

Question No: 27 (Marks: 1) - Please choose one

Which operator can not be overloaded?

- ☐ The relation operator (>=)
- ☐ Assignment operator (=)
- ☐ Script operator ([])
- ☒ Conditional operator (? :)

Question No: 28 (Marks: 1) - Please choose one

Suppose obj1 and obj2 are two objects of a user defined class A. An + operator is overloaded to add obj1 and obj2 using the function call obj1+obj2.

Identify the correct function prototype against the given call?

—

—

_ A operator + (A &obj);

_ int + operator();

_ [int operator \(plus\) \(\);](#)

_ A operator(A &obj3);

Question No: 29 (Marks: 1) - Please choose one

Default constructor is such constructor which either has no -----or if it has some parameters these have ----- values

_ Parameter, temporary

_ Null, Parameter

_ [Parameter, default](#)

_ non of the given

Question No: 30 (Marks: 1) - Please choose one

Public methods of base class can ----- be accessed in its derived class

_ directly

_ [indirectly](#)

_ simultaneously

_ non of the given

Question No: 31 (Marks: 1)

Is Deque a Birectional Container?

[Yes, deque behaves like queue \(line\) such that we can add elements on both sides of it.](#)

Question No: 32 (Marks: 1)

What is meant by Generic Programming?

[Generic programming refers to programs containing generic abstractions general code](#)

[that is same in logic for all data types like printArray function\), then we instantiate that](#)

[generic program abstraction \(function, class\) for a particular data type, such abstractions](#)

[can work with many different types of data.](#)

Question No: 33 (Marks: 2)

Sort the following data in the order in which compiler searches a function?

Complete Specialization, Generic Template, Partial Specialization, Ordinary Function.

Specializations of this function template, instantiations with specific types, can be called

just like an ordinary function:

```
cout << max(3, 7); // outputs 7
```

—

—

The compiler examines the arguments used to call max and determines that this is a call

to max(int, int). It then instantiates a version of the function where the parameterizing

type T is int, making the equivalent of the following function:

```
int max(int x, int y)
```

```
{
```

```
return x < y ? y : x;
```

```
}
```

the C++ Standard Template Library contains the function template max(x, y) which

creates functions that return either x or y, whichever is larger. max() could be defined like

this:

```
template <typename T>
```

```
T max(T x, T y)
```

```
{
```

```
return x < y ? y : x;
```

```
}
```

Question No: 34 (Marks: 2)

State any conflict that may rise due to multiple inheritance?

The conflict may arise is the diamond problem, which our author likes to call the “diamond of doom”. This occurs when a class multiply inherits from two classes which

each inherit from a single base class. This leads to a diamond shaped inheritance pattern.

For example, consider the following set of classes:

```
class PoweredDevice
```

```
{
```

```
};
```

```

classScanner: publicPoweredDevice
{
};
classPrinter: publicPoweredDevice
{
};
classCopier: publicScanner, publicPrinter
{
};

```

—

—
Scanners and printers are both powered devices, so they derived from PoweredDevice.

However, a copy machine incorporates the functionality of both Scanners and Printers.

Ambiguity also cause problem.

Question No: 35 (Marks: 3)

Describe three properties necessary for a container to implement Generic Algorithms.

If you declare a container as holding pointers, you are responsible for managing the

memory for the objects pointed to. The container classes will not automatically free

memory for these objects when an item is erased from the container.

Container classes are expected to implement methods to do the following:

- create a new empty container (constructor),
- report the number of objects it stores (size),
- delete all the objects in the container (clear),
- insert new objects into the container,
- remove objects from it,
- provide access to the stored objects.

Question No: 36 (Marks: 3)

Write three important features of virtual functions.

With virtual functions, derived classes can provide new implementations of functions

from their base classes. When someone calls a virtual function of an object of the derived

class, this new implementation is called, even if the caller uses a pointer to the base class,

and doesn't even know about the particular derived class.

The virtual function is an option, and the language defaults to non virtual, which is the

fastest configuration.

The derived class can completely "override" the implementation or "augment" it (by

explicitly calling the base class implementation in addition to the new things it does).

—

—

Question No: 37 (Marks: 3)

Consider the code below,

```
#include <iostream>
#include <stdlib.h>
using namespace std;
class Shape{
public:
void Draw(){cout<<"shape"<<endl;}
};
class Line : public Shape{
public:
void Draw(){cout<<"Line"<<endl;}
};
class Circle : public Shape{
public:
void Draw(){cout<<"Circle"<<endl;}
};
int main(int argc, char *argv[])
{
Shape * ptr1 = new Shape();
Shape * ptr2 = new Line();
Shape * ptr3 = new Circle();
ptr1->Draw();
ptr2->Draw();
ptr3->Draw();
system("PAUSE");
```

```
return 0;
}
```

This code shows output,

```
Shape
```

```
Shape
```

```
Shape
```

Give the reason for this output

Suppose we want to show the output,

```
Shape
```

```
Line
```

```
Circle
```

—

—

How we can change the code to do that?

```
class shape { public:
```

```
void draw();
```

```
};
```

```
class circle : public shape { };
```

```
int main(int argc, char **argv){
```

```
circle my_circle;
```

```
my_circle.draw();
```

```
}
```

While this has all the usual advantages, e.g., code reuse, the real power of polymorphism

comes into play when draw is declared to be virtual or pure virtual, as follows:

```
class shape{ public:
```

```
virtual void draw()=0;
```

```
};
```

```
class circle : public shape { public:
```

```
void draw();
```

```
}
```

Here, circle has declared its own draw function, which can define behavior appropriate

for a circle. Similarly, we could define other classes derived from shape, which provide

their own versions of draw. Now, because all the classes implement the shape interface,

we can create collections of objects that can provide different behavior invoked in a consistent manner (calling the draw member function). An example of this is shown here.

```
shape *shape_list[3]; // the array that will
// pointer to our shape objects
shape[0] = new shape; // three types of shapes
shape[1] = new line; // we have defined
shape[2] = new circle;
for(int i = 0; i < 3; i++){
shape_list[i].draw();
}
```

When we invoke the draw function for each object on the list, we do not need to know anything about each object; C++ handles the details of invoking the correct version of draw. This is a very powerful technique, allowing us to provide extensibility in our designs. Now we can add new classes derived from shape to provide whatever behavior we desire. The key here is that we have separated the interface (the prototype for shape) from the implementation.

Question No: 38 (Marks: 5)

There are some errors in the code given below, you have to

1. Indicate the line no. with error/s
2. Give the reason for error/s
3. Correct the error/s.

—

—

1. #include <iostream> this will be `#include <iostream.h>`
2. #include <stdlib.h>
3. using namespace std;
4. template <typename T>
5. class MyClass{
6. public:
7. MyClass(){
8. cout<<"This is class1"<<endl;
9. }

```

10. };
11. template <typename T>
12. class MyClass<int*>{
13. public:
14. MyClass(){
15. cout<<"This is class2"<<endl;
16. }
17. };
18. int main(int argc, char *argv[])
19. {
20. MyClass<int> c1;
21. MyClass<int*> c2;
22. system("PAUSE");
23. return 0;
24. }

```

Question No: 39 (Marks: 5)

Given are two classes A and B. class B is inherited from class A. Write a code snippet(for main function) that polymorphically call the method of class B. Also what

changes do you suggest in the given code segment that are required to call the class B method polymorphically.

```

class A
{
public:
void method() { cout<<"A's method \n"; }
};
class B : public A
{
public:
void method() { cout<<"B's method\n"; }

```

—

—

```
};
```

Ans:

```

public class Test
{
public class A {}

```

```

public class B extends A {}
private void test(A a)
{
    System.out.println("test(A)");
}
private void test(B b)
{
    System.out.println("test(B)");
}
public static void main(String[] args)
{
    Test t = new Test();
    A a = t.new A();
    A b = t.new B();
    t.test(a);
    t.test(b);
}
}

```

Question No: 40 (Marks: 10)

Create built-in STL (Standard Template Library) vector class object for strings and add in it some words by taking input from user, then apply the sort() algorithm to array of words stored in this vector class object.

Hint: Use push_back() to add the words in vector class object, and the [] operator and size() to display these sorted words.

The STL is the containers, iterators and algorithms component of the proposed C++

Standard Library [ANSI95]. It represents a novel application of principles which have

their roots in styles of programming other than Object-orientation.

void listWords(istream& in, ostream& out)

—

—

```

{
    string s;
    while (!in.eof() && in >> s) {
        add s to some container
    }
    sort the strings in the container
}

```



```

remove the duplicates
for (each string t in container) {
out << t;
}
}

```

For now, assume that a word is defined as a whitespace-separated string as delivered by the stream extraction operator. Later on we will consider ways of refining this definition.

Given the way this problem is expressed, we can implement this program directly, if

naïvely. The STL container class `vector` will suffice to hold the words: applying the algorithms `sort` and `unique` provides the required result.

```

void listWords(istream& in, ostream& out)
{
string s;
vector<string> v;
while (!in.eof() && in >> s)
v.push_back(s); // (1)
sort(v.begin(), v.end());
vector<string>::iterator e
= unique(v.begin(), v.end()); // (2)
for (vector<string>::iterator b = v.begin();
b != e;
b++) {
out << *b << endl;
}
}

```

At (1) the vector member function `push_back()` is used to add to the end of the vector.

This can also be done using the `insert` member, which takes as a parameter an iterator

identifying the position in the vector at which to place the added element:

```
v.insert(v.end(), s);
```

This allows us to add at any position in the vector. Be aware, though, that adding anywhere other than the end implies the overhead of physically shifting all elements from

the insertion point to the end to make room for the new value. For this reason, and given

the choices made in this example, attempts to optimise this code by maintaining the

—

—
vector in sorted order are unwise. Replace vector with list and this becomes possible -

although in both cases a search over the container will be necessary to determine the

correct position of insertion.

The unique algorithm has the surprising property of not changing the length of the container to which it is applied (it can hardly do this, as it has access not to the underlying container, but only to the pair of iterators it is passed). Instead, it guarantees

that duplicates are removed by moving unique entries towards the beginning of the container, returning an iterator indicating the new end of the container. This can be used

directly (as here, at (2)), conversely it can be passed to the erase member with the old end

iterator, to truncate the container.

Question No: 41 (Marks: 10)

Q. Write a detailed note on Exceptions in Destructors with the help of a coding example.

Exceptions in Destructors:

An object is presumably created to do something. Some of the changes made by an object

should persist after an object dies (is destructed) and some changes should not.

Take an

object implementing a SQL query. If a database field is updated via the SQL object then

that change should persist after the SQL objects dies. To do its work the SQL object

probably created a database connection and allocated a bunch of memory. When the SQL

object dies we want to close the database connection and deallocate the memory, otherwise if a lot of SQL objects are created we will run out of database

connections

and/or memory.

The logic might look like:

```
Sql::~Sql()
```

```
{
delete connection;
delete buffer;
}
```

Let's say an exception is thrown while deleting the database connection. Will the buffer

be deleted? No. Exceptions are basically non-local gotos with stack cleanup. The code for

deleting the buffer will never be executed creating a gaping resource leak.

Special care must be taken to catch exceptions which may occur during object destruction. Special care must also be taken to fully destruct an object when it throws an

exception.

Example code for exception

```
#include<iostream.h>
```

```
#include<conio.c>
```

```
class Exception {
```

```
private:
```

```
—
```

```
—
```

```
char message[30] ;
```

```
public:
```

```
Exception() {strcpy(message,"There is not enough stock");}
```

```
char * get_message() { return message; }
```

```
};
```

```
class Item {
```

```
private:
```

```
int stock ;
```

```
int required_quantity;
```

```
public:
```

```
Item(int stk, int qty)
```

```
{
```

```
stock = stk;
```

```
required_quantity = qty;
```

```
}
```

```
int get_stock()
```

```
{
```

```

return stock;
}
int get_required_quantity()
{
return required_quantity;
}
void order()
{
if (get_stock() < get_required_quantity())
throw Exception();
else
cout << "The required quantity of item is available in the stock";
}
~Item(){}
};
void main()
{
Item obj(10, 20);
try

```

—

—

```

{
obj.order();
}
catch(Exception & exp2)
{
getch();
cout << "Exception: " << exp2.get_message() << endl;
}
getch();

```

FINAL TERM EXAMINATION

Fall 2009

CS304- Object Oriented Programming (Session - 4)

Ref No: 1130772

Time: 120 min

Marks: 75

Question No: 1 (Marks: 1) - Please choose one

A template provides a convenient way to make a family of

☐ variables and data members

☐ functions and classes

☐ classes and exceptions

☐ programs and algorithms

Question No: 2 (Marks: 1) - Please choose one

Which one of the following terms must relate to polymorphism?

☐ Static allocation

☐ Static typing

☒ Dynamic binding

☐ Dynamic allocation

—

—

Question No: 3 (Marks: 1) - Please choose one

What is true about function templates?

☐ The compiler generates only one copy of the function template

☒ The compiler generates a copy of function respective to each type of data

☐ The compiler can only generate copy for the int type data

☐ None of the given.

Question No: 4 (Marks: 1) - Please choose one

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

☒ Templates

☐ Overloading

☐ Data hiding

☐ Encapsulation

Question No: 5 (Marks: 1) - Please choose one
template <>

```
class Vector<char*> { }
```

This is an example of partial specialization.

☐ True

☒ False

Question No: 6 (Marks: 1) - Please choose one

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be

concrete, while classes like Sphere and Cube would normally be abstract.

☒ True

☐ False

Question No: 7 (Marks: 1) - Please choose one

—

—
A non-virtual member function is defined in a base class and overridden in a derived class; if that function is called through a base-class pointer to a derived class object, the derived-class version is used.

— **True**

— False

Question No: 8 (Marks: 1) - Please choose one

Assume a class Derv that is privately derived from class Base. An object of class Derv

located in main() can access

— public members of Derv.

— **protected members of Derv.**

— private members of Derv.

— protected members of Base.

Question No: 9 (Marks: 1) - Please choose one

In order to define a class template, the first line of definition must be:

— **template <typename T>**

— typename <template T>

— Template Class <ClassName>

— Class <Template T>

Question No: 10 (Marks: 1) - Please choose one

If there is a pointer p to objects of a base class, and it contains the address of an object of

a derived class, and both classes contain a nonvirtual member function, ding(), then the

statement p->ding(); will cause the version of ding() in the _____ class to be executed.

— Base

— Derived

—

—

— Abstract

☐ virtual

Question No: 11 (Marks: 1) - Please choose one

When the base class and the derived class have a member function with the same name,

you must be more specific which function you want to call (using _____).

☐ scope resolution operator

☐ dot operator

☒ null operator

☐ Operator overloading

Question No: 12 (Marks: 1) - Please choose one

Non Template Friend functions of a class are friends of _____instance/s of that class.

☐ All

☒ One specific

☐ All instances of one date type

☐ None of the given options

Question No: 13 (Marks: 1) - Please choose one

The find() algorithm

☐ finds matching sequences of elements in two containers.

☒ finds a container that matches a specified container.

☐ takes iterators as its first two arguments.

☐ takes container elements as its first two arguments.

Question No: 14 (Marks: 1) - Please choose one

If you define a vector v with the default constructor, and define another vector w with a

one-argument constructor to a size of 11, and insert 3 elements into each of these vectors

with push_back(), then the size() member function will return _____ for v and _____ for

w.

☐ 11 for v and 3 for w.

☐

☐

☐ 0 for v and 0 for w.

☐ 0 for v and 3 for w.

☒ 3 for v and 11 for w.

Question No: 15 (Marks: 1) - Please choose one

Which of the following may not be an integral part of an object?

- _ State
- _ Behavior
- _ Protected data members

_ **All of given**

Question No: 16 (Marks: 1) - Please choose one

Which is not the Advantage of inheritance?

- _ providing class growth through natural selection.
- _ **facilitating class libraries.**
- _ avoiding the rewriting of code.
- _ providing a useful conceptual framework.

Question No: 17 (Marks: 1) - Please choose one

class DocElement

```
{
public:
virtual void Print() { cout << "Generic element"; }
};
class Heading : public DocElement
{
public:
void Print() { cout << "Heading element"; }
};
class Paragraph : public DocElement
{
public:
void Print() { cout << "Paragraph element"; }
};
```

—

```
—
void main()
{
DocElement * p = new Paragraph();
p->Print();
}
```

When you run this program, it will print out a single line to the console output.

What will be in that line?

Select one correct answer from the following list:

- _ Generic element
- _ Heading element

_ Paragraph element

_ Nothing will be printed.

Question No: 18 (Marks: 1) - Please choose one

When a virtual function is called by referencing a specific object by name and using the

dot member selection operator (e.g., squareObject.draw()), the reference is resolved at compile time.

_ True

_ False

Question No: 19 (Marks: 1) - Please choose one

In case of multiple inheritance a derived class inherits,

_ Only the public member functions of its base classes

_ Only the public data members of its base classes

_ Both public data members and member functions of all its base classes

_ Data members and member functions of any two base classes

Question No: 20 (Marks: 1) - Please choose one

When we write a class template the first line must be:

_

_

_ template < class class_name>

_ template < class data_type>

_ template < class T >

Here T can be replaced with any name but it is preferable.

_ class class-name()

class template<class_name>

Question No: 21 (Marks: 1) - Please choose one

Which of the following is incorrect line regarding function template?

_ template<class T>

_ template <typename U>

_ Class<template T>

_ template < class T, class U>

Question No: 22 (Marks: 1) - Please choose one

An STL container can not be used to,

_ hold objects of class employee.

_ store elements in a way that makes them quickly accessible.

_ compile c++ programs.

_ organize the way objects are stored in memory

Question No: 23 (Marks: 1) - Please choose one
Algorithms can only be implemented using STL containers.

☒ True

☐ False

Question No: 24 (Marks: 1) - Please choose one
Consider a class named Vehicle, which of the following can be the instance of class Vehicle?

☐

☐

- 1. Car
- 2. Computer
- 3. Desk
- 4. Ahmed
- 5. Bicycle
- 6. Truck

☐ 1, 4, 5

☐ 2, 5, 6

☐ 1, 2, 3, 6

☒ 1, 5, 6

Question No: 25 (Marks: 1) - Please choose one
Consider the code below,

```
class Fred {  
public:  
    Fred();  
    ...  
};  
int main()  
{  
    Fred a[10];  
    Fred* p = new Fred[10];  
    ...  
}
```

Select the best option,

☐ Fred a[10]; calls the default constructor 09 times

☐ Fred* p = new Fred[10]; calls the default constructor 10 times

☐ Produce an error

☒ Fred a[10]; calls the default constructor 11 times

Fred* p = new Fred[10]; calls the default constructor 11 times

☐ Fred a[10]; calls the default constructor 10 times

☐ Fred* p = new Fred[10]; calls the default constructor 10 times

Question No: 26 (Marks: 1) - Please choose one

When a variable is define as static in a class then all object of this class,

☐

☐

☐ Have different copies of this variable

☒ **Have same copy of this variable**

☐ Can not access this variable

☐ None of given

Question No: 27 (Marks: 1) - Please choose one

The life of sub object is dependant on the life of master class in _____.

☐ Separation

☒ **Composition**

☐ Aggregation

☐ None of the given

Question No: 28 (Marks: 1) - Please choose one

_____, which means if A declares B as its friend it does NOT mean that A can

access private data of B. It only means that B can access all data of A.

☒ **Friendship is one way only**

☐ Friendship is two way only

☐ NO Friendship between classes

☐ Any kind of friendship

Question No: 29 (Marks: 1) - Please choose one

Which of the following operators always takes no argument if overloaded?

☐ /

☐ -

☐ +

☒ **++**

☐

☐

Question No: 30 (Marks: 1) - Please choose one

In Private ----- only member functions and friend classes or functions of a

derived class can convert pointer or reference of derived object to that of parent object

_ specialization

_ inheritance

_ abstraction

_ composition

Question No: 31 (Marks: 1)

Write the syntax of declaring a pure virtual function in a class?

Ans:

Pure Virtual Function is a Virtual function with no body.

Declaration of Pure Virtual Function:

Since pure virtual function has no body, the programmer must add the notation =0 for

declaration of the pure virtual function in the base class.

General Syntax of Pure Virtual Function takes the form:

class classname //This denotes the base class of C++ virtual function

{

public:

virtual void virtualfunctionname() = 0 //This denotes the pure virtual function in C++

};

Question No: 32 (Marks: 1)

What is meant by direct base class ?

Ans

When a class-type is included in the class-base, it specifies the direct base class of the

class being declared. If a class declaration has no class-base, or if the class-base lists only

interface types, the direct base class is assumed to be object. A class inherits members

from its direct base class,

Deriving a class from more than one direct base class is called multiple inheritance.

Question No: 33 (Marks: 2)

Describe the way to declare a template class as a friend class of any other class.

Ans

The following example is use of a class template:

—

```

—
template<class L> class Key
{
L k;
L* kptr;
int length;
public:
Key(L);
// ...
};

```

Suppose the following declarations appear later:

```

Key<int> i;
Key<char*> c;
Key<mytype> m;

```

The compiler would create three objects.

Question No: 34 (Marks: 2)

What is the purpose of template parameter?

Ans:

There are three kinds of template parameters:

- type
- non-type
- template

You can interchange the keywords class and typename in a template parameter declaration. You cannot use storage class specifiers (static and auto) in a template parameter declaration.

Question No: 35 (Marks: 3)

Describe in simple words how we can use template specialization to enforce case sensitive specialization in String class.

Ans”

The act of creating a new definition of a function, class, or member of a class from a template declaration and one or more template arguments is called template instantiation.

The definition created from a template instantiation is called a specialization. A primary

template is the template that is being specialized.

create function objects to do the case-insensitive compares, and then reuse them when also wanting to do case-insensitive sorting or searching.

Question No: 36 (Marks: 3)

—

—
Can we use compiler generated default assignment operator in case our class is using dynamic memory? Justify your answer.

Ans:

the compiler does not make a separate copy of the object. Even if the types are not the same, the compiler is usually able to do a better job with initialization lists than with assignments.

Consider the following constructor that initializes member object x_ using an initialization list: square::square() : x_(whatever) { }. The most common benefit of doing this is improved performance. For example, if the expression whatever is the same type as member variable x_, the result of the whatever expression is constructed directly inside

x_ — the compiler does not make a separate copy of the object. Even if the types are not the same, the compiler is usually able to do a better job with initialization lists than with assignments.

As if that wasn't bad enough, there's another source of inefficiency when using assignment in a constructor: the member object will get fully constructed by its default constructor, and this might, for example, allocate some default amount of memory or open some default file. All this work could be for naught if the whatever expression and/or assignment operator causes the object to close that file and/or release that memory (e.g., if the default constructor didn't allocate a large enough pool of memory or if it opened the wrong file).

Question No: 37 (Marks: 3)

Give the names of three ways to handle errors in a program.

Ans:

The function will throw `DivideByZero` as an exception that can then be caught by an exception-handling catch statement that catches exceptions of type `int`. The necessary construction for catching exceptions is a try catch system. If you wish to have your program check for exceptions, you must enclose the code that may have exceptions thrown in a try block.

The catch statement catches exceptions that are of the proper type. You can, for example, throw objects of a class to differentiate between several different exceptions. As well, once a catch statement is executed, the program continues to run from the end of the catch.

the errors can be handled outside of the regular code. This means that it is easier to structure the program code, and it makes dealing with errors more centralized. Finally, because the exception is passed back up the stack of calling functions, you can handle errors at any place you choose.

Question No: 38 (Marks: 5)

Consider the following code,

```

—
—
class Base{
private:
void base1();
protected:
void base2();
public:
void base3();
};
class Derived: public Base{
private:
void derived1();
protected:
void derived2();
public:

```

```

void derived3();
};
int main(){
Derived * derived = new Derived();
return 0;
}

```

Fill the table below to tell which member functions of Base and Derived classes we can access using the Derived pointer in the code indicated in bold.

Ans:

Function Name **Availability (Yes / No)?**

base2() **no**

base3() **yes**

derived1() **No**

derived2() **No**

derived3() **Yes**

Question No: 39 (Marks: 5)

What is the output produced by the following program?

```
#include<iostream.h>
```

```
void sample_function(double test) throw (int);
```

—

—

```

int main()
{
try
{
cout << "Trying.\n";
sample_function(98.6);
cout << "Trying after call.\n";
}
catch(int)
{
cout << "Catching.\n";
}
cout << "End program.\n";
return 0;
}
void sample_function(double test) throw (int)
{

```



```
cout << "Starting sample_function.\n";
if(test < 100)
throw 42;
}
```

Ans:

Starting sample_function

Trying

Trying after call

Catching

End program

Question No: 40 (Marks: 10)

Write a publicly derived class “Employee” that is derived from base class named “Company”. Both classes will have function “create()”. Make virtual function of base class and override same function in derived class. Function create will have an

output statement of your own choice.

In “main” Create an object of base class and call both functions with same object type.

—

—

Question No: 41 (Marks: 10)

Write a program in C++ which creates three classes named as

1. Equation
2. Linear
3. Quadratic

Where Linear and Quadratic are inherited from Equation

Each class has the method Graph. Graph method should be pure virtual in Equation class.

This method should be overridden in both the inherited classes. It is meant to display the

Graph shape of its respective class. Graph method of Linear will display the message;

Straight line

Similarly, the Graph method of Quadratic will display the message;

Parabola

In main, call the Graph method of both the Linear and Quadratic equations polymorphically through the parent class (Equation).

Ans:

```

#include "fraction.h"
#include <iostream>
#include <string>
#include <string.h>
#include <stdlib.h>
class equation;
class equation {
int a, b;
public:
int c ()
{return (c);}
void convert (Cequation);
};
class linear {
private:
int side;
public:
void set_side (int a)
{side=a;}
friend class equation;
};

—

—
void equation::convert (Cequation) {
a = 23;
b = 45;
}
int main () {
cequation sqr;
CRectangle rect;
sqr.set_side(4);
rect.convert(sqr);
cout << rect.area();
return 0;
}

```

Question No: 1 (Marks: 1) - Please choose one

Which one of the following terms must relate to **polymorphism**?

- ▶ Static allocation
- ▶ Static typing
- ▶ **Dynamic binding**
- ▶ Dynamic allocation

Question No: 2 (Marks: 1) - Please choose one

Multiple inheritance can be of type

- ▶ Public
- ▶ Private
- ▶ Protected
- ▶ **All of the given**

Question No: 3 (Marks: 1) - Please choose one

When a subclass specifies an alternative definition for an attribute or method of its superclass, it is _____ the definition in the superclass.

- ▶ overload
- ▶ **overriding**
- ▶ copy riding
- ▶ none of given

Question No: 4 (Marks: 1) - Please choose one

Like template functions, a class template may not handle all the types successfully.

► True

► False

Question No: 5 (Marks: 1) - Please choose one

It is sometimes useful to specify a class from which no objects will ever be created.

► True

► False

Question No: 6 (Marks: 1) - Please choose one

Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- ▶ public members of Derv.
- ▶ protected members of Derv.
- ▶ private members of Derv.
- ▶ **protected members of Base.**

Question No: 7 (Marks: 1) - Please choose one

_____ A
pointer to a base class can point to objects of a derived class.

- ▶ **True**
- ▶ False

Question No: 8 (Marks: 1) - Please choose one

_____ A
copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ an argument is passed by value.
- ▶ **a function returns by reference.**
- ▶ an argument is passed by reference.

Question No: 9 (Marks: 1) - Please choose one

_____ A
function call is resolved at run-time in _____

- ▶ non-virtual member function.
- ▶ **virtual member function.**
- ▶ Both non-virtual member and virtual member function.
- ▶ None of given

Question No: 10 (Marks: 1) - Please choose one

When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using _____).

- ▶ **scope resolution operator**
- ▶ dot operator
- ▶ null operator
- ▶ Operator overloading

Question No: 11 (Marks: 1) - Please choose one

Each try block can have _____ no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ **As many as necessary.**

Question No: 12 (Marks: 1) - Please choose one

Two important STL associative containers are _____ and _____.

- ▶ set,map
- ▶ **sequence,mapping**
- ▶ setmet,multipule
- ▶ sit,mat

Question No: 13 (Marks: 1) - Please choose one

The mechanism of selecting function at run time according to the nature of calling object is called,

- ▶ late binding
- ▶ static binding
- ▶ virtual binding
- ▶ **None of the given options**

Question No: 14 (Marks: 1) - Please choose one

abstract class is useful when, An

- ▶ We do not derive any class from it.
- ▶ There are multiple paths from one derived class to another.
- ▶ **We do not want to instantiate its object.**
- ▶ You want to defer the declaration of the class.

Question No: 15 (Marks: 1) - Please choose one

Which of the following is incorrect line regarding function template?

- ▶ `template<class T>`
- ▶ `template <typename U>`
- ▶ **`Class<template T>`**
- ▶ `template < class T, class U>`

Question No: 16 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability

► Maintainability

► **All of given**

Question No: 17 (Marks: 1) - Please choose one

_____ By
default the vector data items are initialized to _____

► **0**

► 0.0

► 1

► null

Question No: 18 (Marks: 1) - Please choose one

Which one of the following functions returns the total number of elements in a vector.

► length();

► **size();**

► ele();

► veclen();

Question No: 19 (Marks: 1) - Please choose one

Suppose you create an uninitialized vector as follows:

```
vector<int> evec;
```

After adding the statment,

```
evec.push_back(21);
```

what will happen?

- ▶ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.
- ▶ The following statement will add an element to the center of evec and will reinitialize it with the value 21.
- ▶ The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.
- ▶ **The following statement will add an element to the end (the back) of evec and initialize it with the value 21.**

Question No: 20 (Marks: 1) - Please choose one

STL container can not be used to, An

- ▶ hold objects of class employee.

- ▶ store elements in a way that makes them quickly accessible.
- ▶ **compile c++ programs.**
- ▶ organize the way objects are stored in memory

Question No: 21 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

- ▶ True
- ▶ **False**

Question No: 22 (Marks: 1) - Please choose one

The main function of scope resolution operator (::) is,

- ▶ **To define an object**
- ▶ To define a data member
- ▶ To link the definition of an identifier to its declaration
- ▶ To make a class private

Question No: 23 (Marks: 1) - Please choose one

When is a constructor called?

- ▶ Each time the constructor identifier is used in a program statement
- ▶ **During the instantiation of a new object**
- ▶ During the construction of a new class
- ▶ At the beginning of any program execution

Question No: 24 (Marks: 1) - Please choose one

Consider the code below,

```
class Fred {  
public:  
Fred();  
...  
};  
int main()
```

```
{  
Fred a[10];  
Fred* p = new Fred[10];  
...  
}
```

Select the best option,

► Fred a[10]; calls the default constructor 09 times

Fred* p = new Fred[10]; calls the default constructor 10 times

► **Produce an error**

► Fred a[10]; calls the default constructor 11 times

Fred* p = new Fred[10]; calls the default constructor 11 times

► Fred a[10]; calls the default constructor 10 times

Fred* p = new Fred[10]; calls the default constructor 10 times

Question No: 25 (Marks: 1) - Please choose one

Associativity can be changed in operator overloading.

▶ True

▶ **False**

Question No: 26 (Marks: 1) - Please choose one

A normal C++ operator that acts in special ways on newly defined data types is said to be

▶ glorified.

▶ encapsulated.

▶ **classified.**

▶ overloaded.

Question No: 27 (Marks: 1) - Please choose one

Which operator can not be overloaded?

- ▶ The relation operator ($\geq$)
- ▶ Assignment operator (=)
- ▶ Script operator ([])
- ▶ **Conditional operator (? :)**

Question No: 28 (Marks: 1) - Please choose one

Suppose obj1 and obj2 are two objects of a user defined class A. An + operator is overloaded to add obj1 and obj2 using the function call obj1+obj2.

Identify the correct function prototype against the given call?

- ▶ A operator + (A &obj);
- ▶ int + operator();
- ▶ **int operator (plus) ();**
- ▶ A operator(A &obj3);

Question No: 29 (Marks: 1) - Please choose one

Default constructor is such constructor which either has no -----or if it has some parameters these have ----- values

- ▶ Parameter, temporary
- ▶ Null, Parameter
- ▶ **Parameter, default**
- ▶ non of the given

Question No: 30 (Marks: 1) - Please choose one

Public methods of base class can ----- be accessed in its derived class

- ▶ directly
- ▶ **indirectly**
- ▶ simultaneously
- ▶ non of the given

Question No: 31 (Marks: 1)

Deque a Birectional Container? Is

Yes, deque behaves like queue (line) such that we can add elements on both sides of it.

Question No: 32 (Marks: 1)

What is meant by Generic Programming?

Generic programming refers to programs containing generic abstractions general code that is same in logic for all data types like printArray function), then we instantiate that generic program abstraction (function, class) for a particular data type, such abstractions can work with many different types of data.

Question No: 33 (Marks: 2)

Sort the following data in the order in which compiler searches a function?

Complete Specialization, Generic Template, Partial Specialization, Ordinary Function.

Specializations of this function template, instantiations with specific types, can be called just like an ordinary function:

```
cout << max(3, 7); // outputs 7
```

The compiler examines the arguments used to call max and determines that this is a call to max(int, int). It then instantiates a version of the function where the parameterizing type T is int, making the equivalent of the following function:

```
int max(int x, int y)
{
    return x < y ? y : x;
}
```

the C++ Standard Template Library contains the function template `max(x, y)` which creates functions that return either `x` or `y`, whichever is larger. `max()` could be defined like this:

```
template <typename T>
T max(T x, T y)
{
    return x < y ? y : x;
}
```

Question No: 34 (Marks: 2)

State any conflict that may rise due to multiple inheritance?

The conflict may arise is the diamond problem, which our author likes to call the “diamond of doom”. This occurs when a class multiply inherits from two classes which each inherit from a single base class. This leads to a diamond shaped inheritance pattern.

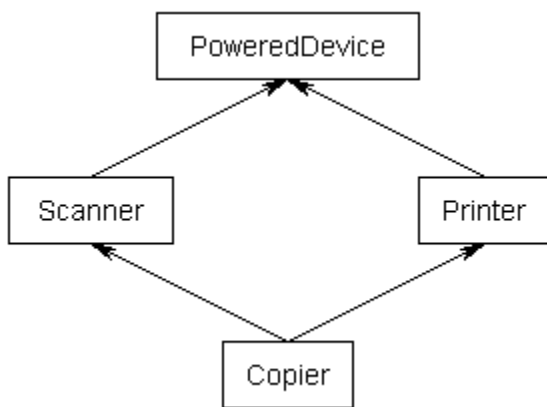
For example, consider the following set of classes:

```
class PoweredDevice
{
};

class Scanner: public PoweredDevice
{
};
```

```
classPrinter: publicPoweredDevice
{
};

classCopier: publicScanner, publicPrinter
{
};
```



Scanners and printers are both powered devices, so they derived from PoweredDevice. However, a copy machine incorporates the functionality of both Scanners and Printers.

Ambiguity also cause problem.

Question No: 35 (Marks: 3)

Describe three properties necessary for a container to implement Generic Algorithms.

If you declare a container as holding pointers, you are responsible for managing the memory for the objects pointed to. The container classes will not automatically free memory for these objects when an item is erased from the container.

Container classes are expected to implement methods to do the following:

- create a new empty container (constructor),
- report the number of objects it stores (size),
- delete all the objects in the container (clear),
- insert new objects into the container,
- remove objects from it,
- provide access to the stored objects.

Question No: 36 (Marks: 3)

Write three important features of virtual functions.

With virtual functions, derived classes can provide new implementations of functions from their base classes. When someone calls a virtual function of an object of the derived class, this new implementation is called, even if the caller uses a pointer to the base class, and doesn't even know about the particular derived class.

The virtual function is an option, and the language defaults to non virtual, which is the fastest configuration.

The derived class can completely "override" the implementation or "augment" it (by explicitly calling the base class implementation in addition to the new things it does).

Question No: 37 (Marks: 3)

Consider the code below,

```
#include <iostream>

#include <stdlib.h>

using namespace std;

class Shape{

    public:

    void Draw(){cout<<"shape"<<endl;}

};

class Line : public Shape{

    public:

    void Draw(){cout<<"Line"<<endl;}

};

class Circle : public Shape{

    public:

    void Draw(){cout<<"Circle"<<endl;}

};

int main(int argc, char *argv[])

{

    Shape * ptr1 = new Shape();

    Shape * ptr2 = new Line();

    Shape * ptr3 = new Circle();
```



```
ptr1->Draw();  
ptr2->Draw();  
ptr3->Draw();  
system("PAUSE");  
return 0;  
}
```

This code shows output,

Shape

Shape

Shape

Give the reason for this output

Suppose we want to show the output,

Shape

Line

Circle

How we can change the code to do that?

```

class shape { public:
    void draw();
};
class circle : public shape { };
int main(int argc, char **argv){
    circle my_circle;
    my_circle.draw();
}

```

While this has all the usual advantages, e.g., code reuse, the real power of polymorphism comes into play when draw is declared to be virtual or pure virtual, as follows:

```

class shape{ public:
    virtual void draw()=0;
};
class circle : public shape { public:
    void draw();
}

```

Here, circle has declared its own draw function, which can define behavior appropriate for a circle. Similarly, we could define other classes derived from shape, which provide their own versions of draw. Now, because all the classes implement the shape interface, we can create collections of objects that can provide different behavior invoked in a consistent manner (calling the draw member function). An example of this is shown here.

```

shape *shape_list[3]; // the array that will
                      // pointer to our shape objects
shape[0] = new shape; // three types of shapes
shape[1] = new line; // we have defined
shape[2] = new circle;
for(int i = 0; i < 3; i++){
    shape_list[i].draw();
}

```

When we invoke the draw function for each object on the list, we do not need to know anything about each object; C++ handles the details of invoking the correct version of draw. This is a very powerful technique, allowing us to provide extensibility in our designs. Now we can add new classes derived from shape to provide whatever behavior we desire. The key here is that we have separated the interface (the prototype for shape) from the implementation.

Question No: 38 (Marks: 5)

There are some errors in the code given below, you have to

1. Indicate the line no. with error/s
2. Give the reason for error/s
3. Correct the error/s.

1. `#include <iostream>` this will be `#include <iostream.h>`
2. `#include <stdlib.h>`

3. `using namespace std;`
4. `template <typename T>`
5. `class MyClass{`
6. `public:`
7. `MyClass(){`
8. `cout<<"This is class1"<<endl;`
9. `}`
10. `};`
11. `template <typename T>`
12. `class MyClass<int*>{`
13. `public:`
14. `MyClass(){`
15. `cout<<"This is class2"<<endl;`
16. `}`
17. `};`
18. `int main(int argc, char *argv[])`
19. `{`
20. `MyClass<int> c1;`
21. `MyClass<int*> c2;`
22. `system("PAUSE");`
23. `return 0;`
24. `}`

Question No: 39 (Marks: 5)

Given are two classes A and B. class B is inherited from class A. Write a code snippet(for main function) that polymorphically call the method of class B. Also what changes do you suggest in the given code segment that are required to call the class B method polymorphically.

```
class A
{
public:
void method() { cout<<"A's method \n"; }

};

class B : public A
{

public:
void method() { cout<<"B's method\n"; }

};
```

Ans:

```
public class Test
{
    public class A {}

    public class B extends A {}

    private void test(A a)
    {
        System.out.println("test(A)");
    }

    private void test(B b)
    {
        System.out.println("test(B)");
    }

    public static void main(String[] args)
    {
        Test t = new Test();
        A a = t.new A();
        A b = t.new B();

        t.test(a);
        t.test(b);
    }
}
```

Question No: 40 (Marks: 10)

Create built-in STL (Standard Template Library) **vector class object** for **strings** and add in it some words by taking input from user, then apply the `sort()` algorithm to array of words stored in this vector class object.

Hint: Use `push_back()` to add the words in vector class object, and the `[]` operator and `size()` to display these sorted words.

The STL is the containers, iterators and algorithms component of the proposed C++ Standard Library [ANSI95]. It represents a novel application of principles which have their roots in styles of programming other than Object-orientation.

```
void listWords(istream& in, ostream& out)
{
    string s;

    while (!in.eof() && in >> s) {
        add s to some container
    }

    sort the strings in the container
    remove the duplicates

    for (each string t in container) {
        out << t;
    }
}
```

For now, assume that a word is defined as a whitespace-separated string as delivered by the stream extraction operator. Later on we will consider ways of refining this definition.

Given the way this problem is expressed, we can implement this program directly, if naïvely. The STL container class `vector` will suffice to hold the words: applying the algorithms `sort` and `unique` provides the required result.

```
void listWords(istream& in, ostream& out)
{
    string s;
    vector<string> v;

    while (!in.eof() && in >> s)
        v.push_back(s);           // (1)
```

```

sort(v.begin(), v.end());

vector<string>::iterator e
    = unique(v.begin(), v.end()); // (2)

for (vector<string>::iterator b = v.begin();
     b != e;
     b++) {
    out << *b << endl;
}
}

```

At (1) the vector member function `push_back()` is used to add to the end of the vector. This can also be done using the `insert` member, which takes as a parameter an iterator identifying the position in the vector at which to place the added element:

```
v.insert(v.end(), s);
```

This allows us to add at any position in the vector. Be aware, though, that adding anywhere other than the end implies the overhead of physically shifting all elements from the insertion point to the end to make room for the new value. For this reason, and given the choices made in this example, attempts to optimise this code by maintaining the vector in sorted order are unwise. Replace vector with list and this becomes possible - although in both cases a search over the container will be necessary to determine the correct position of insertion.

The `unique` algorithm has the surprising property of not changing the length of the container to which it is applied (it can hardly do this, as it has access not to the underlying container, but only to the pair of iterators it is passed). Instead, it guarantees that duplicates are removed by moving unique entries towards the beginning of the container, returning an iterator indicating the new end of the container. This can be used directly (as here, at (2)), conversely it can be passed to the `erase` member with the old end iterator, to truncate the container.

Question No: 41 (Marks: 10)

Q.

Write a detailed note on Exceptions in Destructors with the help of a coding example.

Exceptions in Destructors:

An object is presumably created to do something. Some of the changes made by an object should persist after an object dies (is destructed) and some changes should not. Take an object implementing a SQL query. If a database field is updated via the SQL object then that change should persist after the SQL objects dies. To do its work the SQL object probably created a database connection and allocated a bunch of memory. When the SQL object dies we want to close the database connection and deallocate the memory, otherwise if a lot of SQL objects are created we will run out of database connections and/or memory.

The logic might look like:

```
Sql::~~Sql()
{
    delete connection;
    delete buffer;
}
```

Let's say an exception is thrown while deleting the database connection. Will the buffer be deleted? No. Exceptions are basically non-local gotos with stack cleanup. The code for deleting the buffer will never be executed creating a gaping resource leak.

Special care must be taken to catch exceptions which may occur during object destruction. Special care must also be taken to fully destruct an object when it throws an exception.

Example code for exception

```
#include<iostream.h>
#include<conio.c>
class Exception {
private:
```



```

char message[30] ;
public:

Exception() {strcpy(message,"There is not enough stock");}
char * get_message() { return message; }
};
class Item {
private:

int stock ;
int required_quantity;
public:

Item(int stk, int qty)
{
    stock = stk;
    required_quantity = qty;
}
int get_stock()
{
    return stock;
}

int get_required_quantity()
{
    return required_quantity;
}

void order()
{
    if (get_stock()< get_required_quantity())

        throw Exception();
        else
            cout<<"The required quantity of item is available in the stock";
}

~Item(){}
};

```

```
void main()
{
    Item obj(10, 20);

    try
    {
        obj.order();
    }
    catch(Exception & exp2)
    {
        getch();
        cout << "Exception: " << exp2.get_message() << endl;
    }
    getch();
}
```

FINAL TERM EXAMINATION

SPRING 2010

CS304- OBJECT ORIENTED PROGRAMMING

Shared by IMRAN JEE

Question No: 1 (Marks: 1) - Please choose one

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

► True

► False

Question No: 2 (Marks: 1) - Please choose one

Each try block can have _____ no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ **As many as necessary.**

Question No: 3 (Marks: 1) - Please choose one

Function templates should be used where code and behavior must be identical.

- ▶ True
- ▶ **False**

Question No: 4 (Marks: 1) - Please choose one

Consider the following statement

Cupboard has books

What is the relationship between Cupboard and books?

- ▶ Composition
- ▶ **Aggregation**
- ▶ Inheritance
- ▶ None of the given options

Question No: 5 (Marks: 1) - Please choose one

Identify the correct way of declaring an object of user defined template class **A** for char type members?

- ▶ **A< char > obj;**
- ▶ A obj;

► A obj;

Imrangee

► Obj A;

Question No: 6 (Marks: 1) - Please choose one

The user must define the operation of the copy constructor.

► **True**

► False

Question No: 7 (Marks: 1) - Please choose one

Default constructor is such constructor which either has no -----or if it has some parameters these have ----- values

► Parameter, temporary

► Null, Parameter

► **Parameter, default**

► non of the given

Question No: 8 (Marks: 1) - Please choose one

The type that is used to declare a reference or pointer is called its -----

- ▶ default type
- ▶ **static type**
- ▶ abstract type
- ▶ reference type

Question No: 9 (Marks: 1) - Please choose one

How the information hidden within an object can be accessed?

- ▶ Through its interface
- ▶ Through its private data members
- ▶ **Through its private member functions**
- ▶ Through both public and private members

Question No: 10 (Marks: 1) - Please choose one

The sub-object's life is not dependant on the life of master class in _____.

- ▶ Separation
- ▶ Composition
- ▶ **Aggregation**
- ▶ None of the given

Question No: 11 (Marks: 1) - Please choose one

Encapsulation means

Select correct option:

Imrangee

- ▶ Extending the behaviour of class in another class

- ▶ **Data and behaviour are tightly coupled within an entity**
- ▶ One entity takes all the attributes and operations of the other
- ▶ Taking out the common features and put those in a separate class

Question No: 12 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

- ▶ True
- ▶ **False**

Question No: 13 (Marks: 1) - Please choose one

When we write a class template the first line must be:

- ▶ `template < class class_name>`
- ▶ `template < class data_type>`
- ▶ **`template < class T >`**

Here T can be replaced with any name but it is preferable.

- ▶ `class class-name()`

`class template<class_name>`

Question No: 14 (Marks: 1) - Please choose one

An STL container can not be used to,

- ▶ **hold objects of class employee.**
- ▶ store elements in a way that makes them quickly accessible.
- ▶ compile c++ programs.
- ▶ organize the way objects are stored in memory

Question No: 15 (Marks: 1) - Please choose one

_____, which means if A declares B as its friend it does NOT mean that

A can access private data of B. It only means that B can access all data of A.

- ▶ **Friendship is one way only**
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

Question No: 16 (Marks: 1) - Please choose one

Which of the following may not be an integral part of an object?

- ▶ State

Imrangee

- ▶ Behavior
- ▶ **Protected data members**
- ▶ All of given

Question No: 17 (Marks: 1) - Please choose one

Public methods of base class can ----- be accessed in its derived class

- ▶ **directly**
- ▶ inderectly
- ▶ simultaniously
- ▶ non of the given

Question No: 18 (Marks: 1) - Please choose one

If a class D has been derived using protected inheritance from class B (If B is a protected base and D is derived class) then public and protected members of B - ----- accessed by member functions and friends of class D and classes derived

from D

- ▶ can be
- ▶ cannot be
- ▶ does restrict to be
- ▶ not given

Question No: 19 (Marks: 1) - Please choose one

What is true about function templates?

- ▶ The compiler generates only one copy of the function template
- ▶ **The compiler generates a copy of function respective to each type of data**
- ▶ The compiler can only generate copy for the int type data
- ▶ None of the given.

Question No: 20 (Marks: 1) - Please choose one

Which of the following is an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ Unique identity
- ▶ **All of the given**

Question No: 21 (Marks: 1) - Please choose one

When the base class and the derived class have a member function with the

Imrangee

same name, you must be more specific which function you want to call (using

_____).

- ▶ scope resolution operator
- ▶ **dot operator**
- ▶ null operator
- ▶ Operator overloading

Question No: 22 (Marks: 2)

Describe the way to declare a template function as a friend of any class.

Question No: 23 (Marks: 2)

Explain two benefits of **constructor**.

Question No: 24 (Marks: 2)

Can a constructor throws an exception. How to handle error when the constructor fails?

Question No: 25 (Marks: 2)

Write the code for a function template

Question No: 26 (Marks: 3)

Write three advantages of Iterator.

Question No: 27 (Marks: 3)

What is the difference between Virtual and Multiple Inheritance?

Question No: 28 (Marks: 5)

What is random_iterator? What is relation between random_iterator and Vector?

Question No: 29 (Marks: 5)

The code given below has one template function as a friend of a template class,

1. You have to identify any error/s in this code and describe the reason for error/s.
2. Give the correct code after removing the error/s.

template

```
void Test(U);
```

```
template< class T >
```

```
class B {
```

```
Imrangee
```

```
int data;
```

```
public:
```

```
friend void Test<>( T );
```

```
};
```

```
template
```

```
void Test(U u){
```

```
B < int> b1;
```

```
b1.data = 7;
```

```
}  
  
int main(int argc, char *argv[])  
{  
    char i;  
    Test(i);  
    system("PAUSE");  
    return 0;  
}
```

FINALTERM EXAMINATION

Spring 2010

CS304- Object Oriented Programming

Time: 90 min

Marks: 58

Question No: 1 (Marks: 1) - Please choose one

A template argument is preceded by the keyword _____.

- ▶ vector
- ▶ class
- ▶ template
- ▶ type*

Question No: 2 (Marks: 1) - Please choose one

Which of the following causes run time binding?

- ▶ Declaring object of abstract class
- ▶ Declaring pointer of abstract class
- ▶ Declaring overridden methods as non-virtual
- ▶ None of the given

Question No: 3 (Marks: 1) - Please choose one

A function template can not be overloaded by another function template.

▶ True

▶ False

Question No: 4 (Marks: 1) - Please choose one

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

▶ Templates

▶ Overloading

▶ Data hiding

▶ Encapsulation

Question No: 5 (Marks: 1) - Please choose one

Identify the correct way of declaring an object of user defined template class **A** for char type members?

▶ `A< char > obj;`

▶ `<char>A obj;`

▶ A obj<char>;

▶ Obj <char> A;

Question No: 6 (Marks: 1) - Please choose one

The user must define the operation of the copy constructor.

▶ True

▶ False

Question No: 7 (Marks: 1) - Please choose one

Template functions use _____ than ordinary functions.

▶ Greater Memory

▶ Lesser Memory

▶ Equal Memory

▶ None of the given options

Question No: 8 (Marks: 1) - Please choose one

The find() algorithm

- ▶ finds matching sequences of elements in two containers.
- ▶ finds a container that matches a specified container.
- ▶ takes iterators as its first two arguments.
- ▶ takes container elements as its first two arguments.

Question No: 9 (Marks: 1) - Please choose one

Compiler performs _____ type checking to diagnose type errors,

- ▶ Static
- ▶ Dynamic
- ▶ Bound
- ▶ Unbound

Question No: 10 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ All of given

Question No: 11 (Marks: 1) - Please choose one

Vectors contain contiguous elements stored as a[an] ____.

- ▶ variable
- ▶ array
- ▶ function
- ▶ datatype

Question No: 12 (Marks: 1) - Please choose one

Suppose you create an uninitialized vector as follows:

```
vector<int> evec;
```

After adding the statment,

```
evec.push_back(21);
```

what will happen?

► The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.

► The following statement will add an element to the center of evec and will reinitialize it with the value 21.

► The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.

► The following statement will add an element to the end (the back) of evec and initialize it with the value 21.

Question No: 13 (Marks: 1) - Please choose one

In a de-queue, (chose the best option)

- data can be quickly inserted or deleted at any arbitrary location.
- data can be inserted or deleted at any arbitrary location, but the process is relatively slow.
- data can not be quickly inserted or deleted at either end.
- data can be inserted or deleted at either end, but the process is relatively slow.

Question No: 14 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

▶ True

▶ False

Question No: 15 (Marks: 1) - Please choose one

What is a class?

▶ A class is a section of computer memory containing objects.

▶ A class is a section of the hard disk reserved for object oriented programs

▶ A class is the part of an object that contains the variables.

▶ A class is a description of a kind of object.

Question No: 16 (Marks: 1) - Please choose one

Inheritance is a way to

- ▶ organize data.
- ▶ pass arguments to objects of classes.
- ▶ add features to existing classes without rewriting them.
- ▶ improve data-hiding and encapsulation.

Question No: 17 (Marks: 1) - Please choose one

We can use "**this**" pointer in the constructor in the body and even in the initialization list of any class if we are careful,

- ▶ True
- ▶ False

Question No: 18 (Marks: 1) - Please choose one

_____ and _____ methods may not be declared abstract.

- ▶ private,static
- ▶ private,public
- ▶ static,public
- ▶ none of given

Question No: 19 (Marks: 1) - Please choose one

Default constructor is such constructor which either has no -----or if it has some parameters these have ----- values

- ▶ Parameter, temporary
- ▶ Null, Parameter
- ▶ Parameter, default
- ▶ non of the given

Question No: 20 (Marks: 1) - Please choose one

Public methods of base class can ----- be accessed in its derived class

- ▶ directly
- ▶ inderectly
- ▶ simultaniously

- ▶ non of the given

Question No: 21 (Marks: 1) - Please choose one

The type that is used to declare a reference or pointer is called its -----

- ▶ default type
- ▶ static type
- ▶ abstract type
- ▶ reference type

Question No: 22 (Marks: 1) - Please choose one

----- members are somewhere between public and private members. They are used in inheritance

- ▶ protected
- ▶ public
- ▶ private
- ▶ global

Question No: 23 (Marks: 1) - Please choose one

Which of these are examples of error handling techniques ?

- ▶ Abnormal Termination
- ▶ Graceful Termination
- ▶ Return the illegal
- ▶ all of the given

Question No: 24 (Marks: 1) - Please choose one

----- follow try block to catch the object thrown

- ▶ catch block
- ▶ throw block
- ▶ main block
- ▶ non of the given

Question No: 25 (Marks: 1) - Please choose one

Graphical representation of the classes and objects is called object model it shows -

- ▶ Class Name only
- ▶ Class Name and attributes
- ▶ Relationships of the objects and classes
- ▶ all of the given

Question No: 26 (Marks: 1) - Please choose one

Destructor can be overloaded

- ▶ True
- ▶ False

Question No: 27 (Marks: 2)

Describe the way to declare a template function as a friend of any class.

Template templatename

Class calssname

{

Friend void friend templatename (classname <templatename> astric const prt
classname);

}

Question No: 28 (Marks: 2)

State any two reasons why the virtual methods can not be static?

1-virtual method can not be static as it is dynamic

2-as virtual method is dynamic so it works automatically that is also another reason

That virtual method can not be static.

Question No: 29 (Marks: 2)

Explain the statement below,

```
vector<int> ivec(4, 3);
```

Question No: 30 (Marks: 2)

Explain two benefits of **setter functions**.

1- It minimize the changes to move the objects in inconsistent states

2- You can write checks in your setter functions to check the validity of data entered by the user, for example age functions to check to calculate the age from date entered.

Question No: 31 (Marks: 3)

Consider the code below,

```
template< typename T >
```

```
class T1 {
```



```

public:
    T i;

protected:
    T j;

private:
    T k;

    friend void Test();

};

```

This code has a template class **T1** with three members **i,j and k** and a friend function **Test()**, you have to describe which member/s of **T1** will be available in function **Test()**.

```

public:
    T i;

protected:
    T j;

```

Question No: 32 (Marks: 3)

What do you mean by Stack unwinding?

When we want to check what happens actually to the local variables in the try block when then an exception is thrown this concept is called stack unwinding.

Question No: 33 (Marks: 3)

Give the c++ code of case sensitive comparison function of string class.

Question No: 34 (Marks: 5)

What is random_iterator? What is relation between random_iterator and Vector?

Random_iterator: it provided both increment and decrement and also provide constant time methods for moving forward and backward in arbitrary sized steps. Random iterator provide essentially all of the operations of ordinary c pointer arithmetic.

Vector class provide an stl style random access iterator for use with generic algorithm since neither the vector nor the matrix classes are container classes in actual. The iterator class is really an iterator of data object that is viewed by vector or matrix.

Question No: 35 (Marks: 5)

What would be the output of this code?

```
class mother {
    public:
        mother ()
        { cout << "mother: no parameters\n"; }
        mother (int a)
        { cout << "mother: int parameter\n"; }
};

class daughter : public mother {
    public:
        daughter (int a)
        { cout << "daughter: int parameter\n\n"; }
```

```

};

class son : public mother {

public:

    son (int a) : mother (a)

        { cout << "son: int parameter\n\n"; }

};

int main () {

    daughter rabia (0);

    son salman(0);


    return 0;

}

```

Output will be

Mother

Daughter: rabia

Son: salman

Question No: 36 (Marks: 5)

The code given below has one template function as a friend of a template class,

1. You have to identify any error/s in this code and describe the reason for error/s.
2. Give the correct code after removing the error/s.

```
template<typename U>
```

```
void Test(U);  
template< class T >  
class B {  
    int data;  
    public:  
    friend void Test<>( T );  
};  
template<typename U>  
void Test(U u){  
    B < int> b1;  
    b1.data = 7;  
}  
int main(int argc, char *argv[])  
{  
    char i;  
    Test(i);  
    system("PAUSE");  
    return 0;  
}
```

FINALTERM EXAMINATION

Spring 2010

CS304- Object Oriented Programming

Ref No: 1563375

Time: 90 min

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▶ vector

▶ class

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Identify the correct way of declaring an object of user defined template class **A** for char type members?

▶ `A< char > obj;`

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▶ `A obj<char>;`

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The user must define the operation of the copy constructor.

▶ True

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Class calssname

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vector<int> ivec(4, 3);
```

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class T1 {
    public:
        T i;
    protected:
        T j;
    private:
        T k;
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This code has a template class **T1** with three members **i,j and k** and a friend function **Test()**, you have to describe which member/s of **T1** will be available in function **Test()**.

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T j;

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Vector class provide an STL style random access iterator for use with generic algorithm since neither the vector nor the matrix classes are container classes in actual. The iterator class is really an iterator of data object that is viewed by vector or matrix.

Question No: 35 (Marks: 5)

What would be the output of this code?

```
class mother {  
    public:  
        mother ()  
        { cout << "mother: no parameters\n"; }  
        mother (int a)  
        { cout << "mother: int parameter\n"; }
```

```
};
```

```
class daughter : public mother {  
    public:  
        daughter (int a)  
        { cout << "daughter: int parameter\n\n"; }  
};
```

```
class son : public mother {  
    public:  
        son (int a) : mother (a)  
        { cout << "son: int parameter\n\n"; }  
};
```

```
int main () {  
    daughter rabia (0);  
    son salman(0);  
  
    return 0;  
}
```

Output will be

Mother

Daughter: rabia

Son: salman

Question No: 36 (Marks: 5)

The code given below has one template function as a friend of a template class,

1. You have to identify any error/s in this code and describe the reason for error/s.
2. Give the correct code after removing the error/s.

```
template<typename U>
```

```
void Test(U);
```

```
template< class T >
```

```
class B {
```

```
    int data;
```

```
    public:
```

```
    friend void Test<>( T );
```

```
};
```

```
template<typename U>
void Test(U u){
    B < int> b1;
    b1.data = 7;
}

int main(int argc, char *argv[])
{
    char i;
    Test(i);
    system("PAUSE");
    return 0;
}
```

MUHAMMAD AKRAM ZAHID

FINALTERM EXAMINATION

14 Feb, 2011

CS304- Object Oriented Programming (Session - 3)

Time: 120 min

Question No: 1 (Marks: 1) - Please choose one

Which of the following causes run time binding?

- ▶ Declaring object of abstract class
- ▶ Declaring pointer of abstract class

- ▶ Declaring overridden methods as non-virtual
- ▶ None of the given

Question No: 2 (Marks: 1) - Please choose one

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ Templates
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

Question No: 3 (Marks: 1) - Please choose one

A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ an argument is passed by value.

▶ a function returns by reference.

▶ an argument is passed by reference.

Question No: 4 (Marks: 1) - Please choose one

Like template functions, a class template may not handle all the types successfully.

▶ True

▶ False

Question No: 5 (Marks: 1) - Please choose one

Question No: 6 (Marks: 1) - Please choose one

_____ A
class template may inherit from another class template.

▶ True

▶ False

Question No: 7 (Marks: 1) - Please choose one

By default the vector data items are initialized to ____

- ▶ 0
- ▶ 0.0
- ▶ 1
- ▶ null

Question No: 8 (Marks: 1) - Please choose one

In Private ----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ▶ specialization
- ▶ inheritance
- ▶ abstraction
- ▶ composition

Question No: 9 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability

- ▶ Maintainability
- ▶ All of given

Question No: 10 (Marks: 1) - Please choose one

Template functions use _____ than ordinary functions.

- ▶ Greater Memory
- ▶ Lesser Memory
- ▶ Equal Memory
- ▶ None of the given options

Question No: 11 (Marks: 1) - Please choose one

Non Template Friend functions of a class are friends of _____instance/s of that class.

- ▶ All
- ▶ One specific
- ▶ All instances of one data type
- ▶ None of the given options

Question No: 12 (Marks: 1) - Please choose one

A copy constructor is invoked when

- ▶ a function do not returns by value.

- ▶ an argument is passed by value.
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

Question No: 13 (Marks: 1) - Please choose one

A pointer to a base class can point to objects of a derived class.

- ▶ True
- ▶ False

Question No: 14 (Marks: 1) - Please choose one

A template argument is preceded by the keyword _____.

- ▶ vector
- ▶ class
- ▶ template
- ▶ type*

Question No: 15 (Marks: 1) - Please choose one

Which one of the following terms must relate to **polymorphism**?

- ▶ Static allocation
- ▶ Static typing
- ▶ Dynamic binding

- ▶ Dynamic allocation

Question No: 16 (Marks: 1) - Please choose one

Multiple inheritance can be of type

- ▶ Public
- ▶ Private
- ▶ Protected
- ▶ **All of the given**

Question No: 17 (Marks: 1) - Please choose one

Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- ▶ public members of Derv.
- ▶ protected members of Derv.
- ▶ private members of Derv.
- ▶ **protected members of Base.**

Question No: 18 (Marks: 1) - Please choose one

A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ an argument is passed by value.
- ▶ **a function returns by reference.**

- ▶ an argument is passed by reference.

Question No: 19 (Marks: 1) - Please choose one

A function call is resolved at run-time in_____

- ▶ non-virtual member function.
- ▶ **virtual member function.**
- ▶ Both non-virtual member and virtual member function.
- ▶ None of given

Question No: 20 (Marks: 1) - Please choose one

Two important STL associative containers are _____ and _____.

- ▶ set,map
- ▶ **sequence,mapping**
- ▶ setmet,multipule
- ▶ sit,mat

Question No: 21 (Marks: 1) - Please choose one

An abstract class is useful when,

- ▶ We do not derive any class from it.
- ▶ There are multiple paths from one derived class to another.
- ▶ **We do not want to instantiate its object.**

- ▶ You want to defer the declaration of the class.

Question No: 22 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ **All of given**

Question No: 23 (Marks: 1) - Please choose one

By default the vector data items are initialized to _____

- ▶ **0**
- ▶ 0.0
- ▶ 1
- ▶ null

Question No: 24 (Marks: 1) - Please choose one

Suppose you create an uninitialized vector as follows:

```
vector<int> evec;
```

After adding the statment,

```
evec.push_back(21);
```


what will happen?

▶ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.

▶ The following statement will add an element to the center of evec and will reinitialize it with the value 21.

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Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

▶ **Templates**

▶ Overloading

▶ Data hiding

- ▶ Encapsulation

Question No: 27 (Marks: 1)

Describe the way to declare a template class as a friend of any class.

Question No: 28 (Marks: 1)

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

- ▶ **True**

- ▶ False

Question No: 29 (Marks: 1)

In order to define a class template, the first line of definition must be:

- ▶ **template <typename T>**

- ▶ typename <template T>

- ▶ Template Class <ClassName>

- ▶ Class <Template T>

Question No: 30 (Marks: 1)

In case of multiple inheritance a derived class inherits,

- ▶ Only the public member functions of its base classes

- ▶ Only the public data members of its base classes

► **Both public data members and member functions of all its base classes**

► Data members and member functions of any two base classes

Question No: 31 (Marks: 1)

In Private ----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

► specialization

► inheritance

► abstraction

► composition

Question No: 32 (Marks: 1)

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

► **Templates**

► Overloading

► Data hiding

► Encapsulation

Question No: 33 (Marks: 1)

If we declare a function as friend of a template class will it be a friend for a particular data type or for all data types of that class.

Question No: 34 (Marks: 1)

Question No: 35 (Marks: 1)

Question No: 36 (Marks: 1)

Question No: 37 (Marks: 2)

Write three important features of virtual functions.

Question No: 38 (Marks: 5)

There are some errors in the code given below, you have to Indicate the line no. with error/s

Give the reason for error/s

Correct the error/s.

1. #include <iostream>
2. #include <stdlib.h>

3. using namespace std;
4. template <typename T>
5. class MyClass{
6. public:
7. MyClass(){
8. cout<<"This is class1"<<endl;
9. }

```

10.};
11.template <typename T>
12.class MyClass<int*>{
13.public:
14.MyClass(){
15.cout<<"This is class2"<<endl;
16.}
17.};
18.int main(int argc, char *argv[])
19.{
20.MyClass<int> c1;
21.MyClass<int*> c2;
22.system("PAUSE");
23.return 0;
24.}

```

Question No: 39 (Marks: 5)

What is random iterator? What is its relation with vectors?

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CS304 Solved By Ms.Shazia

Monday, 08 November 2010 10:40 Zubair Hussain

Date: 08-11-2010

Question # 1 of 10

Information hiding can be achieved through_____.

- 1.Encapsulation, Inheritance
- 2.Encapsulation, Polymorphism
- 3.Encapsulation, Abstraction**
- 4.Overloading

Question # 2 of 10 (Start time: 01:11:21 AM) Total Marks: 1

A good model is related to a real life problem.

Select correct option:

- 1.Loosely
- 2.Openly
- 3.Closely**

Question # 3 of 10 (Start time: 01:12:33 AM) Total Marks: 1

Which of the following features of OOP is used to derive a class from another?

Select correct option:

- 1.Encapsulation
- 2.Polymorphism

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3.Data hiding

4. Inheritance

Question # 4 of 10 (Start time: 01:13:51 AM) Total Marks: 1

Which of the following is a weak relationship between two objects?

Select correct option:

1. Inheritance
- 2.Composition

3.Aggregation

4.None of given

Question # 5 of 10 (Start time: 01:14:56 AM) Total Marks: 1

Data items in a class must be private.

Select correct option:

1.True

2.False

Question # 6 of 10 (Start time: 01:15:52 AM) Total Marks: 1

Which one is a class association

Select correct option:

1.Simple Association

2. Inheritance

3.Composition

4.Aggregation

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Question # 7 of 10 (Start time: 01:16:55 AM) Total Marks: 1

Suppose there is an object of type Person, which of the following can be considered as one of its attributes

Select correct option:

1.Name

2.Age

3.Work()

4.Both Name and Age

Question # 8 of 10 (Start time: 01:17:52 AM) Total Marks: 1

Which one is not an object association?

Select correct option:

1.Simple association

2. Inheritance

3.Aggregation

4.Association

Question # 9 of 10 (Start time: 01:18:50 AM) Total Marks: 1

Using encapsulation we can achieve

Select correct option:

1. Information hiding

2.Least interdependencies among modules

3. Implementation independence

4.All of given options

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Question # 10 of 10 (Start time: 01:19:43 AM) Total Marks: 1

In constant member function the type of this pointer is:

Select correct option:

1.Constant pointer

2.Constant pointer to object

3.Constant pointer to class

4.Constant pointer to constant object

Question # 1 of 10

Which of the following is the way to extract common behavior and attributes from the given classes and make a separate class of those common behaviors and attributes?

1.Generalization

2.Sub-typing

3.Specialization

4.Extension

Question # 2 of 10

The ability to derive a class from more than one class is called

1.Single inheritance

2.Encapsulation

3.Multiple inheritance

4.Polymorphism

Question # 3 of 10:

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If MyClass has a destructor what is the destructor named?

1.MyClass

2.~MyClass

3.My~Class

4.MyClass~

Question # 4 of 10:

Class abc{ ----- }; Is a valid class declaration?

1.yes

2.no

Question # 5 of 10:

Without using Deep copy constructor, A _____ problem can occur

1.System crash

2.Memory Leakage

3.Dangling pointer

4.All of the given

Question # 6 of 10:

If only one behaviour of a derived class is incompatible with base class, then it is:

1.Generalization

2.Specialization

3.Extension

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4. Inheritance

Question # 7 of 10:

Which of the following may not be an integral part of an object?

1. state
2. behavior
3. Protected data members

4. All of given

Question # 8 of 10:

Only tangible things can be chosen as an object.

1. True

2. False

- 1.

2. Question # 1 of 10

Information hiding can be achieved through_____.

Encapsulation, Inheritance

Encapsulation, Polymorphism

Encapsulation, Abstraction

Overloading

Question # 2 of 10 (Start time: 01:11:21 AM) Total Marks: 1

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Openly

Closely

Question # 3 of 10 (Start time: 01:12:33 AM) Total Marks: 1

Which of the following features of OOP is used to derive a class from another?

Select correct option:

Encapsulation

Polymorphism

Data hiding

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Inheritance

Question # 4 of 10 (Start time: 01:13:51 AM) Total Marks: 1

Which of the following is a weak relationship between two objects?

Select correct option:

Inheritance

Composition

Aggregation

None of given

Question # 5 of 10 (Start time: 01:14:56 AM) Total Marks: 1

Data items in a class must be private.

Select correct option:

True

False

Question # 6 of 10 (Start time: 01:15:52 AM) Total Marks: 1

Which one is a class association

Select correct option:

Simple Association

Inheritance

Composition

Aggregation

Question # 7 of 10 (Start time: 01:16:55 AM) Total Marks: 1

Suppose there is an object of type Person, which of the following can be considered as

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Select correct option:

Name

Age

Work()

Both Name and Age

Question # 8 of 10 (Start time: 01:17:52 AM) Total Marks: 1

Which one is not an object association?

Select correct option:

Simple association

Inheritance

Aggregation

Association

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Question # 9 of 10 (Start time: 01:18:50 AM) Total Marks: 1

Using encapsulation we can achieve

Select correct option:

Information hiding

Least interdependencies among modules

Implementation independence

All of given options

Question # 10 of 10 (Start time: 01:19:43 AM) Total Marks: 1

In constant member function the type of this pointer is:

Select correct option:

Constant pointer

Constant pointer to object

Constant pointer to class

Constant pointer to constant object

Question # 1 of 10

Which of the following is the way to extract common behavior and attributes from the given classes and make a separate class of those common behaviors and attributes?

Generalization

Sub-typing

Specialization

Extension

Question # 2 of 10

The ability to derive a class from more than one class is called

Single inheritance

Encapsulation

Multiple inheritance

Polymorphism

Question # 3 of 10:

If MyClass has a destructor what is the destructor named?

MyClass

~MyClass

My~Class

MyClass~

Question # 4 of 10:

Class abc{ ----- }; Is a valid class declaration?

yes

no

Question # 5 of 10:

Without using Deep copy constructor, A _____ problem can occur

System crash

Memory Leakage

Dangling pointer

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All of the given

Question # 6 of 10:

If only one behaviour of a derived class is incompatible with base class, then it is:

Generalization

Specialization

Extension

Inheritance

Question # 7 of 10:

Which of the following may not be an integral part of an object?

state

behavior

Protected data members

All of given

Question # 8 of 10:

Only tangible things can be chosen as an object.

True

False

Class is not a mechanism to create objects and define user data types.

1. true

2. false

Memory is allocated to non static members only, when:

1. Class is created

2. Object is defined

3. Object is initialized

4. Object is created

The sub-object's life is not dependent on the life of master class in _____.

1. Composition

2. Aggregation

3. Separation

4. non of the given

Unary operators and assignment operator are right associative.

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1. true

2. false

The >= operator can't be overloaded.

1. true

2. false

_____ is creating obj

1. Association

2. Composition

3. Aggregation

4. Inheritance

ects of one class inside another class.

If we are create array of objects through new operator, then

1. We can call overloaded constructor through new

2. We can't call overloaded constructor through new

3. We can call default constructor through new

4. None of the given

Object can be declared constant with the use of Constant keyword.

1. true
2. false

_____ Operator will take only one operand.

1. New
2. int
3. object
4. none of the given

Which of the following operator(s)

1. ++
2. *
3. %

take(s) one or no argument if overloaded?

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4. All of the given choices

this pointer does not pass implicitly to _____ functions.

1. Static Member
2. Non-Static Member
3. Instance Number
4. None of the given

Operator overloading is

1. making C++ operators work with objects.
2. giving C++ operators more than they can handle.
3. giving new meanings to existing Class members.
4. making new C++ operators

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 1 of 10 (Start time: 09:57:41 AM) Total Marks: 1

Consider the code below, class class1{ public: void func1(); }; class class2 :

private

class1 { }; Function func1 of class1 is

_____ in class2,

Select correct option:

public

protected

private

none of the given options

[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 2 of 10 (Start time: 09:59:01 AM) Total Marks: 1

User can make virtual table explicitly.

Select correct option:

True

False

[Click here to Save Answer & Move to Next Question](#)

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MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 3 of 10 (Start time: 10:00:15 AM) Total Marks: 1

In private inheritance derived class pointer can be assigned to base class pointer in,

Select correct option:

Main function

In derived class member and friend functions

In base class member and friend functions

None of the given options

[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 4 of 10 (Start time: 10:01:15 AM) Total Marks: 1

In C++, we declare a function virtual by preceding the function header with keyword

“Inline”

Select correct option:

True

False

[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 5 of 10 (Start time: 10:02:45 AM) Total Marks: 1

Outside world can access only _____ members of a class using its object.

Select correct option:

Public

Private

Protected

No member is accessible.

[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 6 of 10 (Start time: 10:03:10 AM) Total Marks: 1

Friend Functions of a class are _____ members of that class.

Select correct option:

Public

Private

Protected

None of the given options.

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[Click here to Save Answer & Move to Next Question](#)

MC100202262 : Muhammad Naveed Anjum

Time Left

Quiz Start Time: 09:57 AM

Question # 7 of 10 (Start time: 10:03:54 AM) Total Marks: 1

Consider the following two lines of code written for a class Student, 1. Student subj1,subj2; 2. subj2 = subj1; In line No.2 what constructor of Student class will be called,

Select correct option:

Default constructor of Student class.

Copy constructor of student class

Both default and copy constructor of Student class

No constructor will be called.

Click here to Save Answer & Move to Next Question

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left Class is not a mechanism to create objects and define user data types.

1. true
2. false

Memory is allocated to non static members only, when:

1. Class is created
2. Object is defined
3. Object is initialized
4. **Object is created**

The sub-object's life is not dependent on the life of master class in _____.

1. Composition
2. **Aggregation**
3. Separation
4. non of the given

Unary operators and assignment operator are right associative.

1. true
2. false

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The >= operator can't be overloaded.

1. true
2. false

_____ is creating objects of one class inside another class.

1. Association
2. **Composition**
3. Aggregation
4. Inheritance

If we are create array of objects through new operator, then

1. We can call overloaded constructor through new
2. We can't call overloaded constructor through new
3. We can call default constructor through new
4. **None of the given**

Object can be declared constant with the use of Constant keyword.

1. true
2. false

_____ Operator will take only one operand.

1. New
2. int
3. object

4. none of the given

Which of the following operator(s) take(s) one or no argument if overloaded?

1. ++
2. *
3. %

4. All of the given choices

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this pointer does not pass implicitly to _____ functions.

1. Static Member

2. Non-Static Member

3. Instance Number

4. None of the given

Operator overloading is

1. making C++ operators work with objects.

2. giving C++ operators more than they can handle.

3. giving new meanings to existing Class members.

4. making new C++ operators

BC080400849 : Nimra Qamar

Time Left 88

sec(s)

Quiz Start Time: 10:39 PM

Question # 1 of 8 (Start time: 10:39:47 PM) Total Marks: 1

Which of the following operator(s) take(s) one or no argument if overloaded?

Select correct option:

*

%

++

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ck here to Save Answ er & Move to Next Quest on

BC080400849 : Nimra Qamar Time Left 89 sec(s)

Quiz Start Time: 10:39 PM

Question # 2 of 8 (Start time: 10:40:38 PM) Total Marks: 1

Object can be declared constant with the use of Constant keyword.

Select correct option:

l

Cli i

True

False

Click here to Save Answer & Move to Next Question

BC080400849 : Nimra Qamar

Time Left 89

sec(s)

Quiz Start Time: 10:39 PM

Question # 3 of 8 (Start time: 10:41:41 PM) Total Marks: 1

Static data members are called _____ variable

Select correct option:

Class

Instance

Association

Object

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Incorrect

Class

Structure

None of the given

Click here to Save Answer & Move to Next Question

BC080400849 : Nimra Qamar Time Left 89 sec(s)

Quiz Start Time: 10:39 PM

Question # 4 of 8 (Start time: 10:42:35 PM) Total Marks: 1

Associativity can be change in operator overloading.

Select correct option:

Left

Class

True

False

Click here to Save Answer & Move to Next Question

BC080400849 : Nimra Qamar 89 Time Left sec(s)

Quiz Start Time: 10:39 PM

Question # 5 of 8 (Start time: 10:43:56 PM) Total Marks: 1

_____ and _____ methods may not be declared abstract.

Select correct option:

Private,static

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Interface

Interface

Interface

Class

private, public

static, public

none of the given

Click here to Save Answer & Move to Next Question

BC080400849 : Nimra Qamar

Time Left 89

sec(s)

Quiz Start Time: 10:39 PM

Question # 6 of 8 (Start time: 10:45:17 PM) Total Marks: 1

Let Suppose a class Student with objects std1, std2, and std3. For the statement std3 =

std1 - std2 to work correctly, if the overloaded - operator must

Select correct option:

ii

il

Click

take two arguments.

None of the given choices

taking one argument

take three arguments

Click here to Save Answer & Move to Next Question

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BC080400849 : Nimra Qamar Time Left 88 sec(s)

Quiz Start Time: 10:39 PM

Question # 7 of 8 (Start time: 10:46:48 PM) Total Marks: 1

To initialize an array of objects, only _____ will be called

Select correct option:

l

l

lj

Click

Default Constructor

Overloaded Constructor

Default Object

None of the above

Click here to Save Answer & Move to Next Question

BC080400849 : Nimra Qamar 87 Time Left sec(s)

Quiz Start Time: 10:39 PM

Question # 8 of 8 (Start time: 10:47:49 PM) Total Marks: 1

_____ provide the facility to access the data member.

Select correct option:

accesser function

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i i

inli i

i

cli i

private function

ne function

None of the given

Click here to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 10:04:41 AM) Total Marks: 1

Consider the following two lines of code written for a class Student, 1. Student
obj1; 2.

Student obj2 = obj1; In line No.1

what constructor of student class will be called,

Select correct option:

Default constructor of Student class.

Copy constructor of student class

Both default and copy constructor of student class

None the given options

Click here to Save Answer & Move to Next Question

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

Question # 9 of 10 (Start time: 10:05:09 AM) Total Marks: 1

Consider the code below, class class1{ protected: void func1(); }; class class2 :
protected

class1 { }; Function func1 of class1 is

_____ in class2,

Select correct option:

public

protected

private

none of the given options

Click here to Save Answer & Move to Next Question

MC100202262 : Muhammad Naveed Anjum

Quiz Start Time: 09:57 AM

Time Left

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Question # 10 of 10 (Start time: 10:05:50 AM) Total Marks: 1

Virtual functions allow you to

Select correct option:

create an array of type pointer-to-base class that can hold pointers to derived classes.

create functions that can never be accessed.

group objects of different classes so they can all be accessed by the same function code.

use the same function call to execute member functions of objects from different classes.

[Click here to Save Answer & Move to Next Question](#)

Question # 1 of 10

Information hiding can be achieved through_____.

1. Encapsulation, Inheritance
2. Encapsulation, Polymorphism
- 3. Encapsulation, Abstraction**
4. Overloading

Question # 2 of 10 (Start time: 01:11:21 AM) Total Marks: 1

A good model is related to a real life problem.

Select correct option:

1. Loosely
2. Openly

3. Closely

Question # 3 of 10 (Start time: 01:12:33 AM) Total Marks: 1

Which of the following features of OOP is used to derive a class from another?

Select correct option:

1. Encapsulation
2. Polymorphism
3. Data hiding

4. Inheritance

Question # 4 of 10 (Start time: 01:13:51 AM) Total Marks: 1

Which of the following is a weak relationship between two objects?

Select correct option:

1. Inheritance
2. Composition

3. Aggregation

4. None of given

Question # 5 of 10 (Start time: 01:14:56 AM) Total Marks: 1

Data items in a class must be private.

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Select correct option:

1. True

2. False

Question # 6 of 10 (Start time: 01:15:52 AM) Total Marks: 1

Which one is a class association

Select correct option:

1. Simple Association

2. Inheritance

3. Composition

4. Aggregation

Question # 7 of 10 (Start time: 01:16:55 AM) Total Marks: 1

Suppose there is an object of type Person, which of the following can be considered as

one of its attributes

Select correct option:

1. Name

2. Age

3. Work()

4. Both Name and Age

Question # 8 of 10 (Start time: 01:17:52 AM) Total Marks: 1

Which one is not an object association?

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1. Simple association

2. Inheritance

3. Aggregation

4. Association

Question # 9 of 10 (Start time: 01:18:50 AM) Total Marks: 1

Using encapsulation we can achieve

Select correct option:

1. Information hiding

2. Least interdependencies among modules

3. Implementation independence

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4. All of given options

Question # 10 of 10 (Start time: 01:19:43 AM) Total Marks: 1

In constant member function the type of this pointer is:

Select correct option:

1. Constant pointer
- 2. Constant pointer to object**
3. Constant pointer to class
4. Constant pointer to constant object

Question # 1 of 10

Which of the following is the way to extract common behavior and attributes from the

given classes and make a separate class of those common behaviors and attributes?

1. Generalization

2. Sub-typing
3. Specialization
4. Extension

Question # 2 of 10

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1. Single inheritance
2. Encapsulation

3. Multiple inheritance

4. Polymorphism

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If MyClass has a destructor what is the destructor named?

1. MyClass
- 2. ~MyClass**
3. My~Class
4. MyClass~

Question # 4 of 10:

Class abc{ ----- }; Is a valid class declaration?

1. yes

2. no

Question # 5 of 10:

Without using Deep copy constructor, A _____ problem can occur

1. System crash
2. Memory Leakage
3. Dangling pointer

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4. All of the given

Question # 6 of 10:

If only one behaviour of a derived class is incompatible with base class, then it is:

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2. Specialization

3. Extension

4. Inheritance

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Which of the following may not be an integral part of an object?

1. state

2. behavior

3. Protected data members

4. All of given

Question # 8 of 10:

Only tangible things can be chosen as an object.

1. True

2. False

BC080400849 : Nimra Qamar

Time Left 79

sec(s)

Quiz Start Time: 08:14 PM

Question # 1 of 10 (Start time: 08:14:14 PM) Total Marks: 1

When we create objects, then space is allocated to:

Select correct option:

i

ifi

Member funct ons

Access spec er

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Data members

None of the g ven

ck here to Save Answ er & Move to Next Quest on

BC080400849 : Nimra Qamar

Time Left 87

sec(s)

Quiz Start Time: 08:14 PM

Question # 2 of 10 (Start time: 08:15:18 PM) Total Marks: 1

Constructor and destructor can be declared constant

Select correct option:

l

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True

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ck here to Save Answ er & Move to Next Quest on

BC080400849 : Nimra Qamar Time Left 88 sec(s)

Quiz Start Time: 08:14 PM

Question # 3 of 10 (Start time: 08:16:04 PM) Total Marks: 1

Information hiding can be achieved through_____.

Select correct option:

Encapsulation, Inheritance

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BC080400849 : Nimra Qamar

Time Left 88

sec(s)

Quiz Start Time: 08:14 PM

Question # 4 of 10 (Start time: 08:16:47 PM) Total Marks: 1

A real world object can be transformed into programming entity by defining its
respective

Select correct option:

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BC080400849 : Nimra Qamar

Time Left 88

sec(s)

Quiz Start Time: 08:14 PM

Question # 5 of 10 (Start time: 08:17:50 PM) Total Marks: 1

Which of the following is a weak relationship between two objects?

Select correct option:

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Inher tance

Compos on

Aggregat on

None of g ven

ck here to Save Answ er & Move to Next Quest on

BC080400849 : Nimra Qamar

Time Left 88

sec(s)

Quiz Start Time: 08:14 PM

Question # 6 of 10 (Start time: 08:18:45 PM) Total Marks: 1

Which of the following is a necessary ingredient in an object model?

Select correct option:

Class

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BC080400849 : Nimra Qamar

Time Left 88

sec(s)

Quiz Start Time: 08:14 PM

Question # 7 of 10 (Start time: 08:19:48 PM) Total Marks: 1

If a class A inherits from class B, then class A is called.

Select correct option:

il l

i l

l

il i l

Cli i

Ch d c ass

Der ved c ass

Parent c ass

Ch d and der ved c ass

ck here to Save Answ er & Move to Next Quest on

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BC080400849 : Nimra Qamar

Time Left 88

sec(s)

Quiz Start Time: 08:14 PM

Question # 8 of 10 (Start time: 08:20:54 PM) Total Marks: 1

Which of the following is the way to extract common behaviour and attributes from the

given classes and make a separate class of those common behaviours and attributes?

Select correct option:

li i

i

iali i

i

Cli i

Genera zat on

Sub-typ ng

Spec zat on

Extens on

ck here to Save Answ er & Move to Next Quest on

BC080400849 : Nimra Qamar

Time Left 87

sec(s)

Quiz Start Time: 08:14 PM

Question # 9 of 10 (Start time: 08:21:27 PM) Total Marks: 1

The _____ keyword tells the compiler to substitute the code within the function definition for every instance of a function call

Select correct option:

virtual

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inli

i

i

Click

ne

instance

none of the given

Click here to Save Answer & Move to Next Question

BC080400849 : Nimra Qamar

Time Left 88

sec(s)

Quiz Start Time: 08:14 PM

Question # 10 of 10 (Start time: 08:22:16 PM) Total Marks: 1

The process of hiding unwanted details from users is called _____.

Select correct option:

i

Click

i

i

Click

Protection

Encapsulation

Argumentation

Abstraction

Click here to Save Answer & Move to Next Question

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BC090401541 : Azmat Ur Rehman

Quiz Start Time: 12:40 PM

Question # 1 of 10 (Start time: 12:40:20 PM) Total Marks: 1

Time Left

Which of the following is the way to extract common behaviour and attributes from the

given classes and make a separate class of those common behaviours and attributes?

Click

i

Click

i

Click

Select correct option:

Generalization

Sub-typ ng

Spec zat on

Extens on

ck here to Save Answ er & Move to Next Quest on

BC090401541 : Azmat Ur Rehman

Quiz Start Time: 12:40 PM

Question # 2 of 10 (Start time: 12:41:52 PM) Total Marks: 1

Time Left

“A fan has wings”. Which type of relation exists between fan and wings in this sentence?

i

Select correct option:

Aggregat on

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i i

li i

iti

Assoc at on

Genera zat on

Compos on

Click here to Save Answ er & Move to Next Question

BC090401541 : Azmat Ur Rehman

Time Left

Quiz Start Time: 12:40 PM

Question # 3 of 10 (Start time: 12:42:46 PM) Total Marks: 1

A good model is related to a real life problem.

ly

ly

Cl

Cli i

Select correct option:

Loose

Open

osely

Not

ck here to Save Answ er & Move to Next Quest on

Come & Join Us at www.vustudents.net

BC090401541 : Azmat Ur Rehman

Time Left

Quiz Start Time: 12:40 PM

Question # 5 of 10 (Start time: 12:44:45 PM) Total Marks: 1

When we create objects, then space is allocated to:

i

ifi

i

Cl i

Select correct option:

Member funct ons

Access spec er

Data members

None of the g ven

ck here to Save Answ er & Move to Next Quest on

BC090401541 : Azmat Ur Rehman

Time Left

Quiz Start Time: 12:40 PM

Question # 6 of 10 (Start time: 12:45:21 PM) Total Marks: 1

There is only one form of copy constructor.

Select correct option:

True

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l

Cl i

Fa se

ck here to Save Answ er & Move to Next Quest on

BC090401541 : Azmat Ur Rehman

Time Left 22

sec(s)

Quiz Start Time: 12:40 PM

Question # 7 of 10 (Start time: 12:45:38 PM) Total Marks: 1

Which of the following features of OOP is used to deal with only relevant details?

Select correct option:

i

i idi

j

i

Click

Abstract on

Information

Object

Inheritance

Click here to Save Answer & Move to Next Question

BC090401541 : Azmat Ur Rehman

Time Left 59

sec(s)

Quiz Start Time: 12:40 PM

Question # 8 of 10 (Start time: 12:48:26 PM) Total Marks: 1

Suppose there is an object of type Person, which of the following can be considered as

one of its attributes

Select correct option:

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()

Age

Work

Both Name and Age

Click here to Save Answer & Move to Next Question

BC090401541 : Azmat Ur Rehman

Time Left

Quiz Start Time: 12:40 PM

Question # 9 of 10 (Start time: 12:56:04 PM) Total Marks: 1

Through interface we access object_____.

i

i

Click

Select correct option:

States

Data members

Behaviour

None of the given

Click here to Save Answer & Move to Next Question

Come & Join Us at www.vustudents.net

BC090401541 : Azmat Ur Rehman

Time Left

Quiz Start Time: 12:40 PM

Question # 10 of 10 (Start time: 12:57:00 PM) Total Marks: 1

If a class A inherits from class B, then class A is called.

il l

i l

l

il i l

Cli i

Select correct option:

Ch d c ass

Der ved c ass

Parent c ass

Ch d and der ved c ass

ck here to Save Answ er & Move to Next Quest on

[MC090405816 : Sohail Aslam](#)

Time Left

Quiz Start Time: 12:59 PM

Question # 1 of 10 (Start time: 12:59:51 PM) Total Marks: 1

If some of objects exhibit identical characteristics, then they belong to:

Di l

Select correct option:

fferent c asses

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l ti l

l

i

Cli i

Mu ple c asses

Same c ass

None of the g ven

ck here to Save Answ er & Move to Next Quest on

MC090405816 : Sohail Aslam Time Left 82 sec(s)

Quiz Start Time: 12:59 PM

Question # 2 of 10 (Start time: 01:00:41 PM) Total Marks: 1

_____ is automatically called when the object is created.

Select correct option:

i

j

i
C++
member function
object
constructor
None of the given
Click here to Save Answer & Move to Next Question
Come & Join Us at www.vustudents.net
MC090405816 : Sohail Aslam

Time Left

Quiz Start Time: 12:59 PM

Question # 3 of 10 (Start time: 01:03:09 PM) Total Marks: 1

Which is true about sub-typing in case of inheritance?

1. A base class is a subclass of a derived class.

2. A derived class is a subclass of a base class.

3. A base class is a subclass of a base class.

4. A derived class is a subclass of a derived class.

5. None of the given.

Click here to Save Answer & Move to Next Question

Select correct option:

In sub-typing a new class is derived from existing with extended behavior of its parent.

In sub-typing a new class is derived from existing with

In sub-typing a class is derived from existing one which

None of the given.

Click here to Save Answer & Move to Next Question

MC090405816 : Sohail Aslam

Time Left

Quiz Start Time: 12:59 PM

Question # 4 of 10 (Start time: 01:04:28 PM) Total Marks: 1

If a class involves dynamic memory allocation, then:

1. It is a good practice to use a destructor.

Select correct option:

Default copy constructor, shallow copy implementation

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1. It is a good practice to use a destructor.

2. It is a good practice to use a destructor.

i is i l

User def ned copy constructor, sha ow copy mp

Defau t copy constructor, deep copy mp emented

User def ned copy constructor, deep copy mp em

Click here to Save Answ er & Move to Next Question

MC090405816 : Sohail Aslam

Time Left

Quiz Start Time: 12:59 PM

Question # 5 of 10 (Start time: 01:05:37 PM) Total Marks: 1

Which one is a class association

Si l i i

i

iti

i

Cli i

Select correct option:

mp e Assoc at on

Inher tance

Compos on

Aggregat on

ck here to Save Answ er & Move to Next Quest on

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MC090405816 : Sohail Aslam

Time Left

Quiz Start Time: 12:59 PM

Question # 6 of 10 (Start time: 01:06:50 PM) Total Marks: 1

Data items in a class must be private.

l

Cli i

Select correct option:

True

Fa se

ck here to Save Answ er & Move to Next Quest on

MC090405816 : Sohail Aslam

Time Left

Quiz Start Time: 12:59 PM

Question # 7 of 10 (Start time: 01:07:16 PM) Total Marks: 1

Three main characteristics of "Object Oriented programming" are,

l i i l i

l i l i i d i

l i i i i i i

Select correct option:

Encapsu at on,dynamc b nding,po ymarhp shm

po ymorph sm, over oad ng, overr ng

encapsu at on, nher tance, dynamc b nd ng

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encapsulation, inheritance, polymorphism

Click here to Save Answer & Move to Next Question

MC090405816 : Sohail Aslam

Time Left

Quiz Start Time: 12:59 PM

Question # 8 of 10 (Start time: 01:08:14 PM) Total Marks: 1

Which of the following is the way to extract common behaviour and attributes from the

given classes and make a separate class of those common behaviours and attributes?

l i i

i

iali i

i

Cli i

Select correct option:

Genera zat on

Sub-typ ng

Spec zat on

Extens on

ck here to Save Answer & Move to Next Quest on

MC090405816 : Sohail Aslam

Time Left 69

sec(s)

Quiz Start Time: 12:59 PM

Question # 9 of 10 (Start time: 01:09:04 PM) Total Marks: 1

The sentence "Object Oriented Programming book in bookshelf" is an example of:

Select correct option:

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i i

l t i p l i i

i

Assoc at on

Mu e assoc at on

Aggregat on

Click here to Save Answer & Move to Next Question

MC090405816 : Sohail Aslam

Time Left

Quiz Start Time: 12:59 PM

Question # 10 of 10 (Start time: 01:16:05 PM) Total Marks: 1

Data members are the attributes of objects.

l

Cli i

Select correct option:

True

False

Click here to Save Answer & Move to Next Question

MC090406317 : Aamer Abbas

Time Left

Quiz Start Time: 01:18 PM

Question # 1 of 10 (Start time: 01:18:48 PM) Total Marks: 1

Constructor have same name as the class name.

Select correct option:

True

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False

Click here to Save Answer & Move to Next Question

MC090406317 : Aamer Abbas

Time Left

Quiz Start Time: 01:18 PM

Question # 2 of 10 (Start time: 01:19:03 PM) Total Marks: 1

Which of the following features of OOP is used to derive a class from another?

l i

l i

idi

i

Class

Select correct option:

Encapsulation

Polyorphism

Data hiding

Inheritance

[Click here to Save Answer & Move to Next Question](#)

MC090406317 : Aamer Abbas 81 Time Left sec(s)

Quiz Start Time: 01:18 PM

Question # 3 of 10 (Start time: 01:19:29 PM) Total Marks: 1

Class abc{ ----- }; Is a valid class declaration?

Select correct option:

Yes

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No

[Click here to Save Answer & Move to Next Question](#)

MC090406317 : Aamer Abbas

Time Left

Quiz Start Time: 01:18 PM

Question # 6 of 10 (Start time: 01:22:47 PM) Total Marks: 1

Which of the following is a weak relationship between two objects?

i

iti

i

i

Class

Select correct option:

Inheritance

Composition

Aggregation

None of given

[Click here to Save Answer & Move to Next Question](#)

MC090406317 : Aamer Abbas

Time Left

Quiz Start Time: 01:18 PM

Question # 4 of 10 (Start time: 01:20:47 PM) Total Marks: 1

Without using Deep copy constructor, A _____ problem can occur

Select correct option:

Systemcrash

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li i

All i

Memory Leakage

Dang ng po nter

of the g ven

Click here to Save Answ er & Move to Next Question

MC090406317 : Aamer Abbas

Time Left

Quiz Start Time: 01:18 PM

Question # 5 of 10 (Start time: 01:21:20 PM) Total Marks: 1

An abstract class shows _____ behaviour.

idi

ific

l

i

Cli i

Select correct option:

Overr ng

Spec

Genera

None of the g ven

ck here to Save Answ er & Move to Next Quest on

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MC090406317 : Aamer Abbas

Time Left

Quiz Start Time: 01:18 PM

Question # 7 of 10 (Start time: 01:22:59 PM) Total Marks: 1

Which of the following are benefits of encapsulation?

All i l i l j i

i l l i l

i l i l i

i l l i l

Cli i

Select correct option:

variables can be manipulated as Objects instead
by making a variable protected they are protected
The implementation of a class can be changed with
Making a method protected prevents accidental
click here to Save Answer & Move to Next Question

MC090406317 : Aamer Abbas

Time Left

Quiz Start Time: 01:18 PM

Question # 8 of 10 (Start time: 01:24:19 PM) Total Marks: 1

If a class A inherits from class B, then class A is called.

il l

Select correct option:

Child class

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il l

l

il il

Derived class

Parent class

Child and derived class

Click here to Save Answer & Move to Next Question

MC090406317 : Aamer Abbas

Time Left

Quiz Start Time: 01:18 PM

Question # 9 of 10 (Start time: 01:24:44 PM) Total Marks: 1

Consider the statement "room has chair" Which of the following type of association

exists between room and chair?

i

iti

ii ion

i

Cl i

Select correct option:

Inheritance

Composition

There is no association

Aggregat on
ck here to Save Answ er & Move to Next Quest on
Come & Join Us at www.vustudents.net

MC090406317 : Aamer Abbas

Quiz Start Time: 01:18 PM

Question # 10 of 10 (Start time: 01:25:05 PM) Total Marks: 1

Time Left

The dot operator (or class member access operator) connects the following two entities

(reading from left to right):

l l j

l j l

l l

l j l

Cl i

Select correct option:

A c ass member and a c ass ob ect

A c ass ob ect and a c ass

A c ass and a member of that c ass

A c ass ob ect and a member of that c ass

ck here to Save Answ er & Move to Next Quest on

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FINALTERM EXAMINATION

CS304- Object Oriented Programming (Session - 4)

Question No: 1 (Marks: 1) - Please choose one

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be

concrete, while classes like Sphere and Cube would normally be abstract.

☐ True

☐ False

Question No: 2 (Marks: 1) - Please choose one

Virtual functions allow you to

☐ create an array of type pointer-to-base class that can hold pointers to derived classes.

☐ create functions that can never be accessed.

☐ group objects of different classes so they can all be accessed by the same function code.

☐ use the same function call to execute member functions of objects from different classes

Question No: 3 (Marks: 1) - Please choose one

A pointer to a base class can point to objects of a derived class.

☐ True

☐ False

Question No: 4 (Marks: 1) - Please choose one

A copy constructor is invoked when

☐ a function do not returns by value.

☐ an argument is passed by value.

☐ a function returns by reference.

☐ an argument is passed by reference.

Question No: 5 (Marks: 1) - Please choose one

Each try block can have _____ no. of catch blocks.

☐ 1

☐ 2

☐ 3

☐ As many as necessary.

Question No: 6 (Marks: 1) - Please choose one

Non Template Friend functions of a class are friends of _____instance/s of that class.

☐ All

☐ One specific

☐ All instances of one date type

☐ None of the given options

Question No: 7 (Marks: 1) - Please choose one

Template functions use _____ than ordinary functions.

☐ Greater Memory

☐ Lesser Memory

☐ Equal Memory

☐ None of the given options

Question No: 8 (Marks: 1) - Please choose one

The find() algorithm

☐ finds matching sequences of elements in two containers.

☐ finds a container that matches a specified container.

- _ takes iterators as its first two arguments.
- _ takes container elements as its first two arguments.

Question No: 9 (Marks: 1) - Please choose one

The copy() algorithm returns an iterator to

- _ the last element copied from.
- _ the last element copied to.
- _ the element one past the last element copied from.
- _ the element one past the last element copied to.

Question No: 10 (Marks: 1) - Please choose one

If you define a vector v with the default constructor, and define another vector w with a

one-argument constructor to a size of 11, and insert 3 elements into each of these vectors

with push_back(), then the size() member function will return _____ for v and _____ for

w.

- _ 11 for v and 3 for w.
- _ 0 for v and 0 for w.
- _ 0 for v and 3 for w.
- _ 3 for v and 11 for w.

Question No: 11 (Marks: 1) - Please choose one

Which is not the Advantage of inheritance?

- _ providing class growth through natural selection.
- _ facilitating class libraries.
- _ avoiding the rewriting of code.
- _ providing a useful conceptual framework.

Question No: 12 (Marks: 1) - Please choose one

class DocElement

```
{
public:
virtual void Print() { cout << "Generic element"; }
};
class Heading : public DocElement
{
public:
void Print() { cout << "Heading element"; }
};
```

```

class Paragraph : public DocElement
{
public:
void Print() { cout << "Paragraph element"; }
};
void main()
{
DocElement * p = new Paragraph();
p->Print();
}

```

When you run this program, it will print out a single line to the console output.
What will be in that line?

Select one correct answer from the following list:

- ☐ Generic element
- ☐ Heading element
- ☐ Paragraph element
- ☐ Nothing will be printed.

Question No: 13 (Marks: 1) - Please choose one

Which type of inheritance is being represented by the following statement,
class X : public A, public B { };

- ☐ Single inheritance
- ☐ Multiple inheritance
- ☐ Double inheritance
- ☐ None of the given options

Question No: 14 (Marks: 1) - Please choose one

When we write a class template the first line must be:

- ☐ template < class class_name>
- ☐ template < class data_type>
- ☐ template < class T >

Here T can be replaced with any name but it is preferable.

- ☐ class class-name()
- ☐ class template<class_name>

Question No: 15 (Marks: 1) - Please choose one

Function templates should be used where code and behavior must be identical.

- ☐ True
- ☐ False

Question No: 16 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- ☐ Reusability
- ☐ Writability
- ☐ Maintainability
- ☐ All of given

Question No: 17 (Marks: 1) - Please choose one

The specialization pattern <T*> after the name says that this specialization is to be used

for every,

- ☐ data type
- ☐ meta type
- ☐ virtual type

- ☐ pointer type

Question No: 18 (Marks: 1) - Please choose one

A range is often supplied to an algorithm by two _____ values.

- ☐ italic
- ☐ iteration
- ☐ iterator
- ☐ None of given

Question No: 19 (Marks: 1) - Please choose one

Which of the following is an integral part of an object?

- ☐ State
- ☐ Behavior
- ☐ Unique identity
- ☐ All of the given

Question No: 20 (Marks: 1) - Please choose one

Consider the following statement

Cupboard has books

What is the relationship between Cupboard and books?

- ☐ Composition
- ☐ Aggregation
- ☐ Inheritance
- ☐ None of the given options

Question No: 21 (Marks: 1) - Please choose one

Which sentence clearly defines an object?

- ☐ one instance of a class.

- _ another word for a class.
- _ a class with static methods.
- _ a method that accesses class attributes.

Question No: 22 (Marks: 1) - Please choose one

_____, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.

- _ Friendship is one way only
- _ Friendship is two way only
- _ NO Friendship between classes
- _ Any kind of friendship

Question No: 23 (Marks: 1) - Please choose one

The statement `objA=objB;` will cause a compiler error if the objects are of different classes.

- _ True
- _ False

Question No: 24 (Marks: 1) - Please choose one

Consider the call given below of an overloaded operator "+",

`Rational_number_1 + Rational_number_2`

Where `Rational_number_1` and `Rational_number_2` are the two objects of `Rational_number` class (a user defined class). Identify which of the above two objects

will be passed as an argument to the overloaded operator function?

- _ `Rational_number_1`
- _ `Rational_number_2`
- _ Both `Rational_number_1` & `Rational_number_2`
- _ any of the two objects, randomly

Question No: 25 (Marks: 1) - Please choose one

If a class D has been derived using protected inheritance from class B (If B is a protected

base and D is derived class) then public and protected members of B ----- accessed by

member functions and friends of class D and classes derived from D

- _ can be
- _ cannot be
- _ does restrict to be
- _ not given

Question No: 26 (Marks: 1) - Please choose one

In Private ----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

_ specialization

_ inheritance

_ abstraction

_ composition

Question No: 27 (Marks: 2)

Give two uses of a destructor.

Question No: 28 (Marks: 2)

Describe the way to declare a template class as a friend class of any other class.

Question No: 29 (Marks: 2)

Give the name of two basic types of containers collectively called First class containers?

Question No: 30 (Marks: 2)

State any conflict that may rise due to multiple inheritance?

Question No: 31 (Marks: 3)

What will be the output after executing the following code?

```
class c1{
public:
virtual void function(){
cout<<"I am in c1"<<endl;
```

```
}
```

```
};
```

```
class c2: public c1{
```

```
public:
```

```
void function(){
```

```
cout<<"I am in c2"<<endl;
```

```
}
```

```
};
```

```
class c3: public c1 {
```

```
public:
```

```
void function(){
```

```
cout<<"I am in c3"<<endl;
```

```
}
```

```
};
```

```
int main(){
```

```
c1 * test1 = new c2();
```

```
c1 * test2 = new c3();
```

```
test1->function();
test2->function();
system("PAUSE");
return 0;
}
```

Question No: 32 (Marks: 3)

If we declare a function as friend of a template class will it be a friend for a particular

data type or for all data types of that class.

Question No: 33 (Marks: 3)

Tell the logical error/s in the code given below with reference to resource management;

also describe how we can correct that error/s.

```
class Test{
public:
int function1(){
try{
```

```
FILE *fileptr = fopen("filename.txt","w");
throw exception();
fclose(fileptr);
return 0;
}
catch(Exception e){
...
}
}
};
```

Question No: 34 (Marks: 5)

What is the output produced by the following program?

```
#include<iostream.h>
void sample_function(double test) throw (int);
int main()
{
try
{
cout <<"Trying.\n";
sample_function(98.6);
cout << "Trying after call.\n";
```

```

}
catch(int)
{
cout << "Catching.\n";
}
cout << "End program.\n";
return 0;
}
void sample_function(double test) throw (int)
{
cout << "Starting sample_function.\n";
if(test < 100)
throw 42;
}

```

Question No: 35 (Marks: 5)

The code given below has one template function as a friend of a template class,

1. You have to identify any error/s in this code and describe the reason for error/s.
2. Give the correct code after removing the error/s.

```

template<typename U>
void Test(U);
template< class T >
class B {
int data;
public:
friend void Test<>( T );
};
template<typename U>
void Test(U u){
B < int> b1;
b1.data = 7;
}
int main(int argc, char *argv[])
{
char i;
Test(i);
system("PAUSE");
return 0;
}

```


Question No: 36 (Marks: 5)

Consider the following class,

```
class Base
{
char * p;
public:
Base() { p = new char[10]; }
~Base() { delete [] p; }
};
class Derived : public Base
{
char * q;
public:
Derived() { q = new char[20]; }
~Derived() { delete [] q; }
};
void foo()
{
Base* p = new Derived();

delete p;
}
```

With this program, every time function foo is called, some memory will leak. Explain why memory will leak. Also, explain how to fix this problem.

FINAL TERM EXAMINATION

Fall 2009

CS304- Object Oriented Programming (Session - 1)

Time: 120 min

Marks: 75

Question No: 1 (Marks: 1) - Please choose one

Which one of the following terms must relate to polymorphism?

- ☐ Static allocation
- ☐ Static typing
- ☒ Dynamic binding
- ☐ Dynamic allocation

Question No: 2 (Marks: 1) - Please choose one

Multiple inheritance can be of type

- ☐ Public
- ☐ Private

☐ Protected

☒ All of the given

Question No: 3 (Marks: 1) - Please choose one

When a subclass specifies an alternative definition for an attribute or method of its superclass, it is _____ the definition in the superclass.

☐ overload

☒ overriding

☐ copy riding

☐ none of given

Question No: 4 (Marks: 1) - Please choose one

Like template functions, a class template may not handle all the types successfully.

☒ True

☐ False

Question No: 5 (Marks: 1) - Please choose one

It is sometimes useful to specify a class from which no objects will ever be created.

☐ True

☒ False

Question No: 6 (Marks: 1) - Please choose one

Assume a class Derv that is privately derived from class Base. An object of class Derv

located in main() can access

☐ public members of Derv.

☐ protected members of Derv.

☐ private members of Derv.

☒ protected members of Base.

Question No: 7 (Marks: 1) - Please choose one

A pointer to a base class can point to objects of a derived class.

☒ True

☐ False

Question No: 8 (Marks: 1) - Please choose one

A copy constructor is invoked when

☐ a function do not returns by value.

☐ an argument is passed by value.

☒ a function returns by reference.

☐ an argument is passed by reference.

Question No: 9 (Marks: 1) - Please choose one

A function call is resolved at run-time in_____

- ☐ non-virtual member function.
- ☒ virtual member function.
- ☐ Both non-virtual member and virtual member function.
- ☐ None of given

Question No: 10 (Marks: 1) - Please choose one

When the base class and the derived class have a member function with the same name,

you must be more specific which function you want to call (using _____).

☒ scope resolution operator

- ☐ dot operator
- ☐ null operator
- ☐ Operator overloading

Question No: 11 (Marks: 1) - Please choose one

Each try block can have _____ no. of catch blocks.

- ☐ 1
- ☐ 2
- ☐ 3
- ☒ As many as necessary.

Question No: 12 (Marks: 1) - Please choose one

Two important STL associative containers are _____ and _____.

- ☐ set,map
- ☒ sequence,mapping
- ☐ setmet,multipule
- ☐ sit,mat

Question No: 13 (Marks: 1) - Please choose one

The mechanism of selecting function at run time according to the nature of calling object

is called,

- ☐ late binding
- ☐ static binding
- ☐ virtual binding
- ☒ None of the given options

Question No: 14 (Marks: 1) - Please choose one

An abstract class is useful when,

- ☐ We do not derive any class from it.

- _ There are multiple paths from one derived class to another.
- _ [We do not want to instantiate its object.](#)
- _ You want to defer the declaration of the class.

Question No: 15 (Marks: 1) - Please choose one

Which of the following is incorrect line regarding function template?

- _ `template<class T>`
- _ `template <typename U>`
- _ [Class<template T>](#)
- _ `template < class T, class U>`

Question No: 16 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- _ Reusability
- _ Writability
- _ Maintainability
- _ [All of given](#)

Question No: 17 (Marks: 1) - Please choose one

By default the vector data items are initialized to _____

- _ [0](#)
- _ 0.0
- _ 1
- _ null

Question No: 18 (Marks: 1) - Please choose one

Which one of the following functions returns the total number of elements in a vector.

- _ `length();`
- _ [size\(\);](#)
- _ `ele();`
- _ `veclen();`

Question No: 19 (Marks: 1) - Please choose one

Suppose you create an uninitialized vector as follows:

```
vector<int> evec;
```

After adding the statment,

```
evec.push_back(21);
```

what will happen?

- _ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.
- _ The following statement will add an element to the center of evec and will reinitialize it with the value 21.

_ The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.

_ The following statement will add an element to the end (the back) of evec and initialize it with the value 21.

Question No: 20 (Marks: 1) - Please choose one

An STL container can not be used to,

- _ hold objects of class employee.
- _ store elements in a way that makes them quickly accessible.
- _ compile c++ programs.
- _ organize the way objects are stored in memory

Question No: 21 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

_ True

_ False

Question No: 22 (Marks: 1) - Please choose one

The main function of scope resolution operator (::) is,

_ To define an object

- _ To define a data member
- _ To link the definition of an identifier to its declaration
- _ To make a class private

Question No: 23 (Marks: 1) - Please choose one

When is a constructor called?

- _ Each time the constructor identifier is used in a program statement
- _ During the instantiation of a new object
- _ During the construction of a new class
- _ At the beginning of any program execution

Question No: 24 (Marks: 1) - Please choose one

Consider the code below,

```
class Fred {
public:
    Fred();
    ...
};
int main()
{
    Fred a[10];
    Fred* p = new Fred[10];
    ...
}
```

}

Select the best option,

_ Fred a[10]; calls the default constructor 09 times

Fred* p = new Fred[10]; calls the default constructor 10 times

_ Produce an error

_ Fred a[10]; calls the default constructor 11 times

Fred* p = new Fred[10]; calls the default constructor 11 times

_ Fred a[10]; calls the default constructor 10 times

Fred* p = new Fred[10]; calls the default constructor 10 times

Question No: 25 (Marks: 1) - Please choose one

Associativity can be changed in operator overloading.

_ True

_ False

Question No: 26 (Marks: 1) - Please choose one

A normal C++ operator that acts in special ways on newly defined data types is said to

be

_ glorified.

_ encapsulated.

_ classified.

_ overloaded.

Question No: 27 (Marks: 1) - Please choose one

Which operator can not be overloaded?

_ The relation operator (>=)

_ Assignment operator (=)

_ Script operator ([])

_ Conditional operator (? :)

Question No: 28 (Marks: 1) - Please choose one

Suppose obj1 and obj2 are two objects of a user defined class A. An + operator is overloaded to add obj1 and obj2 using the function call obj1+obj2.

Identify the correct function prototype against the given call?

_ A operator + (A &obj);

_ int + operator();

_ int operator (plus) ();

_ A operator(A &obj3);

Question No: 29 (Marks: 1) - Please choose one

Default constructor is such constructor which either has no -----or if it has some parameters these have ----- values

- _ Parameter, temporary
- _ Null, Parameter
- _ [Parameter, default](#)
- _ non of the given

Question No: 30 (Marks: 1) - Please choose one

Public methods of base class can ----- be accessed in its derived class

- _ directly
- _ [indirectly](#)
- _ simultaneously
- _ non of the given

Question No: 31 (Marks: 1)

Is Deque a Birectional Container?

[Yes, deque behaves like queue \(line\) such that we can add elements on both sides of it.](#)

Question No: 32 (Marks: 1)

What is meant by Generic Programming?

[Generic programming refers to programs containing generic abstractions general code](#)

[that is same in logic for all data types like printArray function\), then we instantiate that](#)

[generic program abstraction \(function, class\) for a particular data type, such abstractions](#)

[can work with many different types of data.](#)

Question No: 33 (Marks: 2)

Sort the following data in the order in which compiler searches a function?

Complete Specialization, Generic Template, Partial Specialization, Ordinary Function.

[Specializations of this function template, instantiations with specific types, can be called](#)

[just like an ordinary function:](#)

[cout << max\(3, 7\); // outputs 7](#)

[The compiler examines the arguments used to call max and determines that this is a call](#)

[to max\(int, int\). It then instantiates a version of the function where the parameterizing](#)

[type T is int, making the equivalent of the following function:](#)

```
int max(int x, int y)
{
return x < y ? y : x;
}
```

the C++ Standard Template Library contains the function template `max(x, y)` which creates functions that return either `x` or `y`, whichever is larger. `max()` could be defined like

this:

```
template <typename T>
T max(T x, T y)
{
return x < y ? y : x;
}
```

Question No: 34 (Marks: 2)

State any conflict that may rise due to multiple inheritance?

The conflict may arise is the diamond problem, which our author likes to call the “diamond of doom”. This occurs when a class multiply inherits from two classes which

each inherit from a single base class. This leads to a diamond shaped inheritance pattern.

For example, consider the following set of classes:

```
class PoweredDevice
{
};
class Scanner: public PoweredDevice
{
};
class Printer: public PoweredDevice
{
};
class Copier: public Scanner, public Printer
{
};
```

Scanners and printers are both powered devices, so they derived from `PoweredDevice`.

However, a copy machine incorporates the functionality of both Scanners and Printers.

Ambiguity also cause problem.

Question No: 35 (Marks: 3)

Describe three properties necessary for a container to implement Generic Algorithms.

If you declare a container as holding pointers, you are responsible for managing the

memory for the objects pointed to. The container classes will not automatically free

memory for these objects when an item is erased from the container.

Container classes are expected to implement methods to do the following:

- create a new empty container (constructor),
- report the number of objects it stores (size),
- delete all the objects in the container (clear),
- insert new objects into the container,
- remove objects from it,
- provide access to the stored objects.

Question No: 36 (Marks: 3)

Write three important features of virtual functions.

With virtual functions, derived classes can provide new implementations of functions

from their base classes. When someone calls a virtual function of an object of the derived

class, this new implementation is called, even if the caller uses a pointer to the base class,

and doesn't even know about the particular derived class.

The virtual function is an option, and the language defaults to non virtual, which is the

fastest configuration.

The derived class can completely "override" the implementation or "augment" it (by

explicitly calling the base class implementation in addition to the new things it does).

Question No: 37 (Marks: 3)

Consider the code below,

```
#include <iostream>
#include <stdlib.h>
using namespace std;
class Shape{
```

```

public:
void Draw(){cout<<"shape"<<endl;}
};
class Line : public Shape{
public:
void Draw(){cout<<"Line"<<endl;}
};
class Circle : public Shape{
public:
void Draw(){cout<<"Circle"<<endl;}
};
int main(int argc, char *argv[])
{
Shape * ptr1 = new Shape();
Shape * ptr2 = new Line();
Shape * ptr3 = new Circle();
ptr1->Draw();
ptr2->Draw();
ptr3->Draw();
system("PAUSE");
return 0;
}

```

This code shows output,

Shape

Shape

Shape

Give the reason for this output

Suppose we want to show the output,

Shape

Line

Circle

How we can change the code to do that?

```

class shape { public:
void draw();
};
class circle : public shape { };
int main(int argc, char **argv){
circle my_circle;

```

```
my_circle.draw();
}
```

While this has all the usual advantages, e.g., code reuse, the real power of polymorphism

comes into play when draw is declared to be virtual or pure virtual, as follows:

```
class shape{ public:
virtual void draw()=0;
};
class circle : public shape { public:
void draw();
}
```

Here, circle has declared its own draw function, which can define behavior appropriate

for a circle. Similarly, we could define other classes derived from shape, which provide

their own versions of draw. Now, because all the classes implement the shape interface,

we can create collections of objects that can provide different behavior invoked in a consistent

manner (calling the draw member function). An example of this is shown here.

```
shape *shape_list[3]; // the array that will
// pointer to our shape objects
shape[0] = new shape; // three types of shapes
shape[1] = new line; // we have defined
shape[2] = new circle;
for(int i = 0; i < 3; i++){
shape_list[i].draw();
}
```

When we invoke the draw function for each object on the list, we do not need to know

anything about each object; C++ handles the details of invoking the correct version of

draw. This is a very powerful technique, allowing us to provide extensibility in our designs. Now we can add new classes derived from shape to provide whatever behavior

we desire. The key here is that we have separated the interface (the prototype for shape)

from the implementation.

Question No: 38 (Marks: 5)

There are some errors in the code given below, you have to

1. Indicate the line no. with error/s
2. Give the reason for error/s
3. Correct the error/s.

```

1. #include <iostream> this will be #include <iostream.h>
2. #include <stdlib.h>
3. using namespace std;
4. template <typename T>
5. class MyClass{
6. public:
7. MyClass(){
8. cout<<"This is class1"<<endl;
9. }
10. };
11. template <typename T>
12. class MyClass<int*>{
13. public:
14. MyClass(){
15. cout<<"This is class2"<<endl;
16. }
17. };
18. int main(int argc, char *argv[])
19. {
20. MyClass<int> c1;
21. MyClass<int*> c2;
22. system("PAUSE");
23. return 0;
24. }

```

Question No: 39 (Marks: 5)

Given are two classes A and B. class B is inherited from class A. Write a code snippet(for main function) that polymorphically call the method of class B. Also what changes do you suggest in the given code segment that are required to call the class B method polymorphically.

```

class A
{
public:
void method() { cout<<"A's method \n"; }

```

```
};
class B : public A
{
public:
void method() { cout<<"B's method\n"; }
```

```
};
Ans:
public class Test
{
public class A {}
public class B extends A {}
private void test(A a)
{
System.out.println("test(A)");
}
private void test(B b)
{
System.out.println("test(B)");
}
public static void main(String[] args)
{
Test t = new Test();
A a = t.new A();
A b = t.new B();
t.test(a);
t.test(b);
}
}
```

Question No: 40 (Marks: 10)

Create built-in STL (Standard Template Library) vector class object for strings and add in it some words by taking input from user, then apply the sort() algorithm to array of words stored in this vector class object.

Hint: Use push_back() to add the words in vector class object, and the [] operator and size() to display these sorted words.

The STL is the containers, iterators and algorithms component of the proposed C++

Standard Library [ANSI95]. It represents a novel application of principles which have

their roots in styles of programming other than Object-orientation.

```
void listWords(istream& in, ostream& out)
```

```
{
string s;
while (!in.eof() && in >> s) {
add s to some container
}
sort the strings in the container
remove the duplicates
for (each string t in container) {
out << t;
}
}
```

For now, assume that a word is defined as a whitespace-separated string as delivered by

the stream extraction operator. Later on we will consider ways of refining this definition.

Given the way this problem is expressed, we can implement this program directly, if

naïvely. The STL container class `vector` will suffice to hold the words: applying the

algorithms `sort` and `unique` provides the required result.

```
void listWords(istream& in, ostream& out)
{
string s;
vector<string> v;
while (!in.eof() && in >> s)
v.push_back(s); // (1)
sort(v.begin(), v.end());
vector<string>::iterator e
= unique(v.begin(), v.end()); // (2)
for (vector<string>::iterator b = v.begin();
b != e;
b++) {
out << *b << endl;
}
}
```

At (1) the vector member function `push_back()` is used to add to the end of the vector.

This can also be done using the `insert` member, which takes as a parameter an iterator

identifying the position in the vector at which to place the added element:

```
v.insert(v.end(), s);
```

This allows us to add at any position in the vector. Be aware, though, that adding anywhere other than the end implies the overhead of physically shifting all elements from

the insertion point to the end to make room for the new value. For this reason, and given

the choices made in this example, attempts to optimise this code by maintaining the

vector in sorted order are unwise. Replace vector with list and this becomes possible -

although in both cases a search over the container will be necessary to determine the

correct position of insertion.

The unique algorithm has the surprising property of not changing the length of the container to which it is applied (it can hardly do this, as it has access not to the underlying container, but only to the pair of iterators it is passed). Instead, it guarantees

that duplicates are removed by moving unique entries towards the beginning of the container, returning an iterator indicating the new end of the container. This can be used

directly (as here, at (2)), conversely it can be passed to the `erase` member with the old end

iterator, to truncate the container.

Question No: 41 (Marks: 10)

Q. Write a detailed note on Exceptions in Destructors with the help of a coding example.

Exceptions in Destructors:

An object is presumably created to do something. Some of the changes made by an object

should persist after an object dies (is destructed) and some changes should not.

Take an

object implementing a SQL query. If a database field is updated via the SQL object then

that change should persist after the SQL objects dies. To do its work the SQL object probably created a database connection and allocated a bunch of memory. When the SQL object dies we want to close the database connection and deallocate the memory, otherwise if a lot of SQL objects are created we will run out of database connections and/or memory.

The logic might look like:

```
Sql::~Sql()
```

```
{
delete connection;
delete buffer;
}
```

Let's say an exception is thrown while deleting the database connection. Will the buffer

be deleted? No. Exceptions are basically non-local gotos with stack cleanup. The code for

deleting the buffer will never be executed creating a gaping resource leak.

Special care must be taken to catch exceptions which may occur during object destruction. Special care must also be taken to fully destruct an object when it throws an exception.

Example code for exception

```
#include<iostream.h>
```

```
#include<conio.c>
```

```
class Exception {
```

```
private:
```

```
char message[30] ;
```

```
public:
```

```
Exception() {strcpy(message,"There is not enough stock");}
```

```
char * get_message() { return message; }
```

```
};
```

```
class Item {
```

```
private:
```

```
int stock ;
```

```
int required_quantity;
```

```
public:
```



```

Item(int stk, int qty)
{
    stock = stk;
    required_quantity = qty;
}
int get_stock()
{
    return stock;
}
int get_required_quantity()
{
    return required_quantity;
}
void order()
{
    if (get_stock() < get_required_quantity())
        throw Exception();
    else
        cout << "The required quantity of item is available in the stock";
}
~Item(){}
};
void main()
{
    Item obj(10, 20);
    try

    {
        obj.order();
    }
    catch(Exception & exp2)
    {
        getch();
        cout << "Exception: " << exp2.get_message() << endl;
    }
    getch();
}

```

FINAL TERM EXAMINATION

Fall 2009

CS304- Object Oriented Programming (Session - 4)

Ref No: 1130772

Time: 120 min

Marks: 75

Question No: 1 (Marks: 1) - Please choose one

A template provides a convenient way to make a family of

- ☒ variables and data members
- ☐ functions and classes
- ☐ classes and exceptions
- ☐ programs and algorithms

Question No: 2 (Marks: 1) - Please choose one

Which one of the following terms must relate to polymorphism?

- ☐ Static allocation
- ☐ Static typing
- ☒ Dynamic binding
- ☐ Dynamic allocation

Question No: 3 (Marks: 1) - Please choose one

What is true about function templates?

- ☐ The compiler generates only one copy of the function template
- ☒ The compiler generates a copy of function respective to each type of data
- ☐ The compiler can only generate copy for the int type data
- ☐ None of the given.

Question No: 4 (Marks: 1) - Please choose one

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ☒ Templates
- ☐ Overloading
- ☐ Data hiding
- ☐ Encapsulation

Question No: 5 (Marks: 1) - Please choose one
template <>

```
class Vector<char*> { }
```

This is an example of partial specialization.

- ☐ True
- ☒ False

Question No: 6 (Marks: 1) - Please choose one

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be

concrete, while classes like Sphere and Cube would normally be abstract.

☒ True

☐ False

Question No: 7 (Marks: 1) - Please choose one

A non-virtual member function is defined in a base class and overridden in a derived class; if that function is called through a base-class pointer to a derived class object, the derived-class version is used.

☒ True

☐ False

Question No: 8 (Marks: 1) - Please choose one

Assume a class Derv that is privately derived from class Base. An object of class Derv

located in main() can access

☐ public members of Derv.

☒ protected members of Derv.

☐ private members of Derv.

☐ protected members of Base.

Question No: 9 (Marks: 1) - Please choose one

In order to define a class template, the first line of definition must be:

☒ `template <typename T>`

☐ `typename <template T>`

☐ `Template Class <ClassName>`

☐ `Class <Template T>`

Question No: 10 (Marks: 1) - Please choose one

If there is a pointer p to objects of a base class, and it contains the address of an object of

a derived class, and both classes contain a nonvirtual member function, ding(), then the

statement `p->ding();` will cause the version of ding() in the _____ class to be executed.

☐ Base

☐ Derived

☐ Abstract

☒ virtual

Question No: 11 (Marks: 1) - Please choose one

When the base class and the derived class have a member function with the same name,

you must be more specific which function you want to call (using _____).

☐ scope resolution operator

☐ dot operator

☒ null operator

☐ Operator overloading

Question No: 12 (Marks: 1) - Please choose one

Non Template Friend functions of a class are friends of _____instance/s of that class.

☐ All

☒ One specific

☐ All instances of one date type

☐ None of the given options

Question No: 13 (Marks: 1) - Please choose one

The find() algorithm

☐ finds matching sequences of elements in two containers.

☒ finds a container that matches a specified container.

☐ takes iterators as its first two arguments.

☐ takes container elements as its first two arguments.

Question No: 14 (Marks: 1) - Please choose one

If you define a vector v with the default constructor, and define another vector w with a

one-argument constructor to a size of 11, and insert 3 elements into each of these vectors

with push_back(), then the size() member function will return _____ for v and _____ for

w.

☐ 11 for v and 3 for w.

☐ 0 for v and 0 for w.

☐ 0 for v and 3 for w.

☒ 3 for v and 11 for w.

Question No: 15 (Marks: 1) - Please choose one

Which of the following may not be an integral part of an object?

☐ State

☐ Behavior

☐ Protected data members

☒ All of given

Question No: 16 (Marks: 1) - Please choose one
Which is not the Advantage of inheritance?

- _ providing class growth through natural selection.
- _ **facilitating class libraries.**
- _ avoiding the rewriting of code.
- _ providing a useful conceptual framework.

Question No: 17 (Marks: 1) - Please choose one
class DocElement

```
{
public:
virtual void Print() { cout << "Generic element"; }
};
class Heading : public DocElement
{
public:
void Print() { cout << "Heading element"; }
};
class Paragraph : public DocElement
{
public:
void Print() { cout << "Paragraph element"; }
};
```

```
void main()
{
DocElement * p = new Paragraph();
p->Print();
}
```

When you run this program, it will print out a single line to the console output.
What will be in that line?

Select one correct answer from the following list:

- _ Generic element
- _ Heading element
- _ **Paragraph element**
- _ Nothing will be printed.

Question No: 18 (Marks: 1) - Please choose one

When a virtual function is called by referencing a specific object by name and using the

dot member selection operator (e.g., squareObject.draw()), the reference is resolved at compile time.

☒ True

☐ False

Question No: 19 (Marks: 1) - Please choose one

In case of multiple inheritance a derived class inherits,

☐ Only the public member functions of its base classes

☐ Only the public data members of its base classes

☒ Both public data members and member functions of all its base classes

☐ Data members and member functions of any two base classes

Question No: 20 (Marks: 1) - Please choose one

When we write a class template the first line must be:

☐ template < class class_name >

☐ template < class data_type >

☒ template < class T >

Here T can be replaced with any name but it is preferable.

☐ class class-name()

☐ class template<class_name>

Question No: 21 (Marks: 1) - Please choose one

Which of the following is incorrect line regarding function template?

☐ template<class T>

☐ template <typename U>

☒ Class<template T>

☐ template < class T, class U>

Question No: 22 (Marks: 1) - Please choose one

An STL container can not be used to,

☒ hold objects of class employee.

☐ store elements in a way that makes them quickly accessible.

☐ compile c++ programs.

☐ organize the way objects are stored in memory

Question No: 23 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

☒ True

☐ False

Question No: 24 (Marks: 1) - Please choose one

Consider a class named Vehicle, which of the following can be the instance of class

Vehicle?

1. Car
2. Computer
3. Desk
4. Ahmed
5. Bicycle
6. Truck

- _ 1, 4, 5
- _ 2, 5, 6
- _ 1, 2, 3, 6
- _ 1, 5, 6**

Question No: 25 (Marks: 1) - Please choose one

Consider the code below,

```
class Fred {
public:
    Fred();
    ...
};
int main()
{
    Fred a[10];
    Fred* p = new Fred[10];
    ...
}
```

Select the best option,

- _ Fred a[10]; calls the default constructor 09 times
- Fred* p = new Fred[10]; calls the default constructor 10 times
- _ Produce an error
- _ Fred a[10]; calls the default constructor 11 times**
- Fred* p = new Fred[10]; calls the default constructor 11 times**
- _ Fred a[10]; calls the default constructor 10 times
- Fred* p = new Fred[10]; calls the default constructor 10 times

Question No: 26 (Marks: 1) - Please choose one

When a variable is define as static in a class then all object of this class,

- _ Have different copies of this variable
- _ Have same copy of this variable**
- _ Can not access this variable
- _ None of given

Question No: 27 (Marks: 1) - Please choose one

The life of sub object is dependant on the life of master class in _____.

- ☐ Separation
- ☒ Composition
- ☐ Aggregation
- ☐ None of the given

Question No: 28 (Marks: 1) - Please choose one

_____, which means if A declares B as its friend it does NOT mean that A can

access private data of B. It only means that B can access all data of A.

- ☒ Friendship is one way only
- ☐ Friendship is two way only
- ☐ NO Friendship between classes
- ☐ Any kind of friendship

Question No: 29 (Marks: 1) - Please choose one

Which of the following operators always takes no argument if overloaded?

- ☐ /
- ☐ -
- ☐ +
- ☒ ++

Question No: 30 (Marks: 1) - Please choose one

In Private ----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ☒ specialization
- ☐ inheritance
- ☐ abstraction
- ☐ composition

Question No: 31 (Marks: 1)

Write the syntax of declaring a pure virtual function in a class?

Ans:

Pure Virtual Function is a Virtual function with no body.

Declaration of Pure Virtual Function:

Since pure virtual function has no body, the programmer must add the notation =0 for

declaration of the pure virtual function in the base class.

General Syntax of Pure Virtual Function takes the form:

class classname //This denotes the base class of C++ virtual function


```
{
public:
virtual void virtualfunctionname() = 0 //This denotes the pure virtual function in
C++
};
```

Question No: 32 (Marks: 1)

What is meant by direct base class ?

Ans

When a class-type is included in the class-base, it specifies the direct base class of the

class being declared. If a class declaration has no class-base, or if the class-base lists only

interface types, the direct base class is assumed to be object. A class inherits members

from its direct base class,

Deriving a class from more than one direct base class is called multiple inheritance.

Question No: 33 (Marks: 2)

Describe the way to declare a template class as a friend class of any other class.

Ans

The following example is use of a class template:

```
template<class L> class Key
{
L k;
L* kptr;
int length;
public:
Key(L);
// ...
};
```

Suppose the following declarations appear later:

```
Key<int> i;
```

```
Key<char*> c;
```

```
Key<mytype> m;
```

The compiler would create three objects.

Question No: 34 (Marks: 2)

What is the purpose of template parameter?

Ans:

There are three kinds of template parameters:

- type
- non-type
- template

You can interchange the keywords `class` and `typename` in a template parameter declaration. You cannot use storage class specifiers (`static` and `auto`) in a template parameter declaration.

Question No: 35 (Marks: 3)

Describe in simple words how we can use template specialization to enforce case sensitive specialization in `String` class.

Ans”

The act of creating a new definition of a function, class, or member of a class from a

template declaration and one or more template arguments is called template instantiation.

The definition created from a template instantiation is called a specialization. A primary

template is the template that is being specialized.

create function objects to do the case-insensitive compares, and then reuse them when also wanting to do case-insensitive sorting or searching.

Question No: 36 (Marks: 3)

Can we use compiler generated default assignment operator in case our class is using dynamic memory? Justify your answer.

Ans:

the compiler does not make a separate copy of the object. Even if the types are not the

same, the compiler is usually able to do a better job with initialization lists than with

assignments.

Consider the following constructor that initializes member object `x_` using an initialization list: `square::square() : x_(whatever) { }`. The most common benefit of doing

this is improved performance. For example, if the expression `whatever` is the same type

as member variable `x_`, the result of the `whatever` expression is constructed directly inside

x_ — the compiler does not make a separate copy of the object. Even if the types are not the same, the compiler is usually able to do a better job with initialization lists than with assignments.

As if that wasn't bad enough, there's another source of inefficiency when using assignment in a constructor: the member object will get fully constructed by its default constructor, and this might, for example, allocate some default amount of memory or open some default file. All this work could be for naught if the whatever expression and/or assignment operator causes the object to close that file and/or release that memory (e.g., if the default constructor didn't allocate a large enough pool of memory or if it opened the wrong file).

Question No: 37 (Marks: 3)

Give the names of three ways to handle errors in a program.

Ans:

The function will throw DivideByZero as an exception that can then be caught by an exception-handling catch statement that catches exceptions of type int. The necessary construction for catching exceptions is a try catch system. If you wish to have your program check for exceptions, you must enclose the code that may have exceptions thrown in a try block.

The catch statement catches exceptions that are of the proper type. You can, for example, throw objects of a class to differentiate between several different exceptions. As well, once a catch statement is executed, the program continues to run from the end of the catch.

the errors can be handled outside of the regular code. This means that it is easier to structure the program code, and it makes dealing with errors more centralized.

Finally, because the exception is passed back up the stack of calling functions, you can handle errors at any place you choose.

Question No: 38 (Marks: 5)
Consider the following code,

```
class Base{
private:
void base1();
protected:
void base2();
public:
void base3();
};
class Derived: public Base{
private:
void derived1();
protected:
void derived2();
public:
void derived3();
};
int main(){
Derived * derived = new Derived();
return 0;
}
```

Fill the table below to tell which member functions of Base and Derived classes we can access using the Derived pointer in the code indicated in bold.

Ans:

Function Name **Availability (Yes / No)?**

base2() **no**

base3() **yes**

derived1() **No**

derived2() **No**

derived3() **Yes**

Question No: 39 (Marks: 5)

What is the output produced by the following program?

```
#include<iostream.h>
```

```
void sample_function(double test) throw (int);
```

```
int main()
```

```
{
```

```

try
{
cout << "Trying.\n";
sample_function(98.6);
cout << "Trying after call.\n";
}
catch(int)
{
cout << "Catching.\n";
}
cout << "End program.\n";
return 0;
}
void sample_function(double test) throw (int)
{
cout << "Starting sample_function.\n";
if(test < 100)
throw 42;
}

```

Ans:

Starting sample_function

Trying

Trying after call

Catching

End program

Question No: 40 (Marks: 10)

Write a publicly derived class “Employee” that is derived from base class named “Company”. Both classes will have function “create()”. Make virtual function of base class and override same function in derived class. Function create will have an

output statement of your own choice.

In “main” Create an object of base class and call both functions with same object type.

Question No: 41 (Marks: 10)

Write a program in C++ which creates three classes named as

1. Equation
2. Linear
3. Quadratic

Where Linear and Quadratic are inherited from Equation

Each class has the method Graph. Graph method should be pure virtual in Equation class.

This method should be overridden in both the inherited classes. It is meant to display the

Graph shape of its respective class. Graph method of Linear will display the message;

Straight line

Similarly, the Graph method of Quadratic will display the message;

Parabola

In main, call the Graph method of both the Linear and Quadratic equations polymorphically through the parent class (Equation).

Ans:

```
#include "fraction.h"
#include <iostream>
#include <string>
#include <string.h>
#include <stdlib.h>
class equation;
class equation {
int a, b;
public:
int c ()
{return (c);}
void convert (Cequation);
};
class linear {
private:
int side;
public:
void set_side (int a)
{side=a;}
friend class equation;
};

void equation::convert (Cequation) {
a = 23;
b = 45;
}
int main () {
```

```
cequation sqr;  
CRectangle rect;  
sqr.set_side(4);  
rect.convert(sqr);  
cout << rect.area();  
return 0;  
}
```

FINAL TERM EXAMINATION

CS304- Object Oriented Programming (Session - 1)

3

Question No: 1 (Marks: 1) - Please choose one

Which one of the following terms must relate to **polymorphism**?

- ▶ Static allocation
- ▶ Static typing
- ▶ **Dynamic binding**
- ▶ Dynamic allocation

Question No: 2 (Marks: 1) - Please choose one

Multiple inheritance can be of type

- ▶ Public
- ▶ Private
- ▶ Protected
- ▶ **All of the given**

Question No: 3 (Marks: 1) - Please choose one

When a subclass specifies an alternative definition for an attribute or method of its superclass, it is _____ the definition in the superclass.

- ▶ overload
- ▶ **overriding**
- ▶ copy riding
- ▶ none of given

Question No: 4 (Marks: 1) - Please choose one

Like template functions, a class template may not handle all the types successfully.

- ▶ **True**
- ▶ False

Question No: 5 (Marks: 1) - Please choose one

It is sometimes useful to specify a class from which no objects will ever be created.

- ▶ True

► **False**

Question No: 6 (Marks: 1) - Please choose one

Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- public members of Derv.
- protected members of Derv.
- private members of Derv.
- **protected members of Base.**

Question No: 7 (Marks: 1) - Please choose one

_____ A
pointer to a base class can point to objects of a derived class.

► **True**

► False

Question No: 8 (Marks: 1) - Please choose one

_____ A
copy constructor is invoked when

- a function do not returns by value.
- an argument is passed by value.
- **a function returns by reference.**
- an argument is passed by reference.

Question No: 9 (Marks: 1) - Please choose one

_____ A
function call is resolved at run-time in_____

- ▶ non-virtual member function.
- ▶ **virtual member function.**
- ▶ Both non-virtual member and virtual member function.
- ▶ None of given

Question No: 10 (Marks: 1) - Please choose one

When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using _____).

- ▶ **scope resolution operator**
- ▶ dot operator
- ▶ null operator
- ▶ Operator overloading

Question No: 11 (Marks: 1) - Please choose one

Each try block can have _____ no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ **As many as necessary.**

Question No: 12 (Marks: 1) - Please choose one

Two important STL associative containers are _____ and _____.

- ▶ set,map
- ▶ **sequence,mapping**
- ▶ setmet,multipule
- ▶ sit,mat

Question No: 13 (Marks: 1) - Please choose one

The mechanism of selecting function at run time according to the nature of calling object is called,

- ▶ late binding
- ▶ static binding
- ▶ virtual binding
- ▶ **None of the given options**

Question No: 14 (Marks: 1) - Please choose one

abstract class is useful when, An

- ▶ We do not derive any class from it.

- ▶ There are multiple paths from one derived class to another.
- ▶ **We do not want to instantiate its object.**
- ▶ You want to defer the declaration of the class.

Question No: 15 (Marks: 1) - Please choose one

Which of the following is incorrect line regarding function template?

- ▶ `template<class T>`
- ▶ `template <typename U>`
- ▶ **`Class<template T>`**
- ▶ `template < class T, class U>`

Question No: 16 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ **All of given**

Question No: 17 (Marks: 1) - Please choose one

_____ By
default the vector data items are initialized to _____

- ▶ **0**
- ▶ 0.0
- ▶ 1
- ▶ null

Question No: 18 (Marks: 1) - Please choose one

Which one of the following functions returns the total number of elements in a vector.

- ▶ length();
- ▶ **size();**
- ▶ ele();
- ▶ veclen();

Question No: 19 (Marks: 1) - Please choose one

Suppose you create an uninitialized vector as follows:

```
vector<int> evec;
```

After adding the statment,

```
evec.push_back(21);
```

what will happen?

- ▶ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.
- ▶ The following statement will add an element to the center of evec and will reinitialize it with the value 21.
- ▶ The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.
- ▶ **The following statement will add an element to the end (the back) of evec and initialize it with the value 21.**

Question No: 20 (Marks: 1) - Please choose one

STL container can not be used to, An

- ▶ hold objects of class employee.
- ▶ store elements in a way that makes them quickly accessible.
- ▶ **compile c++ programs.**
- ▶ organize the way objects are stored in memory

Question No: 21 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

- ▶ True
- ▶ **False**

Question No: 22 (Marks: 1) - Please choose one

The main function of scope resolution operator (::) is,

- ▶ **To define an object**
- ▶ To define a data member
- ▶ To link the definition of an identifier to its declaration
- ▶ To make a class private

Question No: 23 (Marks: 1) - Please choose one

When is a constructor called?

- ▶ Each time the constructor identifier is used in a program statement
- ▶ **During the instantiation of a new object**
- ▶ During the construction of a new class
- ▶ At the beginning of any program execution

Question No: 24 (Marks: 1) - Please choose one

Consider the code below,

```
class Fred {  
public:  
Fred();  
...  
};  
int main()  
{  
Fred a[10];  
Fred* p = new Fred[10];  
...  
}
```

Select the best option,

► Fred a[10]; calls the default constructor 09 times

Fred* p = new Fred[10]; calls the default constructor 10 times

► **Produce an error**

► Fred a[10]; calls the default constructor 11 times

Fred* p = new Fred[10]; calls the default constructor 11 times

► Fred a[10]; calls the default constructor 10 times

Fred* p = new Fred[10]; calls the default constructor 10 times

Question No: 25 (Marks: 1) - Please choose one

Associativity can be changed in operator overloading.

► True

► **False**

Question No: 26 (Marks: 1) - Please choose one

A normal C++ operator that acts in special ways on newly defined data types is said to be

- ▶ glorified.
- ▶ encapsulated.
- ▶ **classified.**
- ▶ overloaded.

Question No: 27 (Marks: 1) - Please choose one

Which operator can not be overloaded?

- ▶ The relation operator ($\geq$)
- ▶ Assignment operator (=)
- ▶ Script operator ([])
- ▶ **Conditional operator (? :)**

Question No: 28 (Marks: 1) - Please choose one

Suppose obj1 and obj2 are two objects of a user defined class A. An + operator is overloaded to add obj1 and obj2 using the function call obj1+obj2.

Identify the correct function prototype against the given call?

- ▶ A operator + (A &obj);
- ▶ int + operator();
- ▶ **int operator (plus) ();**
- ▶ A operator(A &obj3);

Question No: 29 (Marks: 1) - Please choose one

Default constructor is such constructor which either has no -----or if it has some parameters these have ----- values

- ▶ Parameter, temporary
- ▶ Null, Parameter
- ▶ **Parameter, default**
- ▶ non of the given

Question No: 30 (Marks: 1) - Please choose one

Public methods of base class can ----- be accessed in its derived class

- ▶ directly
- ▶ **indirectly**
- ▶ simultaneously
- ▶ non of the given

Question No: 31 (Marks: 1)

Is **Deque** a Birectional Container?

Yes, deque behaves like queue (line) such that we can add elements on both sides of it.

Question No: 32 (Marks: 1)

What is meant by Generic Programming?

Generic programming refers to programs containing generic abstractions general code that is same in logic for all data types like printArray function), then we instantiate that generic program abstraction (function, class) for a particular data type, such abstractions can work with many different types of data.

Question No: 33 (Marks: 2)

Sort the following data in the order in which compiler searches a function?

Complete Specialization, Generic Template, Partial Specialization, Ordinary Function.

Specializations of this function template, instantiations with specific types, can be called just like an ordinary function:

```
cout << max(3, 7); // outputs 7
```

The compiler examines the arguments used to call max and determines that this is a call to max(int, int). It then instantiates a version of the function where the parameterizing type T is int, making the equivalent of the following function:

```
int max(int x, int y)
{
    return x < y ? y : x;
}
```

the C++ Standard Template Library contains the function template max(x, y) which creates functions that return either x or y, whichever is larger. max() could be defined like this:

```
template <typename T>
T max(T x, T y)
{
    return x < y ? y : x;
}
```

Question No: 34 (Marks: 2)

State any conflict that may rise due to multiple inheritance?

The conflict may arise is the diamond problem, which our author likes to call the “diamond of doom”. This occurs when a class multiply inherits from two classes

which each inherit from a single base class. This leads to a diamond shaped inheritance pattern.

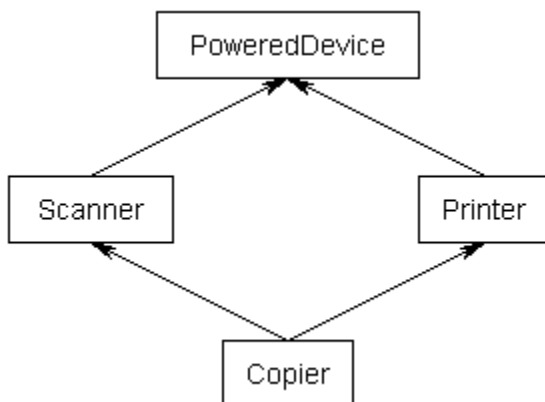
For example, consider the following set of classes:

```
class PoweredDevice
{
};

class Scanner: public PoweredDevice
{
};

class Printer: public PoweredDevice
{
};

class Copier: public Scanner, public Printer
{
};
```



Scanners and printers are both powered devices, so they derived from PoweredDevice. However, a copy machine incorporates the functionality of both Scanners and Printers.

Ambiguity also cause problem.

Question No: 35 (Marks: 3)

Describe three properties necessary for a container to implement Generic Algorithms.

If you declare a container as holding pointers, you are responsible for managing the memory for the objects pointed to. The container classes will not automatically free memory for these objects when an item is erased from the container.

Container classes are expected to implement methods to do the following:

- create a new empty container (constructor),
- report the number of objects it stores (size),
- delete all the objects in the container (clear),
- insert new objects into the container,
- remove objects from it,
- provide access to the stored objects.

Question No: 36 (Marks: 3)

Write three important features of virtual functions.

With virtual functions, derived classes can provide new implementations of functions from their base classes. When someone calls a virtual function of an object of the derived class, this new implementation is called, even if the caller uses a pointer to the base class, and doesn't even know about the particular derived class.

The virtual function is an option, and the language defaults to non virtual, which is the fastest configuration.

The derived class can completely "override" the implementation or "augment" it (by explicitly calling the base class implementation in addition to the new things it does).

Question No: 37 (Marks: 3)

Consider the code below,

```
#include <iostream>

#include <stdlib.h>

using namespace std;

class Shape{
    public:
        void Draw(){cout<<"shape"<<endl;}
};

class Line : public Shape{
    public:
        void Draw(){cout<<"Line"<<endl;}
};
```

```
class Circle : public Shape{
    public:
    void Draw(){cout<<"Circle"<<endl;}
};

int main(int argc, char *argv[])
{
    Shape * ptr1 = new Shape();
    Shape * ptr2 = new Line();
    Shape * ptr3 = new Circle();

    ptr1->Draw();
    ptr2->Draw();
    ptr3->Draw();
    system("PAUSE");
    return 0;
}
```

This code shows output,

Shape

Shape

Shape

Give the reason for this output

Suppose we want to show the output,

Shape

Line

Circle

How we can change the code to do that?

```
class shape { public:
    void draw();
};
class circle : public shape { };
int main(int argc, char **argv){
    circle my_circle;
    my_circle.draw();
}
```

While this has all the usual advantages, e.g., code reuse, the real power of polymorphism comes into play when draw is declared to be virtual or pure virtual, as follows:

```
class shape{ public:
    virtual void draw()=0;
};
class circle : public shape { public:
    void draw();
}
```

Here, circle has declared its own draw function, which can define behavior appropriate for a circle. Similarly, we could define other classes derived from shape, which provide their own versions of draw. Now, because all the classes implement the shape interface, we can create collections of objects that can provide different behavior invoked in a consistent manner (calling the draw member

function). An example of this is shown here.

```
shape *shape_list[3]; // the array that will
                      // pointer to our shape objects
shape[0] = new shape; // three types of shapes
shape[1] = new line;  // we have defined
shape[2] = new circle;
for(int i = 0; i < 3; i++){
    shape_list[i].draw();
}
```

When we invoke the draw function for each object on the list, we do not need to know anything about each object; C++ handles the details of invoking the correct version of draw. This is a very powerful technique, allowing us to provide extensibility in our designs. Now we can add new classes derived from shape to provide whatever behavior we desire. The key here is that we have separated the interface (the prototype for shape) from the implementation.

Question No: 38 (Marks: 5)

There are some errors in the code given below, you have to

4. Indicate the line no. with error/s
5. Give the reason for error/s
6. Correct the error/s.

```
25.#include <iostream>           this will be #include <iostream.h>
26.#include <stdlib.h>
```

```
27.using namespace std;
28.template <typename T>
29.class MyClass{
30.public:
31.MyClass(){
32.cout<<"This is class1"<<endl;
```

```

33.}
34.};
35.template <typename T>
36.class MyClass<int*>{
37.public:
38.MyClass(){
39.cout<<"This is class2"<<endl;
40.}
41.};
42.int main(int argc, char *argv[])
43.{
44.MyClass<int> c1;
45.MyClass<int*> c2;
46.system("PAUSE");
47.return 0;
48.}

```

Question No: 39 (Marks: 5)

Given are two classes A and B. class B is inherited from class A. Write a code snippet(for main function) that polymorphically call the method of class B. Also what changes do you suggest in the given code segment that are required to call the class B method polymorphically.

```

class A
{
public:
void method() { cout<<"A's method \n"; }

};

```



```
class B : public A
{

public:
void method() { cout<<"B's method\n"; }

};
```

Ans:

```
public class Test
{
public class A {}

public class B extends A {}

private void test(A a)
{
System.out.println("test(A)");
}

private void test(B b)
{
System.out.println("test(B)");
}

public static void main(String[] args)
```

```

{
Test t = new Test();
A a = t.new A();
A b = t.new B();

t.test(a);
t.test(b);
}
}

```

Question No: 40 (Marks: 10)

Create built-in STL (Standard Template Library) **vector class object** for **strings** and add in it some words by taking input from user, then apply the `sort()` algorithm to array of words stored in this vector class object.

Hint: Use `push_back()` to add the words in vector class object, and the `[]` operator and `size()` to display these sorted words.

The STL is the containers, iterators and algorithms component of the proposed C++ Standard Library [ANSI95]. It represents a novel application of principles which have their roots in styles of programming other than Object-orientation.

```
void listWords(istream& in, ostream& out)
```

```

{
    string s;

    while (!in.eof() && in >> s) {
        add s to some container
    }

    sort the strings in the container
    remove the duplicates
}

```

```

        for (each string t in container) {
            out << t;
        }
    }

```

For now, assume that a word is defined as a whitespace-separated string as delivered by the stream extraction operator. Later on we will consider ways of refining this definition.

Given the way this problem is expressed, we can implement this program directly, if naïvely. The STL container class `vector` will suffice to hold the words: applying the algorithms `sort` and `unique` provides the required result.

```

void listWords(istream& in, ostream& out)
{

```

```

    string s;
    vector<string> v;

```

```

    while (!in.eof() && in >> s)
        v.push_back(s);           // (1)

```

```

    sort(v.begin(), v.end());

```

```

    vector<string>::iterator e
        = unique(v.begin(), v.end()); // (2)

```

```

    for (vector<string>::iterator b = v.begin();
        b != e;
        b++) {
        out << *b << endl;
    }

```

```

}

```

At (1) the vector member function `push_back()` is used to add to the end of the vector. This can also be done using the `insert` member, which takes as a parameter an iterator identifying the position in the vector at which to place the added element:

```

    v.insert(v.end(), s);

```

This allows us to add at any position in the vector. Be aware, though, that adding anywhere other than the end implies the overhead of physically shifting all elements from the insertion point to the end to make room for the new value. For this reason, and given the choices made in this example, attempts to optimise this code by maintaining the vector in sorted order are unwise. Replace `vector` with `list`

and this becomes possible - although in both cases a search over the container will be necessary to determine the correct position of insertion.

The unique algorithm has the surprising property of not changing the length of the container to which it is applied (it can hardly do this, as it has access not to the underlying container, but only to the pair of iterators it is passed). Instead, it guarantees that duplicates are removed by moving unique entries towards the beginning of the container, returning an iterator indicating the new end of the container. This can be used directly (as here, at (2)), conversely it can be passed to the erase member with the old end iterator, to truncate the container.

Question No: 41 (Marks: 10)

Q.

Write a detailed note on Exceptions in Destructors with the help of a coding example.

Exceptions in Destructors:

An object is presumably created to do something. Some of the changes made by an object should persist after an object dies (is destructed) and some changes should not. Take an object implementing a SQL query. If a database field is updated via the SQL object then that change should persist after the SQL objects dies. To do its work the SQL object probably created a database connection and allocated a bunch of memory. When the SQL object dies we want to close the database connection and deallocate the memory, otherwise if a lot of SQL objects are created we will run out of database connections and/or memory.

The logic might look like:

```
Sql::~Sql()
{
    delete connection;
    delete buffer;
}
```

Let's say an exception is thrown while deleting the database connection. Will the buffer be deleted? No. Exceptions are basically non-local gotos with stack cleanup. The code for deleting the buffer will never be executed creating a gaping resource leak.

Special care must be taken to catch exceptions which may occur during object destruction. Special care must also be taken to fully destruct an object when it throws an exception.

Example code for exception

```
#include<iostream.h>
#include<conio.c>
class Exception {
private:

    char message[30] ;
public:

    Exception() {strcpy(message,"There is not enough stock");}
    char * get_message() { return message; }
};
class Item {
private:

    int stock ;
    int required_quantity;
public:

    Item(int stk, int qty)
    {
        stock = stk;
        required_quantity = qty;
    }
    int get_stock()
    {
        return stock;
    }
};
```

```
int get_required_quantity()
{
    return required_quantity;
}

void order()
{
    if (get_stock() < get_required_quantity())

        throw Exception();
    else
        cout << "The required quantity of item is available in the stock";
}

~Item(){}
};

void main()
{
    Item obj(10, 20);

    try
    {
        obj.order();
    }
    catch(Exception & exp2)
    {
        getch();
        cout << "Exception: " << exp2.get_message() << endl;
    }
    getch();
}
```

Class is not a mechanism to create objects and define user data types.

true

false

Memory is allocated to non static members only, when:

Object is created

The sub-object's life is not dependent on the life of master class in _____.

Aggregation

Unary operators and assignment operator are right associative.

true

The $\geq$ operator can't be overloaded.

false

_____ is creating objects of one class inside another class.

Composition

If we are create array of objects through new operator, then

None of the given

Object can be declared constant with the use of Constant keyword.

false

_____ Operator will take only one operand.

none of the given

Which of the following operator(s) take(s) one or no argument if overloaded?

All of the given choices

this pointer does not pass implicitly to _____ functions.

Static Member

Operator overloading is

giving C++ operators more than they can handle.

Date: 08-11-2010

Question # 1 of 10

Information hiding can be achieved through_____.

Encapsulation, Abstraction

Question # 2 of 10 (Start time: 01:11:21 AM) Total Marks: 1

A good model is related to a real life problem.

Select correct option:

Closely

Question # 3 of 10 (Start time: 01:12:33 AM) Total Marks: 1

Which of the following features of OOP is used to derive a class from another?

Select correct option:

Inheritance

Question # 4 of 10 (Start time: 01:13:51 AM) Total Marks: 1

Which of the following is a weak relationship between two objects?

Select correct option:

Aggregation

Question # 5 of 10 (Start time: 01:14:56 AM) Total Marks: 1

Data items in a class must be private.

Select correct option:

False

Question # 6 of 10 (Start time: 01:15:52 AM) Total Marks: 1

Which one is a class association

Select correct option:

Inheritance

Question # 7 of 10 (Start time: 01:16:55 AM) Total Marks: 1

Suppose there is an object of type Person, which of the following can be considered as one of its attributes

Select correct option:

Both Name and Age

Question # 8 of 10 (Start time: 01:17:52 AM) Total Marks: 1

Which one is not an object association?

Select correct option:

Inheritance

Question # 9 of 10 (Start time: 01:18:50 AM) Total Marks: 1

Using encapsulation we can achieve

Select correct option:

Information hiding

Question # 10 of 10 (Start time: 01:19:43 AM) Total Marks: 1

In constant member function the type of this pointer is:

Select correct option:

CConstant pointer to object

Question # 1 of 10

Which of the following is the way to extract common behavior and attributes from the given classes and make a separate class of those common behaviors and attributes?

Generalization

Question # 2 of 10

The ability to derive a class from more than one class is called

Multiple inheritance

Question # 3 of 10:

If MyClass has a destructor what is the destructor named?

~MyClass

Question # 4 of 10:

Class abc{ ----- }; Is a valid class declaration?

yes

Question # 5 of 10:

Without using Deep copy constructor, A _____ problem can occur

All of the given

Question # 6 of 10:

If only one behaviour of a derived class is incompatible with base class, then it is:

Specialization

Question # 7 of 10:

Which of the following may not be an integral part of an object?

All of given

Question # 8 of 10:

Only tangible things can be chosen as an object.

False

CS304 LATEST 2 QUIZZES SOLVED

Question # 1 of 10

Information hiding can be achieved through_____.

Encapsulation, Inheritance

Encapsulation, Polymorphism

Encapsulation, Abstraction

Overloading

Question # 2 of 10 (Start time: 01:11:21 AM) Total Marks: 1

A good model is related to a real life problem.

Select correct option:

Loosely

Openly

Closely

Question # 3 of 10 (Start time: 01:12:33 AM) Total Marks: 1

Which of the following features of OOP is used to derive a class from another?

Select correct option:

Encapsulation

Polymorphism

Data hiding

Inheritance

Question # 4 of 10 (Start time: 01:13:51 AM) Total Marks: 1

Which of the following is a weak relationship between two objects?

Select correct option:

Inheritance

Composition

Aggregation

None of given

Question # 5 of 10 (Start time: 01:14:56 AM) Total Marks: 1

Data items in a class must be private.

Select correct option:

True

False

Question # 6 of 10 (Start time: 01:15:52 AM) Total Marks: 1

Which one is a class association

Select correct option:

Simple Association

Inheritance

Composition

Aggregation

Question # 7 of 10 (Start time: 01:16:55 AM) Total Marks: 1

Suppose there is an object of type Person, which of the following can be considered as one of its attributes

Select correct option:

Name

Age

Work()

Both Name and Age

Question # 8 of 10 (Start time: 01:17:52 AM) Total Marks: 1

Which one is not an object association?

Select correct option:

Simple association

Inheritance

Aggregation

Association

Question # 9 of 10 (Start time: 01:18:50 AM) Total Marks: 1

Using encapsulation we can achieve

Select correct option:

Information hiding

Least interdependencies among modules

Implementation independence

All of given options

Question # 10 of 10 (Start time: 01:19:43 AM) Total Marks: 1

In constant member function the type of this pointer is:

Select correct option:

Constant pointer

Constant pointer to object

Constant pointer to class

Constant pointer to constant object

Question # 1 of 10

Which of the following is the way to extract common behavior and attributes from the given classes and make a separate class of those common behaviors and attributes?

Generalization

Sub-typing

Specialization

Extension

Question # 2 of 10

The ability to derive a class from more than one class is called

Single inheritance

Encapsulation

Multiple inheritance

Polymorphism

Question # 3 of 10:

If MyClass has a destructor what is the destructor named?

MyClass

~MyClass

My~Class

MyClass~

Question # 4 of 10:

Class abc{ ----- }; Is a valid class declaration?

yes

no

Question # 5 of 10:

Without using Deep copy constructor, A _____ problem can occur

System crash

Memory Leakage

Dangling pointer

All of the given

Question # 6 of 10:

If only one behaviour of a derived class is incompatible with base class, then it is:

Generalization

Specialization

Extension

Inheritance

Question # 7 of 10:

Which of the following may not be an integral part of an object?

state

behavior

Protected data members

All of given

Question # 8 of 10:

Only tangible things can be chosen as an object.

True

False

CS304 TODAYS QUIZ BY SHAZIA 24-11-2010

Class is not a mechanism to create objects and define user data types.

1. true
2. false

Memory is allocated to non static members only, when:

1. Class is created
2. Object is defined
3. Object is initialized
4. Object is created

The sub-object's life is not dependent on the life of master class in _____.

1. Composition
2. Aggregation
3. Separation
4. non of the given

Unary operators and assignment operator are right associative.

1. true
2. false

The >= operator can't be overloaded.

1. true
2. false

_____ is creating objects of one class inside another class.

1. Association
2. Composition
3. Aggregation
4. Inheritance

If we are create array of objects through new operator, then

1. We can call overloaded constructor through new
2. We can't call overloaded constructor through new
3. We can call default constructor through new
4. None of the given

Object can be declared constant with the use of Constant keyword.

1. true
2. false

_____ Operator will take only one operand.

1. New
2. int
3. object
4. none of the given

Which of the following operator(s) take(s) one or no argument if overloaded?

1. ++
2. *
3. %
4. All of the given choices

this pointer does not pass implicitly to _____ functions.

1. Static Member
2. Non-Static Member
3. Instance Number
4. None of the given

Operator overloading is

1. making C++ operators work with objects.
2. giving C++ operators more than they can handle.
3. giving new meanings to existing Class members.

4. making new C++ operators

Information hiding can be achieved through_____.

1. Encapsulation, Inheritance
2. Encapsulation, Polymorphism
3. **Encapsulation, Abstraction**
4. Overloading

A good model is related to a real life problem.
Select correct option:

1. Loosely
2. Openly
3. **Closely**

Which of the following features of OOP is used to derive a class from another?
Select correct option:

1. Encapsulation
2. Polymorphism
3. Data hiding
4. **Inheritance**

Which of the following is a weak relationship between two objects?
Select correct option:

1. Inheritance
2. Composition
3. **Aggregation**
4. None of given

Data items in a class must be private.

Select correct option:

1. True
2. **False**

Which one is a class association

Select correct option:

1. Simple Association
2. **Inheritance**
3. Composition
4. Aggregation

Suppose there is an object of type Person, which of the following can be considered as one of its attributes

Select correct option:

1. Name
2. Age
3. Work()
4. **Both Name and Age**

Which one is not an object association?

Select correct option:

1. Simple association
2. **Inheritance**
3. Aggregation
4. Association

Using encapsulation we can achieve

Select correct option:

1. **Information hiding**
2. Least interdependencies among modules
3. Implementation independence
4. All of given options

In constant member function the type of this pointer is:
Select correct option:

1. Constant pointer
2. **Constant pointer to object**
3. Constant pointer to class
4. Constant pointer to constant object

Which of the following is the way to extract common behavior and attributes from the given classes and make a separate class of those common behaviors and attributes?

1. **Generalization**
2. Sub-typing
3. Specialization
4. Extension

The ability to derive a class from more than one class is called

1. Single inheritance
2. Encapsulation
3. **Multiple inheritance**
4. Polymorphism

If MyClass has a destructor what is the destructor named?

1. MyClass
2. **~MyClass**
3. My~Class
4. MyClass~

Class abc{ ----- }; Is a valid class declaration?

1. **yes**
2. no

Without using Deep copy constructor, A _____ problem can occur

1. System crash
2. Memory Leakage
3. Dangling pointer
4. **All of the given**

If only one behaviour of a derived class is incompatible with base class, then it is:

1. Generalization
2. **Specialization**
3. Extension
4. Inheritance

Which of the following may not be an integral part of an object?

1. state
2. behavior
3. Protected data members
4. All of given

Only tangible things can be chosen as an object.

1. True
2. **False**

السلام عليكم الله ورحمته وبركاته

الله "وَتُعِزُّ مَنْ تَشَاءُ وَتُذِلُّ مَنْ تَشَاءُ"

الله "جسے جابے عزت دے اور جسے جابے ذلیل کرے"

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Methodologies to the development of reusable software relate to

- a) Structure programming
- b) Procedural programming
- c) Generic programming**
- d) None of the given

In case of template specialization, if compiler cannot find required complete specialization, t

a. General template

- b. Complete template
- c. Partial specialization

Which of the following may inherit from the ordinary class?

- a. Class template
- b. Partial specialization
- c. Complete specialization

d. All of given option

Which of the following is not type of inheritance in c++?

- a) Public
- b) Protected
- c) Restricted**

Which of the following statement is true about partial specialization?

- a. Function templates cannot have partial specialization
- b. Class template cannot have partial specialization

c. Both class templates and function template can have partial specialization. d. None of the given

In resolution order of function template, compiler searches for in the end

- a. Complete specialization

b. Partial specialization

c. Generic template

Which of the following is known as
Dereference operator in C++?

a. +

b. *

c. &

When we want to have exactly identical
operations on different data type, are used

a. Function Template

b. Function Overloading

c. Function Overriding

When we specialize a function template, it is
called

a. Function overloading

b. Function overriding

c. Function template overloading

A class template may inherit from another class template.

True

False

Target of a _____ function call is determined at run time.
instance

virtual

operator none of given

Sender of the message does not need to know the exact class of receiver in_____.

Abstraction

Polymorphism

A function call is resolved at run-time in_____ .

non-virtual member

function virtual member

function

Both non-virtual member and virtual member function.

User can make virtual table explicitly.

True

False

Binding means that target function for a call is selected at compile time.

Static

Dynamic

Which line will produce error. Class phone: private Transmit, private Receiver { } 1. int main() 2. { 3. phone obj; 4. Transmit* obj1 = &obj; 5. Received obj2 = &obj; 6. }.

3rd line will produce error

4th line will produce error

3rd and 4th line will produce error.

5th line will produce error

Function overriding is done in context of,

Single class

Derived and base classes

Methodologies to the development of reusable software relate to_____.

Structure programming

generic programming

Function template must have a parameter.

True

False

The default inheritance mode is,

Public inheritance

Private inheritance

None of these options

Two functions with same names, parameters and return type can exist in,

Function overloading

Function overriding

Consider the code below, class c1{ }; class c2 : public c1 { };

class c3 : public c2 { }; Then c2 is,

Direct base class of c3

Direct child class of c3

User can make virtual table explicitly.

True

False

In order to define a class template the first line of definition must be:

template <typename T>

typename <template T>

Consider the following statements: 1) int iArray[5]; 2) int *pArr = iArray;

These statements will compile successfully

Error in second statement

In type in depended function template should be use where code and behavior must be identical.

True

False

It is illegal to make objects of one class members of another class.

True

False

In resolution order compiler search firstly _____.

Complete Specialization

Ordinary function

template<> class Vector {void p; //.... void*& operator[]
((int i); };**

This specialization can then be used as the common implementation for all Vectors of pointers.

This specialization can then be used as the all type implementation for one type classes.

This specialization can then be used double type pointers.

This specialization should be used for Vectors of all type int types.

**Consider the code below, class class1{ private: int i; };
class class2 : private class1 { }; Then int member i of
class1 is ____ in class2, public protected
private**

none of the given options

If there is a pointer, p, to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a virtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the ____ class to be executed.

base

derived

A class template may inherit from another class template.

True

False

Derived class can inherit from public base class as well as private and protected base classes

True

False

Consider the code below, class class1{ private: int i; }; class class2 : public class1 { }; Then int member i of class1 is ____ in class2,

None of the given options

Binding means that target function for a call is selected at run time

Automatic

Dynamic

When we create objects, then space is allocated to:

Member function

Access specifier

Data member

None of given

There is only one form of copy constructor.

True

False

Which of the following features of OOP is used to deal with only relevant details? Abstraction

Information hiding

Object

_____ **Binding means that targets function for a call is selected at compile time.** Static

Dynamic

Automatic

None of given

A Class hierarchy

Shows the same relationships as an organization chart Describes “has a” relationships.

Describes “is a kind of” relationships.

Shows the same relationships as a family tree

In C++, we declare a function virtual by preceding the function header with keyword “Inline”.

True

False

It is illegal to make objects of one class members of another class.

True

False

Derived class can inherit from public base class as well as private and protected base classes

True

False

Which line will produce error? Class phone: Private

Transmit, private Receiver { }

**1.int main () 2. { 3.phone obj; 4. Transmit*obj1 = &obj;
5.Received obj2 = &obj; 6.}**

3rd line will produce

error 4th line will

produce error

3rd and 4th line will

produce error. 5th line will

produce error.

Methodologies to the development of reusable software relate to ____:

Friends are used exactly the same for template and non-template classes.

True

False

A function template must have a parameter

True

False

Child class can call constructor of its,

Direct base class

Indirect base class

Both direct and indirect base
classes None of these.

Which statement will be true for concrete class?

It implements an virtual concept.

It can be instantiated

It cannot be instantiated

None of given

**A class D can be derived from a class C, which is derived
from class B, which is derived from a class A**

True

False

**Adding a derived class to a base class requires fundamental
changes to the base class.**

True

False

**A Class or class template can have member _____
that are themselves templates.**

Variable

Function

Objects

None of given

**Which will be the Primary task or tasks of generic
programming?**

Categorize the abstractions in a domain into concepts

Implement generic algorithms based on the concepts

Build concrete models of the concepts

All of given

The default inheritance mode is,

Public inheritance

Protected Inheritance

Private Inheritance

None of these options

If there is a pointer, p, to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a virtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the _____ class to be executed.

Base

Derived

Virtual

Implemented

Child class can call constructor of its,

Direct base class

Indirect base class

Both direct and indirect base classes

None of these

A pure virtual function is a virtual function that causes its class to be abstract.

returns nothing. is
used in a derived
class. takes no
arguments.

1: Consider the code below, class class1{ protected: int i; };
class class2 :
protected class1 { }; Then int member i of class1 is _____ in
class2,

- + public
- + **protected**

2: By default, assignment operator (=) performs,

- + **Shallow copy**
- + Deep copy

3: The default inheritance mode is,

- + **Public inheritance**

4: In case of dynamic memory allocation in our class we
should use

- + Default assignment operator
- + **copy constructor**

5: Static casting is,?

- + **Implicit way of calling base class functions from derived class**
- + Explicit way of calling base class functions from derived class

6: In case of dynamic memory allocation in a class we should
use,

- + User defined default constructor
- + **user defined copy constructor (true)**

7: Which of the following types of Inheritance is used to model “Implemented in terms of” relationship?

- + Public
- + **private (true)**

8: Virtual functions allow you to

- + create an array of type pointer-to-base class that can hold pointers to derived classes.
- + **group objects of different classes so they can all be accessed by the same function code. (true)**

9: A class D can be derived from a class C, which is derived from a class B, which is derived from a class A.

- + **T (true ans)**
- + F

10: Function overriding is done in context of,

- + Explicit way of calling base class functions from derived class
- + None of these
- + both 1 and 2

1. An abstract class is useful when

- a. no classes should be derived from it.
- b. there are multiple paths from one derived class to another.
- c. no objects should be instantiated from its.**
- d. you want to defer the declaration of the class.

2. In order to define a class template the first line of definition must be:

- a. template <typename T>**

- b. typename <template T>
- c Template Class <Class Name>
- d Class <Template T>

3. Select correct line of code for composition relationship between “Keyboard” class and “Keys” class

- a class keyboard {Keys type;};
- b class keys {KeyBoard type; };
- c class keyboard: private keys
- d class keys: private KeyBoard**

4. Select correct line of code for inheritance relationship between “Keys” class and “Special Keys” class. “Keys” is parent class while “Special Keys” is child class

- A class Special Keys: public Keys**
- B class Keys: public Special Keys
- C class Special Keys: public Keys
- D class Keys: public Special Keys

5 Methodologies to the development of reusable software relate to_____.

- A Structure programming
- B procedural programming
- C generic programming**
- D None of the given

6 User can make virtual table explicitly.

- A True**
- B false

7 A function call is resolved at run-time in_____

- A non-virtual member function
- B virtual member function**
- C Both non-virtual member and virtual member function.
- D None of given

8 Derived class can inherit from public base class as well as private and protected base classes

A true

B false

9 Which of the following function can convert a class into an abstract class?

A Abstract function

B Concrete function

C Virtual function

D Pure virtual function

CS304 Question No: 2

Which of the following causes run time binding?

Declaring object of abstract class

Declaring pointer of abstract class

Declaring overridden methods as non-virtual

None of the given

CS304 Question No: 3

A function template can not be overloaded by another function template.

True

False

CS304 Question No: 4

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

Templates

Overloading

Data hiding

Encapsulation

CS304 Question No: 5

Identify the correct way of declaring an object of user defined template class A for char type members?

A obj;

A obj;

A obj;

Obj A;

CS304 Question No: 6

The user must define the operation of the copy constructor.

True

False

CS304 Question No: 7

Template functions use _____ than ordinary functions.

Greater Memory

Lesser Memory

Equal Memory

None of the given options

CS304 Question No: 8

The find() algorithm

finds matching sequences of elements in two containers.

finds a container that matches a specified container.

takes iterators as its first two arguments.

takes container elements as its first two arguments.

CS304 Question No: 9

Compiler performs _____ type checking to diagnose type errors,

Static

Dynamic

Bound

Unbound

CS304 Question No: 10

Which of the following is/are advantage[s] of generic programming?

Reusability

Writability

Maintainability

All of given

CS304 Question No: 11

Vectors contain contiguous elements stored as a[an] ____.
variable

array

function

datatype

CS304 Question No: 12

**Suppose you create an uninitialized vector as follows:
vector eve;**

After adding the statement,

eve.pushback(21);

what will happen?

The following statement will add an element to the start (the back) of eve and will initialize it with the value 21.

The following statement will add an element to the center of eve and will reinitialize it with the value 21.

In a de-queue, (chose the best option)

data can be quickly inserted or deleted at any arbitrary location.

data can be inserted or deleted at any arbitrary location, but the process is relatively slow.

data cannot be quickly inserted or deleted at either end.

data can be inserted or deleted at either end, but the process is relatively slow.

CS304 Question No: 14

Algorithms can only be implemented using STL containers.

True

False

CS304 Question No: 15

What is a class?

A class is a section of computer memory containing objects.

A class is a section of the hard disk reserved for object oriented programs

A class is the part of an object that contains the variables.

A class is a description of a kind of object.

CS304 Question No: 16

**Inheritance is a way to
organize data.**

pass arguments to objects of classes.

add features to existing classes without rewriting them.

improve data-hiding and encapsulation.

CS304 Question No: 17

We can use "this" pointer in the constructor in the body and even in the initialization list of any class if we are careful,

True

False

CS304 Question No: 18

_____ and _____ methods may not be declared abstract.

private, static

private, public

static, public

none of given

CS304 Question No: 19

**Default constructor is such constructor which either has no -
-----or if it has some parameters these have ----- values**

Parameter, temporary

Null, Parameter

Parameter, default

none of the given

CS304 Question No: 20

**Public methods of base class can ----- be accessed in its
derived class**

directly

indirectly

simultaneously

none of the given

CS304 Question No: 21

The type that is used to declare a reference or pointer is called its -----

default type

static type

abstract type

reference type

CS304 Question No: 22

----- members are somewhere between public and private members. They are used in inheritance protected

public

private

global

CS304 Question No: 23

Which of these are examples of error handling techniques ?

Abnormal Termination

Graceful Termination

Return the illegal

all of the given

CS304 Question No: 24

----- follow try block to catch the object thrown.

catch block

throw block

main block

non of the given

CS304 Question No: 25

Graphical representation of the classes and objects is called object model it shows -----

Class Name only

Class Name and attributes

Relationships of the objects and classes

all of the given

CS304 Question No: 26

Destructor can be overloaded

True

False

CS304 Question No: 27 (M - 2)

Describe the way to declare a template function as a friend of any class.

Template name

Class classmate

{

Friend void friend template name (class name gastric const part class name);

}

CS304 Question No: 28 (M - 2)

State any two reasons why the virtual methods cannot be static?

1-virtual method can not be static as it is dynamic

FINALTERM EXAMINATION
Spring 2010
CS304- Object Oriented Programming (Session - 4)

Question No: 1 (Marks: 1) - Please choose one

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

- ▶ True
- ▶ False

Question No: 2 (Marks: 1) - Please choose one

Virtual functions allow you to

- ▶ create an array of type pointer-to-base class that can hold pointers to derived classes.
- ▶ create functions that can never be accessed.
- ▶ group objects of different classes so they can all be accessed by the same function code.
- ▶ use the same function call to execute member functions of objects from different classes

Question No: 3 (Marks: 1) - Please choose one

A pointer to a base class can point to objects of a derived class.

- ▶ True
- ▶ False

Question No: 4 (Marks: 1) - Please choose one

A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ an argument is passed by value.

- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

Question No: 5 (Marks: 1) - Please choose one

Each try block can have _____ no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ As many as necessary.

Question No: 6 (Marks: 1) - Please choose one

Non Template Friend functions of a class are friends of _____instance/s of that class.

- ▶ All
- ▶ One specific
- ▶ All instances of one data type
- ▶ None of the given options

Question No: 7 (Marks: 1) - Please choose one

Template functions use _____ than ordinary functions.

- ▶ Greater Memory
- ▶ Lesser Memory
- ▶ Equal Memory
- ▶ None of the given options

Question No: 8 (Marks: 1) - Please choose one

The find() algorithm

- ▶ finds matching sequences of elements in two containers.
- ▶ finds a container that matches a specified container.
- ▶ takes iterators as its first two arguments.
- ▶ takes container elements as its first two arguments.

Question No: 9 (Marks: 1) - Please choose one

The copy() algorithm returns an iterator to

- ▶ the last element copied from.
- ▶ the last element copied to.
- ▶ the element one past the last element copied from.
- ▶ the element one past the last element copied to.

Question No: 10 (Marks: 1) - Please choose one

If you define a vector v with the default constructor, and define another vector w with a one-argument constructor to a size of 11, and insert 3 elements into each of these vectors with push_back(), then the size() member function will return _____ for v and _____ for w.

- ▶ 11 for v and 3 for w.
- ▶ 0 for v and 0 for w.
- ▶ 0 for v and 3 for w.
- ▶ 3 for v and 11 for w.

Question No: 11 (Marks: 1) - Please choose one

Which is not the Advantage of inheritance?

- ▶ providing class growth through natural selection.
- ▶ facilitating class libraries.
- ▶ avoiding the rewriting of code.
- ▶ providing a useful conceptual framework.

Question No: 12 (Marks: 1) - Please choose one

```
class DocElement
{
public:
    virtual void Print() { cout << "Generic element"; }
};
class Heading : public DocElement
{
public:
    void Print() { cout << "Heading element"; }
};
```



```
class Paragraph : public DocElement
{
public:
    void Print() { cout << "Paragraph element"; }
};
void main()
{
    DocElement * p = new Paragraph();

    p->Print();
}
```

When you run this program, it will print out a single line to the console output.

What will be in that line?

Select one correct answer from the following list:

- ▶ Generic element
- ▶ Heading element
- ▶ Paragraph element
- ▶ Nothing will be printed.

Question No: 13 (Marks: 1) - Please choose one

Which type of inheritance is being represented by the following statement,
class X : public A, public B { };

- ▶ Single inheritance
- ▶ Multiple inheritance
- ▶ Double inheritance
- ▶ None of the given options

Question No: 14 (Marks: 1) - Please choose one

When we write a class template the first line must be:

- ▶ `template < class class_name>`
- ▶ `template < class data_type>`
- ▶ `template < class T >`

Here T can be replaced with any name but it is preferable.

- ▶ `class class-name()`

`class template<class_name>`

Question No: 15 (Marks: 1) - Please choose one

Function templates should be used where code and behavior must be identical.

- ▶ True
- ▶ False

Question No: 16 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ All of given

Question No: 17 (Marks: 1) - Please choose one

The specialization pattern `<T*>` after the name says that this specialization is to be used for every,

- ▶ data type
- ▶ meta type
- ▶ virtual type

- ▶ pointer type

Question No: 18 (Marks: 1) - Please choose one

A range is often supplied to an algorithm by two _____ values.

- ▶ italic
- ▶ iteration
- ▶ iterator
- ▶ None of given

Question No: 19 (Marks: 1) - Please choose one

Which of the following is an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ Unique identity
- ▶ All of the given

Question No: 20 (Marks: 1) - Please choose one

Consider the following statement

Cupboard has books

What is the relationship between Cupboard and books?

- ▶ Composition
- ▶ Aggregation
- ▶ Inheritance
- ▶ None of the given options

Question No: 21 (Marks: 1) - Please choose one

Which sentence clearly defines an object?

- ▶ one instance of a class.
- ▶ another word for a class.
- ▶ a class with static methods.
- ▶ a method that accesses class attributes.

Question No: 22 (Marks: 1) - Please choose one

_____, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.

- ▶ Friendship is one way only
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

Question No: 23 (Marks: 1) - Please choose one

The statement `objA=objB;` will cause a compiler error if the objects are of different classes.

- ▶ True
- ▶ False

Question No: 24 (Marks: 1) - Please choose one

Consider the call given below of an overloaded operator "+",

Rational_number_1 + Rational_number_2

Where `Rational_number_1` and `Rational_number_2` are the two objects of `Rational_number` class (a user defined class). Identify which of the above two objects will be passed as an argument to the overloaded operator function?

- ▶ Rational_number_1
- ▶ Rational_number_2
- ▶ Both Rational_number_1 & Rational_number_2
- ▶ any of the two objects, randomly

Question No: 25 (Marks: 1) - Please choose one

If a class D has been derived using protected inheritance from class B (If B is a protected base and D is derived class) then public and protected members of B ----- accessed by member functions and friends of class D and classes derived from D

- ▶ can be
- ▶ cannot be
- ▶ does restrict to be
- ▶ not given

Question No: 26 (Marks: 1) - Please choose one

In Private ----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ▶ specialization
- ▶ inheritance
- ▶ abstraction
- ▶ composition

Question No: 27 (Marks: 2)

Give two uses of a destructor.

Question No: 28 (Marks: 2)

Describe the way to declare a template class as a friend class of any other class.

Question No: 29 (Marks: 2)

Give the name of two basic types of containers collectively called First class containers?

Question No: 30 (Marks: 2)

State any conflict that may rise due to multiple inheritance?

Question No: 31 (Marks: 3)

What will be the output after executing the following code?

```
class c1{
public:
virtual void function(){
cout<<"I am in c1"<<endl;
```

```
}

};
class c2: public c1{
public:
void function(){
cout<<"I am in c2"<<endl;

}

};
class c3: public c1 {
public:
void function(){
cout<<"I am in c3"<<endl;
}

};

int main(){

c1 * test1 = new c2();
c1 * test2 = new c3();
test1->function();
test2->function();
system("PAUSE");
return 0;
}
```

Question No: 32 (Marks: 3)

If we declare a function as friend of a template class will it be a friend for a particular data type or for all data types of that class.

Question No: 33 (Marks: 3)

Tell the logical error/s in the code given below with reference to resource management; also describe how we can correct that error/s.

```
class Test{

public:
int function1(){
    try{
```

```
        FILE *fileptr = fopen("filename.txt","w");
        throw exception();
        fclose(fileptr);
        return 0;
    }
    catch(Exception e){
        ...
    }
}
};
```

Question No: 34 (Marks: 5)

What is the output produced by the following program?

```
#include<iostream.h>

void sample_function(double test) throw (int);

int main()
{
    try
    {
        cout << "Trying.\n";
        sample_function(98.6);
        cout << "Trying after call.\n";
    }
    catch(int)
    {
        cout << "Catching.\n";
    }

    cout << "End program.\n";
    return 0;
}

void sample_function(double test) throw (int)
{
    cout << "Starting sample_function.\n";
    if(test < 100)
        throw 42;
}
```

Question No: 35 (Marks: 5)

The code given below has one template function as a friend of a template class,

1. You have to identify any error/s in this code and describe the reason for error/s.
2. Give the correct code after removing the error/s.

```
template<typename U>
void Test(U);
template< class T >

class B {
    int data;
    public:
    friend void Test<>( T );
};

template<typename U>
void Test(U u){
    B < int> b1;
    b1.data = 7;
}
int main(int argc, char *argv[])
{
    char i;
    Test(i);
    system("PAUSE");
    return 0;
}
```

Question No: 36 (Marks: 5)

Consider the following class,

```
class Base
{
    char * p;
public:
    Base() { p = new char[10]; }

    ~Base() { delete [] p; }
};
class Derived : public Base
{
    char * q;
public:
    Derived() { q = new char[20]; }

    ~Derived() { delete [] q; }
};
void foo()
{
    Base* p = new Derived();
}
```



```
        delete p;  
    }
```

With this program, every time function foo is called, some memory will leak.
Explain why memory will leak. Also, explain how to fix this problem.

FINALTERM EXAMINATION

Fall 2009

CS304- Object Oriented Programming (Session - 1)

Time: 120 min

Marks: 75

Question No: 1 (Marks: 1) - Please choose one

Which one of the following terms must relate to **polymorphism**?

- ▶ Static allocation
- ▶ Static typing
- ▶ **Dynamic binding**
- ▶ Dynamic allocation

Question No: 2 (Marks: 1) - Please choose one

Multiple inheritance can be of type

- ▶ Public
- ▶ Private

- ▶ Protected
- ▶ **All of the given**

Question No: 3 (Marks: 1) - Please choose one

When a subclass specifies an alternative definition for an attribute or method of its superclass, it is _____ the definition in the superclass.

- ▶ overload
- ▶ **overriding**
- ▶ copy riding
- ▶ none of given

Question No: 4 (Marks: 1) - Please choose one

Like template functions, a class template may not handle all the types successfully.

- ▶ **True**
- ▶ False

Question No: 5 (Marks: 1) - Please choose one

It is sometimes useful to specify a class from which no objects will ever be created.

- ▶ True
- ▶ **False**

Question No: 6 (Marks: 1) - Please choose one

Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- ▶ public members of Derv.
- ▶ protected members of Derv.

- ▶ private members of Derv.
- ▶ **protected members of Base.**

Question No: 7 (Marks: 1) - Please choose one

A pointer to a base class can point to objects of a derived class.

- ▶ **True**
- ▶ False

Question No: 8 (Marks: 1) - Please choose one

A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ an argument is passed by value.
- ▶ **a function returns by reference.**
- ▶ an argument is passed by reference.

Question No: 9 (Marks: 1) - Please choose one

A function call is resolved at run-time in_____

- ▶ non-virtual member function.
- ▶ **virtual member function.**
- ▶ Both non-virtual member and virtual member function.
- ▶ None of given

Question No: 10 (Marks: 1) - Please choose one

When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using _____).

- ▶ scope resolution operator
- ▶ dot operator
- ▶ null operator
- ▶ Operator overloading

Question No: 11 (Marks: 1) - Please choose one

Each try block can have _____ no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ As many as necessary.

Question No: 12 (Marks: 1) - Please choose one

Two important STL associative containers are _____ and _____.

- ▶ set,map
- ▶ sequence,mapping
- ▶ setmet,multipule
- ▶ sit,mat

Question No: 13 (Marks: 1) - Please choose one

The mechanism of selecting function at run time according to the nature of calling object is called,

- ▶ late binding
- ▶ static binding
- ▶ virtual binding
- ▶ None of the given options

Question No: 14 (Marks: 1) - Please choose one

An abstract class is useful when,

- ▶ We do not derive any class from it.
- ▶ There are multiple paths from one derived class to another.
- ▶ **We do not want to instantiate its object.**
- ▶ You want to defer the declaration of the class.

Question No: 15 (Marks: 1) - Please choose one

Which of the following is incorrect line regarding function template?

- ▶ `template<class T>`
- ▶ `template <typename U>`
- ▶ **`Class<template T>`**
- ▶ `template < class T, class U>`

Question No: 16 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ **All of given**

Question No: 17 (Marks: 1) - Please choose one

By default the vector data items are initialized to ____

- ▶ **0**
- ▶ 0.0
- ▶ 1
- ▶ null

Question No: 18 (Marks: 1) - Please choose one

Which one of the following functions returns the total number of elements in a vector.

- ▶ length();
- ▶ **size();**
- ▶ ele();
- ▶ veclen();

Question No: 19 (Marks: 1) - Please choose one

Suppose you create an uninitialized vector as follows:

```
vector<int> evec;
```

After adding the statment,

```
evec.push_back(21);
```

what will happen?

- ▶ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.
- ▶ The following statement will add an element to the center of evec and will reinitialize it with the value 21.
- ▶ The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.
- ▶ **The following statement will add an element to the end (the back) of evec and initialize it with the value 21.**

Question No: 20 (Marks: 1) - Please choose one

An STL container can not be used to,

- ▶ hold objects of class employee.
- ▶ store elements in a way that makes them quickly accessible.
- ▶ **compile c++ programs.**
- ▶ organize the way objects are stored in memory

Question No: 21 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

- ▶ True
- ▶ **False**

Question No: 22 (Marks: 1) - Please choose one

The main function of scope resolution operator (::) is,

- ▶ **To define an object**
- ▶ To define a data member
- ▶ To link the definition of an identifier to its declaration
- ▶ To make a class private

Question No: 23 (Marks: 1) - Please choose one

When is a constructor called?

- ▶ Each time the constructor identifier is used in a program statement
- ▶ **During the instantiation of a new object**
- ▶ During the construction of a new class
- ▶ At the beginning of any program execution

Question No: 24 (Marks: 1) - Please choose one

Consider the code below,

```
class Fred {  
public:  
Fred();  
...  
};  
int main()  
{  
Fred a[10];  
Fred* p = new Fred[10];  
...  
}
```

Select the best option,

- ▶ Fred a[10]; calls the default constructor 09 times
- Fred* p = new Fred[10]; calls the default constructor 10 times

▶ **Produce an error**

- ▶ Fred a[10]; calls the default constructor 11 times
- Fred* p = new Fred[10]; calls the default constructor 11 times
- ▶ Fred a[10]; calls the default constructor 10 times
- Fred* p = new Fred[10]; calls the default constructor 10 times

Question No: 25 (Marks: 1) - Please choose one

Associativity can be changed in operator overloading.

- ▶ True
- ▶ **False**

Question No: 26 (Marks: 1) - Please choose one

A normal C++ operator that acts in special ways on newly defined data types is said to be

- ▶ glorified.
- ▶ encapsulated.
- ▶ **classified.**
- ▶ overloaded.

Question No: 27 (Marks: 1) - Please choose one

Which operator can not be overloaded?

- ▶ The relation operator (>=)
- ▶ Assignment operator (=)
- ▶ Script operator ([])
- ▶ **Conditional operator (? :)**

Question No: 28 (Marks: 1) - Please choose one

Suppose obj1 and obj2 are two objects of a user defined class A. An + operator is overloaded to add obj1 and obj2 using the function call obj1+obj2. Identify the correct function prototype against the given call?

- ▶ A operator + (A &obj);
- ▶ int + operator();
- ▶ **int operator (plus) ();**
- ▶ A operator(A &obj3);

Question No: 29 (Marks: 1) - Please choose one

Default constructor is such constructor which either has no -----or if it has some parameters these have ----- values

- ▶ Parameter, temporary
- ▶ Null, Parameter
- ▶ **Parameter, default**
- ▶ non of the given

Question No: 30 (Marks: 1) - Please choose one

Public methods of base class can ----- be accessed in its derived class

- ▶ directly
- ▶ **indirectly**
- ▶ simultaneously
- ▶ non of the given

Question No: 31 (Marks: 1)

Is **Deque** a Birectional Container?

Yes, deque behaves like queue (line) such that we can add elements on both sides of it.

Question No: 32 (Marks: 1)

What is meant by Generic Programming?

Generic programming refers to programs containing generic abstractions general code that is same in logic for all data types like printArray function), then we instantiate that generic program abstraction (function, class) for a particular data type, such abstractions can work with many different types of data.

Question No: 33 (Marks: 2)

Sort the following data in the order in which compiler searches a function?

Complete Specialization, Generic Template, Partial Specialization, Ordinary Function.

Specializations of this function template, instantiations with specific types, can be called just like an ordinary function:

```
cout << max(3, 7); // outputs 7
```

The compiler examines the arguments used to call max and determines that this is a call to max(int, int). It then instantiates a version of the function where the parameterizing type T is int, making the equivalent of the following function:

```
int max(int x, int y)
{
    return x < y ? y : x;
}
```

the C++ Standard Template Library contains the function template max(x, y) which creates functions that return either x or y, whichever is larger. max() could be defined like this:

```
template <typename T>
T max(T x, T y)
{
    return x < y ? y : x;
}
```

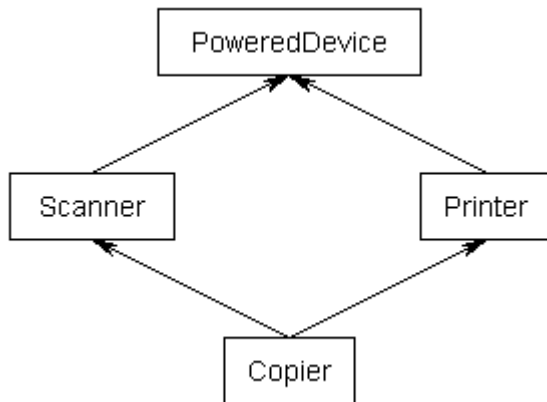
Question No: 34 (Marks: 2)

State any conflict that may rise due to multiple inheritance?

The conflict may arise is the diamond problem, which our author likes to call the “diamond of doom”. This occurs when a class multiply inherits from two classes which each inherit from a single base class. This leads to a diamond shaped inheritance pattern.

For example, consider the following set of classes:

```
class PoweredDevice
{
};
class Scanner: public PoweredDevice
{
};
class Printer: public PoweredDevice
{
};
class Copier: public Scanner, public Printer
{
};
```



Scanners and printers are both powered devices, so they derived from PoweredDevice. However, a copy machine incorporates the functionality of both Scanners and Printers.

Ambiguity also cause problem.

Question No: 35 (Marks: 3)

Describe three properties necessary for a container to implement Generic Algorithms.

If you declare a container as holding pointers, you are responsible for managing the memory for the objects pointed to. The container classes will not automatically free memory for these objects when an item is erased from the container.

Container classes are expected to implement methods to do the following:

- create a new empty container (constructor),
- report the number of objects it stores (size),
- delete all the objects in the container (clear),
- insert new objects into the container,
- remove objects from it,
- provide access to the stored objects.

Question No: 36 (Marks: 3)

Write three important features of virtual functions.

With virtual functions, derived classes can provide new implementations of functions from their base classes. When someone calls a virtual function of an object of the derived class, this new implementation is called, even if the caller uses a pointer to the base class, and doesn't even know about the particular derived class.

The virtual function is an option, and the language defaults to non virtual, which is the fastest configuration.

The derived class can completely "override" the implementation or "augment" it (by explicitly calling the base class implementation in addition to the new things it does).

Question No: 37 (Marks: 3)

Consider the code below,

```
#include <iostream>
#include <stdlib.h>
using namespace std;
class Shape{
    public:
    void Draw(){cout<<"shape"<<endl;}
};
class Line : public Shape{
    public:
    void Draw(){cout<<"Line"<<endl;}
};
class Circle : public Shape{
    public:
    void Draw(){cout<<"Circle"<<endl;}
};
int main(int argc, char *argv[])
{
    Shape * ptr1 = new Shape();
    Shape * ptr2 = new Line();
    Shape * ptr3 = new Circle();

    ptr1->Draw();
    ptr2->Draw();
    ptr3->Draw();
    system("PAUSE");
    return 0;
}
```

This code shows output,

Shape
Shape
Shape

Give the reason for this output

Suppose we want to show the output,

Shape
Line
Circle

How we can change the code to do that?

```
class shape { public:  
    void draw();  
};  
class circle : public shape { };  
int main(int argc, char **argv){  
    circle my_circle;  
    my_circle.draw();  
}
```

While this has all the usual advantages, e.g., code reuse, the real power of polymorphism comes into play when draw is declared to be virtual or pure virtual, as follows:

```
class shape{ public:  
    virtual void draw()=0;  
};  
class circle : public shape { public:  
    void draw();  
}
```

Here, circle has declared its own draw function, which can define behavior appropriate for a circle. Similarly, we could define other classes derived from shape, which provide their own versions of draw. Now, because all the classes implement the shape interface, we can create collections of objects that can provide different behavior invoked in a consistent manner (calling the draw member function). An example of this is shown here.

```
shape *shape_list[3]; // the array that will  
                      // pointer to our shape objects  
shape[0] = new shape; // three types of shapes  
shape[1] = new line; // we have defined  
shape[2] = new circle;  
for(int i = 0; i < 3; i++){  
    shape_list[i].draw();  
}
```

When we invoke the draw function for each object on the list, we do not need to know anything about each object; C++ handles the details of invoking the correct version of draw. This is a very powerful technique, allowing us to provide extensibility in our designs. Now we can add new classes derived from shape to provide whatever behavior we desire. The key here is that we have separated the interface (the prototype for shape) from the implementation.

Question No: 38 (Marks: 5)

There are some errors in the code given below, you have to

1. Indicate the line no. with error/s
2. Give the reason for error/s
3. Correct the error/s.

```
1. #include <iostream>           this will be #include <iostream.h>
2. #include <stdlib.h>

3. using namespace std;
4. template <typename T>
5. class MyClass{
6. public:
7. MyClass(){
8. cout<<"This is class1"<<endl;
9. }
10. };
11. template <typename T>
12. class MyClass<int*>{
13. public:
14. MyClass(){
15. cout<<"This is class2"<<endl;
16. }
17. };
18. int main(int argc, char *argv[])
19. {
20. MyClass<int> c1;
21. MyClass<int*> c2;
22. system("PAUSE");
23. return 0;
24. }
```

Question No: 39 (Marks: 5)

Given are two classes A and B. class B is inherited from class A. Write a code snippet(for main function) that polymorphically call the method of class B. Also what changes do you suggest in the given code segment that are required to call the class B method polymorphically.

```
class A
{
public:
void method() { cout<<"A's method \n"; }

};

class B : public A
{

public:
void method() { cout<<"B's method\n"; }
```

```
};
```

Ans:

```
public class Test
{
    public class A {}

    public class B extends A {}

    private void test(A a)
    {
        System.out.println("test(A)");
    }

    private void test(B b)
    {
        System.out.println("test(B)");
    }

    public static void main(String[] args)
    {
        Test t = new Test();
        A a = t.new A();
        A b = t.new B();

        t.test(a);
        t.test(b);
    }
}
```

Question No: 40 (Marks: 10)

Create built-in STL (Standard Template Library) **vector class object** for **strings** and add in it some words by taking input from user, then apply the sort() algorithm to array of words stored in this vector class object.

Hint: Use push_back() to add the words in vector class object, and the [] operator and size() to display these sorted words.

The STL is the containers, iterators and algorithms component of the proposed C++ Standard Library [ANSI95]. It represents a novel application of principles which have their roots in styles of programming other than Object-orientation.

void listWords(istream& in, ostream& out)

```

{
    string s;

    while (!in.eof() && in >> s) {
        add s to some container
    }

    sort the strings in the container
    remove the duplicates

    for (each string t in container) {
        out << t;
    }
}

```

For now, assume that a word is defined as a whitespace-separated string as delivered by the stream extraction operator. Later on we will consider ways of refining this definition. Given the way this problem is expressed, we can implement this program directly, if naïvely. The STL container class `vector` will suffice to hold the words: applying the algorithms `sort` and `unique` provides the required result.

```

void listWords(istream& in, ostream& out)
{
    string s;
    vector<string> v;

    while (!in.eof() && in >> s)
        v.push_back(s);           // (1)

    sort(v.begin(), v.end());

    vector<string>::iterator e
        = unique(v.begin(), v.end()); // (2)

    for (vector<string>::iterator b = v.begin();
        b != e;
        b++) {
        out << *b << endl;
    }
}

```

At (1) the vector member function `push_back()` is used to add to the end of the vector. This can also be done using the `insert` member, which takes as a parameter an iterator identifying the position in the vector at which to place the added element:

```
v.insert(v.end(), s);
```

This allows us to add at any position in the vector. Be aware, though, that adding anywhere other than the end implies the overhead of physically shifting all elements from the insertion point to the end to make room for the new value. For this reason, and given the choices made in this example, attempts to optimise this code by maintaining the

vector in sorted order are unwise. Replace vector with list and this becomes possible - although in both cases a search over the container will be necessary to determine the correct position of insertion.

The unique algorithm has the surprising property of not changing the length of the container to which it is applied (it can hardly do this, as it has access not to the underlying container, but only to the pair of iterators it is passed). Instead, it guarantees that duplicates are removed by moving unique entries towards the beginning of the container, returning an iterator indicating the new end of the container. This can be used directly (as here, at (2)), conversely it can be passed to the erase member with the old end iterator, to truncate the container.

Question No: 41 (Marks: 10)

Q. Write a detailed note on Exceptions in Destructors with the help of a coding example.

Exceptions in Destructors:

An object is presumably created to do something. Some of the changes made by an object should persist after an object dies (is destructed) and some changes should not. Take an object implementing a SQL query. If a database field is updated via the SQL object then that change should persist after the SQL objects dies. To do its work the SQL object probably created a database connection and allocated a bunch of memory. When the SQL object dies we want to close the database connection and deallocate the memory, otherwise if a lot of SQL objects are created we will run out of database connections and/or memory.

The logic might look like:

```
Sql::~Sql()  
{  
    delete connection;  
    delete buffer;  
}
```

Let's say an exception is thrown while deleting the database connection. Will the buffer be deleted? No. Exceptions are basically non-local gotos with stack cleanup. The code for deleting the buffer will never be executed creating a gaping resource leak.

Special care must be taken to catch exceptions which may occur during object destruction. Special care must also be taken to fully destruct an object when it throws an exception.

Example code for exception

```
#include<iostream.h>  
#include<conio.c>  
class Exception {  
private:
```

```
char message[30] ;
public:

Exception() {strcpy(message,"There is not enough stock");}
char * get_message() { return message; }
};
class Item {
private:

int stock ;
int required_quantity;
public:

Item(int stk, int qty)
{
    stock = stk;
    required_quantity = qty;
}
int get_stock()
{
    return stock;
}

int get_required_quantity()
{
    return required_quantity;
}

void order()
{
    if (get_stock()< get_required_quantity())

        throw Exception();
        else
            cout<<"The required quantity of item is available in the stock";
}

~Item(){}
};

void main()
{

    Item obj(10, 20);

    try
```

```
{  
    obj.order();  
}  
catch(Exception & exp2)  
{  
    getch();  
    cout << "Exception: " << exp2.get_message() << endl;  
}  
getch();
```

FINALTERM EXAMINATION

Fall 2009

CS304- Object Oriented Programming (Session - 4)

Ref No: 1130772

Time: 120 min

Marks: 75

Question No: 1 (Marks: 1) - Please choose one

A template provides a convenient way to make a family of

▶ **variables and data members**

▶ functions and classes

▶ classes and exceptions

▶ programs and algorithms

Question No: 2 (Marks: 1) - Please choose one

Which one of the following terms must relate to **polymorphism**?

▶ Static allocation

▶ Static typing

▶ **Dynamic binding**

▶ Dynamic allocation

Question No: 3 (Marks: 1) - Please choose one

What is true about function templates?

- ▶ The compiler generates only one copy of the function template
- ▶ **The compiler generates a copy of function respective to each type of data**
- ▶ The compiler can only generate copy for the int type data
- ▶ None of the given.

Question No: 4 (Marks: 1) - Please choose one

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ **Templates**
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

Question No: 5 (Marks: 1) - Please choose one

```
template <>
class Vector<char*> { }
```

This is an example of partial specialization.

- ▶ True
- ▶ **False**

Question No: 6 (Marks: 1) - Please choose one

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

- ▶ **True**
- ▶ False

Question No: 7 (Marks: 1) - Please choose one

A non-virtual member function is defined in a base class and overridden in a derived class; if that function is called through a base-class pointer to a derived class object, the derived-class version is used.

► **True**

► False

Question No: 8 (Marks: 1) - Please choose one

Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- public members of Derv.
- **protected members of Derv.**
- private members of Derv.
- protected members of Base.

Question No: 9 (Marks: 1) - Please choose one

In order to define a class template, the first line of definition must be:

- **template <typename T>**
- typename <template T>
- Template Class <ClassName>
- Class <Template T>

Question No: 10 (Marks: 1) - Please choose one

If there is a pointer p to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a nonvirtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the _____ class to be executed.

- Base
- Derived

- ▶ Abstract
- ▶ **virtual**

Question No: 11 (Marks: 1) - Please choose one

When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using _____).

- ▶ scope resolution operator
- ▶ dot operator
- ▶ **null operator**
- ▶ Operator overloading

Question No: 12 (Marks: 1) - Please choose one

Non Template Friend functions of a class are friends of _____instance/s of that class.

- ▶ All
- ▶ **One specific**
- ▶ All instances of one data type
- ▶ None of the given options

Question No: 13 (Marks: 1) - Please choose one

The find() algorithm

- ▶ finds matching sequences of elements in two containers.
- ▶ **finds a container that matches a specified container.**
- ▶ takes iterators as its first two arguments.
- ▶ takes container elements as its first two arguments.

Question No: 14 (Marks: 1) - Please choose one

If you define a vector v with the default constructor, and define another vector w with a one-argument constructor to a size of 11, and insert 3 elements into each of these vectors with push_back(), then the size() member function will return _____ for v and _____ for w.

- ▶ 11 for v and 3 for w.

- ▶ 0 for v and 0 for w.
- ▶ 0 for v and 3 for w.
- ▶ **3 for v and 11 for w.**

Question No: 15 (Marks: 1) - Please choose one

Which of the following may not be an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ Protected data members
- ▶ **All of given**

Question No: 16 (Marks: 1) - Please choose one

Which is not the Advantage of inheritance?

- ▶ providing class growth through natural selection.
- ▶ **facilitating class libraries.**
- ▶ avoiding the rewriting of code.
- ▶ providing a useful conceptual framework.

Question No: 17 (Marks: 1) - Please choose one

```
class DocElement
{
public:
    virtual void Print() { cout << "Generic element"; }
};
class Heading : public DocElement
{
public:
    void Print() { cout << "Heading element"; }
};
class Paragraph : public DocElement
{
public:
    void Print() { cout << "Paragraph element"; }
};
```

```
void main()
{
    DocElement * p = new Paragraph();

    p->Print();
}
```

When you run this program, it will print out a single line to the console output.

What will be in that line?

Select one correct answer from the following list:

- ▶ Generic element
- ▶ Heading element
- ▶ **Paragraph element**
- ▶ Nothing will be printed.

Question No: 18 (Marks: 1) - Please choose one

When a virtual function is called by referencing a specific object by name and using the dot member selection operator (e.g., squareObject.draw()), the reference is resolved at compile time.

- ▶ **True**
- ▶ False

Question No: 19 (Marks: 1) - Please choose one

In case of multiple inheritance a derived class inherits,

- ▶ Only the public member functions of its base classes
- ▶ Only the public data members of its base classes
- ▶ **Both public data members and member functions of all its base classes**
- ▶ Data members and member functions of any two base classes

Question No: 20 (Marks: 1) - Please choose one

When we write a class template the first line must be:

▶ template < class *class_name*>

▶ template < class *data_type*>

▶ **template < class T >**

Here T can be replaced with any name but it is preferable.

▶ class class-name()

class template<*class_name*>

Question No: 21 (Marks: 1) - Please choose one

Which of the following is incorrect line regarding function template?

▶ template<class T>

▶ template <typename U>

▶ **Class<template T>**

▶ template < class T, class U>

Question No: 22 (Marks: 1) - Please choose one

An STL container can not be used to,

▶ **hold objects of class employee.**

▶ store elements in a way that makes them quickly accessible.

▶ compile c++ programs.

▶ organize the way objects are stored in memory

Question No: 23 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

▶ **True**

▶ False

Question No: 24 (Marks: 1) - Please choose one

Consider a class named Vehicle, which of the following can be the instance of class Vehicle?

1. Car
2. Computer
3. Desk
4. Ahmed
5. Bicycle
6. Truck

▶ 1, 4, 5

▶ 2, 5, 6

▶ 1, 2, 3, 6

▶ **1, 5, 6**

Question No: 25 (Marks: 1) - Please choose one

Consider the code below,

```
class Fred {  
public:  
Fred();  
...  
};  
int main()  
{  
Fred a[10];  
Fred* p = new Fred[10];  
...  
}
```

Select the best option,

▶ Fred a[10]; calls the default constructor 09 times

Fred* p = new Fred[10]; calls the default constructor 10 times

▶ Produce an error

▶ **Fred a[10]; calls the default constructor 11 times**

Fred* p = new Fred[10]; calls the default constructor 11 times

▶ Fred a[10]; calls the default constructor 10 times

Fred* p = new Fred[10]; calls the default constructor 10 times

Question No: 26 (Marks: 1) - Please choose one

When a variable is define as **static** in a class then all object of this class,

- ▶ Have different copies of this variable
- ▶ **Have same copy of this variable**
- ▶ Can not access this variable
- ▶ None of given

Question No: 27 (Marks: 1) - Please choose one

The life of sub object is dependant on the life of master class in _____.

- ▶ Separation
- ▶ **Composition**
- ▶ Aggregation
- ▶ None of the given

Question No: 28 (Marks: 1) - Please choose one

_____, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.

- ▶ **Friendship is one way only**
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

Question No: 29 (Marks: 1) - Please choose one

Which of the following operators always takes no argument if overloaded?

- ▶ /
- ▶ -
- ▶ +
- ▶ **++**

Question No: 30 (Marks: 1) - Please choose one

In Private ----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ▶ specialization
- ▶ inheritance
- ▶ abstraction
- ▶ composition

Question No: 31 (Marks: 1)

Write the syntax of declaring a pure virtual function in a class?

Ans:

Pure Virtual Function is a Virtual function with no body.

Declaration of Pure Virtual Function:

Since pure virtual function has no body, the programmer must add the notation =0 for declaration of the pure virtual function in the base class.

General Syntax of Pure Virtual Function takes the form:

```
class classname //This denotes the base class of C++ virtual function
{
public:
virtual void virtualfunctionname() = 0 //This denotes the pure virtual function in C++
};
```

Question No: 32 (Marks: 1)

What is meant by direct base class ?

Ans

When a class-type is included in the class-base, it specifies the direct base class of the class being declared. If a class declaration has no class-base, or if the class-base lists only interface types, the direct base class is assumed to be object. A class inherits members from its direct base class,

Deriving a class from more than one *direct base class* is called multiple inheritance.

Question No: 33 (Marks: 2)

Describe the way to declare a template class as a friend class of any other class.

Ans

The following example is use of a class template:

```
template<class L> class Key
{
    L k;
    L* kptr;
    int length;
public:
    Key(L);
    // ...
};
```

Suppose the following declarations appear later:

```
Key<int> i;
Key<char*> c;
Key<mytype> m;
```

The compiler would create three objects.

Question No: 34 (Marks: 2)

What is the purpose of template parameter?

Ans:

There are three kinds of template parameters:

- type
- non-type
- template

You can interchange the keywords **class** and **typename** in a template parameter declaration. You cannot use storage class specifiers (**static** and **auto**) in a template parameter declaration.

Question No: 35 (Marks: 3)

Describe in simple words how we can use template specialization to enforce case sensitive specialization in String class.

Ans”

The act of creating a new definition of a function, class, or member of a class from a template declaration and one or more template arguments is called template instantiation. The definition created from a template instantiation is called a specialization. A primary template is the template that is being specialized.

create function objects to do the case-insensitive compares, and then reuse them when also wanting to do case-insensitive sorting or searching.

Question No: 36 (Marks: 3)

Can we use compiler generated default assignment operator in case our class is using dynamic memory? Justify your answer.

Ans:

the compiler does not make a separate copy of the object. Even if the types are not the same, the compiler is usually able to do a better job with initialization lists than with assignments.

Consider the following constructor that initializes member object `x_` using an initialization list: `square::square() : x_(whatever) { }`. The most common benefit of doing this is improved performance. For example, if the expression *whatever* is the same type as member variable `x_`, the result of the *whatever* expression is constructed directly inside `x_` — the compiler does not make a separate copy of the object. Even if the types are not the same, the compiler is usually able to do a better job with initialization lists than with assignments.

As if that wasn't bad enough, there's another source of inefficiency when using assignment in a constructor: the member object will get fully constructed by its default constructor, and this might, for example, allocate some default amount of memory or open some default file. All this work could be for naught if the *whatever* expression and/or assignment operator causes the object to close that file and/or release that memory (e.g., if the default constructor didn't allocate a large enough pool of memory or if it opened the wrong file).

Question No: 37 (Marks: 3)

Give the names of three ways to handle errors in a program.

Ans:

The function will throw `DivideByZero` as an exception that can then be caught by an exception-handling catch statement that catches exceptions of type `int`. The necessary construction for catching exceptions is a try catch system. If you wish to have your program check for exceptions, you must enclose the code that may have exceptions thrown in a try block.

The catch statement catches exceptions that are of the proper type. You can, for example, throw objects of a class to differentiate between several different exceptions. As well, once a catch statement is executed, the program continues to run from the end of the catch.

the errors can be handled outside of the regular code. This means that it is easier to structure the program code, and it makes dealing with errors more centralized. Finally, because the exception is passed back up the stack of calling functions, you can handle errors at any place you choose.

Question No: 38 (Marks: 5)

Consider the following code,

```

class Base{
    private:
        void base1();
    protected:
        void base2();
    public:
        void base3();
};

class Derived: public Base{
    private:
        void derived1();
    protected:
        void derived2();
    public:
        void derived3();
};

int main(){
Derived * derived = new Derived();
return 0;
}

```

Fill the table below to tell which member functions of Base and Derived classes we can access using the **Derived** pointer in the code indicated in bold.

Ans:

Function Name	Availability (Yes / No)?
base2()	no
base3()	yes
derived1()	No
derived2()	No
derived3()	Yes

Question No: 39 (Marks: 5)

What is the output produced by the following program?

```
#include<iostream.h>
```

```
void sample_function(double test) throw (int);
```

```
int main()
{
    try
    {
        cout << "Trying.\n";
        sample_function(98.6);
        cout << "Trying after call.\n";
    }
    catch(int)
    {
        cout << "Catching.\n";
    }

    cout << "End program.\n";
    return 0;
}

void sample_function(double test) throw (int)
{
    cout << "Starting sample_function.\n";
    if(test < 100)
        throw 42;
}
```

Ans:

Starting sample_function

Trying

Trying after call

Catching

End program

Question No: 40 (Marks: 10)

Write a publicly derived class “**Employee**” that is derived from base class named “**Company**”. Both classes will have function “**create()**”. Make virtual function of base class and override same function in derived class. Function create will have an output statement of your own choice.

In “**main**” Create an object of base class and call both functions with same object type.

Question No: 41 (Marks: 10)

Write a program in C++ which creates three classes named as

- 1. Equation**
- 2. Linear**
- 3. Quadratic**

Where Linear and Quadratic are inherited from Equation

Each class has the method Graph. Graph method should be pure virtual in Equation class.

This method should be overridden in both the inherited classes. It is meant to display the Graph shape of its respective class. Graph method of Linear will display the message;

Straight line

Similarly, the Graph method of Quadratic will display the message;

Parabola

In main, call the Graph method of both the Linear and Quadratic equations polymorphically through the parent class (Equation).

Ans:

```
#include "fraction.h"
#include <iostream>
#include <string>
#include <string.h>
#include <stdlib.h>
class equation;

class equation {
    int a, b;
public:
    int c ()
    {return (c);}
    void convert (Cequation);
};

class linear {
private:
    int side;
public:
    void set_side (int a)
    {side=a;}
    friend class equation;
};
```

```
void equation::convert (Cequation) {  
    a = 23;  
    b = 45;  
}
```

```
int main () {  
    cequation sqr;  
    CRectangle rect;  
    sqr.set_side(4);  
    rect.convert(sqr);  
    cout << rect.area();  
    return 0;  
}
```

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ALL answers are verified if found any mistake then Correct ACCORDINGLY

1. Function func1 of class1 is _____ in class2.
 - ▶ Public
 - ▶ **Protected**
 - ▶ Private
 - ▶ None of the given
2. The default inheritance mode is,
 - a) Public inheritance
 - b) Protected inheritance
 - c) Private inheritance**
 - d) None of the given
3. A template provides a convenient way to make a family of
 - a) Variables and data members
 - b) Functions and classes**
 - c) Classes and exceptions
 - d) Programs and algorithms
4. Then int member i of base class is accessible in class,
 - a) derived 1 only
 - b) derived 2 only**
 - c) both derived 1 derived 2
 - d) none of the given options
5. In case of public inheritance, protected members of base class will be _____ in derived class?
 - a) Private
 - b) Public
 - c) Protected**
 - d) hidden
6. By default assignment operator (=) performs,
 - a) Shallow copy**
 - b) Deep copy
 - c) Both of these
 - d) None of these
7. In protected inheritance the public members of base class become _____ in derived class.
 - a) Public
 - b) Private
 - c) Protected**
 - d) None of the given
8. We can access public members of the class from outside the class using _____ operator with its object pointer
 - a) ->

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- b) .
 - c) &
 - d) @
9. Compiler can automatically generates the following constructor/s for a class,
- a) Default constructor
 - b) Copy constructor
 - c) Both of these
 - d) None of these
10. Friend's functions of class are _____members of that class.
- a) Public
 - b) Private
 - c) Protected
 - d) None of the given
11. Methodologies to the development of reusable software relate to_____.
- a) Structure programming
 - b) Procedural programming
 - c) Generic programming
 - d) None of the given
12. Function overriding is done in context of,
- a) Single class
 - b) Single derived class
 - c) Single base class
 - d) Derived and base classes
13. Which of the following is not type of inheritance in c++?
- a) Public
 - b) Protected
 - c) Restricted
 - d) Private
14. If there is a pointer ,p, to objects of a base class and it contains the address of an object of a derived class and both classes contain a virtual member function, ding(),then the statement p->ding(); will cause the version of ding in the_____class to be executed.
- a) Base
 - b) Derived
 - c) Virtual
 - d) Implemented
15. Target class of a____function call is determined t run time.
- a) Instance
 - b) Virtual
 - c) Operator
 - d) None of the given
16. In c++ dynamic binding and polymorphism will be achieved when the member function will be_____

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- a) Private
 - b) Public
 - c) Virtual
 - d) Inline
17. To make a pure virtual, we need to give ____ after() of this function.
- a) =0;
 - b) =1;
 - c) Null;
 - d) None of the given
18. If we have not given any constructor for the class, compiler generates which of the following constructors?
- a) Explicit Default Constructor
 - b) Implicit Default Constructor
 - c) Explicit Parameterized Constructor
 - d) Implicit Parameterized Constructor
19. In c++ by default access of classes is ____.
- a) Private
 - b) Protected
 - c) Public
 - d) None of the given
20. In public inheritance the public members of base class become ____ in derived class
- a) Private
 - b) Protected
 - c) Public
 - d) None of the given
21. A class hierarchy
- a) Shows the same relationships as a organization chart
 - b) Describe "has a" relationship.
 - c) Describe "is a kind of" relationship.
 - d) Shows the same relationships as a family tree.
22. In private Inheritance the public members of the base class become ____ in derived class
- a) Public
 - b) private
 - c) Protected
 - d) None of the given option
23. In the line No.2, which constructor of student class will be
- a) Default constructor of student class
 - b) Copy constructor of student class
 - c) Both Default and Copy constructor of student class
 - d) No constructor will be called
24. Which of the following is Not casting operator in c++ standard?
- a) Static-cast

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- b) **Var-cast**
 - c) Dynamic-cast
 - d) Reinterpret-cast
25. Which of the following type of inheritance is used to model “Implemented in terms of” relationship?
- a) Public
 - b) **Private**
 - c) Protected
 - d) Confidential
26. When we want to implement one class in terms of another class then we use, **a)**
- Public inheritance**
- b) Protected inheritance
 - c) Private inheritance
 - d) None of these options
27. Public inheritance represents;
- a) “IS A “relationship**
 - b) “Has A” relationship
 - c) “IS Special kind of” relationship
 - d) None of these option
28. Suppose we have defined derived class copy constructor but have not defined base class copy constructor then compiler will,
- a) Use base class default constructor
 - b) **Generate base class copy constructor itself**
 - c) Use base class assignment operator
 - d) Nine of these option is correct
29. Child class can call constructor of its, **a) Direct base class**
- b) Indirect base class
 - c) Both direct and indirect base classes
 - d) None of these
30. We can have _____ type of member functions in a class.
- a) Public
 - b) Private
 - c) Protected
 - d) **All of these options**
31. Compiler generated copy constructor performs _____
- a) Shallow copy**
 - b) Deep copy
 - c) Both Shallow and Deep copy
 - d) None of these options
32. In c++ generic programming is done using _____
- a) Procedures

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- b) Packages
 - c) **Templates**
 - d) None of the given
33. Target class of a ____ function call is determined at run time.
- a) Instance
 - b) **Virtual**
 - c) Operator
 - d) None of the given
34. Suppose person is a user defined class. In statement "Person*pPtr", static type of pPtr is ----.
- a) pPtr
 - b) **pointer**
 - c) person
 - d) None of the given
35. Which statement is true for concrete class?
- a) It cannot be inherit from an abstract class
 - b) **It can be instantiated**
 - c) It cannot be instantiated
 - d) None of given
36. In c++, compiler can generate which of the following operators 'code,
- a) **==**
 - b) =
 - c) &
 - d) &&
37. In specialization we can,
- a) Replace child class with its base class
 - b) **Replace base class with its child class**
 - c) Replace both child and base classes interchangeably
 - d) None of the given option
38. In private inheritance derived class pointer can be assigned to base class pointer in,
- a) **Main function**
 - b) In derived class member and friend class function
 - c) In base class member and friend functions
 - d) None of the given option
39. When we access private members of the class from outside the class using ____ operator with its object pointer
- a. **->**
 - b. .
 - c. &
 - d. None of above
40. Which of the following access specifier ensure that base class member is accessible in derived class and Not outside of this class
- a. Public

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- b. Private
 - c. Protected
 - d. All of above
41. Static casting is
- a. C++ way of calling base class function from derived class
 - b. C way of calling base class function from derived class
 - c. Both of these
 - d. None of these
42. Suppose both derive and base classes have compilers generated copy constructors then derived class copy construct is
- a. Compiler generated default constructor
 - b. Compiler generated copy constructor
 - c. Both of these options are correct
 - d. None of these operators are correct
43. In case of protected inheritance, public members of the base class will be _____ in derived class
- a. Private
 - b. Public
 - c. Protected
 - d. Hidden
44. Methodologies to the development of reusable software related to _____
- a. Structured programming
 - b. Procedural programming
 - c. Generic programming
 - d. None of the given
45. A child class can call constructor of the parent call through
- a. Its constructor initialization list
 - b. Its constructor body
 - c. Both from its constructor initialization list or body
 - d. Cannot call the constructor of its parent class
46. In case of dynamic memory allocation in or class we should use
- a. Default assignment operator
 - b. User defined assignment operator
 - c. Default copy constructor
 - d. None of these
47. Consider the code below.
- ```
Class class1 {
Protected:
Int l;
};
Class class2: private calls 1{
};
```
- a) Public

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- b) Protected
  - c) Private
  - d) None of the given options
48. Consider the following two lines of code written for class student
- 1. Student subj1;
  - 2. Student subj2(subj1)
- In line 2 which constructor of student class will be called?
- a) Default constructor of student class
  - b) Copy constructor of student class
  - c) Both default and copy constructor of student class
  - d) No constructor will be called
49. Which of the following is /are casting operator(s) inc++?
- a. Static\_cast
  - b. Const\_cast
  - c. Dynamic\_cast
  - d. All of the given options
50. In case of private inheritance, private members of base class will be \_\_\_\_\_ in derived class?
- a. Private
  - b. Public
  - c. Protected
  - d. Hidden
51. Consider the code below
- ```
class class1 {  
    Private:  
    Void func1();  
};  
Class class2 : public class1{  
};
```
- Function func1 of class 1 is _____ in class2.
- a. Public
 - b. Protected
 - c. Private
 - d. None of the given options
52. We can call base class assignment operator in derived class user defined assignment operator
- a. Implicitly
 - b. Explicitly
 - c. Using both of these options
 - d. Using none of these options
53. A function call is resolved at run time in

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a. Non virtual members function **b. Virtual member function**

- c. Both none-virtual and virtual member function
- d. None of given

54. If the user does not specify the type of inheritance, then the default type of inheritance is _____

a. Public inheritance **b. Private inheritance**

- c. Protected inheritance
- d. None of given

55. Consider the code below

```
class class1 {  
public:  
Void func1();  
};  
Class class2: protected class1{  
};
```

Function func1 of class 1 is _____ in class2.

a. Public

b. **Protected**

c. Private

d. None of the given options

56. Which of the following causes run time binding?

- ▶ Declaring object of abstract class
- ▶ Declaring pointer of abstract class
- ▶ **Declaring overridden methods as non-virtual** ▶ (Page 226)
- None of the given

57. Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ **Templates** (Page 256)
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

58. A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ **an argument is passed by value.** (Page 78)
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

59. Like template functions, a class template may not handle all the types successfully.

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- ▶ True (Page 258)
- ▶ False

60. A class template may inherit from another class template.

- ▶ True (Page 288)
- ▶ False

61. By default the vector data items are initialized to ____

- ▶ 0 google
- ▶ 0.0
- ▶ 1
- ▶ null

62. In Private ----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ▶ specialization
- ▶ inheritance (Page 216)
- ▶ abstraction
- ▶ composition

63. Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ All of given (Page 256)

64. Template functions use _____ than ordinary functions.

- ▶ Greater Memory
- ▶ Lesser Memory
- ▶ Equal Memory
- ▶ None of the given options

65. Non Template Friend functions of a class are friends of _____ instance/s of that class.

- ▶ All google
- ▶ One specific
- ▶ All instances of one date type
- ▶ None of the given options

66. A pointer to a base class can point to objects of a derived class.

- ▶ True google
- ▶ False

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67. A template argument is preceded by the keyword _____.
▶ vector
▶ **class**
▶ template
▶ type*
68. Which one of the following terms must relate to polymorphism?
▶ Static allocation
▶ Static typing
▶ **Dynamic binding** (Page 239)
▶ Dynamic allocation
69. Multiple inheritance can be of type
▶ Public
▶ Private
▶ Protected
▶ **All of the given**
70. Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access
▶ **public members of Derv**
▶ protected members of Derv.
▶ private members of Derv.
▶ protected members of Base.
71. A function call is resolved at run-time in _____.
▶ non-virtual member function.
▶ **virtual member function.** (Page 239)
▶ Both non-virtual member and virtual member function.
▶ None of given
72. Two important STL associative containers are _____ and _____.
▶ **set,map**
▶ sequence,mapping
▶ setmet,multipule
▶ sit,mat
73. An abstract class is useful when,
▶ We do not derive any class from it.
▶ There are multiple paths from one derived class to another.
▶ **We do not want to instantiate its object**
▶ You want to defer the declaration of the class.

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74. Suppose you create an uninitialized vector as follows:

```
vector<int> evec;
```

After adding the statement,

```
evec.push_back(21);
```

what will happen?

- ▶ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.
- ▶ The following statement will add an element to the center of evec and will reinitialize it with the value

21.

- ▶ The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.
- ▶ The following statement will add an element to the end (the back) of evec and initialize it with the value 21.

75. Default constructor is such constructor which either has no -----or if it has some parameters these have -----

--- values

- ▶ Parameter, temporary
- ▶ Null, Parameter
- ▶ Parameter, default (Page 75)
- ▶ non of the given

76. Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

- ▶ True
- ▶ False

77. In order to define a class template, the first line of definition must be:

- ▶ `template <typename T>` (Page 257)
- ▶ `typename <template T>`
- ▶ `Template Class <ClassName>`
- ▶ `Class <Template T>`

78. A function template can not be overloaded by another function template.

- ▶ True

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► False

79. Identify the correct way of declaring an object of user defined template class A for char type members?

- `A< char > obj;`
- `<char>A obj;`
- `A obj<char>;`
- `Obj <char> A;`

80. The user must define the operation of the copy constructor.

- True
- False

81. The find() algorithm

- finds matching sequences of elements in two containers.
- finds a container that matches a specified container.
- takes iterators as its first two arguments. (Object-Oriented Programming in C++)
- takes container elements as its first two arguments.

82. Compiler performs _____ type checking to diagnose type errors,

- Static (Page 261)
- Dynamic
- Bound
- Unbound

83. Vectors contain contiguous elements stored as a[an] ____.

- variable
- array (Page 306)
- function
- datatype

84. In a de-queue, (chose the best option)

- data can be quickly inserted or deleted at any arbitrary location.
- data can be inserted or deleted at any arbitrary location, but the process is relatively slow.
- data can not be quickly inserted or deleted at either end.
- data can be inserted or deleted at either end, but the process is relatively slow.

85. Algorithms can only be implemented using STL containers.

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- ▶ True
- ▶ False(Object-Oriented Programming in C++)

86. What is a class?

- ▶ A class is a section of computer memory containing objects.
- ▶ A class is a section of the hard disk reserved for object oriented programs
- ▶ A class is the part of an object that contains the variables.
- ▶ A class is a description of a kind of object. [Click here for detail](#)

87. Inheritance is a way to

- ▶ organize data.
- ▶ pass arguments to objects of classes.
- ▶ add features to existing classes without rewriting them. (Page 27)
- ▶ improve data-hiding and encapsulation.

88. We can use "this" pointer in the constructor in the body and even in the initialization list of any class if we are careful,

- ▶ True
- ▶ False

89. _____ and _____ methods may not be declared abstract.

- ▶ private,static[Click here for detail](#)
- ▶ private,public
- ▶ static,public
- ▶ none of given

90. ----- members are somewhere between public and private members. They are used in inheritance

- ▶ protected
- ▶ public
- ▶ private
- ▶ global (Page 187)

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91. Which of these are examples of error handling techniques?

- ▶ Abnormal Termination
- ▶ Graceful Termination
- ▶ Return the illegal
- ▶ all of the given (Page 329)

92. ----- follow try block to catch the object thrown

- ▶ catch block (Page 333)
 - ▶ throw block
 - ▶ main block
 - ▶ non of the given

93. Graphical representation of the classes and objects is called object model it shows -----

- ▶ Class Name only
- ▶ Class Name and attributes
- ▶ Relationships of the objects and classes
- ▶ all of the given

94. Destructor can be overloaded

- ▶ True
- ▶ False(Page 92)

95. Which of the following is the best approach to implement generic algorithms with minimum number of coding lines?

- ▶ **Templates** (Page 256)
- ▶ Overloading
- ▶ Overriding
- ▶ Friend function/class

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96. Each try block can have _____ no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ As many as necessary.

97. class DocElement

```
{
public:
    virtual void Print() { cout << "Generic element"; }
};
class Heading : public DocElement
{
public:
    void Print() { cout << "Heading element"; }
};
class Paragraph : public DocElement
{
public:
    void Print() { cout << "Paragraph element"; }
};
void main()
{
    DocElement * p = new Paragraph();
```

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```
p->Print();  
}
```

98. When you run this program, it will print out a single line to the console output.

What will be in that line?

Select one correct answer from the following list:

- ▶ Generic element ▶
- Heading element
- ▶ Paragraph element
- ▶ Nothing will be printed.

99. Suppose we have two derived classes from a single class, can we write a method with same name in both these derived classes? Choose the best option.

- ▶ No
- ▶ Only if the two classes have the same name (Page 204)
- ▶ Only if the main program does not declare both kinds
- ▶ Yes

100. When a virtual function is called by referencing a specific object by name and using the dot member selection operator (e.g., squareObject.draw()), the reference is resolved at compile time.

- ▶ True
- ▶ False

101. Considering the resolution order in which compiler search for functions in a program; the first priority is given to;

- ▶ general template
- ▶ partial specialization
- ▶ complete specialization
- ▶ ordinary function (Page 287)

102. One purpose of an iterator in the STL is to connect algorithms and containers.

- ▶ True
- ▶ False

103. In _____, a base class can be replaced by its derived class,

- ▶ Sub-typing (Page 31)
- ▶ Super-typing
- ▶ Multiple-typing
- ▶ Restricted-typing

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104. one this pointer does not point to current object of any class,

- ▶ True
- ▶ False

105. Which of the following operator(s) take(s) one or no argument if overloaded?

- ▶ ++(Page 162)
- ▶ -
- ▶ +
- ▶ All of the given

106. Which of the following operators can not be overloaded?

- ▶ Scope resolution operator (::) (Page 141)
- ▶ Insertion operator (<<)
- ▶ Extraction operator (>>)
- ▶ The relation operator (>)

107. Which of these are examples of error handling techniques ?

- ▶ Abnormal Termination
- ▶ Graceful Termination
- ▶ Return the illegal
- ▶ all of the given (Page 329)

108. _____ “is a” relationship

- ▶ Inheritance (Page 25)

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- ▶ Polymorphism
- ▶ abstraction
- ▶ encapsulation

109. Virtual functions allow you to

- ▶ create an array of type pointer-to-base class that can hold pointers to derived classes.
- ▶ create functions that can never be accessed.
- ▶ group objects of different classes so they can all be accessed by the same function code.
- ▶ use the same function call to execute member functions of objects from different classes

110. The copy() algorithm returns an iterator to

- ▶ the last element copied from.
- ▶ the last element copied to.
- ▶ the element one past the last element copied from.
- ▶ the element one past the last element copied to.

111. If you define a vector v with the default constructor, and define another vector w with a one-argument constructor to a size of 11, and insert 3 elements into each of these vectors with push_back(), then the size() member function will return _____ for v and _____ for w.

- ▶ 11 for v and 3 for w.
- ▶ 0 for v and 0 for w.
- ▶ 0 for v and 3 for w.
- ▶ 3 for v and 11 for w.

112. Which is not the Advantage of inheritance?

- ▶ providing class growth through natural selection. (Object-Oriented Programming in C++)
- ▶ facilitating class libraries.
- ▶ avoiding the rewriting of code.
- ▶ providing a useful conceptual framework.

113. Which type of inheritance is being represented by the following statement, class X : public A, public B { };

- ▶ Single inheritance
- ▶ Multiple inheritance (Page 41)

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- ▶ Double inheritance
- ▶ None of the given options

114. When we write a class template the first line must be:

- ▶ `template < class class_name>`
- ▶ `template < class data_type>`
- ▶ `template < class T >` (Page 257)

115. Function templates should be used where code and behavior must be identical.

- ▶ True (Page 262)
- ▶ False

116. The specialization pattern `<T*>` after the name says that this specialization is to be used for every,

- ▶ data type
- ▶ meta type
- ▶ virtual type
- ▶ pointer type (Page 286)

117. A range is often supplied to an algorithm by two _____ values.

- ▶ italic
- ▶ iteration (Object-Oriented Programming in C++)
- ▶ iterator
- ▶ None of given

118. Which of the following is an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ Unique identity
- ▶ All of the given (Page 12)

119. Consider the following statement

Cupboard has books

What is the relationship between Cupboard and books?

- ▶ Composition
- ▶ Aggregation
- ▶ Inheritance
- ▶ None of the given options

120. Which sentence clearly defines an object?

- ▶ one instance of a class. (Page 23)

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- ▶ another word for a class.
- ▶ a class with static methods.
- ▶ a method that accesses class attributes.

121. _____, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.

- ▶ Friendship is one way only
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

122. The statement `objA=objB;` will cause a compiler error if the objects are of different classes.

- ▶ True
- ▶ False

123. Consider the call given below of an overloaded operator "+", *Rational_number_1* + *Rational_number_2*

Where *Rational_number_1* and *Rational_number_2* are the two objects of *Rational_number* class (a user defined class). Identify which of the above two objects will be passed as an argument to the overloaded operator function?

- ▶ *Rational_number_1*
- ▶ *Rational_number_2*
- ▶ Both *Rational_number_1* & *Rational_number_2*
- ▶ any of the two objects, randomly

124. If a class D has been derived using protected inheritance from class B (If B is a protected base and D is derived class) then public and protected members of B ----- accessed by member functions and friends of class D and classes derived from D

- ▶ can be
- ▶ cannot be
- ▶ does restrict to be
- ▶ not given

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The type that is used to declare a reference or pointer is called its _____

- ▶ abstract type
- ▶ reference type
- ▶ ▶ static type
- ▶ defaulttype

125. How the information hidden within an object can be accessed?

- ▶ Through its interface
- ▶ Through its private data members
- ▶ ▶ Through its private member functions
- ▶ Through both public and private members

126. The sub-object's life is not dependant on the life of master class in _____.

- ▶ Separation
- ▶ Composition
- ▶ None of the given

127. Encapsulation means

Select correct option:

- ▶ Extending the behaviour of class in another class
- ▶ ▶ Data and behaviour are tightly coupled within an entity (Page 16)
- ▶ One entity takes all the attributes and operations of the other
- ▶ Taking out the common features and put those in a separate class

128. An STL container can not be used to,

- ▶ hold objects of class employee.
- ▶ store elements in a way that makes them quickly accessible.
- ▶ ▶ compile c++ programs. (Object-Oriented Programming in C++)
- ▶ organize the way objects are stored in memory

129. Which of the following may not be an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ ▶ Protected data members (Page 12)
- ▶ All of given

130. Public methods of base class can ----- be accessed in its derived class

- ▶ ▶ directly (Page 179) rep
- ▶ inderectly
- ▶ simultaneously
- ▶ non of the given

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131. What is true about function templates?

- ▶ The compiler generates only one copy of the function template
- ▶ The compiler generates a copy of function respective to each type of data (Page 256)
- ▶ The compiler can only generate copy for the int type data
- ▶ None of the given.

132. When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using _____).

- ▶ scope resolution operator
- ▶ dot operator
- ▶ null operator
- ▶ Operator overloading

133. A template provides a convenient way to make a family of

- ▶ variables and data members
- ▶ functions and classes (Object-Oriented Programming in C++)
- ▶ classes and exceptions
- ▶ programs and algorithms

134. A non-virtual member function is defined in a base class and overridden in a derived class; if that function is called through a base-class pointer to a derived class object, the derived-class version is used.

- ▶ True
- ▶ False

135. If there is a pointer p to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a nonvirtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the _____ class to be executed.

- ▶ Base (Object-Oriented Programming in C++)
- ▶ Derived
- ▶ Abstract
- ▶ virtual

136. In case of multiple inheritance a derived class inherits,

- ▶ Only the public member functions of its base classes
- ▶ Only the public data members of its base classes
- ▶ Both public data members and member functions of all its base classes
- ▶ Data members and member functions of any two base classes

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137. Which of the following is incorrect line regarding function template?

- ▶ `template<class T>`
- ▶ `template <typename U>`
- ▶ `Class<template T>` (Page 257)
- ▶ `template < class T, class U>`

138. Consider a class named Vehicle, which of the following can be the instance of class Vehicle?

- i. Car
 - ii. Computer
 - iii. Desk
 - iv. Ahmed
 - v. Bicycle
 - vi. Truck
- 1,4,5
 - 2,5,6
 - 1,2,3,6
 - 1, 5, 6 Conceptual

139. Consider the code below,

```
class Fred {  
public:  
Fred();  
...  
};  
int main()  
{  
Fred a[10];  
Fred* p = new Fred[10];  
...  
}
```

Select the best option,

- ▶ `Fred a[10];` calls the default constructor 09 times `Fred* p = new Fred[10];` calls the default constructor 10 times
- ▶ Produce an error

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- ▶ Fred a[10]; calls the default constructor 11 times

Fred* p = new Fred[10]; calls the default constructor 11 times

- ▶ Fred a[10]; calls the default constructor 10 times

Fred* p = new Fred[10]; calls the default constructor 10 times

140. When a variable is defined as static in a class then all objects of this class,

- ▶ Have different copies of this variable
- ▶ Have same copy of this variable (Page 110)
- ▶ Can not access this variable
- ▶ None of given

141. Which of the following operators always takes no argument if overloaded?

- ▶ /
- ▶ -
- ▶ +
- ▶ ++ (Page 162)

142. It is sometimes useful to specify a class from which no objects will ever be created.

- ▶ True(Object-Oriented Programming in C++)
- ▶ False

143. When a subclass specifies an alternative definition for an attribute or method of its superclass, it is _____ the definition in the superclass.

- ▶ overload
- ▶ overriding (Page 34)
- ▶ copy riding
- ▶ none of given

144. The mechanism of selecting function at run time according to the nature of calling object is called,

- ▶ late binding
- ▶ static binding
- ▶ virtual binding
- ▶ None of the given options (Page 227)

145. Dynamic binding means that target function for a call is selected at run time

Which one of the following functions returns the total number of elements in a vector.

- ▶ length();
- ▶ size(); (Page 318)
- ▶ ele();
- ▶ veclen();

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146. The main function of scope resolution operator (::) is,

- ▶ To define an object
- ▶ To define a data member
- ▶ [To link the definition of an identifier to its declaration](#)
- ▶ To make a class private

Google

147. When is a constructor called?

- ▶ Each time the constructor identifier is used in a program statement
- ▶ [During the instantiation of a new object \(Object-Oriented Programming in C++\)](#)
- ▶ During the construction of a new class
- ▶ At the beginning of any program execution

148. Associativity can be changed in operator overloading.

- ▶ True
- ▶ [False \(Page 141\)](#)

149. A normal C++ operator that acts in special ways on newly defined data types is said to be

- ▶ glorified.
- ▶ encapsulated.
- ▶ classified.
- ▶ [overloaded.](#)

150. Which operator can not be overloaded?

- ▶ The relation operator (>=)
- ▶ Assignment operator (=)
- ▶ Script operator ([])
- ▶ [Conditional operator \(? : \) \(Page 141\)](#)

151. Suppose obj1 and obj2 are two objects of a user defined class A. An + operator is overloaded to add obj1 and obj2 using the function call obj1+obj2.

Identify the correct function prototype against the given call?

- ▶ A operator + (A &obj);
- ▶ [int + operator\(\); \(Page 143\)](#)
- ▶ int operator (plus) ();
- ▶ A operator(A &obj3);

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152. Suppose that the Test class does not have an overloaded assignment operator. What happens when an assignment $a=b$; is given for two Test objects a and b?

- ▶ The automatic assignment operator is used
- ▶ **The copy constructor is used**
- ▶ Compiler error
- ▶ Run-time error

153. A static member function can be called, even when a class is not _____.

- ▶ Declared
- ▶ Define
- ▶ **Instantiated**
- ▶ Called

154. Identify which of the following overloaded operator function's declaration is appropriate for the given call?

Rational_number_1 + 2.325

Where Rational_number_1 is an object of user defined class Rational_number.

- ▶ Rational_number operator+(Rational_number & obj);
- ▶ Rational_number operator+(double& obj);
- ▶ Rational_number operator+(Rational_number &obj, double& num);
- ▶ operator+(double& obj);

155. What problem(s) may occur when we copy objects without using deep copy constructor?

- ▶ Dangling pointer
- ▶ Memory Leakage
- ▶ **All of the given**
- ▶ System crash

156. _____ provide the facility to access the data member.

- ▶ **accesser function**
- ▶ private function
- ▶ inline function
- ▶ None of the given

157. Constant objects cannot change their state,

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▶ **True**

▶ False

158. The _____ relationship indicates that an object contains other objects.

▶ None of given

▶ **'has-a'**

▶ 'is-a'

▶ 'be-

159. Which one of the following features of OOP is used to derive a class from another?

▶ Encapsulation

▶ Polymorphism

▶ Data hiding

▶ **Inheritance**

160. _____ satisfy the condition of polymorphism

▶ Carbon

▶ Diamond

▶ Coal

▶ **all of the given**

161. A generalization-specialization relation between classes are implemented using

▶ data hiding

▶ friend classes

▶ encapsulation

▶ **inheritance**

162. The $\geq$ operator can be overloaded.

▶ **True**

▶ False

163. In order to free the memory occupied by the object, we use -----

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- ▶ Constructor
- ▶ **Destructor**
- ▶ Shallow Copy
- ▶ Deep Copy

164. Which of the following is not an example of multiple inheritances?

- ▶ Mermaid
- ▶ **Woman**
- ▶ None of the given
- ▶ Amphibious Vehicle

165. Static variable can be initialized more than once.

- ▶ True
- ▶ **False**

166. A generic class showing all the common attributes and a behavior of other classes represents a very important feature in oop called -----

- ▶ **Inheritance**
- ▶ Encapsulation
- ▶ Polymorphism
- ▶ Abstraction

167. The parameters given in template definition other than those used for mentioning templates types are called ____

- a. Type Parameters
- b. **Non Type parameters**
- c. Default Type Parameters
- d. None of the given option

168. Which of the following is the correct way to define a template class X?

- e. Class < typename T> class X {}
- f. Template class C {}
- g. **Template < typename T> class X {}**
- h. Typename < class T > class X {}

169. In resolution order of function template, compiler searches for _____ in the end

- a. Complete specialization
- b. Partial specialization
- c. Ordinary function

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d. Generic template

170. _____ may inherit from a complete specialization
- a. Partial specialization
 - b. Complete specialization
 - c. Ordinary class
 - d. **All of given options**
171. Which of the following may inherit from the ordinary class?
- a. Class template
 - b. Partial specialization
 - c. Complete specialization
 - d. **All of given option**
172. Which of the following is a sequence container?
- a. Map
 - b. **Vector**
 - c. Set
 - d. multiset
173. Each _____ of a template class by default becomes function template
- a. Data member
 - b. **Member function**
 - c. Type parameter
 - d. Object
174. When we specialize a function template, it is called _____
- a. Function overloading
 - b. Function overriding
 - c. **Function template overloading**
 - d. Function template overriding
175. Which of the following is NOT an associative container?
- a. Set
 - b. Multiset
 - c. Map
 - d. **Group**
176. A _____ is a pointer which is declared outside the _____
- a. Container, cursor
 - b. **Cursor, container**
 - c. Container, class
 - d. This, cursor
177. Which of the following are collectively referred to as the first class containers?
- a. Associative containers and container adapters
 - b. **Sequence containers and associative containers**
 - c. Sequence containers and container adapters

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- d. None of the given options
178. _____ class is a single class that provides functionality to operate on different type of data
- a. Friend
 - b. **Template**
 - c. Ordinary
 - d. None of the give options
179. Which of the following can be passed as type argument to template?
- a. Primitive type
 - b. User defined types
 - c. **Both primitive type and user defined types**
 - d. None of the given options
180. Which of the following is known as Dereference operator in C++?
- a. **+b.**
 - ***
 - c. &
 - d. ::
181. We can change behavior of template using _____
- a. **Template parameters**
 - b. Function parameters
 - c. Class Templates
 - d. none of the given options
182. Which of the following represents partial specialization?
- a. Template < int, char >
 - b. Template < class T, Type t>
 - c. **Template<class T, Class U, int>**
 - d. Template< typename T, class W>
183. When we declare template function as friend of any class, then which of the followingspecialization(s) of template function also become friend of the class granting friendship?
- a. Explicit
 - b. Implicit
 - c. Partial
 - d. **All of the given options**
184. In statement “template <class T, class U, int I = 5>” then non-type parameters is
- a. Class T
 - b. Class U
 - c. **int I**
 - d. All of the given option
185. Which of the following is the correct syntax for passing two type argument to atemplate?

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- a. `template < type T, type U>`
 - b. `template < typename T, typename U>`
 - c. `template <type T,U>`
 - d. `template Typename < T,U >`
186. Which of the following represents complete specialization?
- a. `Template < int, char >`
 - b. `Template < class T, float>`
 - c. `Template<class T, Class U , int>`
 - d. `Template< class , class W>`
187. Which of the following is/are the key component of standard template library?
- a. Containers
 - b. Iterators
 - c. Algorithm
 - d. **All of the given option**
188. Which of the following statement is true about partial specialization?
- a. Function templates cannot have partial specialization
 - b. Class template cannot have partial specialization
 - c. **Both class templates and function template can have partial specialization.**
 - d. None of the given
189. Which of the following is correct code portion to initialize static data member "value" of int type for class X?
- a. `Template <class T> X::value =0;`
 - b. `Template<class T>int X::value =0;`
 - c. **`Template<class T> int X < T >::value = 0;`**
 - d. `Template<class T> X < T >::value =0;`
190. When we want to have exactly identical operations on different data type, _____ are used
- a. **Function Template**
 - b. Function Overloading
 - c. Function Overriding
 - d. None of the given option
191. In order to define a class template the first line of definition must be:
- a. **`Template <typename T>`**
 - b. `typename < template T>`
 - c. `Template class <ClassName>`
 - d. `Class <Template T>`
192. In case of template specialization, if compiler cannot find required complete specialization then it searches for some _
- a. General template
 - b. Complete template
 - c. **Partial specialization**

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d. None of the given

193. Which of the following is the correct code to instantiate the object of given templatevector class of int type?

```
Template <class T>  
class vector  
{  
};
```

- a. Vector obj <int>;
- b. Vector obj <> int;
- c. Vector <int>obj;
- c. Vector int obj;

194. Which of the following types of container is NOT provided by standard TemplateLibrary?

- a. Sequence containers
- b. Structured containers
- c. Associative containers
- d. Container adapters

195. A class template _____

- a. Facilitates rescue of class
- b. Dose not facilitates reuse of class
- c. Does not support generic methods
- d. Does not support static members

196. Suppose a template class “test” has a static data members, how many copies of thisstatic data member will be created when template class is instantiated by writing following line of C++ code?

```
Test <char>x,y,z
```

- a. Zero
- b. One Correct
- c. Two

197. STL stands for:

- a. Stack Template Library
- b. Standard Template Library
- c. Structured Template Library
- d. Secure Template Library

198. Which of the following is not a sequence container?

- a. Vector
- b. Deque
- c. Map
- d. List

199. Suppose a template class “test” has a static data members, how many copies of thisstatic data member will be created when template class is instantiated by writing following line of C++ code?

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Test <int>x; Test <char>y;

- a. Zero
- b. One
- c. Two**
- d. Three

200. Suppose we have a template class “Test” which have a static data member “data” which is initialized with 0. What will be output when following code is written in main() function for given class?

```
Test<int> a, b; a.data = 5;
```

```
b.data = 7; cout<<a.data;
```

- a. 0
- b. 1
- c. 5**
- d. 7 Correct**

201. In resolution order, highest priority is given to _____ in template specialization.

- a. General template
- b. **Complete specialization**
- c. Partial specialization
- d. None of the given options

202. Which of the following is a constrained version of some first-class container?

- a. Sequence containers
- b. Associative containers
- c. Container Adapter**
- d. None of the given options

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CS304- Object Oriented Programming
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FINALTERM EXAMINATION

14 Feb, 2011

CS304- Object Oriented Programming (Session - 3)

Question No: 1 (Marks: 1) - Please choose one

Which of the following causes run time binding?

- ▶ Declaring object of abstract class
- ▶ Declaring pointer of abstract class
- ▶ **Declaring overridden methods as non-virtual (Page 226)**
- ▶ None of the given

Question No: 2 (Marks: 1) - Please choose one

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ **Templates (Page 256)**
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

Question No: 3 (Marks: 1) - Please choose one

A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ **an argument is passed by value. (Page 78)**
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

Question No: 4 (Marks: 1) - Please choose one

Like template functions, a class template may not handle all the types successfully.

- ▶ **True (Page 258)**
- ▶ False

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Question No: 6 (Marks: 1) - Please choose one

A class template may inherit from another class template.

- ▶ **True** (Page 288)
- ▶ False

Question No: 7 (Marks: 1) - Please choose one

By default the vector data items are initialized to _____

- ▶ **0** [Click here for detail](#)
- ▶ 0.0
- ▶ 1
- ▶ null

Question No: 8 (Marks: 1) - Please choose one

In Private ----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ▶ specialization
- ▶ **inheritance** (Page 216)
- ▶ abstraction
- ▶ composition

Question No: 9 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ **All of given** (Page 256)

Question No: 10 (Marks: 1) - Please choose one

Template functions use _____ than ordinary functions.

- ▶ Greater Memory
- ▶ **Lesser Memory**
- ▶ Equal Memory
- ▶ None of the given options

Question No: 11 (Marks: 1) - Please choose one

Non Template Friend functions of a class are friends of _____ instance/s of that class.

- ▶ **All** [Click here for detail](#)
- ▶ One specific
- ▶ All instances of one data type
- ▶ None of the given options

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Question No: 12 (Marks: 1) - Please choose one

A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ **an argument is passed by value.** (Page 78)
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

Question No: 13 (Marks: 1) - Please choose one

A pointer to a base class can point to objects of a derived class.

- ▶ **True** [Click here for detail](#)
- ▶ False

Question No: 14 (Marks: 1) - Please choose one

A template argument is preceded by the keyword _____.

- ▶ vector
- ▶ **class** [Click here for Detail](#)
- ▶ template
- ▶ type*

Question No: 15 (Marks: 1) - Please choose one

Which one of the following terms must relate to **polymorphism**?

- ▶ Static allocation
- ▶ Static typing
- ▶ **Dynamic binding** (Page 239)
- ▶ Dynamic allocation

Question No: 16 (Marks: 1) - Please choose one

Multiple inheritance can be of type

- ▶ Public
- ▶ Private
- ▶ Protected
- ▶ **All of the given** [Click here for detail](#)

Question No: 17 (Marks: 1) - Please choose one

Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- ▶ **public members of Derv.** (Object-Oriented Programming in C++)
- ▶ protected members of Derv.
- ▶ private members of Derv.
- ▶ protected members of Base.

Question No: 18 (Marks: 1) - Please choose one

A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ **an argument is passed by value. (Page 78) (rep)**
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

Question No: 19 (Marks: 1) - Please choose one

A function call is resolved at run-time in _____

- ▶ non-virtual member function.
- ▶ **virtual member function. (Page 239)**
- ▶ Both non-virtual member and virtual member function.
- ▶ None of given

Question No: 20 (Marks: 1) - Please choose one

Two important STL associative containers are _____ and _____.

- ▶ **set,map (Object-Oriented Programming in C++)**
- ▶ sequence,mapping
- ▶ setmet,multipule
- ▶ sit,mat

Question No: 21 (Marks: 1) - Please choose one

An abstract class is useful when,

- ▶ We do not derive any class from it.
- ▶ There are multiple paths from one derived class to another.
- ▶ **We do not want to instantiate its object. (Object-Oriented Programming in C++)**
- ▶ You want to defer the declaration of the class.

Question No: 22 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ **All of given (Page 256) rep**

Question No: 23 (Marks: 1) - Please choose one

By default the vector data items are initialized to _____

- ▶ **0 Click here for Detail rep**
- ▶ 0.0
- ▶ 1
- ▶ null

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Question No: 24 (Marks: 1) - Please choose one

Suppose you create an uninitialized vector as follows:

```
vector<int> evec;
```

After adding the statment,

```
evec.push_back(21);
```

what will happen?

- ▶ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.
- ▶ The following statement will add an element to the center of evec and will reinitialize it with the value 21.
- ▶ The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.
- ▶ **The following statement will add an element to the end (the back) of evec and initialize it with the value 21. [Click here for detail](#)**

Question No: 25 (Marks: 1) - Please choose one

Default constructor is such constructor which either has no -----or if it has some parameters these have ---- values

- ▶ Parameter, temporary
- ▶ Null, Parameter
- ▶ **Parameter, default (Page 75)**
- ▶ non of the given

Question No: 26 (Marks: 1) - Please choose one

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ **Templates (Page 256) rep**
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

Question No: 27 (Marks: 1)

Describe the way to declare a template class as a friend of any class.

Question No: 28 (Marks: 1)

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

- ▶ **True [Click here for detail](#)**
- ▶ False

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Question No: 29 (Marks: 1)

In order to define a class template, the first line of definition must be:

- ▶ **template <typename T> (Page 257)**
- ▶ typename <template T>
- ▶ Template Class <ClassName>
- ▶ Class <Template T>

Question No: 30 (Marks: 1)

In case of multiple inheritance a derived class inherits,

- ▶ Only the public member functions of its base classes
 - ▶ Only the public data members of its base classes
 - ▶ **Both public data members and member functions of all its base classes**
- [Click here for detail](#)
- ▶ Data members and member functions of any two base classes

Question No: 31 (Marks: 1)

In Private ----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ▶ specialization
- ▶ **inheritance (Page 216) rep**
- ▶ abstraction
- ▶ composition

Question No: 32 (Marks: 1)

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ **Templates (Page 256) rep**
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

FINAL TERM EXAMINATION
Spring 2010
CS304- Object Oriented Programming

Question No: 1 (Marks: 1) - Please choose one

A template argument is preceded by the keyword _____.

- ▶ vector
- ▶ **class (Object-Oriented Programming in C++)**
- ▶ template
- ▶ type*

Question No: 2 (Marks: 1) - Please choose one

Which of the following causes run time binding?

- ▶ Declaring object of abstract class
- ▶ Declaring pointer of abstract class
- ▶ **Declaring overridden methods as non-virtual (Page 226)**
- ▶ None of the given

Question No: 3 (Marks: 1) - Please choose one

A function template can not be overloaded by another function template.

- ▶ **True (Object-Oriented Programming in C++)**
- ▶ False

Question No: 4 (Marks: 1) - Please choose one

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ **Templates (Page 256) rep**
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

Question No: 5 (Marks: 1) - Please choose one

Identify the correct way of declaring an object of user defined template class **A** for char type members?

- ▶ **A< char > obj; (Object-Oriented Programming in C++)**
- ▶ <char>A obj;
- ▶ A obj<char>;
- ▶ Obj <char> A;

Question No: 6 (Marks: 1) - Please choose one

The user must define the operation of the copy constructor.

- ▶ **True** [Click here for detail](#)
- ▶ False

Question No: 7 (Marks: 1) - Please choose one

Template functions use _____ than ordinary functions.

- ▶ Greater Memory
- ▶ **Lesser Memory**
- ▶ Equal Memory
- ▶ None of the given options

Question No: 8 (Marks: 1) - Please choose one

The find() algorithm

- ▶ finds matching sequences of elements in two containers.
- ▶ finds a container that matches a specified container.
- ▶ **takes iterators as its first two arguments. (Object-Oriented Programming in C++)**
- ▶ takes container elements as its first two arguments.

Question No: 9 (Marks: 1) - Please choose one

Compiler performs _____ type checking to diagnose type errors,

- ▶ **Static (Page 261)**
- ▶ Dynamic
- ▶ Bound
- ▶ Unbound

Question No: 10 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ **All of given (Page 256) rep**

Question No: 11 (Marks: 1) - Please choose one

Vectors contain contiguous elements stored as a[an] ____.

- ▶ variable
- ▶ **array (Page 306)**
- ▶ function
- ▶ datatype

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Question No: 12 (Marks: 1) - Please choose one

Suppose you create an uninitialized vector as follows:

```
vector<int> evec;
```

After adding the statment,

```
evec.push_back(21);
```

what will happen?

- ▶ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.
- ▶ The following statement will add an element to the center of evec and will reinitialize it with the value 21.
- ▶ The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.
- ▶ **The following statement will add an element to the end (the back) of evec and initialize it with the value 21. [Click here for detail](#)**

Question No: 13 (Marks: 1) - Please choose one

In a de-queue, (chose the best option)

- ▶ data can be quickly inserted or deleted at any arbitrary location.
 - ▶ **data can be inserted or deleted at any arbitrary location, but the process is relatively slow.**
- (Object-Oriented Programming in C++)**
- ▶ data can not be quickly inserted or deleted at either end.
 - ▶ data can be inserted or deleted at either end, but the process is relatively slow.

Question No: 14 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

- ▶ True
- ▶ **False (Object-Oriented Programming in C++)**

Question No: 15 (Marks: 1) - Please choose one

What is a class?

- ▶ A class is a section of computer memory containing objects.
- ▶ A class is a section of the hard disk reserved for object oriented programs
- ▶ A class is the part of an object that contains the variables.
- ▶ **A class is a description of a kind of object. [Click here for detail](#)**

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Question No: 16 (Marks: 1) - Please choose one
Inheritance is a way to

- ▶ organize data.
- ▶ pass arguments to objects of classes.
- ▶ **add features to existing classes without rewriting them. (Page 27)**
- ▶ improve data-hiding and encapsulation.

Question No: 17 (Marks: 1) - Please choose one

We can use "**this**" pointer in the constructor in the body and even in the initialization list of any class if we are careful,

- ▶ **True** [Click here for detail](#)
- ▶ False

Question No: 18 (Marks: 1) - Please choose one
_____ and _____ methods may not be declared abstract.

- ▶ **private,static** [Click here for detail](#)
- ▶ private,public
- ▶ static,public
- ▶ none of given

Question No: 19 (Marks: 1) - Please choose one

Default constructor is such constructor which either has no -----or if it has some parameters these have ----- values

- ▶ Parameter, temporary
- ▶ Null, Parameter
- ▶ **Parameter, default** (Page 75) rep
- ▶ non of the given

Question No: 20 (Marks: 1) - Please choose one

Public methods of base class can ----- be accessed in its derived class

- ▶ **directly** (Page 179)
- ▶ indirectly
- ▶ simultaneously
- ▶ non of the given

Question No: 21 (Marks: 1) - Please choose one

The type that is used to declare a reference or pointer is called its -----

- ▶ default type
- ▶ **static type** (Page 185)
- ▶ abstract type
- ▶ reference type

Question No: 22 (Marks: 1) - Please choose one

----- members are somewhere between public and private members. They are used in inheritance

- ▶ **protected** (Page 187)
- ▶ public
- ▶ private
- ▶ global

Question No: 23 (Marks: 1) - Please choose one

Which of these are examples of error handling techniques?

- ▶ Abnormal Termination
- ▶ Graceful Termination
- ▶ Return the illegal
- ▶ **all of the given** (Page 329)

Question No: 24 (Marks: 1) - Please choose one

----- follow try block to catch the object thrown

- ▶ **catch block** (Page 333)
- ▶ throw block
- ▶ main block
- ▶ non of the given

Question No: 25 (Marks: 1) - Please choose one

Graphical representation of the classes and objects is called object model it shows -----

- ▶ Class Name only
- ▶ Class Name and attributes
- ▶ Relationships of the objects and classes
- ▶ **all of the given** [Click here for detail](#)

Question No: 26 (Marks: 1) - Please choose one

Destructor can be overloaded

- ▶ True
- ▶ **False** (Page 92)

FINALTERM EXAMINATION
Spring 2010
CS304- Object Oriented Programming (Session - 3)

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Question No: 1 (Marks: 1) - Please choose one

Which one of the following terms must relate to **polymorphism**?

- ▶ Static allocation
- ▶ Static typing
- ▶ **Dynamic binding (Page 239)**
- ▶ Dynamic allocation

Question No: 2 (Marks: 1) - Please choose one

Which of the following causes run time binding?

- ▶ Declaring object of abstract class
- ▶ Declaring pointer of abstract class
- ▶ **Declaring overridden methods as non-virtual (Page 226)**
- ▶ None of the given

Question No: 3 (Marks: 1) - Please choose one

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ **Templates (Page 256) rep**
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

Question No: 4 (Marks: 1) - Please choose one

Which of the following is the best approach to implement generic algorithms with minimum number of coding lines?

- ▶ **Templates (Page 256)**
- ▶ Overloading
- ▶ Overriding
- ▶ Friend function/class

Question No: 5 (Marks: 1) - Please choose one

Like template functions, a class template may not handle all the types successfully.

- ▶ **True (Page 258) rep**
- ▶ False

Question No: 6 (Marks: 1) - Please choose one

A class template may inherit from another class template.

- ▶ **True (Page 288) rep**
- ▶ False

Question No: 7 (Marks: 1) - Please choose one

Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- ▶ **public members of Derv. (Object-Oriented Programming in C++) rep**
- ▶ protected members of Derv.
- ▶ private members of Derv.
- ▶ protected members of Base.

Question No: 8 (Marks: 1) - Please choose one

A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ **an argument is passed by value. (Page 78) rep**
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

Question No: 9 (Marks: 1) - Please choose one

Each try block can have _____ no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ **As many as necessary. [Click here for detail](#)**

A single try block can have multiple catch blocks but only one finally block.

Question No: 10 (Marks: 1) - Please choose one

class DocElement

```
{
public:
    virtual void Print() { cout << "Generic element"; }
};
class Heading : public DocElement
{
public:
    void Print() { cout << "Heading element"; }
};
class Paragraph : public DocElement
{
public:
    void Print() { cout << "Paragraph element"; }
};
void main()
{
    DocElement * p = new Paragraph();
```

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```
p->Print();  
}
```

When you run this program, it will print out a single line to the console output.

What will be in that line?

Select one correct answer from the following list:

- ▶ Generic element
- ▶ Heading element
- ▶ Paragraph element
- ▶ **Nothing will be printed.**

Question No: 11 (Marks: 1) - Please choose one

Suppose we have two derived classes from a single class, can we write a method with same name in both these derived classes? Choose the best option.

- ▶ No
- ▶ **Only if the two classes have the same name (Page 204)**
- ▶ Only if the main program does not declare both kinds
- ▶ Yes

Question No: 12 (Marks: 1) - Please choose one

When a virtual function is called by referencing a specific object by name and using the dot member selection operator (e.g., squareObject.draw()), the reference is resolved at compile time.

- ▶ True
- ▶ **False (Object-Oriented Programming in C++)**
(the function invocation is resolved at compile time)

Question No: 13 (Marks: 1) - Please choose one

Considering the resolution order in which compiler search for functions in a program; the first priority is given to;

- ▶ general template
- ▶ partial specialization
- ▶ complete specialization
- ▶ **ordinary function (Page 287)**

Question No: 14 (Marks: 1) - Please choose one

Vectors contain contiguous elements stored as a[an] ____.

- ▶ variable
- ▶ **array (Page 306)**
- ▶ function
- ▶ datatype

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Question No: 15 (Marks: 1) - Please choose one

By default the vector data items are initialized to ____

- ▶ 0 [Click here for detail](#) rep
- ▶ 0.0
- ▶ 1
- ▶ null

Question No: 16 (Marks: 1) - Please choose one

One purpose of an iterator in the STL is to connect algorithms and containers.

- ▶ True (Object-Oriented Programming in C++)
- ▶ False

Question No: 17 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

- ▶ True
- ▶ False (Object-Oriented Programming in C++) rep

Question No: 18 (Marks: 1) - Please choose one

In _____, a base class can be replaced by its derived class,

- ▶ Sub-typing (Page 31)
- ▶ Super-typing
- ▶ Multiple-typing
- ▶ Restricted-typing

Question No: 19 (Marks: 1) - Please choose one

this pointer does not point to current object of any class,

- ▶ True
- ▶ False [Click here for detail](#)

The **this pointer** is a hidden pointer inside every class member function that points to the class object the member function is working with.

Question No: 20 (Marks: 1) - Please choose one

Which of the following operator(s) take(s) one or no argument if overloaded?

- ▶ ++ (Page 162)
- ▶ -
- ▶ +
- ▶ All of the given

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Question No: 21 (Marks: 1) - Please choose one

Which of the following operators can not be overloaded?

- ▶ **Scope resolution operator (::)** (Page 141)
- ▶ Insertion operator (<<)
- ▶ Extraction operator (>>)
- ▶ The relation operator (>)

Question No: 22 (Marks: 1) - Please choose one

The type that is used to declare a reference or pointer is called its -----

- ▶ default type
- ▶ **static type** (Page 185) rep
- ▶ abstract type
- ▶ reference type

Question No: 23 (Marks: 1) - Please choose one

----- members are somewhere between public and private members. They are used in inheritance

- ▶ **protected** (Page 187) rep
- ▶ public
- ▶ private
- ▶ global

Question No: 24 (Marks: 1) - Please choose one

Which of these are examples of error handling techniques ?

- ▶ Abnormal Termination
- ▶ Graceful Termination
- ▶ Return the illegal
- ▶ **all of the given** (Page 329)

Question No: 25 (Marks: 1) - Please choose one

_____ "is a" relationship

- ▶ **Inheritance** (Page 25)
- ▶ Polymorphism
- ▶ abstraction
- ▶ encapsulation

Question No: 26 (Marks: 1) - Please choose one

Graphical representation of the classes and objects is called object model it shows -----

- ▶ Class Name only
- ▶ Class Name and attributes
- ▶ Relationships of the objects and classes
- ▶ **all of the given** [Click here for detail](#) rep

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FINALTERM EXAMINATION
Spring 2010
CS304- Object Oriented Programming (Session - 4)

Question No: 1 (Marks: 1) - Please choose one

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

- ▶ **True** [Click here for detail](#) rep
- ▶ False

Question No: 2 (Marks: 1) - Please choose one

Virtual functions allow you to

- ▶ create an array of type pointer-to-base class that can hold pointers to derived classes.
- ▶ create functions that can never be accessed.
- ▶ group objects of different classes so they can all be accessed by the same function code.
- ▶ **use the same function call to execute member functions of objects from different classes (Object-Oriented Programming in C++)**

Question No: 3 (Marks: 1) - Please choose one

- ▶ **True** [Click here for detail](#) rep
- ▶ False

Question No: 4 (Marks: 1) - Please choose one

A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ **an argument is passed by value. (Page 78) rep**
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

Question No: 5 (Marks: 1) - Please choose one

Each try block can have _____ no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ **As many as necessary.** [Click here for detail](#) rep

Question No: 6 (Marks: 1) - Please choose one

Non Template Friend functions of a class are friends of _____ instance/s of

- ▶ **All** [Click here for detail](#) rep
- ▶ One specific
- ▶ All instances of one data type
- ▶ None of the given options

Question No: 7 (Marks: 1) - Please choose one

Template functions use _____ than ordinary functions.

- ▶ Greater Memory
- ▶ **Lesser Memory**
- ▶ Equal Memory
- ▶ None of the given options

Question No: 8 (Marks: 1) - Please choose one

The find() algorithm

- ▶ finds matching sequences of elements in two containers.
- ▶ finds a container that matches a specified container.
- ▶ **takes iterators as its first two arguments. (Object-Oriented Programming in C++)**
- ▶ takes container elements as its first two arguments.

Question No: 9 (Marks: 1) - Please choose one

The copy() algorithm returns an iterator to

- ▶ the last element copied from.
- ▶ the last element copied to.
- ▶ the element one past the last element copied from.
- ▶ **the element one past the last element copied to. (Object-Oriented Programming in C++)**

Question No: 10 (Marks: 1) - Please choose one

If you define a vector v with the default constructor, and define another vector w with a one-argument constructor to a size of 11, and insert 3 elements into each of these vectors with push_back(), then the size() member function will return _____ for v and _____ for w.

- ▶ 11 for v and 3 for w.
- ▶ 0 for v and 0 for w.
- ▶ 0 for v and 3 for w.
- ▶ **3 for v and 11 for w. (Object-Oriented Programming in C++)**

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Question No: 11 (Marks: 1) - Please choose one

Which is not the Advantage of inheritance?

- ▶ **providing class growth through natural selection.** (Object-Oriented Programming in C++)
- ▶ facilitating class libraries.
- ▶ avoiding the rewriting of code.
- ▶ providing a useful conceptual framework.

Question No: 12 (Marks: 1) - Please choose one

class DocElement

```
{
public:
    virtual void Print() { cout << "Generic element"; }
};
class Heading : public DocElement
{
public:
    void Print() { cout << "Heading element"; }
};
class Paragraph : public DocElement
{
public:
    void Print() { cout << "Paragraph element"; }
};
void main()
{
    DocElement * p = new Paragraph();

    p->Print();
}
```

When you run this program, it will print out a single line to the console output.

What will be in that line?

Select one correct answer from the following list:

- ▶ Generic element
- ▶ Heading element
- ▶ Paragraph element
- ▶ **Nothing will be printed.**

Question No: 13 (Marks: 1) - Please choose one

Which type of inheritance is being represented by the following statement,
`class X : public A, public B { ... ... };`

- ▶ Single inheritance
- ▶ **Multiple inheritance (Page 41)**
- ▶ Double inheritance
- ▶ None of the given options

Question No: 14 (Marks: 1) - Please choose one

When we write a class template the first line must be:

- ▶ `template < class class_name >`
- ▶ `template < class data_type >`
- ▶ **`template < class T >` (Page 257)**

Here T can be replaced with any name but it is preferable.

▶ `class class-name()`
`class template < class_name >`

Question No: 15 (Marks: 1) - Please choose one

Function templates should be used where code and behavior must be identical.

- ▶ **True (Page 262)**
- ▶ False

Question No: 16 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ **All of given (Page 256) rep**

Question No: 17 (Marks: 1) - Please choose one

The specialization pattern <T*> after the name says that this specialization is to be used for every,

- ▶ data type
- ▶ meta type
- ▶ virtual type
- ▶ **pointer type (Page 286)**

Question No: 18 (Marks: 1) - Please choose one

A range is often supplied to an algorithm by two _____ values.

- ▶ italic
- ▶ **iteration (Object-Oriented Programming in C++)**
- ▶ iterator
- ▶ None of given

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Question No: 19 (Marks: 1) - Please choose one

Which of the following is an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ Unique identity
- ▶ **All of the given (Page 12)**

Question No: 20 (Marks: 1) - Please choose one

Consider the following statement

Cupboard has books

What is the relationship between Cupboard and books?

- ▶ Composition
- ▶ **Aggregation**
- ▶ Inheritance
- ▶ None of the given options

Question No: 21 (Marks: 1) - Please choose one

Which sentence clearly defines an object?

- ▶ **one instance of a class. (Page 23)**
- ▶ another word for a class.
- ▶ a class with static methods.
- ▶ a method that accesses class attributes.

Question No: 22 (Marks: 1) - Please choose one

_____, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.

- ▶ **Friendship is one way only [Click here for detail](#)**
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

Question No: 23 (Marks: 1) - Please choose one

The statement `objA=objB;` will cause a compiler error if the objects are of different classes.

- ▶ True
- ▶ **False (Object-Oriented Programming in C++)**

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Question No: 24 (Marks: 1) - Please choose one

Consider the call given below of an overloaded operator "+",

Rational_number_1 + Rational_number_2

Where Rational_number_1 and Rational_number_2 are the two objects of Rational_number class (a user defined class). Identify which of the above two objects will be passed as an argument to the overloaded operator function?

- ▶ Rational_number_1
- ▶ Rational_number_2
- ▶ Both Rational_number_1 & Rational_number_2
- ▶ **any of the two objects, randomly**

Question No: 25 (Marks: 1) - Please choose one

If a class D has been derived using protected inheritance from class B (If B is a protected base and D is derived class) then public and protected members of B ----- accessed by member functions and friends of class D and classes derived from D

- ▶ **can be** [Click here for detail](#)
- ▶ cannot be
- ▶ does restrict to be
- ▶ not given

Question No: 26 (Marks: 1) - Please choose one

In Private ----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ▶ specialization
- ▶ **inheritance** (Page 216) rep
- ▶ abstraction
- ▶ composition

FINAL TERM EXAMINATION SPRING 2010

Question No: 1 (Marks: 1) - Please choose one

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

- ▶ **True** [Click here for detail](#) rep
- ▶ False

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Question No: 2 (Marks: 1) - Please choose one

Each try block can have _____ no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ **As many as necessary.** [Click here detail](#)

Question No: 3 (Marks: 1) - Please choose one

Function templates should be used where code and behavior must be identical.

- ▶ **True** (Page 262)
- ▶ False

Question No: 4 (Marks: 1) - Please choose one

Consider the following statement

Cupboard has books

What is the relationship between Cupboard and books?

- ▶ Composition
- ▶ **Aggregation**
- ▶ Inheritance
- ▶ None of the given options

Question No: 5 (Marks: 1) - Please choose one

Identify the correct way of declaring an object of user defined template class **A** for char type members?

- ▶ **A< char > obj;** (Object-Oriented Programming in C++)
- ▶ <char>A obj;
- ▶ A obj<char>;
- ▶ Obj <char> A;

Question No: 6 (Marks: 1) - Please choose one

The user must define the operation of the copy constructor.

- ▶ **True** [Click here for detail](#)
- ▶ False

Question No: 7 (Marks: 1) - Please choose one

Default constructor is such constructor which either has no -----or if it has some parameters these have ----- values

- ▶ Parameter, temporary
- ▶ Null, Parameter
- ▶ **Parameter, default** (Page 75) rep
- ▶ non of the given

Question No: 8 (Marks: 1) - Please choose one

The type that is used to declare a reference or pointer is called its -----

- ▶ default type
- ▶ **static type** (Page 185)
- ▶ abstract type
- ▶ reference type

Question No: 9 (Marks: 1) - Please choose one

How the information hidden within an object can be accessed?

- ▶ Through its interface
- ▶ Through its private data members
- ▶ **Through its private member functions**
- ▶ Through both public and private members

Question No: 10 (Marks: 1) - Please choose one

The sub-object's life is not dependant on the life of master class in _____.

- ▶ Separation
- ▶ Composition
- ▶ **Aggregation (Page 134)**
- ▶ None of the given

Question No: 11 (Marks: 1) - Please choose one

Encapsulation means

Select correct option:

- ▶ Extending the behaviour of class in another class
- ▶ **Data and behaviour are tightly coupled within an entity (Page 16)**
- ▶ One entity takes all the attributes and operations of the other
- ▶ Taking out the common features and put those in a separate class

Question No: 12 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

- ▶ True
- ▶ **False (Object-Oriented Programming in C++) rep**

Question No: 13 (Marks: 1) - Please choose one

When we write a class template the first line must be:

- ▶ `template < class class_name>`
 - ▶ `template < class data_type>`
 - ▶ **`template < class T >` (Page 257) rep**
Here T can be replaced with any name but it is preferable.
 - ▶ `class class-name()`
- `class template<class_name>`

Question No: 14 (Marks: 1) - Please choose one

An STL container can not be used to,

- ▶ **hold objects of class employee.**
- ▶ store elements in a way that makes them quickly accessible.
- ▶ **compile c++ programs. (Object-Oriented Programming in C++)**
- ▶ organize the way objects are stored in memory

Question No: 15 (Marks: 1) - Please choose one

_____, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.

▶ **Friendship is one way only** [Click here for detail](#)

- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

Question No: 16 (Marks: 1) - Please choose one

Which of the following may not be an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ **Protected data members (Page 12)**
- ▶ All of given

Question No: 17 (Marks: 1) - Please choose one

Public methods of base class can ----- be accessed in its derived class

- ▶ **directly (Page 179) rep**
- ▶ indirectly
- ▶ simultaneously
- ▶ non of the given

Question No: 18 (Marks: 1) - Please choose one

If a class D has been derived using protected inheritance from class B (If B is a protected base and D is derived class) then public and protected members of B ----- accessed by member functions and friends of class D and classes derived from D

- ▶ **can be** [Click here for detail](#)
- ▶ cannot be
- ▶ does restrict to be
- ▶ not given

Question No: 19 (Marks: 1) - Please choose one

What is true about function templates?

- ▶ The compiler generates only one copy of the function template
- ▶ **The compiler generates a copy of function respective to each type of data (Page 256)**
- ▶ The compiler can only generate copy for the int type data
- ▶ None of the given.

Question No: 20 (Marks: 1) - Please choose one

Which of the following is an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ Unique identity
- ▶ **All of the given (Page 12) rep**

Question No: 21 (Marks: 1) - Please choose one

When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using _____).

- ▶ scope resolution operator
- ▶ dot operator
- ▶ null operator
- ▶ **Operator overloading** [Click here for detail](#)

FINALTERM EXAMINATION

Fall 2009

CS304- Object Oriented Programming (Session - 4)

Question No: 1 (Marks: 1) - Please choose one

A template provides a convenient way to make a family of

- ▶ variables and data members
- ▶ **functions and classes** **(Object-Oriented Programming in C++)**
- ▶ classes and exceptions
- ▶ programs and algorithms

Question No: 2 (Marks: 1) - Please choose one

Which one of the following terms must relate to **polymorphism**?

- ▶ Static allocation
- ▶ Static typing
- ▶ **Dynamic binding** **(Page 239) rep**
- ▶ Dynamic allocation

Question No: 3 (Marks: 1) - Please choose one

What is true about function templates?

- ▶ The compiler generates only one copy of the function template
- ▶ **The compiler generates a copy of function respective to each type of data (Page 256) rep**
- ▶ The compiler can only generate copy for the int type data
- ▶ None of the given.

Question No: 4 (Marks: 1) - Please choose one

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ **Templates** **(Page 256) rep**
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

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Question No: 5 (Marks: 1) - Please choose one

template <>

```
class Vector<char*> { }
```

This is an example of partial specialization.

▶ **True** (Page 287)

▶ False

Question No: 6 (Marks: 1) - Please choose one

Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

▶ **True** [Click here for detail](#) rep

▶ False

Question No: 7 (Marks: 1) - Please choose one

A non-virtual member function is defined in a base class and overridden in a derived class; if that function is called through a base-class pointer to a derived class object, the derived-class version is used.

▶ True

▶ False

Question No: 8 (Marks: 1) - Please choose one

Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

▶ **public members of Derv.** (Object-Oriented Programming in C++) rep

▶ protected members of Derv.

▶ private members of Derv.

▶ protected members of Base.

Question No: 9 (Marks: 1) - Please choose one

In order to define a class template, the first line of definition must be:

▶ **template <typename T>** (Page 257) rep

▶ typename <template T>

▶ Template Class <ClassName>

▶ Class <Template T>

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Question No: 10 (Marks: 1) - Please choose one

If there is a pointer p to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a nonvirtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the _____ class to be executed.

- ▶ **Base (Object-Oriented Programming in C++)**
- ▶ Derived
- ▶ Abstract
- ▶ virtual

Question No: 11 (Marks: 1) - Please choose one

When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using _____).

- ▶ scope resolution operator
- ▶ dot operator
- ▶ null operator
- ▶ **Operator overloading** [Click here for detail](#)

Question No: 12 (Marks: 1) - Please choose one

Non Template Friend functions of a class are friends of _____instance/s of that class.

- ▶ **All** [Click here for detail](#)
- ▶ One specific
- ▶ All instances of one date type
- ▶ None of the given options

Question No: 13 (Marks: 1) - Please choose one

The find() algorithm

- ▶ finds matching sequences of elements in two containers.
- ▶ finds a container that matches a specified container.
- ▶ **takes iterators as its first two arguments. (Object-Oriented Programming in C++) rep**
- ▶ takes container elements as its first two arguments.

Question No: 14 (Marks: 1) - Please choose one

If you define a vector v with the default constructor, and define another vector w with a one-argument constructor to a size of 11, and insert 3 elements into each of these vectors with push_back(), then the size() member function will return _____ for v and _____ for w.

- ▶ 11 for v and 3 for w.
- ▶ 0 for v and 0 for w.
- ▶ 0 for v and 3 for w.
- ▶ **3 for v and 11 for w. (Object-Oriented Programming in C++) rep**

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Question No: 15 (Marks: 1) - Please choose one

Which of the following may not be an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ **Protected data members (Page 12)**
- ▶ All of given

Question No: 16 (Marks: 1) - Please choose one

Which is not the Advantage of inheritance?

- ▶ **providing class growth through natural selection. (Object-Oriented Programming in C++) rep**
- ▶ facilitating class libraries.
- ▶ avoiding the rewriting of code.
- ▶ providing a useful conceptual framework.

Question No: 17 (Marks: 1) - Please choose one

1class DocElement

```
{
public:
    virtual void Print() { cout << "Generic element"; }
};
```

2class Heading : public DocElement

```
{
public:
    void Print() { cout << "Heading element"; }
};
```

3class Paragraph : public DocElement

```
{
public:
    void Print() { cout << "Paragraph element"; }
};
```

void main()

```
{
    DocElement * p = new Paragraph();

    p->Print();
}
```

When you run this program, it will print out a single line to the console output.

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What will be in that line?

Select one correct answer from the following list:

- ▶ Generic element
- ▶ Heading element
- ▶ Paragraph element
- ▶ **Nothing will be printed.**

Question No: 18 (Marks: 1) - Please choose one

When a virtual function is called by referencing a specific object by name and using the dot member selection operator (e.g., `squareObject.draw()`), the reference is resolved at compile time.

- ▶ True
- ▶ **False (Object-Oriented Programming in C++)**
(the function invocation is resolved at compile time)

Question No: 19 (Marks: 1) - Please choose one

In case of multiple inheritance a derived class inherits,

- ▶ Only the public member functions of its base classes
- ▶ Only the public data members of its base classes
- ▶ **Both public data members and member functions of all its base classes**
[Click here for detail](#)
- ▶ Data members and member functions of any two base classes

Question No: 20 (Marks: 1) - Please choose one

When we write a class template the first line must be:

- ▶ `template < class class_name >`
- ▶ `template < class data_type >`
- ▶ **`template < class T >` (Page 257) rep**

Here T can be replaced with any name but it is preferable.

- ▶ `class class-name()`

`class template<class_name>`

Question No: 21 (Marks: 1) - Please choose one

Which of the following is incorrect line regarding function template?

- ▶ `template<class T>`
- ▶ `template <typename U>`
- ▶ **`Class<template T>` (Page 257)**
- ▶ `template < class T, class U>`

Question No: 22 (Marks: 1) - Please choose one

An STL container can not be used to,

- ▶ hold objects of class employee.
- ▶ store elements in a way that makes them quickly accessible.
- ▶ **compile c++ programs. (Object-Oriented Programming in C++) rep**
- ▶ organize the way objects are stored in memory

Question No: 23 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

- ▶ True
- ▶ **False (Object-Oriented Programming in C++) rep**

Question No: 24 (Marks: 1) - Please choose one

Consider a class named Vehicle, which of the following can be the instance of class Vehicle?

1. Car
 2. Computer
 3. Desk
 4. Ahmed
 5. Bicycle
 6. Truck
- ▶ 1, 4, 5
 - ▶ 2, 5, 6
 - ▶ 1, 2, 3, 6
 - ▶ **1, 5, 6 (correct)**

Question No: 25 (Marks: 1) - Please choose one

Consider the code below,

```
class Fred {  
public:  
Fred();  
...  
};  
int main()  
{  
Fred a[10];  
Fred* p = new Fred[10];  
...  
}
```

Select the best option,

- ▶ Fred a[10]; calls the default constructor 09 times
- Fred* p = new Fred[10]; calls the default constructor 10 times
- ▶ Produce an error

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▶ Fred a[10]; calls the default constructor 11 times
Fred* p = new Fred[10]; calls the default constructor 11 times

▶ **Fred a[10]; calls the default constructor 10 times**
Fred* p = new Fred[10]; calls the default constructor 10 times [Click here for detail](#)

Question No: 26 (Marks: 1) - Please choose one

When a variable is define as **static** in a class then all object of this class,

- ▶ Have different copies of this variable
- ▶ **Have same copy of this variable (Page 110)**
- ▶ Can not access this variable
- ▶ None of given

Question No: 27 (Marks: 1) - Please choose one

The life of sub object is dependant on the life of master class in _____.

- ▶ Separation
- ▶ **Composition (Page 53)**
- ▶ Aggregation
- ▶ None of the given

Question No: 28 (Marks: 1) - Please choose one

_____, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.

- ▶ **Friendship is one way only** [Click here for detail](#)
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

Question No: 29 (Marks: 1) - Please choose one

Which of the following operators always takes no argument if overloaded?

- ▶ /
- ▶ -
- ▶ +
- ▶ **++ (Page 162)**

Question No: 30 (Marks: 1) - Please choose one

In Private ----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ▶ specialization
- ▶ **inheritance (Page 216) rep**
- ▶ abstraction
- ▶ composition

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**Campus:- Institute of E-Learning & Modern
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FINAL TERM EXAMINATION
Fall 2009
CS304- Object Oriented Programming (Session - 1)

Question No: 1 (Marks: 1) - Please choose one

Which one of the following terms must relate to **polymorphism**?

- ▶ Static allocation
- ▶ Static typing
- ▶ **Dynamic binding (Page 239) rep**
- ▶ Dynamic allocation

Question No: 2 (Marks: 1) - Please choose one

Multiple inheritance can be of type

- ▶ Public
- ▶ Private
- ▶ Protected
- ▶ **All of the given [Click here for detail](#)**

Question No: 3 (Marks: 1) - Please choose one

When a subclass specifies an alternative definition for an attribute or method of its superclass, it is _____ the definition in the superclass.

- ▶ overload
- ▶ **overriding (Page 34)**
- ▶ copy riding
- ▶ none of given

Question No: 4 (Marks: 1) - Please choose one

Like template functions, a class template may not handle all the types successfully.

- ▶ **True (Page 258) rep**
- ▶ False

Question No: 5 (Marks: 1) - Please choose one

It is sometimes useful to specify a class from which no objects will ever be created.

- ▶ **True (Object-Oriented Programming in C++)**
- ▶ False

Question No: 6 (Marks: 1) - Please choose one

Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access

- ▶ **public members of Derv. (Object-Oriented Programming in C++) rep**
- ▶ protected members of Derv.
- ▶ private members of Derv.
- ▶ protected members of Base.

Question No: 7 (Marks: 1) - Please choose one

A pointer to a base class can point to objects of a derived class.

- ▶ **True** [Click here for detail](#)
- ▶ False

Question No: 8 (Marks: 1) - Please choose one

A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ **an argument is passed by value.** (Page 78) rep
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

Question No: 9 (Marks: 1) - Please choose one

A function call is resolved at run-time in_____

- ▶ non-virtual member function.
- ▶ **virtual member function.** (Page 239) rep
- ▶ Both non-virtual member and virtual member function.
- ▶ None of given

Question No: 10 (Marks: 1) - Please choose one

When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using _____).

- ▶ scope resolution operator
- ▶ dot operator
- ▶ null operator
- ▶ **Operator overloading** [Click here for detail](#)

Question No: 11 (Marks: 1) - Please choose one

Each try block can have _____ no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ **As many as necessary.** [Click here for detail](#) rep

Question No: 12 (Marks: 1) - Please choose one

Two important STL associative containers are _____ and _____.

- ▶ **set,map** (Object-Oriented Programming in C++) rep
- ▶ sequence,mapping
- ▶ setmet,multipule
- ▶ sit,mat

Question No: 13 (Marks: 1) - Please choose one

The mechanism of selecting function at run time according to the nature of calling object is called,

- ▶ late binding
- ▶ static binding
- ▶ virtual binding
- ▶ **None of the given options (Page 227)**

Dynamic binding means that target function for a call is selected at run time

Question No: 14 (Marks: 1) - Please choose one

An abstract class is useful when

- ▶ We do not derive any class from it.
- ▶ There are multiple paths from one derived class to another.
- ▶ **We do not want to instantiate its object. (Object-Oriented Programming in C++) rep**
- ▶ You want to defer the declaration of the class.

Question No: 15 (Marks: 1) - Please choose one

Which of the following is incorrect line regarding function template?

- ▶ `template<class T>`
- ▶ `template <typename U>`
- ▶ **`Class<template T>` (Page 257) rep**
- ▶ `template < class T, class U>`

Question No: 16 (Marks: 1) - Please choose one

Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ **All of given (Page 256) rep**

Question No: 17 (Marks: 1) - Please choose one

By default the vector data items are initialized to ____

- ▶ **0 [Click here for detail](#) rep**
- ▶ 0.0
- ▶ 1
- ▶ null

Question No: 18 (Marks: 1) - Please choose one

Which one of the following functions returns the total number of elements in a vector.

- ▶ `length();`
- ▶ **`size();` (Page 318)**
- ▶ `ele();`
- ▶ `veclen();`

Question No: 19 (Marks: 1) - Please choose one

Suppose you create an uninitialized vector as follows:

```
vector<int> evec;
```

After adding the statment,

```
evec.push_back(21);
```

what will happen?

- ▶ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.
- ▶ The following statement will add an element to the center of evec and will reinitialize it with the value 21.
- ▶ The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.
- ▶ **The following statement will add an element to the end (the back) of evec and initialize it with the value 21.** [Click here for detail](#)

Question No: 20 (Marks: 1) - Please choose one

An STL container can not be used to,

- ▶ hold objects of class employee.
- ▶ store elements in a way that makes them quickly accessible.
- ▶ **compile c++ programs. (Object-Oriented Programming in C++) rep**
- ▶ organize the way objects are stored in memory

Question No: 21 (Marks: 1) - Please choose one

Algorithms can only be implemented using STL containers.

- ▶ True
- ▶ **False (Object-Oriented Programming in C++) rep**

Question No: 22 (Marks: 1) - Please choose one

The main function of scope resolution operator (::) is,

- ▶ To define an object
- ▶ To define a data member
- ▶ **To link the definition of an identifier to its declaration** [Click here for detail](#)
- ▶ To make a class private

Question No: 23 (Marks: 1) - Please choose one

When is a constructor called?

- ▶ Each time the constructor identifier is used in a program statement
- ▶ **During the instantiation of a new object (Object-Oriented Programming in C++)**
- ▶ During the construction of a new class
- ▶ At the beginning of any program execution

Question No: 24 (Marks: 1) - Please choose one

Consider the code below,

```
class Fred {  
public:  
Fred();  
...  
};  
int main()  
{  
Fred a[10];  
Fred* p = new Fred[10];  
...  
}
```

Select the best option,

- ▶ Fred a[10]; calls the default constructor 09 times
- Fred* p = new Fred[10];
- ▶ Produce an error
- ▶ Fred a[10]; calls the default constructor 11 times
- Fred* p = new Fred[10]; calls the default constructor 11 times

▶ **Fred a[10]; calls the default constructor 10 times**
▶ **Fred* p = new Fred[10]; calls the default constructor 10 times** [Click here for detail](#)

Question No: 25 (Marks: 1) - Please choose one

Associativity can be changed in operator overloading.

- ▶ True
- ▶ **False (Page 141)**

Question No: 26 (Marks: 1) - Please choose one

A normal C++ operator that acts in special ways on newly defined data types is said to be

- ▶ glorified.
- ▶ encapsulated.
- ▶ classified.
- ▶ **overloaded. (Object-Oriented Programming in C++)**

Question No: 27 (Marks: 1) - Please choose one

Which operator can not be overloaded?

- ▶ The relation operator (>=)
- ▶ Assignment operator (=)
- ▶ Script operator ([])
- ▶ **Conditional operator (? :) (Page 141)**

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Question No: 28 (Marks: 1) - Please choose one

Suppose obj1 and obj2 are two objects of a user defined class A. An + operator is overloaded to add obj1 and obj2 using the function call obj1+obj2.

Identify the correct function prototype against the given call?

- ▶ A operator + (A &obj);
- ▶ **int + operator(); (Page 143)**
- ▶ int operator (plus) ();
- ▶ A operator(A &obj3);

Question No: 29 (Marks: 1) - Please choose one

Default constructor is such constructor which either has no -----or if it has some parameters these have ----- values

- ▶ Parameter, temporary
- ▶ Null, Parameter
- ▶ **Parameter, default (Page 75) rep**
- ▶ non of the given

Question No: 30 (Marks: 1) - Please choose one

Public methods of base class can ----- be accessed in its derived class

- ▶ **directly (Page 179) rep**
- ▶ inderectly
- ▶ simultaneously
- ▶ non of the given

Cs304 Important Mcq's For Final Term

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Paid Tasks = LMS Handling + Online Classes + Project

A template provides a convenient way to make a family of.

variables and data members

functions and classes

classes and exceptions

programs and algorithms

A class template may inherit from another class template.

True

False

Target of a _____ function call is determined at run time.

instance

virtual

operator

none of given

A class hierarchy .

shows the same relationships as an organization chart.

describes "has a" relationships.

describes "is a kind of" relationships.

shows the same relationships as a family tree.

Sender of the message does not need to know the exact class of receiver in_____.

Abstraction

Polymorphism

Inheritance

none of the given

A function call is resolved at run-time in_____ .

non-virtual member function

virtual member function

Both non-virtual member and virtual member function.

None of given

Adding a derived class to a base class requires fundamental changes to the base class.

True

False

User can make virtual table explicitly.

True

False

Binding means that target function for a call is selected at compile time.

Static

Dynamic

Automatic

None of given

Target of a _____ function call is determined at run time.

instance

virtual
operator
none of given

Which line will produce error. Class phone: private Transmit, private Receiver { }
1. int main() 2. { 3. phone obj; 4. Transmit* obj1 = &obj; 5. Receiver obj2 = &obj; 6. }

3rd line will produce error
4th line will produce error
3rd and 4th line will produce error.
5th line will produce error

Function overriding is done in context of,
Single class
Single derived class
Single base class
Derived and base classes

**Consider the code below, class class1{ public: void func1(); }; class class2 :
protected class1 { }; Function func1 of class1 is ____ in class2,**
public
protected
private
none of the given options

The following statements: 1) int iArray[5]; 2) int *pArr = iArray;
These statements will compile successfully
Error in first statement
Error in second statement
None of given options

Methodologies to the development of reusable software relate to_____.
Structure programming
procedural programming
generic programming
None of the given

Function template must have a parameter.
True
False

The default inheritance mode is,
Public inheritance
Protected inheritance
Private inheritance
None of these options

Two functions with same names, parameters and return type can exist in,
Function overloading
Function overriding
Operator overloading
None of these options

Consider the code below, class c1{ }; class c2 : public c1 { }; class c3 : public c2 { }; Then c2 is,
Direct base class of c3
Direct child class of c3
Direct base class of c1
None of these

Virtual functions allow you to

create an array of type pointer-to-base class that can hold pointers to derived classes.
create functions that can never be accessed.

group objects of different classes so they can all be accessed by the same function code.

use the same function call to execute member functions of objects from different classes.

User can make virtual table explicitly.

True

False

In order to define a class template the first line of definition must be:

template <typename T>

typename <template T>

Template Class <ClassName>

Class <Template T>

Consider the following statements: 1) int iArray[5]; 2) int *pArr = iArray;

These statements will compile successfully

Error in first statement

Error in second statement

None of given options

In c++ dynamic binding and polymorphism will be achieved when member function will be ____:

private

public

virtual

inline

In type in depended function template should be use where code and behavior must be identical.

True

False

Consider the code below, class class1{ protected: int i; }; class class2 : private class1 { }; Then int member i of class1 is ____ in class2,

public

protected

private

none of the given options

In specialization we can,

Replace child class with its base class

Replace base class with its child class (Not Sure)

Replace both child and base classes interchangeably

None of the given options

Consider the code below, class class1{ public: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is ____ in class2,

public

protected

private

none of the given options

It is illegal to make objects of one class members of another class.

True

False

An abstract class is useful when

no classes should be derived from it.

there are multiple paths from one derived class to another.

no objects should be instantiated from its.

you want to defer the declaration of the class.

In resolution order compiler search firstly _____:

Generic Template

Partial Specialization

Complete Specialization

Ordinary function

template<> class Vector{ void p; //.... void*& operator[] ((int i); };**

This specialization can then be used as the common implementation for all Vectors of pointers.

This specialization can then be used as the all type implementation for one type classes.

This specialization can then be used double type pointers.

This specialization should be used for Vectors of all type int types.

In private inheritance derived class pointer can be assigned to base class pointer in.

Main function

In derived class member and friend functions

In base class member and friend functions

None of the given options

Which statement will be true for concrete class?

it implements an virtual concept.

it can be instantiated

it cannot be instantiated

none of given

Target of a _____ function call is determined at run time.

instance

virtual

operator

none of given

The Specialization pattern after the name says that this specialization is to be used for every_____.

data types

meta types

virtual types

pointers type

c++ dynamic binding and polymorphism will be achieved when member function will be ____.

private

public

virtual

inline

Consider the code below, class class1{ protected: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is _____ in class2,

public

protected

private

none of the given options

Consider the code below, class class1{ protected: int i; }; class class2 :
protected class1 { }; Then int member i of class1 is ____ in class2,
public
protected
private
none of the given options

Consider the code below, class class1{ private: void func1(); }; class class2 :
private class1 { }; Function func1 of class1 is ____ in class2,
public
protected
private
none of the given options

Target of a ____ function call is determined at run time.
instance
virtual
operator
none of given

Consider the following statements: 1) int iArray[5]; 2) int *pArr = iArray;
These statements will compile successfully
Error in first statement
Error in second statement
None of given options

Consider the code below, class class1{ private: int i; }; class class2 : private
class1 { }; Then int member i of class1 is ____ in class2,
public
protected
private
none of the given options

**If there is a pointer, p, to objects of a base class, and it contains the address of
an object of a derived class, and both classes contain a virtual member function,
ding(), then the statement p->ding(); will cause the version of ding() in
the ____ class to be executed.**
base
derived
virtual
implemented

A class template may inherit from another class template.
True
False

**Derived class can inherit from public base class as well as private and protected
base classes**
True
False

Two functions with same names, parameters and return type can exist in,
Function overloading
Function overriding
Operator overloading
None of these options

Consider the code below, class class1{ private: int i; }; class class2 : public
class1 { }; Then int member i of class1 is ____ in class2,

public
protected
private

none of the given options

Target of a _____ function call is determined at run time.

instance

virtual

operator

none of given

A class template may inherit from another class template.

True

False

A function call is resolved at run-time in _____

non-virtual member function

virtual member function

Both non-virtual member and virtual member function.

None of given

A class hierarchy

shows the same relationships as an organization chart.

describes “has a” relationships.

describes “is a kind of” relationships.

shows the same relationships as a family tree.

Consider the code below, class class1{ public: int i; }; class class2 : public class1 { }; Then int member i of class1 is _____ in class2,

public

protected

private

none of the given options

Consider the code below, class c1{ }; class c2 : public c1 { }; class c3 : public c2 { }; Then c1 is,

Direct base class of c3

Direct child class of c3

Direct base class of c2

Direct child class of c2

A class can inherit from more than one class is called.

Simple inheritance

Multiple inheritances

Single inheritance

Double inheritance

template<> class Vector{ void p; //.... void*& operator[] ((int i); };**

This specialization can then be used as the common implementation for all Vectors of pointers.

This specialization can then be used as the all type implementation for one type classes.

This specialization can then be used double type pointers.

This specialization should be used for Vectors of all type int types.

Consider the code below, class class1{ public: int i; }; class class2 : protected class1 { }; Then int member i of class1 is _____ in class2,

public

protected

private
none of the given options

Consider the code below, class class1{ private: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is _____ in class2,

public
protected
private
none of the given options

Templates automatically create different versions of a function, depending on user input.

True
False

_____ **Binding means that target function for a call is selected at run time**

Automatic
Dynamic
Static
Dramatic

When we create objects, then space is allocated to:

Member function
Access specifier
Data member
None of given

There is only one form of copy constructor.

True
False

Which of the following features of OOP is used to deal with only relevant details?

Abstraction
Information hiding
Object

_____ **Binding means that targets function for a call is selected at compile time.**

Static
Dynamic
Automatic
None of given

A Class hierarchy

Shows the same relationships as an organization chart
Describes "has a" relationships.

Describes "is a kind of" relationships.

Shows the same relationships as a family tree

In C++, we declare a function virtual by preceding the function header with keyword "Inline".

True
False

It is illegal to make objects of one class members of another class.

True
False

In Resolution order compiler search firstly_____:

Generic Template

Partial Specification

Complete Specification

Ordinary function

Derived class can inherit from public base class as well as private and protected base classes

True

False

Which line will produce error. Class phone: Private Transmit, private Receiver { }
1.int main () 2. { 3.phone obj; 4.Tranmit*obj1 = &obj; 5.Received obj2 = &obj; 6.}

3rd line will produce error

4th line will produce error

3rd and 4th line will produce error.

5th line will produce error.

Methodologies to the development of reusable software relate to _____:

Structure programming

Procedural programming

Generic programming

None of the given

A template argument is preceded by the keyword_____:

Vector

Class

Template

Type*

Friends are used exactly the same for template and non-template classes.

True

False

A function template must have a parameter

True

False

Child class can call constructor of its,

Direct base class

Indirect base class

Both direct and indirect base classes

None of these.

Which statement will be true for concrete class?

It implements an virtual concept.

It can be instantiated

It cannot be instantiated

None of given

A class D can be derived from a class C, which is derived from a class B, which is derived from a class A

True

False

Adding a derived class to a base class requires fundamental changes to the base class.

True

False

A Class or class template can have member _____ that are themselves templates.

Variable

Function

Objects

None of given

Which will be the Primary task or tasks of generic programming?

Categorize the abstractions in a domain into concepts

Implement generic algorithms based on the concepts

Build concrete models of the concepts

All of given

The default inheritance mode is,

Public inheritance

Protected Inheritance

Private Inheritance

None of these options

If there is a pointer, p, to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a virtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the _____ class to be executed.

Base

Derived

Virtual

Implemented

Sender of the message does not need to know the exact class of receiver in _____.

Abstraction

Polymorphism

Inheritance

none of the given

Child class can call constructor of its,

Direct base class

Indirect base class

Both direct and indirect base classes

None of these

A pure virtual function is a virtual function that causes its class to be abstract.

returns nothing.

is used in a derived class.

takes no arguments.

1: Consider the code below, class class1{ protected: int i; }; class class2 : protected class1 { }; Then int member i of class1 is _____ in class2,

- public
- protected (true)

2: By default, assignment operator (=) performs,

- Shallow copy (true)
- Deep copy

3: The default inheritance mode is,

- Public inheritance(true)

4: In case of dynamic memory allocation in our class we should use

- Default assignment operator
- copy constructor(true)

5: Static casting is,?

- Implicit way of calling base class functions from derived class (true)
- Explicit way of calling base class functions from derived class
- None of these
- both 1 and 2

6: In case of dynamic memory allocation in a class we should use,

- User defined default constructor
- user defined copy constructor (true)

7: Which of the following types of Inheritance is used to model "Implemented in terms of" relationship?

- Public
- private (true)
- Protected

8: Virtual functions allow you to

- create an array of type pointer-to-base class that can hold pointers to derived classes.
- create functions that can never be accessed.
- use the same function call to execute member functions of objects from different classes.
- group objects of different classes so they can all be accessed by the same function code. (true)

9: A class D can be derived from a class C, which is derived from a class B, which is derived from a class A.

- T (true ans)
- F

10: Function overriding is done in context of,

- Single class
- Single derived class
- base and derived class (true)

1: Consider the code below, class class1{ protected: int i; }; class class2 : protected class1 { }; Then int member i of class1 is _____ in class2,

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- protected (true)

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- T (true ans)
- F

10: Function overriding is done in context of,

- Single class
- Single derived class
- base and derived class (true)

Friend Functions of a class are

_____ members of that class .

Select correct option :

Public

Private

Protected

None of the given options .

Question # 2 of 10 (Start time:

02 :17 :30 PM) Total Marks: 1

Consider the code below , class

class1 { protected : int i ; }; class

class2 : protected class1 { }; Then int

member i of class1 is _____ in class 2 ,

Select correct option :

public

protected

private

none of the given options

Question # 3 of 10 (Start time:

02 :17 :56 PM) Total Marks: 1

A Child class can call constructor of its

parent class through ,

Select correct option :

Its constructor initialization list

Its constructor body

Both from its constructor initialization list or body -

Can not call the constructor of its

parent class

Question # 4 of 10 (Start time:

02 :19 :01 PM) Total Marks: 1 We can

access private members of the class

from outside the class using _____

operator with its object pointer .

Select correct option :

- >

.

&

None of the given options

Question # 5 of 10 (Start time:

02 :20 :13 PM) Total Marks: 1

A function call is resolved at run - time

in _____

Select correct option :

non - virtual member function

virtual member function

Both non - virtual member and virtual member function .

None of given

Question # 6 of 10 (Start time:

02 :20 :35 PM) Total Marks: 1

It is sometimes useful to specify a

class from which no objects will ever be created .

Select correct option :

True

Flase

Question # 7 of 10 (Start time:

02 :21 :01 PM) Total Marks: 1

Consider the code below , class

class1 { public: void func 1 (); }; class

class2 : private class 1 { }; Function

func1 of class1 is _____ in class 2 ,

Select correct option :

public

protected

private

none of the given options

Question # 8 of 10 (Start time:

02 :21 :26 PM) Total Marks: 1

Consider the code below , class

class1 { public: int i ; }; class class 2 :

protected class 1 { }; Then int member i

of class 1 is _____ in class2 ,

Select correct option :

public

protected

private

none of the given options

Question # 9 of 10 (Start time:

02 :22 :00 PM) Total Marks: 1

Static casting is,

Select correct option :

Implicit way of calling base class

functions from derived class

Explicit way of calling base class

functions from derived class

None of these

Both of these

A template provides a convenient way to make a family of.

variables and data members

functions and classes

classes and exceptions

programs and algorithms

A class template may inherit from another class template.

True

False

Target of a ____ function call is determined at run time.

instance

virtual

operator

none of given

A class hierarchy .

shows the same relationships as an organization chart.

describes “has a” relationships.

describes “is a kind of” relationships.

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Sender of the message does not need to know the exact class of receiver in_____.

Abstraction

Polymorphism

Inheritance

none of the given

A function call is resolved at run-time in_____ .

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Both non-virtual member and virtual member function.

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Adding a derived class to a base class requires fundamental changes to the base class.

True

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User can make virtual table explicitly.

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Binding means that target function for a call is selected at compile time.

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3rd line will produce error

4th line will produce error

3rd and 4th line will produce error.

5th line will produce error

Function overriding is done in context of,

Single class

Single derived class

Single base class

Derived and base classes

Consider the code below, class class1{ public: void func1(); }; class class2 : protected class1 { }; Function func1 of class1 is ____ in class2,

public

protected

private

none of the given options

The following statements: 1) int iArray[5]; 2) int *pArr = iArray;

These statements will compile successfully

Error in first statement

Error in second statement

None of given options

Methodologies to the development of reusable software relate to_____.

Structure programming

procedural programming

generic programming

None of the given

Function template must have a parameter.

True

False

The default inheritance mode is,

Public inheritance

Protected inheritance

Private inheritance

None of these options

Two functions with same names, parameters and return type can exist in,

Function overloading

Function overriding

Operator overloading

None of these options

Consider the code below, class c1{ }; class c2 : public c1 { }; class c3 : public c2 { }; Then c2 is,

Direct base class of c3

Direct child class of c3

Direct base class of c1

None of these

Virtual functions allow you to

create an array of type pointer-to-base class that can hold pointers to derived classes.

create functions that can never be accessed.

group objects of different classes so they can all be accessed by the same function code.

use the same function call to execute member functions of objects from different classes.

User can make virtual table explicitly.

True

False

In order to define a class template the first line of definition must be:

template <typename T>

typename <template T>

Template Class <ClassName>

Class <Template T>

Consider the following statements: 1) int iArray[5]; 2) int *pArr = iArray;

These statements will compile successfully

Error in first statement

Error in second statement

None of given options

In c++ dynamic binding and polymorphism will be achieved when member function will be __.

private

public

virtual

inline

In type in depended function template should be use where code and behavior must be identical.

True

False

Consider the code below, class class1{ protected: int i; }; class class2 : private class1 { };

Then int member i of class1 is ___ in class2,

public

protected

private

none of the given options

In specialization we can,

Replace child class with its base class

Replace base class with its child class (Not Sure)

Replace both child and base classes interchangeably

None of the given options

Consider the code below, class class1{ public: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is ___ in class2,

public

protected

private

none of the given options

It is illegal to make objects of one class members of another class.

True

False

An abstract class is useful when

no classes should be derived from it.

there are multiple paths from one derived class to another.

no objects should be instantiated from its.

you want to defer the declaration of the class.

In resolution order compiler search firstly ____.

Generic Template

Partial Specialization

Complete Specialization

Ordinary function

template<> class Vector{ void p; //.... void*& operator[] ((int i));};**

This specialization can then be used as the common implementation for all Vectors of pointers.

This specialization can then be used as the all type implementation for one type classes.

This specialization can then be used double type pointers.

This specialization should be used for Vectors of all type int types.

In private inheritance derived class pointer can be assigned to base class pointer in.

Main function

In derived class member and friend functions

In base class member and friend functions

None of the given options

Which statement will be true for concrete class?

it implements an virtual concept.

it can be instantiated

it cannot be instantiated

none of given

Target of a ____ function call is determined at run time.

instance

virtual

operator

none of given

The Specialization pattern after the name says that this specialization is to be used for every__.

data types

meta types

virtual types

pointers type

c++ dynamic binding and polymorphism will be achieved when member function will be __.

private

public

virtual

inline

Consider the code below, class class1{ protected: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is ____ in class2,

public

protected

private

none of the given options

Consider the code below, class class1{ protected: int i; }; class class2 : protected class1 { }; Then int member i of class1 is ____ in class2,

public

protected

private

none of the given options

Consider the code below, class class1{ private: void func1(); }; class class2 : private class1 { }; Function func1 of class1 is ____ in class2,

public

protected

private

none of the given options

Target of a ____ function call is determined at run time.

instance

virtual

operator

none of given

Consider the following statements: 1) int iArray[5]; 2) int *pArr = iArray;

These statements will compile successfully

Error in first statement

Error in second statement

None of given options

Consider the code below, class class1{ private: int i; }; class class2 : private class1 { };

Then int member i of class1 is ___ in class2,

public

protected

private

none of the given options

If there is a pointer, p, to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a virtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the ___ class to be executed.

base

derived

virtual

implemented

A class template may inherit from another class template.

True

False

Derived class can inherit from public base class as well as private and protected base classes

True

False

Two functions with same names, parameters and return type can exist in,

Function overloading

Function overriding

Operator overloading

None of these options

Consider the code below, class class1{ private: int i; }; class class2 : public class1 { };

Then int member i of class1 is ___ in class2,

public

protected

private

none of the given options

Target of a ___ function call is determined at run time.

instance

virtual

operator

none of given

A class template may inherit from another class template.

True

False

A function call is resolved at run-time in _____

non-virtual member function

virtual member function

Both non-virtual member and virtual member function.

None of given

A class hierarchy

shows the same relationships as an organization chart.

describes "has a" relationships.

describes "is a kind of" relationships.

shows the same relationships as a family tree.

Consider the code below, class class1{ public: int i; }; class class2 : public class1 { }; Then int member i of class1 is ____ in class2,

public

protected

private

none of the given options

Consider the code below, class c1{ }; class c2 : public c1 { }; class c3 : public c2 { }; Then c1 is,

Direct base class of c3

Direct child class of c3

Direct base class of c2

Direct child class of c2

A class can inherit from more than one class is called.

Simple inheritance

Multiple inheritances

Single inheritance

Double inheritance

template<> class Vector{ void** p; //.... void*& operator[] ((int i));};

This specialization can then be used as the common implementation for all Vectors of pointers.

This specialization can then be used as the all type implementation for one type classes.

This specialization can then be used double type pointers.

This specialization should be used for Vectors of all type int types.

Consider the code below, class class1{ public: int i; }; class class2 : protected class1 { }; Then int member i of class1 is ____ in class2,

public

protected

private

none of the given options

Consider the code below, class class1{ private: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is ____ in class2,

public

protected

private

none of the given options

Templates automatically create different versions of a function, depending on user input.

True

False

_____ Binding means that target function for a call is selected at run time

Automatic

Dynamic

Static

Dramatic

When we create objects, then space is allocated to:

Member function

Access specifier

Data member

None of given

There is only one form of copy constructor.

True

False

Which of the following features of OOP is used to deal with only relevant details?

Abstraction

Information hiding

Object

_____ Binding means that targets function for a call is selected at compile time.

Static

Dynamic

Automatic

None of given

A Class hierarchy

Shows the same relationships as an organization chart

Describes "has a" relationships.

Describes "is a kind of" relationships.

Shows the same relationships as a family tree

In C++, we declare a function virtual by preceding the function header with keyword "Inline".

True

False

It is illegal to make objects of one class members of another class.

True

False

In Resolution order compiler search firstly_____.

Generic Template

Partial Specification

Complete Specification

Ordinary function

Derived class can inherit from public base class as well as private and protected base classes

True

False

Which line will produce error. Class phone: Private Transmit, private Receiver { } 1.int main () 2. { 3.phone obj; 4.Tranmit*obj1 = &obj; 5.Received obj2 = &obj; 6.}

3rd line will produce error

4th line will produce error

3rd and 4th line will produce error.

5th line will produce error.

Methodologies to the development of reusable software relate to _____.

Structure programming

Procedural programming

Generic programming

None of the given

A template argument is preceded by the keyword_____.

Vector

Class

Template

Type*

Friends are used exactly the same for template and non-template classes.

True

False

A function template must have a parameter

True

False

Child class can call constructor of its,

Direct base class

Indirect base class

Both direct and indirect base classes

None of these.

Which statement will be true for concrete class?

It implements an virtual concept.

It can be instantiated

It cannot be instantiated

None of given

A class D can be derived from a class C, which is derived from a class B, which is derived from a class A

True

False

Adding a derived class to a base class requires fundamental changes to the base class.

True

False

A Class or class template can have member _____ that are themselves templates.

Variable

Function

Objects

None of given

Which will be the Primary task or tasks of generic programming?

Categorize the abstractions in a domain into concepts

Implement generic algorithms based on the concepts

Build concrete models of the concepts

All of given

The default inheritance mode is,

Public inheritance

Protected Inheritance

Private Inheritance

None of these options

If there is a pointer, p, to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a virtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the _____ class to be executed.

Base

Derived

Virtual

Implemented

Sender of the message does not need to know the exact class of receiver in_____.

Abstraction

Polymorphism

Inheritance

none of the given

Question # 1 of 20 (Start time: 08:20:11 PM) Total Marks: 1

A Child class can call constructor of its parent class through,

Select correct option:

Its constructor initialization list (Correct)

Its constructor body

Both from its constructor initialization list or body

Can not call the constructor of its parent class

A class template may inherit from another class template.

Select correct option: True

False (False)

Question # 3 of 20 (Start time: 08:22:17 PM) Total Marks: 1

Consider the code below, class c1{ }; class c2 : public c1 { }; class c3 : public c2 { }; Then c2 is,

Select correct option:

Direct base class of c3

Direct child class of c3 (Correct)

Direct base class of c1

None of these

Question # 4 of 20 (Start time: 08:22:57 PM) Total Marks: 1

Child class can call constructor of its,

Select correct option:

Direct base class (Correct)

Indirect base class

Both direct and indirect base classes

None of these

Question # 5 of 20 (Start time: 08:23:26 PM) Total Marks: 1

Consider the code below, class class1{ public: void func1(); }; class class2 : protected class1 { }; Function func1 of class1 is ____ in class2,

Select correct option:

public

protected

private (Correct)

none of the given options

Question # 6 of 20 (Start time: 08:24:08 PM) Total Marks: 1

A class hierarchy

Select correct option:

shows the same relationships as an organization chart.

describes "has a" relationships.

describes "is a kind of" relationships. (Correct)

shows the same relationships as a family tree.

Question # 7 of 20 (Start time: 08:24:47 PM) Total Marks: 1

Consider the code below, class class1{ public: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is ____ in class2,

Select correct option:

public (Correct)

protected

private

none of the given options

Question # 8 of 20 (Start time: 08:26:03 PM) Total Marks: 1

In C++, we declare a function virtual by preceding the function header with keyword "Inline"

Select correct option:

True (Correct)

False

Question # 9 of 20 (Start time: 08:26:37 PM) Total Marks: 1

Consider the code below, class class1{ private: int i; }; class class2 : protected class1 { }; Then int member i of class1 is ____ in class2,

Select correct option:

public

protected (Correct)

private

none of the given options

Question # 10 of 20 (Start time: 08:27:39 PM) Total Marks: 1

Which will be the Primary task or tasks of generic programming?

Select correct option:

Categorize the abstractions in a domain into concepts

Implement generic algorithms based on the concepts

Build concrete models of the concepts

All of given (Correct)

Question # 11 of 20 (Start time: 08:28:04 PM) Total Marks: 1

When we want to implement one class in terms of another class then we use,

Select correct option:

Public inheritance (Correct)

Protected inheritance

Private inheritance

None of these options

Question # 12 of 20 (Start time: 08:29:20 PM) Total Marks: 1

Virtual functions allow you to

Select correct option:

create an array of type pointer-to-base class that can hold pointers to derived classes.

create functions that can never be accessed.

group objects of different classes so they can all be accessed by the same function code. (Correct)

use the same function call to execute member functions of objects from different classes.

Question # 13 of 20 (Start time: 08:30:05 PM) Total Marks: 1

A class or class template can have member _____ that are themselves templates.

Select correct option:

Variable

Function

Objects

None of given (Correct)

Question # 14 of 20 (Start time: 08:31:34 PM) Total Marks: 1

A class template _____

Select correct option:

Facilitates reuse of class (Correct)

Does not facilitate reuse of class

Does not support generic methods

Does not support static members

Question # 15 of 20 (Start time: 08:32:48 PM) Total Marks: 1

Target of a _____ function call is determined at run time.

Select correct option:

instance

virtual (Correct)

operator

none of given

Question # 16 of 20 (Start time: 08:34:18 PM) Total Marks: 1

The default inheritance mode is,

Select correct option:

Public inheritance (Correct)

Protected inheritance

Private inheritance

None of these options

Question # 17 of 20 (Start time: 08:34:54 PM) Total Marks: 1

A class D can be derived from a class C, which is derived from a class B, which is derived from a class A.

Select correct option:

True (Correct)

False

Question # 18 of 20 (Start time: 08:36:11 PM) Total Marks: 1

A pure virtual function is a virtual function that

Select correct option:

causes its class to be abstract. (Correct)

returns nothing.

is used in a derived class.
takes no arguments.

Question # 19 of 20 (Start time: 08:36:47 PM) Total Marks: 1

A template provides a convenient way to make a family of

Select correct option:

variables and data members (Correct)

functions and classes

classes and exceptions

programs and algorithms

Question # 20 of 20 (Start time: 08:37:17 PM) Total Marks: 1

**Consider the code below, class class1{ protected: int i; }; class class2 : protected class1 {
}; Then int member i of class1 is _____ in class2,**

Select correct option:

public

protected

private (Correct)

none of the given options

Quiz No. 02 of the subject Object Oriented Programming (CS304) is going to upload in a couple of days, before submitting the Quiz, **please read all instructions thoroughly.**

Quiz No. 02 of the subject Object Oriented Programming (CS304) is going to upload in a couple of days, before submitting the Quiz, **please read all instructions thoroughly.**

Solved Today Quiz

Question # 1 of 10 (Start time: 01:59:09 PM)

What will be most appropriate C++ data type for variable "InterestRate"?

Select correct option:

int

long int

double

=>float

Question # 2 of 10 (Start time: 01:59:59 PM)

Select correct line of code for aggregation relationship between "Person" class and "Address" class.

Select correct option:

=>class Person { Address * a; };

class Address { Person * a; };

class Person { Address a; };

class Address { Person a; };

Question # 3 of 10 (Start time: 02:01:24 PM)

_____is creating objects of one class inside another class.

Select correct option:

Aggregation

=>Composition

Inheritance

Association

Question # 4 of 10 (Start time: 02:02:03 PM)

What type of relationship exists between "SavingAccount" class and "CurrentAccount" class?

Select correct option:

Both are child classes

Both contains each other

=>Both are composed of each other

None of the given option

Question # 5 of 10 (Start time: 02:03:28 PM)

In _____ destructor of sub-object is called after destructor of master class.

Select correct option:

Aggregation

=>Composition

Inheritance

Association

Question # 6 of 10 (Start time: 02:04:53 PM)

Select correct line of code for inheritance relationship between "Keys" class and "StandardKeys" class. "Keys" is parent class while "StandardKeys" is child class.

Select correct option:

=>class StandardKeys: public Keys

class Keys: public StandardKeys

class StandardKeys:: public Keys

class Keys:: public StandardKeys

Question # 7 of 10 (Start time: 02:06:07 PM)

Select correct line of code for aggregation relationship between "Person" class and "Phone" class.

Select correct option:

=>class Person { Phone * p; };

class Phone { Person * p; };

class Person { Phone p; };

class Phone { Person p; };

Question # 8 of 10 (Start time: 02:07:09 PM)

"Keyboard" class is composed off "keys" class. What will be the order of execution of destructor of "Keyboard" class and "Keys" class?

Select correct option:

First Destructo::Keys.. Second Destructor::Keyboard..

=>First Destructor:: Keyboard.. Second Destructor:: Keys..

Question # 9 of 10 (Start time: 02:08:38 PM)

There is an object of type Person, which of the following can be considered as one of its attributes:

Select correct option:

Name

Workat()

Age

=>Both age and name

Question # 10 of 10 (Start time: 02:09:27 PM)

Select correct line of code for composition relationship between "Keyboard" class and "Keys" class.

Select correct option:

=>class keyboard { Keys type;};

class keys{ KeyBoard type; };

class keyboard : private keys

class keys: private KeyBoard

In _____ a pointer or reference to an object is created inside a class

Select correct option:

Aggregation

Composition

Seperation

None of the given

Sub-Object can not be initialized using the constructor

Select correct option:

True

False

To initialize an array of objects, only _____ will be called

Select correct option:

Default Constructor

Overloaded Constructor

Default Object

None of the above

Let Suppose a class Student with objects std1, std2, and std3. For the statement $\text{std3} = \text{std1} - \text{std2}$ to work correctly, if the overloaded – operator must

Select correct option:

take two arguments.

None of the given choices

take single argument

take three arguments

Copy constructor is called when:

Select correct option:

An object is created in term of pre existence object

An object is created

An object is initialized

None of the given

Which of the following operator doesn't take any argument if overloaded?

Select correct option:

/

-

+

None of the given choices

Object can be declared constant with the use of Constant keyword.

Select correct option:

True

False

Always make pointer to null, whenever dynamic memory is deleted.

Select correct option:

True

False

_____ provide the facility to access the data member.

Select correct option:

accesser function

private function

inline function

None of the given

Question # 1 of 10 (Start time: 02:16:16 PM) Total Marks: 1

Friend Functions of a class are _____ members of that class.

Select correct option:

Public

Private

Protected

None of the given options.

Question # 2 of 10 (Start time: 02:17:30 PM) Total Marks: 1

Consider the code below, class class1{ protected: int i; }; class class2 : protected class1 { };

Then int member i of class1 is _____ in class2,

Select correct option:

public

protected

private

none of the given options

Question # 3 of 10 (Start time: 02:17:56 PM) Total Marks: 1

A Child class can call constructor of its parent class through,

Select correct option:

Its constructor initialization list

Its constructor body

Both from its constructor initialization list or body -

Can not call the constructor of its parent class

Question # 4 of 10 (Start time: 02:19:01 PM) Total Marks: 1 We can access private members of the class from outside the class using _____ operator with its object pointer.

Select correct option:

->

.

&

None of the given options

Question # 5 of 10 (Start time: 02:20:13 PM) Total Marks: 1

A function call is resolved at run-time in _____

Select correct option:

non-virtual member function

virtual member function

Both non-virtual member and virtual member function.

None of given

Question # 6 of 10 (Start time: 02:20:35 PM) Total Marks: 1

It is sometimes useful to specify a class from which no objects will ever be created.

Select correct option:

True

Flase

Question # 7 of 10 (Start time: 02:21:01 PM) Total Marks: 1

Consider the code below, class class1{ public: void func1(); }; class class2 : private class1 { };

Function func1 of class1 is _____ in class2,

Select correct option:

public

protected

private

none of the given options

Question # 8 of 10 (Start time: 02:21:26 PM) Total Marks: 1

Consider the code below, class class1{ public: int i; }; class class2 : protected class1 { }; Then int member i of class1 is _____ in class2,

Select correct option:

public

protected

private

none of the given options

Question # 9 of 10 (Start time: 02:22:00 PM) Total Marks: 1

Static casting is,

Select correct option:

Implicit way of calling base class functions from derived class

Explicit way of calling base class functions from derived class

None of these

Both of these

Question # 10 of 10 (Start time: 02:23:15 PM) Total Marks: 1

Consider the code below, class class1{ protected: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is _____ in class2,

Function func1 of class1 is _____ in class2,

Select correct option:

public

protected

private

none of the given option

[BS\(CS\)Permalink](#) Reply by [|| 22 || p.m.](#) on June 14, 2013 at 9:50pm

My Quizzzzzzz

_____ Operator will take only one operant.

New

int

object

Unary operators and assignment operator are right associative.

True

False

We can create array of objects if we have a user defined constructor.

True

False

Object can be declared constant with the use of Constant keyword.

True

False

To initialize an array of objects, only _____ will be called

Default Constructor

Overloaded Constructor

Default Object

None of the above

Let Suppose a class Student with objects std1, std2, and std3. For the statement $\text{std3} = \text{std1} - \text{std2}$ to work correctly, if the overloaded – operator must

take two arguments.

None of the given choices

take single argument

take three arguments

_____ provide the facility to access the data member.

accesser function

private function

inline function

None of the given

Static variable can be initialized more than once.

True

False

____ is creating objects of one class inside another class.

Association

Composition

Aggregation

Inheritance

Which of the following operator(s) take(s) one or no argument if overloaded?

++

*

All of the given choices

Question No.1 Time :90 second Marks-1

Sender of the message does not need to know the exact class of receiver in_____.

Please chose one

- 1) Abstraction
- 2) Polymorphism
- 3) Inheritance
- 4) none of the given

Answer: 2

Question No.2 Time :90 second Marks-1

Target of a ____ function call is determined at run time.

Please chose one

- 1) instance
- 2) virtual
- 3) operator
- 4) none of given

Answer: 2

Question No.3 Time :90 second Marks-1

A function call is resolved at run-time in_____ .

Please chose one

non-virtual member function
virtual member function

Both non-virtual member and virtual member function.

None of given

Answer: 3

Question No.4 Time :90 second Marks-1

Consider the code below, class class1{ public: void func1(); }; class class2 : protected class1 { };

Function func1 of class1 is ____ in class2,

Please chose one

public

protected

private

none of the given options

Answer: 2

Question No.5 Time :90 second Marks-1

Consider the code below, class c1{ }; class c2 : public c1 { }; class c3 : public c2 { }; Then c2 is,

Please chose one

Direct base class of c3

Direct child class of c3

Direct base class of c1

None of these

Answer: 1

Question No.6 Time :90 second Marks-1

A template provides a convenient way to make a family of.

Please chose one

1) variables and data members

2) functions and classes

3) classes and exceptions

4) programs and algorithms

Answer: 2

Question No.7 Time :90 second Marks-1

Consider the code below, class c1{ }; class c2 : public c1 { }; class c3 : public c2 { }; Then c1 is,

Please chose one

Direct base class of c3

Direct child class of c3

Direct base class of c2

Direct child class of c2

Answer: 2

Question No.8 Time :90 second Marks-1

User can make virtual table explicitly.

Please chose one

True

False

Answer: 1

Question No.9 Time :90 second Marks-1

In type in depended function template should be use where code and behavior must be identical.

Please chose one

True

False

Answer: 1

Child class can call constructor of its,

[Direct base class](#)

Indirect base class

Both direct and indirect base classes

None of these

A pure virtual function is a virtual function that
[causes its class to be abstract.](#)

returns nothing.

is used in a derived class.

takes no arguments.

<https://www.vuvickypro.com/2018/08/cs304-solved-quizzes-file-1.html>

check this link for important mcqs for final term

CS304 - Object Oriented Programming

Quiz No. 3, Feb 07, 2013

A template provides a convenient way to make a family of

Select correct option:

variables and data members

functions and classes

classes and exceptions

programs and algorithms

A class template may inherit from another class template.

Select correct option:

True

False

Target of a _____ function call is determined at run time.

Select correct option:

instance

virtual

operator

none of given

A class hierarchy

Select correct option:

shows the same relationships as an organization chart.

describes "has a" relationships.

describes "is a kind of" relationships.

shows the same relationships as a family tree.

Sender of the message does not need to know the exact class of receiver in_____.

Select correct option:

Abstraction

Polymorphism

Inheritance

none of the given

A function call is resolved at run-time in_____

Select correct option:

non-virtual member function

virtual member function

Both non-virtual member and virtual member function.

None of given

Adding a derived class to a base class requires fundamental changes to the base class.

Select correct option:

True

False

User can make virtual table explicitly.

Select correct option:

True

False

Binding means that target function for a call is selected at compile time.

Select correct option:

Static

Dynamic

Automatic

None of given

Target of a _____ function call is determined at run time.

Select correct option:

instance

virtual

operator

none of given

Which line will produce error. Class phone: private Transmit, private Receiver { } 1. int main() 2. { 3. phone obj; 4. Transmit* obj1 = &obj; 5. Received obj2 = &obj; 6. }

Select correct option:

3rd line will produce error

4th line will produce error

3rd and 4th line will produce error.

5th line will produce error

Function overriding is done in context of,

Select correct option:

Single class

Single derived class

Single base class

Derived and base classes

Consider the code below, class class1{ public: void func1(); }; class class2 : protected class1 { }; Function func1 of class1 is _____ in class2,

Select correct option:

public

protected

private

none of the given options

the following statements: 1) int iArray[5]; 2) int *pArr = iArray;

Select correct option:

These statements will compile successfully

Error in first statement

Error in second statement

None of given options

Methodologies to the development of reusable software relate to _____.

Select correct option:

Structure programming

procedural programming

generic programming

None of the given

function template must have a parameter.

Select correct option:

True

False

The default inheritance mode is,

Select correct option:

Public inheritance

Protected inheritance

Private inheritance

None of these options

Two functions with same names, parameters and return type can exist in,

Select correct option:

Function overloading

Function overriding

Operator overloading

None of these options

Consider the code below, class c1{ }; class c2 : public c1 { }; class c3 : public c2 { }; Then c2 is,

Select correct option:

Direct base class of c3

Direct child class of c3
Direct base class of c1
None of these

Virtual functions allow you to

Select correct option:

create an array of type pointer-to-base class that can hold pointers to derived classes.
create functions that can never be accessed.

group objects of different classes so they can all be accessed by the same function code.

use the same function call to execute member functions of objects from different classes.

User can make virtual table explicitly.

Select correct option:

True

False

In order to define a class template the first line of definition must be:

Select correct option:

template <typename T>

typename <template T>

Template Class <ClassName>

Class <Template T>

Consider the following statements: 1) int iArray[5]; 2) int *pArr = iArray;

Select correct option:

These statements will compile successfully

Error in first statement

Error in second statement

None of given options

In c++ dynamic binding and polymorphism will be achieved when member function will be ____.

Select correct option:

private

public

virtual

inline

In type in depended function template should be use where code and behavior must be identical.

Select correct option:

True

False

Consider the code below, class class1{ protected: int i; }; class class2 : private class1 { };
Then int member i of class1 is ____ in class2,

Select correct option:

public

protected

private

none of the given options

In specialization we can,

Select correct option:

Replace child class with its base class

Replace base class with its child class (Not Sure)

Replace both child and base classes interchangeably

None of the given options

Consider the code below, class class1{ public: void func1(); }; class class2 : public class1 {
}; Function func1 of class1 is ____ in class2,

Select correct option:

public

protected

private

none of the given options

It is illegal to make objects of one class members of another class.

Select correct option:

True

False

An abstract class is useful when

Select correct option:

no classes should be derived from it.

there are multiple paths from one derived class to another.

no objects should be instantiated from its.

you want to defer the declaration of the class.

In resolution order compiler search firstly _____.

Select correct option:

Generic Template

Partial Specialization

Complete Specialization

Ordinary function

```
template<> class Vector{ void** p; //.... void*& operator[] ((int i); };
```

Select correct option:

This specialization can then be used as the common implementation for all Vectors of pointers.

This specialization can then be used as the all type implementation for one type classes.

This specialization can then be used double type pointers.

This specialization should be used for Vectors of all type int types.

In private inheritance derived class pointer can be assigned to base class pointer in,

Select correct option:

Main function

In derived class member and friend functions

In base class member and friend functions

None of the given options

Which statement will be true for concrete class?

it implements an virtual concept.

it can be instantiated

it cannot be instantiated

none of given

Target of a _____ function call is determined at run time.

Select correct option:

instance

virtual

operator

none of given

The Specialization pattern after the name says that this specialization is to be used for every ____.

Select correct option:

data types

meta types

virtual types

pointers type

c++ dynamic binding and polymorphism will be achieved when member function will be ____.

Select correct option:

private
public
virtual
inline

Consider the code below, class class1{ protected: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is ____ in class2,
Select correct option:

public
protected
private
none of the given options

Consider the code below, class class1{ protected: int i; }; class class2 : protected class1 { }; Then int member i of class1 is ____ in class2,
Select correct option:

public
protected
private
none of the given options

Consider the code below, class class1{ private: void func1(); }; class class2 : private class1 { }; Function func1 of class1 is ____ in class2,
Select correct option:

public
protected
private
none of the given options

Target of a ____ function call is determined at run time.

Select correct option:

instance
virtual
operator
none of given

Consider the following statements: 1) int iArray[5]; 2) int *pArr = iArray;

Select correct option:

These statements will compile successfully

Error in first statement

Error in second statement

None of given options

Consider the code below, class class1{ private: int i; }; class class2 : private class1 { };
Then int member i of class1 is ____ in class2,

Select correct option:

- public
- protected
- private

none of the given options

If there is a pointer, p, to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a virtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the ____ class to be executed.

Select correct option:

- base**
- derived
- virtual
- implemented

A class template may inherit from another class template.

Select correct option:

- True**
- False

Derived class can inherit from public base class as well as private and protected base classes

Select correct option:

- True
- False**

Two functions with same names, parameters and return type can exist in,

Select correct option:

- Function overloading**
- Function overriding
- Operator overloading
- None of these options

Consider the code below, class class1{ private: int i; }; class class2 : public class1 { }; Then int member i of class1 is ____ in class2,

Select correct option:

- public
- protected
- private

none of the given options

Target of a _____ function call is determined at run time.

Select correct option:

instance

virtual

operator

none of given

A class template may inherit from another class template.

Select correct option:

True

False

A function call is resolved at run-time in _____

Select correct option:

non-virtual member function

virtual member function

Both non-virtual member and virtual member function.

None of given

hello

A class hierarchy

Select correct option:

shows the same relationships as an organization chart.

describes "has a" relationships.

describes "is a kind of" relationships.

shows the same relationships as a family tree.

Consider the code below, `class class1{ public: int i; }; class class2 : public class1 { }; Then int member i of class1 is _____ in class2,`

Select correct option:

public

protected

private

none of the given options

Consider the code below, `class c1{ }; class c2 : public c1 { }; class c3 : public c2 { }; Then c1 is,`

Select correct option:

Direct base class of c3

Direct child class of c3

Direct base class of c2

Direct child class of c2

A class can inherit from more than one class is called.

Select correct option:

Simple inheritance

Multiple inheritances

Single inheritance

Double inheritance

```
template<> class Vector{ void** p; //.... void*& operator[] ((int i); );
```

Select correct option:

This specialization can then be used as the common implementation for all Vectors of pointers.

This specialization can then be used as the all type implementation for one type classes.

This specialization can then be used double type pointers.

This specialization should be used for Vectors of all type int types.

Consider the code below, class class1{ public: int i; }; class class2 : protected class1 { };

Then int member i of class1 is ____ in class2,

Select correct option:

public

protected

private

none of the given options

Consider the code below, class class1{ private: void func1(); }; class class2 : public class1

{ }; Function func1 of class1 is ____ in class2,

Select correct option:

public

protected

private

none of the given options

Templates automatically create different versions of a function, depending on user input.

Select correct option:

True

False

_____ Binding means that target function for a call is selected at run time

Select correct option:

Automatic

Dynamic

Static

Dramatic

When we create objects, then space is allocated to:

Member function

Access specifier

Data member

None of given

There is only one form of copy constructor.

True

False

Which of the following features of OOP is used to deal with only relevant details?

Abstraction

Information hiding

Object

_____ Binding means that targets function for a call is selected at compile time.

Static

Dynamic

Automatic

None of given

A Class hierarchy

Shows the same relationships as an organization chart

Describes "has a" relationships.

Describes "is a kind of" relationships.

Shows the same relationships as a family tree

In C++, we declare a function virtual by preceding the function header with keyword "Inline"

True

False

It is illegal to make objects of one class members of another class.

True

False

In Resolution order compiler search firstly_____.

Generic Template

Partial Specification

Complete Specification

Ordinary function

Derived class can inherit from public base class as well as private and protected base classes

True

False

Which line will produce error. Class phone: Private Transmit, private Receiver { } 1.int main () 2. { 3.phone obj; 4.Tranmit*obj1 = &obj; 5.Received obj2 = &obj; 6.}

3rd line will produce error

4th line will produce error

3rd and 4th line will produce error.

5th line will produce error.

Methodologies to the development of reusable software relate to _____.

Structure programming

Procedural programming

Generic programming

None of the given

A template argument is preceded by the keyword _____.

Vector

Class

Template

Type*

Friends are used exactly the same for template and non-template classes.

True

False

A function template must have a parameter

True

False

Child class can call constructor of its,

Direct base class

Indirect base class

Both direct and indirect base classes

None of these.

Which statement will be true for concrete class?

It implements an virtual concept.

It can be instantiated

It cannot be instantiated

None of given

A class D can be derived from a class C, which is derived from a class B, which is derived from a class A

True

False

Adding a derived class to a base class requires fundamental changes to the base class.

True

False

A Class or class template can have member _____ that are themselves templates.

Variable

Function

Objects

None of given

Which will be the Primary task or tasks of generic programming?

Categorize the abstractions in a domain into concepts

Implement generic algorithms based on the concepts

Build concrete models of the concepts

All of given

The default inheritance mode is,

Public inheritance

Protected Inheritance

Private Inheritance

None of these options

If there is a pointer, p, to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a virtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the _____ class to be executed.

Base

Derived

Virtual

Implemented

A class template _____

Facilitates reuse of class

Does not facilitate reuse of class

Sender of the message does not need to know the exact class of receiver in _____.

Abstraction

Polymorphism

Inheritance

none of the given

Friend Functions of a class are _____ members of that class.

Select correct option:

Public

Private ---

Protected

None of the given options.

Consider the code below, class class1{ protected: int i; }; class class2 : protected class1 {
}; Then int member i of class1 is _____ in class2,

Select correct option:

public

protected

private

none of the given options

A Child class can call constructor of its parent class through,

Select correct option:

Its constructor initialization list

Its constructor body

Both from its constructor initialization list or body -----

Can not call the constructor of its parent class

We can access private members of the class from outside the class using _____
operator with its object pointer.

Select correct option:

->

.

&

None of the given options

A function call is resolved at run-time in _____

Select correct option:

non-virtual member function

virtual member function

Both non-virtual member and virtual member function.

None of given

It is sometimes useful to specify a class from which no objects will ever be created.

Select correct option:

True

False

Consider the code below, class class1{ public: void func1(); }; class class2 : private class1 { }; Function func1 of class1 is _____ in class2,

Select correct option:

public

protected

private

none of the given options

Consider the code below, class class1{ public: int i; }; class class2 : protected class1 { }; Then int member i of class1 is _____ in class2,

Select correct option:

public

protected

private

none of the given options

Static casting is,

Select correct option:

Implicit way of calling base class functions from derived class

Explicit way of calling base class functions from derived class

None of these

Both of these

Consider the code below, class class1{ protected: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is _____ in class2,

Select correct option:

public

protected

private

none of the given options

Consider the code below, class class1{ public: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is _____ in class2

Public

Protected
Private
none of the given options

Consider the code below, class class1{ protected: int i; }; class class2 : public class1 { };
Then int member i of class1 is _____ in class2,
public
protected
private
none of the given options

In Private Inheritance the public members of base class become _____ in derived class.
Public
Private
Protected
None of the given options.

Consider the following two lines of code written for a class Student, 1. Student subj1; 2. Student subj2(subj1); In line No.2 what constructor of Student class will be called,
Default constructor of Student class
Copy constructor of student class
Both default and copy constructor of Student class
No constructor will be called

السلام عليكم الله ورحمته وبركاته

الله "وَتُعِزُّ مَنْ تَشَاءُ وَتُذِلُّ مَنْ تَشَاءُ"

الله "جسے جابے عزت دے اور جسے جابے ذلیل کرے"

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Methodologies to the development of reusable software relate to

- a) Structure programming
- b) Procedural programming
- c) Generic programming**
- d) None of the given

In case of template specialization, if compiler cannot find required complete specialization, t

a. General template

- b. Complete template
- c. Partial specialization

Which of the following may inherit from the ordinary class?

- a. Class template
- b. Partial specialization
- c. Complete specialization

d. All of given option

Which of the following is not type of inheritance in c++?

- a) Public
- b) Protected
- c) Restricted**

Which of the following statement is true about partial specialization?

- a. Function templates cannot have partial specialization
- b. Class template cannot have partial specialization

c. Both class templates and function template can have partial specialization. d. None of the given

In resolution order of function template, compiler searches for in the end

- a. Complete specialization

b. Partial specialization

c. Generic template

Which of the following is known as
Dereference operator in C++?

a. +

b. *

c. &

When we want to have exactly identical
operations on different data type, are used

a. Function Template

b. Function Overloading

c. Function Overriding

When we specialize a function template, it is
called

a. Function overloading

b. Function overriding

c. Function template overloading

A class template may inherit from another class template.

True

False

Target of a _____ function call is determined at run time.
instance

virtual

operator none of given

Sender of the message does not need to know the exact class of receiver in_____.

Abstraction

Polymorphism

A function call is resolved at run-time in_____ .

non-virtual member

function virtual member

function

Both non-virtual member and virtual member function.

User can make virtual table explicitly.

True

False

Binding means that target function for a call is selected at compile time.

Static

Dynamic

Which line will produce error. Class phone: private Transmit, private Receiver { } 1. int main() 2. { 3. phone obj; 4. Transmit* obj1 = &obj; 5. Received obj2 = &obj; 6. }.

3rd line will produce error

4th line will produce error

3rd and 4th line will produce error.

5th line will produce error

Function overriding is done in context of,

Single class

Derived and base classes

Methodologies to the development of reusable software relate to_____.

Structure programming

generic programming

Function template must have a parameter.

True

False

The default inheritance mode is,

Public inheritance

Private inheritance

None of these options

Two functions with same names, parameters and return type can exist in,

Function overloading

Function overriding

Consider the code below, class c1{ }; class c2 : public c1 { };

class c3 : public c2 { }; Then c2 is,

Direct base class of c3

Direct child class of c3

User can make virtual table explicitly.

True

False

In order to define a class template the first line of definition must be:

template <typename T>

typename <template T>

Consider the following statements: 1) int iArray[5]; 2) int *pArr = iArray;

These statements will compile successfully

Error in second statement

In type in depended function template should be use where code and behavior must be identical.

True

False

It is illegal to make objects of one class members of another class.

True

False

In resolution order compiler search firstly _____.

Complete Specialization

Ordinary function

template<> class Vector {void p; //.... void*& operator[]
((int i); };**

This specialization can then be used as the common implementation for all Vectors of pointers.

This specialization can then be used as the all type implementation for one type classes.

This specialization can then be used double type pointers.

This specialization should be used for Vectors of all type int types.

**Consider the code below, class class1{ private: int i; };
class class2 : private class1 { }; Then int member i of
class1 is ____ in class2, public protected
private**

none of the given options

If there is a pointer, p, to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a virtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the ____ class to be executed.

base

derived

A class template may inherit from another class template.

True

False

Derived class can inherit from public base class as well as private and protected base classes

True

False

Consider the code below, class class1{ private: int i; }; class class2 : public class1 { }; Then int member i of class1 is ____ in class2,

None of the given options

Binding means that target function for a call is selected at run time

Automatic

Dynamic

When we create objects, then space is allocated to:

Member function

Access specifier

Data member

None of given

There is only one form of copy constructor.

True

False

Which of the following features of OOP is used to deal with only relevant details? Abstraction

Information hiding

Object

_____ **Binding means that targets function for a call is selected at compile time.** Static

Dynamic

Automatic

None of given

A Class hierarchy

Shows the same relationships as an organization chart Describes “has a” relationships.

Describes “is a kind of” relationships.

Shows the same relationships as a family tree

In C++, we declare a function virtual by preceding the function header with keyword “Inline”.

True

False

It is illegal to make objects of one class members of another class.

True

False

Derived class can inherit from public base class as well as private and protected base classes

True

False

Which line will produce error? Class phone: Private

Transmit, private Receiver { }

**1.int main () 2. { 3.phone obj; 4. Transmit*obj1 = &obj;
5.Received obj2 = &obj; 6.}**

3rd line will produce

error 4th line will

produce error

3rd and 4th line will

produce error. 5th line will

produce error.

Methodologies to the development of reusable software relate to ____:

Friends are used exactly the same for template and non-template classes.

True

False

A function template must have a parameter

True

False

Child class can call constructor of its,

Direct base class

Indirect base class

Both direct and indirect base classes
None of these.

Which statement will be true for concrete class?

It implements an virtual concept.

It can be instantiated

It cannot be instantiated

None of given

A class D can be derived from a class C, which is derived from class B, which is derived from a class A

True

False

Adding a derived class to a base class requires fundamental changes to the base class.

True

False

A Class or class template can have member _____ that are themselves templates.

Variable

Function

Objects

None of given

Which will be the Primary task or tasks of generic programming?

Categorize the abstractions in a domain into concepts

Implement generic algorithms based on the concepts

Build concrete models of the concepts

All of given

The default inheritance mode is,

Public inheritance

Protected Inheritance

Private Inheritance

None of these options

If there is a pointer, p, to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a virtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the _____ class to be executed.

Base

Derived

Virtual

Implemented

Child class can call constructor of its,

Direct base class

Indirect base class

Both direct and indirect base classes

None of these

A pure virtual function is a virtual function that causes its class to be abstract.

returns nothing. is
used in a derived
class. takes no
arguments.

1: Consider the code below, class class1{ protected: int i; };
class class2 :
protected class1 { }; Then int member i of class1 is _____ in
class2,

- + public
- + **protected**

2: By default, assignment operator (=) performs,

- + **Shallow copy**
- + Deep copy

3: The default inheritance mode is,

- + **Public inheritance**

4: In case of dynamic memory allocation in our class we
should use

- + Default assignment operator
- + **copy constructor**

5: Static casting is,?

- + **Implicit way of calling base class functions from derived class**
- + Explicit way of calling base class functions from derived class

6: In case of dynamic memory allocation in a class we should
use,

- + User defined default constructor
- + **user defined copy constructor (true)**

7: Which of the following types of Inheritance is used to model “Implemented in terms of” relationship?

- + Public
- + **private (true)**

8: Virtual functions allow you to

- + create an array of type pointer-to-base class that can hold pointers to derived classes.
- + **group objects of different classes so they can all be accessed by the same function code. (true)**

9: A class D can be derived from a class C, which is derived from a class B, which is derived from a class A.

- + **T (true ans)**
- + F

10: Function overriding is done in context of,

- + Explicit way of calling base class functions from derived class
- + None of these
- + both 1 and 2

1. An abstract class is useful when

- a. no classes should be derived from it.
- b. there are multiple paths from one derived class to another.
- c. no objects should be instantiated from its.**
- d. you want to defer the declaration of the class.

2. In order to define a class template the first line of definition must be:

- a. template <typename T>**

- b. typename <template T>
- c Template Class <Class Name>
- d Class <Template T>

3. Select correct line of code for composition relationship between “Keyboard” class and “Keys” class

- a class keyboard {Keys type;};
- b class keys {KeyBoard type; };
- c class keyboard: private keys
- d class keys: private KeyBoard**

4. Select correct line of code for inheritance relationship between “Keys” class and “Special Keys” class. “Keys” is parent class while “Special Keys” is child class

- A class Special Keys: public Keys**
- B class Keys: public Special Keys
- C class Special Keys: public Keys
- D class Keys: public Special Keys

5 Methodologies to the development of reusable software relate to_____.

- A Structure programming
- B procedural programming
- C generic programming**
- D None of the given

6 User can make virtual table explicitly.

- A True**
- B false

7 A function call is resolved at run-time in_____

- A non-virtual member function
- B virtual member function**
- C Both non-virtual member and virtual member function.
- D None of given

8 Derived class can inherit from public base class as well as private and protected base classes

A true

B false

9 Which of the following function can convert a class into an abstract class?

A Abstract function

B Concrete function

C Virtual function

D Pure virtual function

CS304 Question No: 2

Which of the following causes run time binding?

Declaring object of abstract class

Declaring pointer of abstract class

Declaring overridden methods as non-virtual

None of the given

CS304 Question No: 3

A function template can not be overloaded by another function template.

True

False

CS304 Question No: 4

Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

Templates

Overloading

Data hiding

Encapsulation

CS304 Question No: 5

Identify the correct way of declaring an object of user defined template class A for char type members?

A obj;

A obj;

A obj;

Obj A;

CS304 Question No: 6

The user must define the operation of the copy constructor.

True

False

CS304 Question No: 7

Template functions use _____ than ordinary functions.

Greater Memory

Lesser Memory

Equal Memory

None of the given options

CS304 Question No: 8

The find() algorithm

finds matching sequences of elements in two containers.

finds a container that matches a specified container.

takes iterators as its first two arguments.

takes container elements as its first two arguments.

CS304 Question No: 9

Compiler performs _____ type checking to diagnose type errors,

Static

Dynamic

Bound

Unbound

CS304 Question No: 10

Which of the following is/are advantage[s] of generic programming?

Reusability

Writability

Maintainability

All of given

CS304 Question No: 11

Vectors contain contiguous elements stored as a[an] ____.
variable

array

function

datatype

CS304 Question No: 12

**Suppose you create an uninitialized vector as follows:
vector eve;**

After adding the statement,

eve.pushback(21);

what will happen?

The following statement will add an element to the start (the back) of eve and will initialize it with the value 21.

The following statement will add an element to the center of eve and will reinitialize it with the value 21.

In a de-queue, (chose the best option)

data can be quickly inserted or deleted at any arbitrary location.

data can be inserted or deleted at any arbitrary location, but the process is relatively slow.

data cannot be quickly inserted or deleted at either end.

data can be inserted or deleted at either end, but the process is relatively slow.

CS304 Question No: 14

Algorithms can only be implemented using STL containers.

True

False

CS304 Question No: 15

What is a class?

A class is a section of computer memory containing objects.

A class is a section of the hard disk reserved for object oriented programs

A class is the part of an object that contains the variables.

A class is a description of a kind of object.

CS304 Question No: 16

**Inheritance is a way to
organize data.**

pass arguments to objects of classes.

add features to existing classes without rewriting them.

improve data-hiding and encapsulation.

CS304 Question No: 17

We can use "this" pointer in the constructor in the body and even in the initialization list of any class if we are careful,

True

False

CS304 Question No: 18

_____ and _____ methods may not be declared abstract.

private, static

private, public

static, public

none of given

CS304 Question No: 19

**Default constructor is such constructor which either has no -
-----or if it has some parameters these have ----- values**

Parameter, temporary

Null, Parameter

Parameter, default

none of the given

CS304 Question No: 20

**Public methods of base class can ----- be accessed in its
derived class**

directly

indirectly

simultaneously

none of the given

CS304 Question No: 21

The type that is used to declare a reference or pointer is called its -----

default type

static type

abstract type

reference type

CS304 Question No: 22

----- members are somewhere between public and private members. They are used in inheritance protected

public

private

global

CS304 Question No: 23

Which of these are examples of error handling techniques ?

Abnormal Termination

Graceful Termination

Return the illegal

all of the given

CS304 Question No: 24

----- follow try block to catch the object thrown.

catch block

throw block

main block

non of the given

CS304 Question No: 25

Graphical representation of the classes and objects is called object model it shows -----

Class Name only

Class Name and attributes

Relationships of the objects and classes

all of the given

CS304 Question No: 26

Destructor can be overloaded

True

False

CS304 Question No: 27 (M - 2)

Describe the way to declare a template function as a friend of any class.

Template name

Class classmate

{

Friend void friend template name (class name gastric const part class name);

}

CS304 Question No: 28 (M - 2)

State any two reasons why the virtual methods cannot be static?

1-virtual method can not be static as it is dynamic

CS304-Object Oriented Program

QUIZ-2 LEC(31 to 38)

Solved by JUNAID MALIK



AL-Junaid Institute

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ALL answers are verified if found any
mistake then Correct ACCORDINGLY

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1. A function template must have a parameter

- **True**
- False

2. Child class can call constructor of its,

- **Direct base class**
- Indirect base class
- Both direct and indirect base classes
- None of these.

3. Which statement will be true for concrete class?

- It implements an virtual concept.
- **It can be instantiated**
- It cannot be instantiated
- None of given

4. A class D can be derived from a class C, which is derived from a class B, which is derived from a class A

- **True**
- False

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5. Adding a derived class to a base class requires fundamental changes to the base class.

- **True**
- False

6. A Class or class template can have member _____ that are themselves templates.

- Variable
- **Function**
- Objects
- None of given

7. Which will be the Primary task or tasks of generic programming?

- Categorize the abstractions in a domain into concepts
- Implement generic algorithms based on the concepts
- Build concrete models of the concepts
- **All of given**

8. The default inheritance mode is,

- **Public inheritance**
- Protected Inheritance
- Private Inheritance
- None of these options

9. If there is a pointer, p, to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a virtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the _____ class to be executed.

- Base
- Derived
- **Virtual**
- Implemented

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10. Sender of the message does not need to know the exact class of receiver in _____.

- Abstraction
- **Polymorphism**
- Inheritance
- none of the given

11. A template provides a convenient way to make a family of

variables and data members

functions and classes

classes and exceptions

programs and algorithms

12. A class template may inherit from another class template.

Select correct option:

True

False

13. Target of a _____ function call is determined at run time.

instance

virtual

operator

none of given

14. A class hierarchy

shows the same relationships as an organization chart.

describes “has a” relationships.

◀ **describes “is a kind of” relationships.**

shows the same relationships as a family tree.

15. Sender of the message does not need to know the exact class of receiver

in _____.

Abstraction

Polymorphism

Inheritance

none of the given

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16. A function call is resolved at run-time in _____

non-virtual member function

virtual member function

Both non-virtual member and virtual member function.

None of given

17. Adding a derived class to a base class requires fundamental changes to the base class.

True

False

18. User can make virtual table explicitly.

True

False

19. Binding means that target function for a call is selected at compile time.

Static

Dynamic

Automatic

None of given

20. Target of a _____ function call is determined at run time.

instance

virtual

operator

none of given

21. Which line will produce error. Class phone: private Transmit, private Receiver { } 1. int main() 2. { 3. phone obj; 4. Transmit* obj1 = &obj; 5. Received obj2 = &obj; 6. }

3rd line will produce error

4th line will produce error

3rd and 4th line will produce error.

5th line will produce error

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22. Function overriding is done in context of,

Single class

Single derived class

Single base class

Derived and base classes

23. Consider the code below, class class1 { public: void func1(); }; class class2 :
protected class1 { }; Function func1 of class1 is ____ in class2,

public

protected

private

none of the given options

24. the following statements: 1) int iArray[5]; 2) int *pArr = iArray;

These statements will compile successfully

Error in first statement

Error in second statement

None of given options

25. Methodologies to the development of reusable software relate to _____.

Structure programming

procedural programming

generic programming

None of the given

26. function template must have a parameter.

True

False

27. The default inheritance mode is,

Public inheritance

Protected inheritance

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Private inheritance

None of these options

28. Two functions with same names, parameters and return type can exist in,

Function overloading

Function overriding

Operator overloading

None of these options

29. Consider the code below, `class c1 { }; class c2 : public c1 { }; class c3 : public c2 { };` Then c2 is,

Direct base class of c3

Direct child class of c3

Direct base class of c1

None of these

30. Virtual functions allow you to

create an array of type pointer-to-base class that can hold pointers to derived classes.

create functions that can never be accessed.

group objects of different classes so they can all be accessed by the same function code.

use the same function call to execute member functions of objects from different classes.

31. User can make virtual table explicitly.

True

False

32. In order to define a class template the first line of definition must be:

template <typename T>

typename <template T>

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Template Class <ClassName>

Class <Template T>

33. Consider the following statements: 1) `int iArray[5];` 2) `int *pArr = iArray;`

These statements will compile successfully

Error in first statement

Error in second statement

None of given options

34. In c++ dynamic binding and polymorphism will be achieved when member function will be ____.

private

public

virtual

inline

35. In type in depended function template should be use where code and behavior must be identical.

Select correct option:

True

False

36. Consider the code below, `class class1 { protected: int i; }; class class2 : private class1 { }; Then int member i of class1 is ____ in class2,`

public

protected

private

none of the given options

37. In specialization we can,

Replace child class with its base class

Replace base class with its child class (Not Sure)

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Replace both child and base classes interchangeably
None of the given options

38. Consider the code below, `class class1 { public: void func1(); }; class class2 : public class1 { }; Function func1 of class1 is ____ in class2,`

public
protected
private
none of the given options

39. It is illegal to make objects of one class members of another class.

True
False

40. Two functions with same names, parameters and return type can exist in,

Function overloading
Function overriding
Operator overloading
None of these options

41. Consider the code below, `class class1 { private: int i; }; class class2 : public class1 { }; Then int member i of class1 is ____ in class2,`

public
protected
private
none of the given options

42. Target of a ____ function call is determined at run time.

instance
virtual
operator
none of given

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43. A class template may inherit from another class template.

Select correct option:

True

False

44. A function call is resolved at run-time in _____

non-virtual member function

virtual member function

Both non-virtual member and virtual member function.

None of given

45. A class hierarchy

shows the same relationships as an organization chart.

describes “has a” relationships.

describes “is a kind of” relationships.

shows the same relationships as a family tree.

46. Consider the code below, `class class1 { public: int i; }; class class2 : public class1 { }; Then int member i of class1 is ____ in class2,`

public

protected

private

none of the given options

47. Consider the code below, `class c1 { }; class c2 : public c1 { }; class c3 : public c2 { }; Then c1 is,`

Direct base class of c3

Direct child class of c3

Direct base class of c2

Direct child class of c2

48. A class can inherit from more than one class is called.

Simple inheritance

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Multiple inheritances

Single inheritance

Double inheritance

49. `template< class Vector{ void** p; //.... void*& operator[] ((int i); };`

This specialization can then be used as the common implementation for all Vectors of pointers.

This specialization can then be used as the all type implementation for one type classes.

This specialization can then be used double type pointers.

This specialization should be used for Vectors of all type int types.

50. Consider the code below, `class class1 { public: int i; }; class class2 : protected class1 { };` Then int member i of class1 is ____ in class2,

public

protected

private

none of the given options

51. Consider the code below, `class class1 { private: void func1(); }; class class2 : public class1 { };` Function func1 of class1 is ____ in class2,

public

protected

private

none of the given options

52. Templates automatically create different versions of a function, depending on user input.

True

False

53. _____ Binding means that target function for a call is selected at run time

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Automatic
Dynamic
Static
Dramatic

54. When we create objects, then space is allocated to:

- Member function
- Access specifier
- Data member
- **None of given**

55. There is only one form of copy constructor.

- True
- **False**

56. Which of the following features of OOP is used to deal with only relevant details?

- Abstraction
- Information hiding
- **Object**

57. _____ Binding means that targets function for a call is selected at compile time.

- **Static**
- Dynamic
- Automatic
- None of given

58. A Class hierarchy

- Shows the same relationships as an organization chart
- Describes “has a” relationships.
- **Describes “is a kind of” relationships.**
- Shows the same relationships as a family tree

58. In C++, we declare a function virtual by preceding the function header with keyword “Inline”

- **True**
- False

59. It is illegal to make objects of one class members of another class.

- **True**
- False

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60. In Resolution order compiler search firstly_____.

- Generic Template
- Partial Specification
- Complete Specification
- **Ordinary function**

61. Derived class can inherit from public base class as well as private and protected base classes

- **True**
- False

62. Which line will produce error. Class phone: Private Transmit, private Receiver { } 1.int main () 2. { 3.phone obj; 4.Transmit*obj1 = &obj; 5.Received obj2 = &obj; 6.}

- **3rd line will produce error**
- 4th line will produce error
- 3rd and 4th line will produce error.
- 5th line will produce error.

63. Methodologies to the development of reusable software relate to _____.

- Structure programming
- Procedural programming
- **Generic programming**
- None of the given

64. A template argument is preceded by the keyword_____.

- Vector
- **Class**
- Template
- Type*

65. Friends are used exactly the same for template and non-template classes.

- True
- **False**

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CS304 Quiz No.2 Spring 2020 by MCS of Virtuallians

1. The parameters given in temple definition other than those used for mentioning templates types are called _____
 - a. Type Parameters
 - b. Non Type parameters
 - c. Default Type Parameters
 - d. None of the given option
2. Which of the following is the correct way to define a template class X?
 - a. Class < typename T>class X{}
 - b. Template class C{}
 - c. Template < typename T> classX{}
 - d. Typename<class T > class X{}
3. In resolution order of function template, compiler searches for _____ in the end
 - a. Complete specialization
 - b. Partial specialization
 - c. Ordinary function
 - d. Generic template
4. _____ may inherit from a complete specialization
 - a. Partial specialization
 - b. Complete specialization
 - c. Ordinary class
 - d. All of given options
5. Which of the following may inherit form the ordinary class?
 - a. Class template
 - b. Partial specialization
 - c. Complete specialization
 - d. All of given option
6. Which of the following is a sequence container?
 - a. Map
 - b. Vector
 - c. Set
 - d. multiset
7. Each _____ of a template class by default becomes function template
 - a. Date member
 - b. Member function
 - c. Type parameter
 - d. Object
8. When we specialize a function template, it is called _____
 - a. Function overloading
 - b. Function overriding
 - c. Function template overloading
 - d. Function template overriding

9. Which of the following is NOT an associative container?
- a. Set
 - b. Multiset
 - c. Map
 - d. Group
10. A _____ is a pointer which is declared outside the _____
- a. Container, cursor
 - b. Cursor, container
 - c. Container, class
 - d. This, cursor
11. Which of the following are collectively referred to as the first class containers?
- a. Associative containers and containers adapters
 - b. Sequence containers and associative containers
 - c. Sequence containers and container adapters
 - d. None of the given options
12. _____ class is a single class that provides functionality to operate on different type of data
- a. Friend
 - b. Template
 - c. Ordinary
 - d. None of the give options
13. Which of the following can be passed as type argument to template?
- a. Primitive type
 - b. User defined types
 - c. Both primitive type and user defined types
 - d. None of the given options
14. Which of the following is known as Dereference operator in C++?
- a. +
 - b. *
 - c. &
 - d. ::
15. We can change behavior of template using _____
- a. Template parameters
 - b. Function parameters
 - c. Class Templates
 - d. none of the given options
16. Which of the following represents partial specialization?
- a. Template < int, char >
 - b. Template < class T, Type t>
 - c. Template<class T, Class U, int>
 - d. Template<typename T, class W>

17. When we declare template function as friend of any class, then which of the following specialization(s) of template function also become friend of the class granting friendship?

- a. Explicit
- b. Implicit
- c. Partial
- d. All of the given options

18. In statement “template <class T, class U, int I = 5>” then non-type parameters is

- a. Class T
- b. Class U
- c. int I
- d. All of the given option

19. Which of the following is the correct syntax for passing two type argument to a template?

- a. template < type T, type U>
- b. template < typename T, typename U>
- c. template <type T,U>
- d. template Typename < T,U >

20. Which of the following represents complete specialization?

- a. Template < int, char >
- b. Template < class T, float>
- c. Template<class T, Class U , int>
- d. Template< class , class W>

21. Which of the following is/are the key component of standard template library?

- a. Containers
- b. Iterators
- c. Algorithm
- d. All of the given option

22. Which of the following statement is true about partial specialization?

- a. Function templates cannot have partial specialization
- b. Class template cannot have partial specialization
- c. Both class templates and function template can have partial specialization.
- d. None of the given

23. Which of the following is correct code portion to initialize static data member “value” of int type for class X?

- a. Template <class T> X::value =0;
- b. Template<class T>int X::value =0;
- c. Template<class T> int X < T >::value = 0;
- d. Template<class T> X < T >::value =0;

24. When we want to have exactly identical operations on different data type, _____ are used

- a. Function Template

- b. Function Overloading
 - c. Function Overriding
 - d. None of the given option
25. In order to define a class template the first line of definition must be:
- a. **Template <typename T>**
 - b. typename < template T>
 - c. Template class <ClassName>
 - d. Class <Template T>
26. In case of template specialization, if compiler cannot find required complete specialization then it searches for some _____
- a. General template
 - b. Complete template
 - c. **Partial specialization**
 - d. None of the given
27. Which of the following is the correct code to instantiate the object of given template vector class of int type?
- ```
Template <class T>
class vector
{
};
```
- a. Vector obj <int>;
  - b. Vector obj <> int;
  - c. **Vector <int>obj;**
  - d. Vector int obj;
28. Which of the following types of container is NOT provided by standard Template Library?
- a. Sequence containers
  - b. **Structured containers**
  - c. Associative containers
  - d. Container adapters
29. A class template \_\_\_\_\_
- a. **Facilitates reuse of class**
  - b. Does not facilitates reuse of class
  - c. Does not support generic methods
  - d. Does not support static members
30. Suppose a template class “test” has a static data members, how many copies of this static data member will be created when template class is instantiated by writing following line of C++ code?
- ```
Test <char>x,y,z
```
- a. Zero
 - b. One
 - c. Two
 - d. **Three**

31. STL stands for:

- a. Stack Template Library
- b. Standard Template Library
- c. Structured Template Library
- d. Secure Template Library

32. Which of the following is not a sequence container?

- a. Vector
- b. Deque
- c. Map
- d. List

33. Suppose a template class “test” has a static data members, how many copies of this static data member will be created when template class is instantiated by writing following line of C++ code?

```
Test <int>x;
```

```
Test <char>y;
```

- a. Zero
- b. One
- c. Two
- d. Three

34. Suppose we have a template class “Test” which have a static data member “data” which is initialized with 0. What will be output when following code is written in main() function for given class?

```
Test<int> a, b;
```

```
a.data = 5;
```

```
b.data = 7;
```

```
cout<<a.data;
```

- a. 0
- b. 1
- c. 5
- d. 7

35. In resolution order, highest priority is given to _____ in template specialization.

- a. General template
- b. Complete specialization
- c. Partial specialization
- d. None of the given options

36. Which of the following is a constrained version of some first-class container?

- a. Sequence containers
- b. Associative containers
- c. Container Adapter
- d. None of the given options

Cs304 quizz
Solved by Arsl
Shani

BC160201273: Muhammad Atif

CS304: Quiz No. 2

Question # 5 of 10 (Start time: 02:19:24 PM, 31 August 2020)

Which of the following is a Sequence Container?

Select the correct option

☐	map
☒	vector
☐	set
☐	multiset



Type here to search



BC160201273: Muhammad Atif

CS304: Quiz No. 2

Question # 3 of 10 (**Start time: 02:17:03 PM, 31 August 2020**)

Which of the following is NOT an Associative Container?

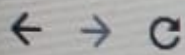
Select the correct option

☐	set
☐	multiset
☐	map
☐	group



Type here to search





quiz.vu.edu.pk/QuizQuestion.aspx?ver=511b9c29-85b8-42c2-81f1-2c0450a

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CS304: Quiz No. 2

Question # 2 of 10 (Start time: 02:16:41 PM, 31 August 2020)

Which of the following is/are key component(s) of Standard Template Library?

Select the correct option

- | | |
|-----------------------|--------------------------|
| ☐ | Containers |
| ☐ | Iterators |
| ☐ | Algorithms |
| ☐ | All of the given options |
- 



Type here to search



Question # 10 of 10 (Start time: 02:06:26 PM, 31 August 2020)

Suppose a template class "Test" has a static data member. How many copies of this static data member will be created when template class is instantiated by writing following lines of C++ code?

```
Test<int> x;  
Test<char> y;
```

Select the correct option

☐

Zero

☐

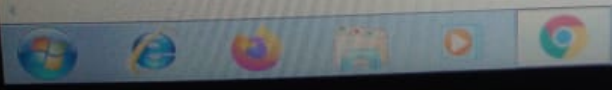
One

☒

Two

☐

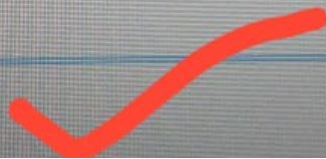
Three



_____ may inherit from a complete specialization.

Select the correct option

- | | |
|-----------------------|--------------------------|
| ☐ | Partial specialization |
| ☐ | Complete specialization |
| ☐ | Ordinary class |
| ☐ | All of the given options |



_____ may inherit from a complete specialization.

Select the correct option

- | | |
|-----------------------|--------------------------|
| ☐ | Partial specialization |
| ☐ | Complete specialization |
| ☐ | Ordinary class |
| ☐ | All of the given options |

A class template _____

Select the correct option

- | | |
|----------------------------------|------------------------------------|
| ☒ | Facilitates reuse of class |
| ☐ | Does not facilitate reuse of class |
| ☐ | Does not support generic methods |
| ☐ | Does not support static members |

Which of the following is/are key component(s) of Standard Template Library?

Select the correct option

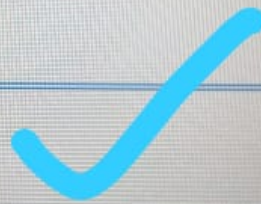
- | | |
|-----------------------|--------------------------|
| ☐ | Containers |
| ☐ | Iterators |
| ☐ | Algorithms |
| ☐ | All of the given options |
- 

STL stands for:

02:04:33 PM, 31 August 2020)

Select the correct option

- | | |
|-----------------------|-----------------------------|
| ☐ | Stack Template Library |
| ☐ | Standard Template Library |
| ☐ | Structured Template Library |
| ☐ | Secure Template Library |



Question # 9 of 10 (Start time: 01:06:05 AM, 31 August 2020)

Each _____ of a template class by default becomes function template.

▶ Select the correct option

☐	Data member
☒	Member function
☐	Type parameter
☐	Object

Question # 10 of 10 (Start time: 01:07:29 AM, 31 August 2020)

When we specialize a function template, it is called _____



Select the correct option

- | | |
|-----------------------|-------------------------------|
| ☐ | function overloading |
| ☐ | function overriding |
| ☐ | function template overloading |
| ☐ | function template overriding |
- 

Question # 4 of 10 (Start time: 01:46:42 PM, 31 August 2020)

Which of the following types of containers is NOT provided by Standard Template Library?

Select the correct option

☐ Sequence Containers

☒ Structured Containers

☐ Associative Containers

☐ Container Adapters

Question # 8 of 10 (Start time: 01:05:15 AM, 31 August 2020)

Which of the following is NOT an Associative Container?

Select the correct option

☐	set
☐	multiset
☐	map
☒	group

CS304:Quiz No. 2

Question # 4 of 10 (**Start time: 01:00:53 AM, 31 August 2020**)

Which of the following is a Sequence Container?

Select the correct option

☐	map
☒	vector
☐	set
☐	multiset

CS304:Quiz No. 2

Question # 7 of 10 (**Start time: 01:04:37 AM, 31 August 2020**)

Which of the following may inherit from an ordinary class?

Select the correct option

- | | |
|----------------------------------|--------------------------|
| ☐ | Class template |
| ☐ | Partial specialization |
| ☐ | Complete specialization |
| ☒ | All of the given options |

Question # 6 of 10 (**Start time: 01:03:38 AM, 31 August 2020**)

_____ may inherit from a complete specialization.

Select the correct option

- | | |
|----------------------------------|--------------------------|
| ☐ | Partial specialization |
| ☐ | Complete specialization |
| ☐ | Ordinary class |
| ☒ | All of the given options |

Question # 3 of 10 (Start time: 12:58:10 AM, 31 August 2020)

Total Marks: 1

Which of the following is correct way to define a template class X?

Select the correct option

- | | | |
|----------------------------------|---|--|
| ☐ | <code>class< typename T > class X {};</code> | |
| ☐ | <code>template class X {};</code> | |
| ☐ | <code>template< typename T > class X {};</code> | |
| ☒ | <code>typename <class T > class X {};</code> | |

▶ Saving and loading next question...

Question # 8 of 10 (Start time: 01:06:46 PM, 31 August 2020)

Total Marks: 1

Which of the following is the correct syntax for passing two type arguments to a template?

Select the correct option

☐`template< type T, type U >`☒`template< typename T, typename U >`☐`template< type T, U >`☐`template Typename< T, U >`

CS304: Quiz No. 2

Question # 5 of 10 (Start time: 01:02:04 AM, 31 August 2020)

In resolution order of function templates, compiler searches for _____ in the end.

▶ Select the correct option



complete specialization



partial specialization



ordinary function



generic template

Option D

CS304:Quiz No. 2

Question # 2 of 10 (Start time: 12:57:36 AM, 31 August 2020)

The parameters given in template definition other than those used for mentioning templates types are called _____.

Select the correct option

- | | |
|----------------------------------|---------------------------|
| ☐ | Type Parameters |
| ☒ | Non-Type Parameters |
| ☐ | Default Type Parameters |
| ☐ | None of the given options |

In statement "template <class T, class U, int I = 5>" ,the non-type parameter is _____.

Select the correct option

☐

class T

☐

class U

☒

int I

☐

All of the given options

Which of the following statement is true about partial specialization?

Select the correct option

- ☐ Function templates cannot have partial specialization.
- ☐ Class templates cannot have partial specialization.
- ☒ Both Class templates and function templates can have partial specialization. 
- ☐ None of the given options

Question # 3 of 10 (Start time: 12:58:10 AM, 31 August 2020)

Which of the following is correct way to define a template class X?

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | <code>class< typename T > class X { };</code> |
| ☐ | <code>template class X { };</code> |
| ☐ | <code>template< typename T > class X { };</code> |
| ☒ | <code>typename <class T > class X { };</code> |

Which of the following represents partial specialization?

Select the correct option



template< int , char >



template< class T, type T>



template< class T, class U, int >



template< typename T, class W >

When we declare a template function as friend of any class, then which of the following specialization(s) of template function also becomes friend of the class granting friendship?

Select the correct option



Explicit



Implicit



Partial



All of the given options



Question # 5 of 10 (Start time: 12:34:33 PM, 31 August 2020)

STL stands for:

Select the correct option

- | | |
|----------------------------------|-----------------------------|
| ☐ | Stack Template Library |
| ☒ | Standard Template Library |
| ☐ | Structured Template Library |
| ☐ | Secure Template Library |

Which of the following represents partial specialization?

Select the correct option

- ☐ `template< int , char >`
- ☐ `template< class T, type T>`
- ☒ `template< class T, class U, int >`
- ☐ `template< typename T, class W >`

Question # 7 of 10 (Start time: 12:35:40 PM, 31 August 2020)

_____ class is a single class that provides functionality to operate on different types of data.

Select the correct option

☐

Friend

☒

Template

☐

Ordinary

☐

None of the given options

When we declare a template function as friend of any class, then w
friendship?

Select the correct option

☐	Explicit
☐	Implicit
☐	Partial
☐	All of the given options



Question # 3 of 10 (Start time: 12:32:50 PM, 31 August 2020)

We can change behavior of a template using _____

Select the correct option

☒ Template Parameters

☐ Function Templates

☐ Class Templates

☐ None of the given options

BC190200086: MUHAMMAD IMRAN KHAN

CS304: Quiz No. 2

Question # 1 of 10 (Start time: 12:31:11 PM, 31 August 2020)

Which of the following may inherit from an ordinary class?

Select the correct option

- | | |
|-----------------------|--------------------------|
| ☐ | Class template |
| ☐ | Partial specialization |
| ☐ | Complete specialization |
| ☐ | All of the given options |
- 

CS304:Quiz No. 2

Question # 8 of 10 (**Start time: 12:19:48 PM. 31 August 2020**)

Which of the following can be passed as type arguments to templates?

Select the correct option

☐	Primitive types
☐	User-defined types
☐	Both Primitive types and User-defined types
☐	None of the given options



Question # 2 of 10 (Start time: 12:32:11 PM, 31 August 2020)

Which of the following is known as Dereference operator in C++?

Select the correct option

☐ +

☒ *

☐ &

☐ ..

Question # 3 of 10 (Start time: 12:10:48 PM, 31 August 2020)

_____ class is a single class that provides functionality to operate on different types of data.

Select the correct option

☐	Friend
☒	Template
☐	Ordinary
☐	None of the given options

Question # 10 of 10 (Start time: 12:23:26 PM, 31 August 2020)

A _____ is a pointer which is declared outside the _____.

Select the correct option

☐	Container, Cursor
☒	Cursor, Container
☐	Container, Class
☐	This, Cursor

CS304: Quiz No. 2

Question # 9 of 10 (Start time: 12:21:25 PM, 31 August 2020)

Which of the following are collectively referred to as the first-class containers?

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | Associative Containers and Container Adapters |
| ☒ | Sequence Containers and Associative Containers |
| ☐ | Sequence Containers and Container Adapters |
| ☐ | None of the given options |

Question # 6 of 10 (**Start time: 01:33:38 PM, 31 August 2020**)

Which of the following is not a Sequence Container?

Select the correct option

☐	vector
☐	deque
☒	map
☐	list

🔍 Type here to search



Question # 9 of 10 (Start time: 01:36:29 PM, 31 August 2020)

When we want to have exactly identical operations on different data types, _____ are used.

Select the correct option

- | | |
|----------------------------------|---------------------------|
| ☒ | Function Templates |
| ☐ | Function Overloading |
| ☐ | Function Overriding |
| ☐ | None of the given options |

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ALL answers are verified if found any mistake then Correct ACCORDINGLY

1. Function func1 of class1 is _____ in class2.
 - ▶ Public
 - ▶ **Protected**
 - ▶ Private
 - ▶ None of the given
2. The default inheritance mode is,
 - a) Public inheritance
 - b) Protected inheritance
 - c) Private inheritance**
 - d) None of the given
3. A template provides a convenient way to make a family of
 - a) Variables and data members
 - b) Functions and classes**
 - c) Classes and exceptions
 - d) Programs and algorithms
4. Then int member i of base class is accessible in class,
 - a) derived 1 only
 - b) derived 2 only**
 - c) both derived 1 derived 2
 - d) none of the given options
5. In case of public inheritance, protected members of base class will be _____ in derived class?
 - a) Private
 - b) Public
 - c) Protected**
 - d) hidden
6. By default assignment operator (=) performs,
 - a) Shallow copy**
 - b) Deep copy
 - c) Both of these
 - d) None of these
7. In protected inheritance the public members of base class become _____ in derived class.
 - a) Public
 - b) Private
 - c) Protected**
 - d) None of the given
8. We can access public members of the class from outside the class using _____ operator with its object pointer
 - a) ->

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- b) .
 - c) &
 - d) @
9. Compiler can automatically generates the following constructor/s for a class,
- a) Default constructor
 - b) Copy constructor
 - c) Both of these
 - d) None of these
10. Friend's functions of class are _____members of that class.
- a) Public
 - b) Private
 - c) Protected
 - d) None of the given
11. Methodologies to the development of reusable software relate to_____.
- a) Structure programming
 - b) Procedural programming
 - c) Generic programming
 - d) None of the given
12. Function overriding is done in context of,
- a) Single class
 - b) Single derived class
 - c) Single base class
 - d) Derived and base classes
13. Which of the following is not type of inheritance in c++?
- a) Public
 - b) Protected
 - c) Restricted
 - d) Private
14. If there is a pointer ,p, to objects of a base class and it contains the address of an object of a derived class and both classes contain a virtual member function, ding(),then the statement p->ding(); will cause the version of ding in the_____class to be executed.
- a) Base
 - b) Derived
 - c) Virtual
 - d) Implemented
15. Target class of a_____function call is determined t run time.
- a) Instance
 - b) Virtual
 - c) Operator
 - d) None of the given
16. In c++ dynamic binding and polymorphism will be achieved when the member function will be_____

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- a) Private
 - b) Public
 - c) Virtual
 - d) Inline
17. To make a pure virtual, we need to give ____ after() of this function.
- a) =0;
 - b) =1;
 - c) Null;
 - d) None of the given
18. If we have not given any constructor for the class, compiler generates which of the following constructors?
- a) Explicit Default Constructor
 - b) Implicit Default Constructor
 - c) Explicit Parameterized Constructor
 - d) Implicit Parameterized Constructor
19. In c++ by default access of classes is ____.
- a) Private
 - b) Protected
 - c) Public
 - d) None of the given
20. In public inheritance the public members of base class become ____ in derived class
- a) Private
 - b) Protected
 - c) Public
 - d) None of the given
21. A class hierarchy
- a) Shows the same relationships as a organization chart
 - b) Describe "has a" relationship.
 - c) Describe "is a kind of" relationship.
 - d) Shows the same relationships as a family tree.
22. In private Inheritance the public members of the base class become ____ in derived class
- a) Public
 - b) private
 - c) Protected
 - d) None of the given option
23. In the line No.2, which constructor of student class will be
- a) Default constructor of student class
 - b) Copy constructor of student class
 - c) Both Default and Copy constructor of student class
 - d) No constructor will be called
24. Which of the following is Not casting operator in c++ standard?
- a) Static-cast

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- b) **Var-cast**
 - c) Dynamic-cast
 - d) Reinterpret-cast
25. Which of the following type of inheritance is used to model “Implemented in terms of” relationship?
- a) Public
 - b) **Private**
 - c) Protected
 - d) Confidential
26. When we want to implement one class in terms of another class then we use, **a)**
- Public inheritance**
- b) Protected inheritance
 - c) Private inheritance
 - d) None of these options
27. Public inheritance represents;
- a) “IS A “relationship**
 - b) “Has A” relationship
 - c) “IS Special kind of” relationship
 - d) None of these option
28. Suppose we have defined derived class copy constructor but have not defined base class copy constructor then compiler will,
- a) Use base class default constructor
 - b) **Generate base class copy constructor itself**
 - c) Use base class assignment operator
 - d) Nine of these option is correct
29. Child class can call constructor of its, **a) Direct base class**
- b) Indirect base class
 - c) Both direct and indirect base classes
 - d) None of these
30. We can have _____ type of member functions in a class.
- a) Public
 - b) Private
 - c) Protected
 - d) **All of these options**
31. Compiler generated copy constructor performs _____
- a) Shallow copy**
 - b) Deep copy
 - c) Both Shallow and Deep copy
 - d) None of these options
32. In c++ generic programming is done using _____
- a) Procedures

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- b) Packages
 - c) **Templates**
 - d) None of the given
33. Target class of a ____ function call is determined at run time.
- a) Instance
 - b) **Virtual**
 - c) Operator
 - d) None of the given
34. Suppose person is a user defined class. In statement "Person*pPtr", static type of pPtr is ----.
- a) pPtr
 - b) **pointer**
 - c) person
 - d) None of the given
35. Which statement is true for concrete class?
- a) It cannot be inherit from an abstract class
 - b) **It can be instantiated**
 - c) It cannot be instantiated
 - d) None of given
36. In c++, compiler can generate which of the following operators 'code,
- a) **==**
 - b) =
 - c) &
 - d) &&
37. In specialization we can,
- a) Replace child class with its base class
 - b) **Replace base class with its child class**
 - c) Replace both child and base classes interchangeably
 - d) None of the given option
38. In private inheritance derived class pointer can be assigned to base class pointer in,
- a) **Main function**
 - b) In derived class member and friend class function
 - c) In base class member and friend functions
 - d) None of the given option
39. When we access private members of the class from outside the class using ____ operator with its object pointer
- a. **->**
 - b. .
 - c. &
 - d. None of above
40. Which of the following access specifier ensure that base class member is accessible in derived class and Not outside of this class
- a. Public

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- b. Private
 - c. Protected
 - d. All of above
41. Static casting is
- a. C++ way of calling base class function from derived class
 - b. C way of calling base class function from derived class
 - c. Both of these
 - d. None of these
42. Suppose both derive and base classes have compilers generated copy constructors then derived class copy construct is
- a. Compiler generated default constructor
 - b. Compiler generated copy constructor
 - c. Both of these options are correct
 - d. None of these operators are correct
43. In case of protected inheritance, public members of the base class will be _____ in derived class
- a. Private
 - b. Public
 - c. Protected
 - d. Hidden
44. Methodologies to the development of reusable software related to _____
- a. Structured programming
 - b. Procedural programming
 - c. Generic programming
 - d. None of the given
45. A child class can call constructor of the parent call through
- a. Its constructor initialization list
 - b. Its constructor body
 - c. Both from its constructor initialization list or body
 - d. Cannot call the constructor of its parent class
46. In case of dynamic memory allocation in or class we should use
- a. Default assignment operator
 - b. User defined assignment operator
 - c. Default copy constructor
 - d. None of these
47. Consider the code below.
- ```
Class class1 {
Protected:
Int l;
};
Class class2: private calls 1{
};
```
- a) Public

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- b) Protected
- c) Private
- d) None of the given options

48. Consider the following two lines of code written for class student

- 1. Student subj1;
- 2. Student subj2(subj1)

In line 2 which constructor of student class will be called?

- a) Default constructor of student class
- b) Copy constructor of student class
- c) Both default and copy constructor of student class
- d) No constructor will be called

49. Which of the following is /are casting operator(s) inc++?

- a. Static\_cast
- b. Const\_cast
- c. Dynamic\_cast
- d. All of the given options

50. In case of private inheritance, private members of base class will be \_\_\_\_\_ in derived class?

- a. Private
- b. Public
- c. Protected
- d. Hidden

51. Consider the code below

```
class class1 {
 Private:
 Void func1();
};
Class class2 : public class1 {
};
```

Function func1 of class 1 is \_\_\_\_\_ in class2.

- a. Public
- b. Protected
- c. Private
- d. None of the given options

52. We can call base class assignment operator in derived class user defined assignment operator

- a. Implicitly
- b. Explicitly
- c. Using both of these options
- d. Using none of these options

53. A function call is resolved at run time in

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a. Non virtual members function **b. Virtual member function**

- c. Both none-virtual and virtual member function
- d. None of given

54. If the user does not specify the type of inheritance, then the default type of inheritance is \_\_\_\_\_

a. Public inheritance **b. Private inheritance**

- c. Protected inheritance
- d. None of given

55. Consider the code below

```
class class1 {
public:
Void func1();
};
Class class2: protected class1{
};
```

Function func1 of class 1 is \_\_\_\_\_ in class2.

- a. Public
- b. **Protected**
- c. Private
- d. None of the given options

56. Which of the following causes run time binding?

- ▶ Declaring object of abstract class
- ▶ Declaring pointer of abstract class
- ▶ **Declaring overridden methods as non-virtual** ▶ (Page 226)
- None of the given

57. Which of the following is the best approach if it is required to have more than one functions having exactly same functionality and implemented on different data types?

- ▶ **Templates** (Page 256)
- ▶ Overloading
- ▶ Data hiding
- ▶ Encapsulation

58. A copy constructor is invoked when

- ▶ a function do not returns by value.
- ▶ **an argument is passed by value.** (Page 78)
- ▶ a function returns by reference.
- ▶ an argument is passed by reference.

59. Like template functions, a class template may not handle all the types successfully.

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- ▶ True (Page 258)
- ▶ False

60. A class template may inherit from another class template.

- ▶ True (Page 288)
- ▶ False

61. By default the vector data items are initialized to \_\_\_\_

- ▶ 0 google
- ▶ 0.0
- ▶ 1
- ▶ null

62. In Private ----- only member functions and friend classes or functions of a derived class can convert pointer or reference of derived object to that of parent object

- ▶ specialization
- ▶ inheritance (Page 216)
- ▶ abstraction
- ▶ composition

63. Which of the following is/are advantage[s] of generic programming?

- ▶ Reusability
- ▶ Writability
- ▶ Maintainability
- ▶ All of given (Page 256)

64. Template functions use \_\_\_\_\_ than ordinary functions.

- ▶ Greater Memory
- ▶ Lesser Memory
- ▶ Equal Memory
- ▶ None of the given options

65. Non Template Friend functions of a class are friends of \_\_\_\_\_ instance/s of that class.

- ▶ All google
- ▶ One specific
- ▶ All instances of one date type
- ▶ None of the given options

66. A pointer to a base class can point to objects of a derived class.

- ▶ True google
- ▶ False

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67. A template argument is preceded by the keyword \_\_\_\_\_.  
▶ vector  
▶ **class**  
▶ template  
▶ type\*
68. Which one of the following terms must relate to polymorphism?  
▶ Static allocation  
▶ Static typing  
▶ **Dynamic binding** (Page 239)  
▶ Dynamic allocation
69. Multiple inheritance can be of type  
▶ Public  
▶ Private  
▶ Protected  
▶ **All of the given**
70. Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access  
▶ **public members of Derv**  
▶ protected members of Derv.  
▶ private members of Derv.  
▶ protected members of Base.
71. A function call is resolved at run-time in \_\_\_\_\_.  
▶ non-virtual member function.  
▶ **virtual member function.** (Page 239)  
▶ Both non-virtual member and virtual member function.  
▶ None of given
72. Two important STL associative containers are \_\_\_\_\_ and \_\_\_\_\_.  
▶ **set,map**  
▶ sequence,mapping  
▶ setmet,multipule  
▶ sit,mat
73. An abstract class is useful when,  
▶ We do not derive any class from it.  
▶ There are multiple paths from one derived class to another.  
▶ **We do not want to instantiate its object**  
▶ You want to defer the declaration of the class.

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74. Suppose you create an uninitialized vector as follows:

```
vector<int> evec;
```

After adding the statment,

```
evec.push_back(21);
```

what will happen?

- ▶ The following statement will add an element to the start (the back) of evec and will initialize it with the value 21.
- ▶ The following statement will add an element to the center of evec and will reinitialize it with the value

21.

- ▶ The following statement will delete an element to the end (the back) of evec and will reinitialize it with the value 21.
- ▶ The following statement will add an element to the end (the back) of evec and initialize it with the value 21.

75. Default constructor is such constructor which either has no -----or if it has some parameters these have -----

--- values

- ▶ Parameter, temporary
- ▶ Null, Parameter
- ▶ Parameter, default (Page 75)
- ▶ non of the given

76. Classes like TwoDimensionalShape and ThreeDimensionalShape would normally be concrete, while classes like Sphere and Cube would normally be abstract.

- ▶ True
- ▶ False

77. In order to define a class template, the first line of definition must be:

- ▶ `template <typename T>` (Page 257)
- ▶ `typename <template T>`
- ▶ `Template Class <ClassName>`
- ▶ `Class <Template T>`

78. A function template can not be overloaded by another function template.

- ▶ True

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- ▶ False

79. Identify the correct way of declaring an object of user defined template class A for char type members?

- ▶ `A< char > obj;`
- ▶ `<char>A obj;`
- ▶ `A obj<char>;`
- ▶ `Obj <char> A;`

80. The user must define the operation of the copy constructor.

- ▶ True
- ▶ False

81. The find() algorithm

- ▶ finds matching sequences of elements in two containers.
- ▶ finds a container that matches a specified container.
- ▶ takes iterators as its first two arguments. (Object-Oriented Programming in C++)
- ▶ takes container elements as its first two arguments.

82. Compiler performs \_\_\_\_\_ type checking to diagnose type errors,

- ▶ Static (Page 261)
- ▶ Dynamic
- ▶ Bound
- ▶ Unbound

83. Vectors contain contiguous elements stored as a[an] \_\_\_\_.

- ▶ variable
- ▶ array (Page 306)
- ▶ function
- ▶ datatype

84. In a de-queue, (chose the best option)

- ▶ data can be quickly inserted or deleted at any arbitrary location.
- ▶ data can be inserted or deleted at any arbitrary location, but the process is relatively slow.
- ▶ data can not be quickly inserted or deleted at either end.
- ▶ data can be inserted or deleted at either end, but the process is relatively slow.

85. Algorithms can only be implemented using STL containers.



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- ▶ True
- ▶ False(Object-Oriented Programming in C++)

86. What is a class?

- ▶ A class is a section of computer memory containing objects.
- ▶ A class is a section of the hard disk reserved for object oriented programs
- ▶ A class is the part of an object that contains the variables.
- ▶ A class is a description of a kind of object. [Click here for detail](#)

87. Inheritance is a way to

- ▶ organize data.
- ▶ pass arguments to objects of classes.
- ▶ add features to existing classes without rewriting them. (Page 27)
- ▶ improve data-hiding and encapsulation.

88. We can use "this" pointer in the constructor in the body and even in the initialization list of any class if we are careful,

- ▶ True
- ▶ False

89. \_\_\_\_\_ and \_\_\_\_\_ methods may not be declared abstract.

- ▶ private,static[Click here for detail](#)
- ▶ private,public
- ▶ static,public
- ▶ none of given

90. ----- members are somewhere between public and private members. They are used in inheritance

- ▶ protected
- ▶ public
- ▶ private
- ▶ global (Page 187)

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91. Which of these are examples of error handling techniques?

- ▶ Abnormal Termination
- ▶ Graceful Termination
- ▶ Return the illegal
- ▶ all of the given (Page 329)

92. ----- follow try block to catch the object thrown

- ▶ catch block (Page 333)
  - ▶ throw block
  - ▶ main block
  - ▶ non of the given

93. Graphical representation of the classes and objects is called object model it shows -----

- ▶ Class Name only
- ▶ Class Name and attributes
- ▶ Relationships of the objects and classes
- ▶ all of the given

94. Destructor can be overloaded

- ▶ True
- ▶ False(Page 92)

95. Which of the following is the best approach to implement generic algorithms with minimum number of coding lines?

- ▶ **Templates** (Page 256)
- ▶ Overloading
- ▶ Overriding
- ▶ Friend function/class

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96. Each try block can have \_\_\_\_\_ no. of catch blocks.

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ As many as necessary.

97. class DocElement

```
{
public:
 virtual void Print() { cout << "Generic element"; }
};
class Heading : public DocElement
{
public:
 void Print() { cout << "Heading element"; }
};
class Paragraph : public DocElement
{
public:
 void Print() { cout << "Paragraph element"; }
};
void main()
{
 DocElement * p = new Paragraph();
```

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```
p->Print();
}
```

98. When you run this program, it will print out a single line to the console output.

What will be in that line?

Select one correct answer from the following list:

- ▶ Generic element ▶
- Heading element
- ▶ Paragraph element
- ▶ Nothing will be printed.

99. Suppose we have two derived classes from a single class, can we write a method with same name in both these derived classes? Choose the best option.

- ▶ No
- ▶ Only if the two classes have the same name (Page 204)
- ▶ Only if the main program does not declare both kinds
- ▶ Yes

100. When a virtual function is called by referencing a specific object by name and using the dot member selection operator (e.g., squareObject.draw()), the reference is resolved at compile time.

- ▶ True
- ▶ False

101. Considering the resolution order in which compiler search for functions in a program; the first priority is given to;

- ▶ general template
- ▶ partial specialization
- ▶ complete specialization
- ▶ ordinary function (Page 287)

102. One purpose of an iterator in the STL is to connect algorithms and containers.

- ▶ True
- ▶ False

103. In \_\_\_\_\_, a base class can be replaced by its derived class,

- ▶ Sub-typing (Page 31)
- ▶ Super-typing
- ▶ Multiple-typing
- ▶ Restricted-typing

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104. one this pointer does not point to current object of any class,

- ▶ True
- ▶ False

105. Which of the following operator(s) take(s) one or no argument if overloaded?

- ▶ ++(Page 162)
- ▶ -
- ▶ +
- ▶ All of the given

106. Which of the following operators can not be overloaded?

- ▶ Scope resolution operator ( :: ) (Page 141)
- ▶ Insertion operator ( << )
- ▶ Extraction operator ( >> )
- ▶ The relation operator ( > )

107. Which of these are examples of error handling techniques ?

- ▶ Abnormal Termination
- ▶ Graceful Termination
- ▶ Return the illegal
- ▶ all of the given (Page 329)

108. \_\_\_\_\_ “is a” relationship

- ▶ Inheritance (Page 25)

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- ▶ Polymorphism
- ▶ abstraction
- ▶ encapsulation

109. Virtual functions allow you to

- ▶ create an array of type pointer-to-base class that can hold pointers to derived classes.
- ▶ create functions that can never be accessed.
- ▶ group objects of different classes so they can all be accessed by the same function code.
- ▶ use the same function call to execute member functions of objects from different classes

110. The copy() algorithm returns an iterator to

- ▶ the last element copied from.
- ▶ the last element copied to.
- ▶ the element one past the last element copied from.
- ▶ the element one past the last element copied to.

111. If you define a vector v with the default constructor, and define another vector w with a one-argument constructor to a size of 11, and insert 3 elements into each of these vectors with push\_back(), then the size() member function will return \_\_\_\_\_ for v and \_\_\_\_\_ for w.

- ▶ 11 for v and 3 for w.
- ▶ 0 for v and 0 for w.
- ▶ 0 for v and 3 for w.
- ▶ 3 for v and 11 for w.

112. Which is not the Advantage of inheritance?

- ▶ providing class growth through natural selection. (Object-Oriented Programming in C++)
- ▶ facilitating class libraries.
- ▶ avoiding the rewriting of code.
- ▶ providing a useful conceptual framework.

113. Which type of inheritance is being represented by the following statement, class X : public A, public B { ... ... };

- ▶ Single inheritance
- ▶ Multiple inheritance (Page 41)

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- ▶ Double inheritance
- ▶ None of the given options

114. When we write a class template the first line must be:

- ▶ `template < class class_name>`
- ▶ `template < class data_type>`
- ▶ `template < class T >` (Page 257)

115. Function templates should be used where code and behavior must be identical.

- ▶ True (Page 262)
- ▶ False

116. The specialization pattern `<T*>` after the name says that this specialization is to be used for every,

- ▶ data type
- ▶ meta type
- ▶ virtual type
- ▶ pointer type (Page 286)

117. A range is often supplied to an algorithm by two \_\_\_\_\_ values.

- ▶ italic
- ▶ iteration (Object-Oriented Programming in C++)
- ▶ iterator
- ▶ None of given

118. Which of the following is an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ Unique identity
- ▶ All of the given (Page 12)

119. Consider the following statement

Cupboard has books

What is the relationship between Cupboard and books?

- ▶ Composition
- ▶ Aggregation
- ▶ Inheritance
- ▶ None of the given options

120. Which sentence clearly defines an object?

- ▶ one instance of a class. (Page 23)

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- ▶ another word for a class.
- ▶ a class with static methods.
- ▶ a method that accesses class attributes.

121. \_\_\_\_\_, which means if A declares B as its friend it does NOT mean that A can access private data of B. It only means that B can access all data of A.

- ▶ Friendship is one way only
- ▶ Friendship is two way only
- ▶ NO Friendship between classes
- ▶ Any kind of friendship

122. The statement `objA=objB;` will cause a compiler error if the objects are of different classes.

- ▶ True
- ▶ False

123. Consider the call given below of an overloaded operator "+", *Rational\_number\_1* + *Rational\_number\_2*

Where *Rational\_number\_1* and *Rational\_number\_2* are the two objects of *Rational\_number* class (a user defined class). Identify which of the above two objects will be passed as an argument to the overloaded operator function?

- ▶ *Rational\_number\_1*
- ▶ *Rational\_number\_2*
- ▶ Both *Rational\_number\_1* & *Rational\_number\_2*
- ▶ any of the two objects, randomly

124. If a class D has been derived using protected inheritance from class B (If B is a protected base and D is derived class) then public and protected members of B ----- accessed by member functions and friends of class D and classes derived from D

- ▶ can be
- ▶ cannot be
- ▶ does restrict to be
- ▶ not given



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The type that is used to declare a reference or pointer is called its \_\_\_\_\_

- ▶ abstract type
- ▶ reference type
- ▶ ▶ static type
- ▶ defaulttype

125. How the information hidden within an object can be accessed?

- ▶ Through its interface
- ▶ Through its private data members
- ▶ ▶ Through its private member functions
- ▶ Through both public and private members

126. The sub-object's life is not dependant on the life of master class in \_\_\_\_\_.

- ▶ Separation
- ▶ Composition
- ▶ None of the given

127. Encapsulation means

Select correct option:

- ▶ Extending the behaviour of class in another class
- ▶ ▶ Data and behaviour are tightly coupled within an entity (Page 16)
- ▶ One entity takes all the attributes and operations of the other
- ▶ Taking out the common features and put those in a separate class

128. An STL container can not be used to,

- ▶ hold objects of class employee.
- ▶ store elements in a way that makes them quickly accessible.
- ▶ ▶ compile c++ programs. (Object-Oriented Programming in C++)
- ▶ organize the way objects are stored in memory

129. Which of the following may not be an integral part of an object?

- ▶ State
- ▶ Behavior
- ▶ ▶ Protected data members (Page 12)
- ▶ All of given

130. Public methods of base class can ----- be accessed in its derived class

- ▶ ▶ directly (Page 179) rep
- ▶ inderectly
- ▶ simultaneously
- ▶ non of the given

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131. What is true about function templates?

- ▶ The compiler generates only one copy of the function template
- ▶ The compiler generates a copy of function respective to each type of data (Page 256)
- ▶ The compiler can only generate copy for the int type data
- ▶ None of the given.

132. When the base class and the derived class have a member function with the same name, you must be more specific which function you want to call (using \_\_\_\_\_).

- ▶ scope resolution operator
- ▶ dot operator
- ▶ null operator
- ▶ Operator overloading

133. A template provides a convenient way to make a family of

- ▶ variables and data members
- ▶ functions and classes (Object-Oriented Programming in C++)
- ▶ classes and exceptions
- ▶ programs and algorithms

134. A non-virtual member function is defined in a base class and overridden in a derived class; if that function is called through a base-class pointer to a derived class object, the derived-class version is used.

- ▶ True
- ▶ False

135. If there is a pointer p to objects of a base class, and it contains the address of an object of a derived class, and both classes contain a nonvirtual member function, ding(), then the statement p->ding(); will cause the version of ding() in the \_\_\_\_\_ class to be executed.

- ▶ Base (Object-Oriented Programming in C++)
- ▶ Derived
- ▶ Abstract
- ▶ virtual

136. In case of multiple inheritance a derived class inherits,

- ▶ Only the public member functions of its base classes
- ▶ Only the public data members of its base classes
- ▶ Both public data members and member functions of all its base classes
- ▶ Data members and member functions of any two base classes

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137. Which of the following is incorrect line regarding function template?

- ▶ `template<class T>`
- ▶ `template <typename U>`
- ▶ `Class<template T>` (Page 257)
- ▶ `template < class T, class U>`

138. Consider a class named Vehicle, which of the following can be the instance of class Vehicle?

- i. Car
  - ii. Computer
  - iii. Desk
  - iv. Ahmed
  - v. Bicycle
  - vi. Truck
- 1,4,5
  - 2,5,6
  - 1,2,3,6
  - 1, 5, 6 Conceptual

139. Consider the code below,

```
class Fred {
public:
Fred();
...
};
int main()
{
Fred a[10];
Fred* p = new Fred[10];
...
}
```

Select the best option,

- ▶ `Fred a[10];` calls the default constructor 09 times `Fred* p = new Fred[10];` calls the default constructor 10 times
- ▶ Produce an error

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- ▶ Fred a[10]; calls the default constructor 11 times

Fred\* p = new Fred[10]; calls the default constructor 11 times

- ▶ Fred a[10]; calls the default constructor 10 times

Fred\* p = new Fred[10]; calls the default constructor 10 times

140. When a variable is defined as static in a class then all objects of this class,

- ▶ Have different copies of this variable
- ▶ Have same copy of this variable (Page 110)
- ▶ Can not access this variable
- ▶ None of given

141. Which of the following operators always takes no argument if overloaded?

- ▶ /
- ▶ -
- ▶ +
- ▶ ++ (Page 162)

142. It is sometimes useful to specify a class from which no objects will ever be created.

- ▶ True(Object-Oriented Programming in C++)
- ▶ False

143. When a subclass specifies an alternative definition for an attribute or method of its superclass, it is \_\_\_\_\_ the definition in the superclass.

- ▶ overload
- ▶ overriding (Page 34)
- ▶ copy riding
- ▶ none of given

144. The mechanism of selecting function at run time according to the nature of calling object is called,

- ▶ late binding
- ▶ static binding
- ▶ virtual binding
- ▶ None of the given options (Page 227)

145. Dynamic binding means that target function for a call is selected at run time

Which one of the following functions returns the total number of elements in a vector.

- ▶ length();
- ▶ size(); (Page 318)
- ▶ ele();
- ▶ veclen();

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146. The main function of scope resolution operator (::) is,

- ▶ To define an object
- ▶ To define a data member
- ▶ [To link the definition of an identifier to its declaration](#)
- ▶ To make a class private

Google

147. When is a constructor called?

- ▶ Each time the constructor identifier is used in a program statement
- ▶ [During the instantiation of a new object \(Object-Oriented Programming in C++\)](#)
- ▶ During the construction of a new class
- ▶ At the beginning of any program execution

148. Associativity can be changed in operator overloading.

- ▶ True
- ▶ [False \(Page 141\)](#)

149. A normal C++ operator that acts in special ways on newly defined data types is said to be

- ▶ glorified.
- ▶ encapsulated.
- ▶ classified.
- ▶ [overloaded.](#)

150. Which operator can not be overloaded?

- ▶ The relation operator ( >= )
- ▶ Assignment operator ( = )
- ▶ Script operator ( [] )
- ▶ [Conditional operator \( ? : \) \(Page 141\)](#)

151. Suppose obj1 and obj2 are two objects of a user defined class A. An + operator is overloaded to add obj1 and obj2 using the function call obj1+obj2.

Identify the correct function prototype against the given call?

- ▶ A operator + ( A &obj );
- ▶ [int + operator\(\); \(Page 143\)](#)
- ▶ int operator (plus) ();
- ▶ A operator(A &obj3);

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152. Suppose that the Test class does not have an overloaded assignment operator. What happens when an assignment  $a=b$ ; is given for two Test objects a and b?

- ▶ The automatic assignment operator is used
- ▶ **The copy constructor is used**
- ▶ Compiler error
- ▶ Run-time error

153. A static member function can be called, even when a class is not \_\_\_\_\_.

- ▶ Declared
- ▶ Define
- ▶ **Instantiated**
- ▶ Called

154. Identify which of the following overloaded operator function's declaration is appropriate for the given call?

**Rational\_number\_1 + 2.325**

Where Rational\_number\_1 is an object of user defined class Rational\_number.

- ▶ Rational\_number operator+( Rational\_number & obj);
- ▶ Rational\_number operator+(double& obj);
- ▶ Rational\_number operator+(Rational\_number &obj, double& num);
- ▶ operator+(double& obj);

155. What problem(s) may occur when we copy objects without using deep copy constructor?

- ▶ Dangling pointer
- ▶ Memory Leakage
- ▶ **All of the given**
- ▶ System crash

156. \_\_\_\_\_ provide the facility to access the data member.

- ▶ **accesser function**
- ▶ private function
- ▶ inline function
- ▶ None of the given

157. Constant objects cannot change their state,

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▶ **True**

▶ False

158. The \_\_\_\_\_ relationship indicates that an object contains other objects.

▶ None of given

▶ **'has-a'**

▶ 'is-a'

▶ 'be-

159. Which one of the following features of OOP is used to derive a class from another?

▶ Encapsulation

▶ Polymorphism

▶ Data hiding

▶ **Inheritance**

160. \_\_\_\_\_ satisfy the condition of polymorphism

▶ Carbon

▶ Diamond

▶ Coal

▶ **all of the given**

161. A generalization-specialization relation between classes are implemented using

▶ data hiding

▶ friend classes

▶ encapsulation

▶ **inheritance**

162. The  $\geq$  operator can be overloaded.

▶ **True**

▶ False

163. In order to free the memory occupied by the object, we use -----

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- ▶ Constructor
- ▶ **Destructor**
- ▶ Shallow Copy
- ▶ Deep Copy

164. Which of the following is not an example of multiple inheritances?

- ▶ Mermaid
- ▶ **Woman**
- ▶ None of the given
- ▶ Amphibious Vehicle

165. Static variable can be initialized more than once.

- ▶ True
- ▶ **False**

166. A generic class showing all the common attributes and a behavior of other classes represents a very important feature in oop called -----

- ▶ **Inheritance**
- ▶ Encapsulation
- ▶ Polymorphism
- ▶ Abstraction

167. The parameters given in template definition other than those used for mentioning templates types are called \_\_\_\_

- a. Type Parameters
- b. **Non Type parameters**
- c. Default Type Parameters
- d. None of the given option

168. Which of the following is the correct way to define a template class X?

- e. Class < typename T> class X {}
- f. Template class C {}
- g. **Template < typename T> class X {}**
- h. Typename < class T > class X {}

169. In resolution order of function template, compiler searches for \_\_\_\_\_ in the end

- a. Complete specialization
- b. Partial specialization
- c. Ordinary function



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## **d. Generic template**

170. \_\_\_\_\_ may inherit from a complete specialization
- a. Partial specialization
  - b. Complete specialization
  - c. Ordinary class
  - d. **All of given options**
171. Which of the following may inherit from the ordinary class?
- a. Class template
  - b. Partial specialization
  - c. Complete specialization
  - d. **All of given option**
172. Which of the following is a sequence container?
- a. Map
  - b. **Vector**
  - c. Set
  - d. multiset
173. Each \_\_\_\_\_ of a template class by default becomes function template
- a. Data member
  - b. **Member function**
  - c. Type parameter
  - d. Object
174. When we specialize a function template, it is called \_\_\_\_\_
- a. Function overloading
  - b. Function overriding
  - c. **Function template overloading**
  - d. Function template overriding
175. Which of the following is NOT an associative container?
- a. Set
  - b. Multiset
  - c. Map
  - d. **Group**
176. A \_\_\_\_\_ is a pointer which is declared outside the \_\_\_\_\_
- a. Container, cursor
  - b. **Cursor, container**
  - c. Container, class
  - d. This, cursor
177. Which of the following are collectively referred to as the first class containers?
- a. Associative containers and container adapters
  - b. **Sequence containers and associative containers**
  - c. Sequence containers and container adapters

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- d. None of the given options
178. \_\_\_\_\_ class is a single class that provides functionality to operate on different type of data
- a. Friend
  - b. **Template**
  - c. Ordinary
  - d. None of the give options
179. Which of the following can be passed as type argument to template?
- a. Primitive type
  - b. User defined types
  - c. **Both primitive type and user defined types**
  - d. None of the given options
180. Which of the following is known as Dereference operator in C++?
- a. **+b.**
  - \***
  - c. &
  - d. ::
181. We can change behavior of template using \_\_\_\_\_
- a. **Template parameters**
  - b. Function parameters
  - c. Class Templates
  - d. none of the given options
182. Which of the following represents partial specialization?
- a. Template < int, char >
  - b. Template < class T, Type t>
  - c. **Template<class T, Class U, int>**
  - d. Template< typename T, class W>
183. When we declare template function as friend of any class, then which of the followingspecialization(s) of template function also become friend of the class granting friendship?
- a. Explicit
  - b. Implicit
  - c. Partial
  - d. **All of the given options**
184. In statement “template <class T, class U, int I = 5>” then non-type parameters is
- a. Class T
  - b. Class U
  - c. **int I**
  - d. All of the given option
185. Which of the following is the correct syntax for passing two type argument to atemplate?

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- a. `template < type T, type U>`
  - b. `template < typename T, typename U>`
  - c. `template <type T,U>`
  - d. `template Typename < T,U >`
186. Which of the following represents complete specialization?
- a. `Template < int, char >`
  - b. `Template < class T, float>`
  - c. `Template<class T, Class U , int>`
  - d. `Template< class , class W>`
187. Which of the following is/are the key component of standard template library?
- a. Containers
  - b. Iterators
  - c. Algorithm
  - d. **All of the given option**
188. Which of the following statement is true about partial specialization?
- a. Function templates cannot have partial specialization
  - b. Class template cannot have partial specialization
  - c. **Both class templates and function template can have partial specialization.**
  - d. None of the given
189. Which of the following is correct code portion to initialize static data member "value" of int type for class X?
- a. `Template <class T> X::value =0;`
  - b. `Template<class T>int X::value =0;`
  - c. **`Template<class T> int X < T >::value = 0;`**
  - d. `Template<class T> X < T >::value =0;`
190. When we want to have exactly identical operations on different data type, \_\_\_\_\_ are used
- a. **Function Template**
  - b. Function Overloading
  - c. Function Overriding
  - d. None of the given option
191. In order to define a class template the first line of definition must be:
- a. **`Template <typename T>`**
  - b. `typename < template T>`
  - c. `Template class <ClassName>`
  - d. `Class <Template T>`
192. In case of template specialization, if compiler cannot find required complete specialization then it searches for some \_
- a. General template
  - b. Complete template
  - c. **Partial specialization**

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d. None of the given

193. Which of the following is the correct code to instantiate the object of given templatevector class of int type?

```
Template <class T>
class vector
{
};
```

- a. Vector obj <int>;
- b. Vector obj <> int;
- c. Vector <int>obj;
- c. Vector int obj;

194. Which of the following types of container is NOT provided by standard TemplateLibrary?

- a. Sequence containers
- b. Structured containers
- c. Associative containers
- d. Container adapters

195. A class template \_\_\_\_\_

- a. Facilitates rescue of class
- b. Dose not facilitates reuse of class
- c. Does not support generic methods
- d. Does not support static members

196. Suppose a template class “test” has a static data members, how many copies of thisstatic data member will be created when template class is instantiated by writing following line of C++ code?

```
Test <char>x,y,z
```

- a. Zero
- b. One Correct
- c. Two

197. STL stands for:

- a. Stack Template Library
- b. Standard Template Library
- c. Structured Template Library
- d. Secure Template Library

198. Which of the following is not a sequence container?

- a. Vector
- b. Deque
- c. Map
- d. List

199. Suppose a template class “test” has a static data members, how many copies of thisstatic data member will be created when template class is instantiated by writing following line of C++ code?

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Test <int>x; Test <char>y;

- a. Zero
- b. One
- c. Two**
- d. Three

200. Suppose we have a template class “Test” which have a static data member “data” which is initialized with 0. What will be output when following code is written in main() function for given class?

```
Test<int> a, b; a.data = 5;
```

```
b.data = 7; cout<<a.data;
```

- a. 0
- b. 1
- c. 5**
- d. 7 Correct**

201. In resolution order, highest priority is given to \_\_\_\_\_ in template specialization.

- a. General template
- b. **Complete specialization**
- c. Partial specialization
- d. None of the given options

202. Which of the following is a constrained version of some first-class container?

- a. Sequence containers
- b. Associative containers
- c. Container Adapter**
- d. None of the given options

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**FINALTERM EXAMINATION 2012 (FEB)**

**Q: 42: Write the names of two types of Templates. MARKS: 2**

**Answer:- (Page 256)**

- a. Function Templates (in case we want to write general function like printArray)
- b. Class Templates (in case we want to write general class like Array class)

**Q: 42: Describe the way to declare a template function as a friend of any class. 2marks**

**Answer:- (Page 294)**

```
template< class T >
class B {
int data;
friend void doSomething(T); // granting friendship to template
doSomething in // class B
friend A< T >; // granting friendship to class A in class B
};
```

**Q: 43: Can a constructor throw an exception? How to handle the errors, when the constructor fails?**

**Answer:-**

Yes, Constructor can throw an exception But In constructor if an exception is thrown than all objects created so far are destroyed. Exception due to constructor of any contained object or the constructor of a parent class can be caught in the member initialization list.

**Q: 44: Why these operators can not be overloaded:**

**. , .\* , ?: , ::**

**Answer:- (Page 141)**

Reason: They take actual current object name, rather than value in their argument

**Q: 46: What is graceful termination method of Error handling; give its example using C++.**

**Answer:- (Page 330)**

Program can be designed in such a way that instead of abnormal termination, that causes the wastage of resources, program performs clean up tasks, mean we add check for expected errors (using if conditions),

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Example – Graceful Termination

```
int Quotient (int a, int b) {
if(b == 0){
cout << "Denominator can't " << " be zero" << endl;
// Do local clean up
exit(1);
}
return a / b;
}
```

**Q: 47: How can you describe Virtual destructor? Why Virtual destructor is used?**

**Answer:- (Page 234)**

When delete operator is applied to a base class pointer, base class destructor is called regardless of the object type. Only base class part of object will be deleted other parts will remain as it is this result in memory leak (wastage of memory), Virtual Destructors make it possible that complete object is being deleted so there is no memory leak (waste of memory).

**Q: 48: Motor car can be called a good candidate for being an object so, this object contains Encapsulation characteristic. You are required to apply the concept of Encapsulation on “Motor Car”.**

**Answer:-**

Yes Motor Car is a good candidate for apply the concept of encapsulation

In Motor Car we have the following Objects which fully describes the phenomena of encapsulation.

1. Steer Wheels
2. Accelerate
3. Change Gear
4. Apply Brakes
5. Turn Lights On/Off

In encapsulation we just provide the user an interface and hide the Full implementation from the user because user don't want to know the implementation he Just want his work should be done perfectly

For example: Breaks of car, we gave just a paddle to the User and encapsulate how it works inside the engine. Same for accelerator, gear, steer wheels and turn on and off of Light.

**Q: 49: Suppose base class and derived class has a member function with same signature. Now call that member function through the pointers to a base class object. Now, what determines that the base class member function or derived class function will be called?**

**Q: 51: What do you know about exception in initialization? Explain with example.**

**Answer:- (Page 345)**

**Exception in Initialization List**

Exception due to constructor of any contained object or the constructor of a parent class can be caught in the member initialization list.

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**Example**

```
Student::Student (String aName) : name(aName)
/*The constructor of String can throw a exception*/
{
...
}
```

**Exception in Initialization List**

The programmer may want to catch the exception and perform some action to rectify the problem

**Example**

```
Student::Student (String aName)
try : name(aName) {
...
}
catch (...) {
}
```

## FINAL TERM EXAMINATION 2012 (FEB)

**1. How can we set the default values for non type parameters?****Answer:- (Page 284)**

We can set default value for this non type parameters, as we do for parameters passed in ordinary functions,

```
template< class T, int SIZE = 10 >
```

```
class Array {
private:
T ptr[SIZE];
public:
void doSomething();
...
}
```

**2. Explain the statement vector <int> ivec (4,3)****Answer:-**

This would create a vector of four elements, each one initialized to 3. ivec looks like this:

```
3 3 3 3
```

**3. Tell the procedure of making a class abstract in C++ by giving an example.****Answer:- (Page 230)**

In C++, we can make a class abstract by making its function(s) pure virtual. Conversely, a class with no pure virtual function is a concrete class (which object can be instantiated)

A class having pure virtual function(s) becomes abstract

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```
class Shape {
...
public:
virtual void draw() = 0;
}
```

**4. Write down a list of four intangible objects?**

**Answer:-**

1. Time
2. Light
3. intelligence
4. Temperature
5. software

**5. Give the pseudo code of non case sensitive comparison function of string class.**

**Answer:- (Page 263)**

```
int caseSencompare(char* str1, char* str2)
{
for (int i = 0; i < strlen(str1) && i < strlen(str2); ++i)
if (str1[i] != str2[i])
return str1[i] - str2[i];
return strlen(str1) - strlen(str2);
}
```

**6. Describe three properties necessary a container to implement generic algorithms.**

**Answer:- (Page 301)**

We claimed that this algorithm is generic, because it works for any aggregate object (container) that defines following three operations

- a. Increment operator (++)
- b. Dereferencing operator (\*)
- c. Inequality operator (!=)

**7. Mobile is a good entity for being an object. Write characteristics, behavior and unique ID.**

**Answer:-**

Characteristics: Buttons, Battery, Screen

Behavior: Calling, Sending Message, Internet, Multimedia

Unique ID: Nokia

**8. What are the container requirements? Write details.**

**Answer:-**

As each container need to perform certain operations on the elements added to it like their copy while creating another instance of container or their comparison while performing certain operation on them like their sorting, so the elements that we are going to add in containers should provide this kind of basic functionality.

Examples of these functionalities are given below,

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- When an element is inserted into a container, a copy of that element is made using,
  - ❖ Copy Constructor
  - ❖ Assignment Operator

So the elements need to be added to any container should provide copy and assignment functionality.

- . Associative containers and many algorithms compare elements so the elements that are added to associative containers should have this functionality,

o Operator ==

o Operator <

c++ does not provide functionality of comparison operator (==) or less than operator (<) by itself so we have to provide this functionality by ourselves in element class if we want to use it in associative containers.

## FINAL TERM EXAMINATION 2012 (FEB)

### 1- Resolve problems in overloading assignment operator in string class.

**Answer:-**

The problem in statement

str1 = str2 = str3 is,

str1=str2=str3 is resolved as:

str1.operator=(str2.operator=(str3))

Assignment operator is being called two times one for part str2 = str3 and then for str1 = (str2 = str3) as assignment operator is right associate so first str2=str3 will be executed, and str2 will become equal to str3, then first overloaded assignment operator execution result will be assigned to s1,

str1.operator=(str2.operator=(str3))

### 2- What is random-iterator? What relation b/w random iterator and vector 5 marks

**Answer:-**

Random Access Iterators:-

They have all the capabilities of bidirectional Iterators plus they can directly access any element of a container.

You can think of vectors as smart arrays. They manage storage allocation for you, expanding and contracting the size of the vector as you insert or erase data. You can use vectors much like arrays, accessing elements with the [] operator. Such random access is very fast with vectors. as we can access any element of vector using its index so we can use random access iterator.

It's also fast to add (or push) a new data item onto the end (the back) of the vector. When this happens, the vector's size is automatically increased to hold the new item.

### 3- Class complex {double real,ing public ..... }; overload the \*= operator for the complex class by writing C++ code 3 marks

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**Answer:- (Page 153)**

```
class Complex{
double real, img;
public:
Complex & operator+=(const Complex & rhs);
Complex & operator+=(count double & rhs);
...
};
```

**4-suppose there is a class Doctor with data member. Name Department and Doctor. you are required to write default copy constructor as well as deep copy constructor in order to initialize object and explain the difference with the help of main() function. Only write up to constructor code and main() function 5 marks**

**FINAL TERM EXAMINATION 2011**

**Question No: 37 ( Marks: 2 )**

Write three important features of virtual functions.

**Answer:-** [Click here for detail](#)

These are one of the core components of C++ and enable different specializations of a base class. An important feature of the virtual function mechanism is that it is truly dynamic with respect to the resolution of the correct method to be called.

**Question No: 38 ( Marks: 5 )**

**There are some errors in the code given below, you have to Indicate the line no. with error/s**

**Give the reason for error/s**

**Correct the error/s.**

1. #include <iostream>
2. #include <stdlib.h>
3. using namespace std;
4. template <typename T>
5. class MyClass{
6. public:
7. MyClass(){
8. cout<<"This is class1"<<endl;
9. }
10. };
- 11.

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```

12. template <typename T>
13. class MyClass<int*>{
14. public:
15. MyClass(){
16. cout<<"This is class2"<<endl;
17. }
18. };
19.
20. int main(int argc, char *argv[])
21. {
22. MyClass<int> c1;
23. MyClass<int*> c2;
24. system("PAUSE");
25. return 0;
26. }

```

**Answer: Correct Code**

```

#include <iostream>
#include <stdlib.h>

using namespace std;
template <typename T>
class MyClass{
public:
MyClass(){
cout<<"This is class1"<<endl;
}
};

template <typename T>
class MyClasss{ // class MyClass<int*>{ templates we do not specify type in definition.
public:
MyClasss(){
cout<<"This is class2"<<endl;
}
};

int main(int argc, char *argv[])
{
MyClass<int> c1;
MyClasss<int> c2;*
system("PAUSE");
return 0;
}

```

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**Question No: 39 ( Marks: 5 )**

**What is random iterator? What is its relation with vectors?**

**Answer:-**

Random Access Iterators:-

They have all the capabilities of bidirectional Iterators plus they can directly access any element of a container.

You can think of vectors as smart arrays. They manage storage allocation for you, expanding and contracting the size of the vector as you insert or erase data. You can use vectors much like arrays, accessing elements with the [] operator. Such random access is very fast with vectors. as we can access any element of vector using its index so we can use random access iterator

It's also fast to add (or push) a new data item onto the end (the back) of the vector. When this happens, the vector's size is automatically increased to hold the new item.

**FINAL TERM EXAMINATION 2011**

**Create a vector array of length 8, and also initialize the elements of this array with values 0 1 2 3 4 5 6 7 (5 mark)**

**Suppose base class and derive class have a member function with same signature. Now call that member function through pointer to a base class object. Now what determines that the base class member function or derived class will be called? (mark 5)**

**Describe the salient feature of abstract class (mrk3)**

**Answer:- (Page 230)**

Abstract class's objects cannot be instantiated they are used for inheriting interface and/or implementation, so that derived classes can give implementation of these concepts.

In C++, we can make a class abstract by making its function(s) pure virtual. Conversely, a class with no pure virtual function is a concrete class

**Give the C++ code of template function to print the values of any type of array I int. this function will take 2 parameters one will be pointer and other will be size of array (mrk3)**

**Answer:- (Page 257)**

```
template< typename T >
void printArray(T* array, int size)
{
 for (int i = 0; i < size; i++)
 cout << array[i] << " , "; // here data type of array is T
}
```

**Give the name of three operation that a cursor or iterator generally provide (mrk3)**

**Answer:- (Page 305, 309)**

- T\* first()
- T\* beyond()
- T\* next( T\* )

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**What do you know about function Template?**

(Answer: Page 262)

Function templates are used when we want to have exactly identical operations on different data types in case of function templates we cannot change implementation from data type to data type however we can specialize implementation for a particular data type.

**Give the basic difference between iterator and cursors?**

**Answer:- (Page 308)**

cursors were external pointer that we accessing internal data of any container like vector, it is against the principle of data hiding as we can access any container data using cursors so it is not good programming practice to given access in container for the use of cursors (first, next, beyond methods) we have alternate to cursors in the form of Iterators which are that traverse a container without exposing its internal representation.

**The least one advantage and Disadvantage of Template**

**Answer:- (Page 300)**

Advantages:

Templates provide

- Reusability
- Writability

Disadvantages:

- Can consume memory if used without care.

## **FINAL TERM EXAMINATION 2011**

**1-define composition and give its example with coding**

**Answer:- (Page 53)**

An object may be composed of other smaller objects, the relationship between the “part” objects and the “whole” object is known as Composition.

Example

Ali is made up of different body parts; They can't exist independent of Ali.

**2-what are container classes? How many types of container classes are there?**

**Answer:- (Page 312)**

Container is an object that contains a collection of data elements like we have studied before now we will study them in detail.

STL provides three kinds of containers,

1. Sequence Containers
2. Associative Containers
3. Container Adapters

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### 3-what is virtual inheritance?

**Answer:- (Page 253)**

In virtual inheritance there is exactly one copy of the anonymous base class object

### 4-can a constructor throw exception? If it fails, how should this error be handled?

**Answer:- [Click here for detail](#)**

One-stage constructors should throw if they fail to fully initialize the object. If the object cannot be initialized, it must not be allowed to exist, so the constructor must throw.

### 5-define inheritance and give its example.

**Answer:- (Page 29)**

- ❖ . Derived class inherits all the characteristics of the base class
- ❖ . Besides inherited characteristics, derived class may have its own unique characteristics
- ❖ . Major benefit of inheritance is reuse

### 7-what is constructor?

**Answer:- (Page 74)**

Constructor is used to **initialize** the objects of a class. Constructor is used to ensure that object is in well defined state at the time of creation.

### 8-define static and dynamic binding.

**Answer:- (Page 227)**

**Static binding** means that target function for a call is selected at compile time

**Dynamic binding** means that target function for a call is selected at run time

## FINAL TERM EXAMINATION 2011

Q1) Fill in the blanks below with public, protected or private keyword.

a. Public members of base class are \_\_\_\_\_ members of derived class

b. Protected members of base class are \_\_\_\_\_ members of derived class.

**Answer:-**

a. Public members of base class are \_\_\_\_\_ **public** \_\_\_\_\_ members of derived class

b. Protected members of base class are \_\_\_\_\_ **protected** \_\_\_\_\_ members of derived class.

Q2) Can a constructor throws an exception? How to handle the error when the constructor fails?

**Answer:- Repeated**

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**Q3) What is the difference (if any) between the two types of function declarations?**

**template function\_declaration;**

**template function\_declaration;**

**Answer:-** [Click here for detail](#)

The format for declaring function templates with type parameters is:

**template <class identifier> function\_declaration;**

**template <typename identifier> function\_declaration;**

The only difference between both prototypes is the use of either the keyword class or the keyword typename. Its use is indistinct, since both expressions have exactly the same meaning and behave exactly the same way.

**Q4) State any two reasons why the virtual methods can not be static?**

**Answer:-** [Click here for detail](#)

The virtual method implies membership, so a virtual function cannot be a nonmember function. Nor can a virtual function be a static member, since a virtual function call relies on a specific object for determining which function to invoke. A virtual function declared in one class can be declared a friend in another class.

**Q5) Give three advantages that Iterators provide over Cursors.**

**Answer: - (Page 311)**

- a. With Iterators more than one traversal can be pending on a single container
- b. Iterators allow to change the traversal strategy without changing the aggregate object
- c. They contribute towards data abstraction by emulating pointers

**Q6) Consider the code given below explain what kind of association exists between class A and class B. Justify your answer as well.**

**class A{**

**private:**

**int a,b,c;**

**public:**

**....**

**};**

**class B{**

**private:**

**int d,e,f;**

**A obj1;**

**public:**

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```
....
};
```

**Q7) If we declare a function as friend of a template class will it be a friend for a particular data type or for all data types of that class.**

**Q8) Is it possible to have Virtual Constructor? Justify your answer.**

**Answer: -** <http://r4r.co.in/answer.php?id=1966&option=C++%20Subjective>

There is nothing like Virtual Constructor. The Constructor can't be virtual as the constructor is a code which is responsible for creating an instance of a class and it can't be delegated to any other object by virtual keyword means.

**Q9) Explain the difference between static member variables with Non-static member variables of a class with the help of example.**

**Q10) There are some errors in the code given below, you have to**

- 1. Indicate the line no. with error/s**
- 2. Give the reason for error/s**
- 3. Correct the error/s.**

```
1. #include (header extension is missing)
2. #include
3. using namespace std;
4. template
5. class MyClass{
6. public:
7. MyClass(){
8. cout<<"This is class1"<<endl;
9. }
10. };
11. template
12. class MyClass{
13. public:
14. MyClass(){
15. cout<<"This is class2"<<endl;
16. }
17. };
18. int main(int argc, char *argv[])
19. {
20. MyClass c1;
21. MyClass c2;
22. system("PAUSE");
23. return 0;
}
```

**Answer: - Repeated**

**Q11) What is a container class? What are the types of container classes?**

**Answer: - Repeated**

**Q12) What is a Virtual Destructor? Why we use Virtual Destructors.**

**Answer: - Repeated**

## **FINAL TERM EXAMINATION 2011**

**1. What are container requirements (5)**

**Answer:- Repeated**

**2. Give the name of two cases when you MUST use initialization list as opposed to assignment in constructors. 5**

**Answer:- [Click here for detail](#)**

Both non-static const data members and reference data members cannot be assigned values; instead, you should use initialization list to initialize them.

**3. In which situation do we need to implement Virtual inheritance? explain with an example (5 marks)**

**Answer:-**

In multiple inheritance while solving diamond problem virtual inheritance need to implement. The solution of avoid this problem is virtual inheritance so that in multiple inheritance only one copy of base class is generated as shown below instead of two separate copies.

In virtual inheritance there is exactly one copy of the anonymous base class object

**Example:**

```
class Vehicle{
protected:
int weight;
};
class LandVehicle :
public virtual Vehicle{
};
class WaterVehicle :
public virtual Vehicle{
};
Example
class AmphibiousVehicle:
public LandVehicle,
public WaterVehicle{
public:
AmphibiousVehicle(){
weight = 10;
}
```

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```
};
```

**4. What is random iterator? what is the relation between Random iterator and vector?**

**Answer:-** rep

**5. Give C++ code of template function to print the values of any type of array.. 3Marks**

**[ Hint: this function will take two parameters, one will be array pointer and other will be size of the array]**

**6. If iter is an iterator to a container. Write an expression that will have the value of the object pointed to by iterator, and will then cause iterator to point to the next element. (3marks)**

**Answer:-**

```
*iter++
```

**7. Describe three problems with multiple inheritance (3 marks)**

**Answer:- (Page 248)**

- ❖ If more than one base class have a function with same signature then the child will have two copies of that function
- ❖ Calling such function will result in ambiguity

**8. Name two types of template (2 Marks)**

**Answer:- (Page 256)**

- a. Function Templates (in case we want to write general function like printArray)
- b. Class Templates (in case we want to write general class like Array class)

**9. Sort data in the order in which compiler searches a function. Complete specialization, generic template, Partial specialization, Ordinary function. (2Marks )**

**Answer:- (Page 286)**

- a. First of all compiler looks for complete specialization
- b. If it can not find any required complete specialization then it searches for some partial specialization
- c. In the end it searches for some general template

**10. Give the names of two types of containers basically known as first class containers.(2 Marks)**

**Answer:- (Page 317)**

Sequence and associative containers are collectively referred to as the first-class containers

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## FINAL TERM EXAMINATION 2010

**Question No: 27 ( Marks: 2 )**

**Describe the way to declare a template function as a friend of any class.**

**Answer:- (Page 294) rep**

**Question No: 28 ( Marks: 2 )**

**State any two reasons why the virtual methods can not be static?**

**Answer:- rep**

**Question No: 29 ( Marks: 2 )**

**Explain the statement below,**

**vector<int> ivec(4, 3);**

**Answer:- rep**

**Question No: 30 ( Marks: 2 )**

**Explain two benefits of setter functions.**

**Answer:- (Page 67)**

- 1- It minimize the changes to move the objects in inconsistent states
- 2- You can write checks in your setter functions to check the validity of data entered by the user, for example age functions to check to calculate the age from date entered.

**Question No: 31 ( Marks: 3 )**

**Consider the code below,**

**template< typename T >**

**class T1 {**

**public:**

**T i;**

**protected:**

**T j;**

**private:**

**T k;**

**friend void Test();**

**};**

**This code has a template class T1 with three members i,j and k and a friend function Test(), you have to describe which member/s of T1 will be available in function Test().**

**Answer:- [Click here for detail](#)**

All of them (i, j, k) will be available in function Test().

**Question No: 32 ( Marks: 3 )**

**What do you mean by Stack unwinding?**

**Answer:- (Page 336)**

The flow control ( the order in which code statements and function calls are made) as a result of throw

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statement is referred as “stack unwinding”

**Question No: 33 ( Marks: 3 )**

**Give the c++ code of case sensitive comparison function of string class.**

**Answer:- (Page 265)**

```
class CaseSenCmp {
public:
static int isEqual(char x, char y) {
return x == y;
}
};
class NonCaseSenCmp {
public:
static int isEqual(char x, char y) {
return toupper(x) == toupper(y);
}
};
template< typename C >
int compare(char* str1, char* str2)
{
for (int i = 0; i < strlen(str1) && i < strlen(str2); i++)
if (!C::isEqual (str1[i], str2[i]))
return str1[i] - str2[i];
return strlen(str1) - strlen(str2);
};
int main() {
int i, j;
char *x = "hello", *y = "HELLO";
i = compare< CaseSenCmp >(x, y);
j = compare< NonCaseSenCmp >(x, y);
cout << "Case Sensitive: " << i;
cout << "\nNon-Case Sensitive: “
<< j << endl;
return 0;
}
```

**Question No: 34 ( Marks: 5 )**

**What is random\_iterator? What is relation between random\_iterator and Vector?**

**Answer:- rep**

**Question No: 35 ( Marks: 5 )**

**What would be the output of this code?**

```
class mother {
public:
mother ()
```

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```

 { cout << "mother: no parameters\n"; }
 mother (int a)
 { cout << "mother: int parameter\n"; }
};
class daughter : public mother {
 public:
 daughter (int a)
 { cout << "daughter: int parameter\n\n"; }
};
class son : public mother {
 public:
 son (int a) : mother (a)
 { cout << "son: int parameter\n\n"; }
};
int main () {
 daughter rabia (0);
 son salman(0);

 return 0;
}

```

**Answer:-** [Click here for detail](#)

mother: no parameters  
daughter: int parameter

mother: int parameter  
son: int parameter

**Question No: 36 ( Marks: 5 )**

The code given below has one template function as a friend of a template class,

1. You have to identify any error/s in this code and describe the reason for error/s.
2. Give the correct code after removing the error/s.

```

template<typename U>void Test(U);
template< class T > class B {
 int data;
 public:
 friend void Test<>(T);
};
template<typename U>
void Test(U u){
 B < int> b1;
 b1.data = 7;
}

```

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```

 }

int main(int argc, char *argv[])
{
 char i;
 Test(i);
 system("PAUSE");
 return 0;
}

```

**Answer:- (correct code)**    [Click here for detail](#)

```

#include <cstdlib>
template<typename U> void Test(U);

template< class T > class B {
 int data;
 public:
 template <typename U > friend void Test(U); // this statement is missing
};

template<typename U>
void Test(U u){
 B < int> b1;
 b1.data = 7;
}

int main(int argc, char *argv[])
{
 char i;
 Test(i);
 system("PAUSE");
 return 0;
}

```

## FINAL TERM EXAMINATION 2010

**Question No: 27 ( Marks: 2 )**

Describe the way to declare a template function as a friend of any class.

**Answer:- rep**

**Question No: 28 ( Marks: 2 )**

Give the names of any two types of template.

**Answer:- rep**

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**Question No: 29 ( Marks: 2 )**

Explain the statement below,

```
vector<int> ivec(4, 3);
```

**Answer:- rep**

**Question No: 30 ( Marks: 2 )**

Q. Enlist the kinds of association w.r.t Cardinality (3)

**Answer:- (page 51)**

With respect to cardinality association has the following types,

- a. Binary Association
- b. Ternary Association
- c. N-ary Association

**Question No: 31 ( Marks: 3 )**

Give three advantages that Iterators provide over Cursors.

**Answer:- rep**

**Question No: 32 ( Marks: 3 )**

Give the differences between virtual inheritance and multiple inheritances.

**Answer:- (Page 248, 253)**

In Multiple Inheritance, If more than one base class have a function with same signature then the child will have two copies of that function; Calling such function will result in ambiguity But In virtual inheritance there is exactly one copy of the anonymous base class object.

**Question No: 33 ( Marks: 3 )**

If we declare a function as friend of a template class will it be a friend for a particular data type or for all data types of that class.

**Question No: 34 ( Marks: 5 )**

See the 5 code snippets below and tell whether these are correct or incorrect also justify your answers in the table given at the end.

**Snippet No.1**

```
template< class T>
class A {
};
template< class T >
class B : public A< T* >
{ ... }
```

**Snippet No.2**

```
template< >
```

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```
class B< int* > : public A< T* >
{ ... }
```

### Snippet No.3

```
class B : public A< T* >
{ ... }
```

### Snippet No.4

```
template< >
class B< char* > : public A
{ ... };
```

### Snippet No.5

```
template< class T >
class B : public A< T* >
{ ... }
```

Table:

| Snippet No. | Is it correct or not (Correct/ Incorrect) | Justification of your answer |
|-------------|-------------------------------------------|------------------------------|
| 1           |                                           |                              |
| 2           |                                           |                              |
| 3           |                                           |                              |
| 4           |                                           |                              |
| 5           |                                           |                              |

### Question No: 35 ( Marks: 5 )

What is the output produced by the following program?

```
#include<iostream.h>
```

```
void sample_function(double test) throw (int);
```

```
int main()
```

```
{
 try
 {
 cout <<"Trying.\n";
 sample_function(98.6);
 cout << "Trying after call.\n";
 }
}
```

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```

 }
 catch(int)
 {
 cout << "Catching.\n";
 }

 cout << "End program.\n";
 return 0;
}
void sample_function(double test) throw (int)
{
 cout << "Starting sample_function.\n";
 if(test < 100)
 throw 42;
}

```

**Answer:-**

Trying.  
Starting sample\_function.  
Catching.  
End program.

**Question No: 36 ( Marks: 5 )**

Suppose the base class and the derived class each have a member function with the same signature. When you have a pointer to a base class object and call a function member through the pointer, discuss what determines which function is actually called, the base class member function or the derived-class function.

## FINAL TERM EXAMINATION 2010

**Question No: 27 ( Marks: 2 )**

Give two uses of a destructor.

**Answer:- (page 92)**

1. Destructor is used to free memory that is allocated through dynamic allocation. We have to free memory allocated using new operator by over self in destructor otherwise it remain occupied even after our program ends.
2. Destructor is used to perform house keeping operations.

**Question No: 28 ( Marks: 2 )**

Describe the way to declare a template class as a friend class of any other class.

**Answer:- rep**

**Question No: 29 ( Marks: 2 )**

Give the name of two basic types of containers collectively called First class containers?

**Answer:- rep**

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**Question No: 30 ( Marks: 2 )**

State any conflict that may rise due to multiple inheritance?

**Answer :- ( page 248)**

If more than one base class has a function with same signature then the child will have two copies of that function. Calling such function will result in ambiguity.

**Question No: 31 ( Marks: 3 )**

What will be the output after executing the following code?

```
class c1{
public:
virtual void function(){
cout<<"I am in c1"<<endl;
}

};
class c2: public c1{
public:
void function(){
cout<<"I am in c2"<<endl;
}

};
class c3: public c1 {
public:
void function(){
cout<<"I am in c3"<<endl;
}

};

int main(){

c1 * test1 = new c2();
c1 * test2 = new c3();
test1->function();
test2->function();
system("PAUSE");
return 0;
}
```

**Answer:-**

am in c2  
I am in c3

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**Question No: 32 ( Marks: 3 )**

If we declare a function as friend of a template class will it be a friend for a particular data type or for all data types of that class.

**Question No: 33 ( Marks: 3 )**

Tell the logical error/s in the code given below with reference to resource management; also describe how we can correct that error/s.

```
class Test{

public:
int function1(){
 try{
 FILE *fileptr = fopen("filename.txt","w");
 throw exception();
 fclose(fileptr);
 return 0;
 }
 catch(Exception e){
 ...
 }
}
};
```

**Answer:- (Page 343)**

In the above code, In case of exception the call to fclose will be ignored and file will remain opened. We can remove this issue in following ways

```
int function1(){
try{
FILE *fileptr = fopen("filename.txt","w");
fwrite("Hello World",1,11,fileptr);
...
throw exception();
fclose(fileptr);
} catch(...) {
fclose(fileptr); // adding fclose in catch handler as well
throw;
}
return 0;
}
```

**Question No: 34 ( Marks: 5 )**

What is the output produced by the following program?

```

#include<iostream.h>
void sample_function(double test) throw (int);

int main()
{
 try
 {
 cout << "Trying.\n";
 sample_function(98.6);
 cout << "Trying after call.\n";
 }
 catch(int)
 {
 cout << "Catching.\n";
 }

 cout << "End program.\n";
 return 0;
}

void sample_function(double test) throw (int)
{
 cout << "Starting sample_function.\n";
 if(test < 100)
 throw 42;
}

```

**Answer:- rep**

**Question No: 35 ( Marks: 5 )**

The code given below has one template function as a friend of a template class,

1. You have to identify any error/s in this code and describe the reason for error/s.
2. Give the correct code after removing the error/s.

```

template<typename U>
void Test(U);
template< class T >

class B {
 int data;
 public:
 friend void Test<>(T);
};

template<typename U>
void Test(U u){

```

```

 B < int> b1;
 b1.data = 7;
}
int main(int argc, char *argv[])
{
 char i;
 Test(i);
 system("PAUSE");
 return 0;
}

```

**Answer:- rep**

**Question No: 36 ( Marks: 5 )**

Consider the following class,

```

class Base
{
 char * p;
public:
 Base() { p = new char[10]; }

 ~Base() { delete [] p; }
};
class Derived : public Base
{
 char * q;
public:
 Derived() { q = new char[20]; }

 ~Derived() { delete [] q; }
};
void foo()
{
 Base* p = new Derived();

 delete p;
}

```

With this program, every time function foo is called, some memory will leak.  
Explain why memory will leak. Also, explain how to fix this problem.

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**Question No: 31 ( Marks: 1 )**

Write the syntax of declaring a pure virtual function in a class?

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**Answer:- (page 230)**

```
class Shape {
...
public:
virtual void draw() = 0;
}
```

**Question No: 32 ( Marks: 1 )**

What is meant by direct base class?

**Answer:- (page 208)**

A direct base class is explicitly listed in a derived class's header with a colon (:)

```
class Child1:public Parent1 { // Here Parent1 is Direct Base Class of Child1
```

```
...
};
```

**Question No: 33 ( Marks: 2 )**

Describe the way to declare a template class as a friend class of any other class.

**Answer:- rep**

**Question No: 34 ( Marks: 2 )**

What is the purpose of template parameter?

**Answer:- (page 263)**

We can change behavior of a template using template parameter.

**Question No: 35 ( Marks: 3 )**

Describe in simple words how we can use template specialization to enforce case sensitive specialization in String class.

**Question No: 36 ( Marks: 3 )**

Can we use compiler generated default assignment operator in case our class is using dynamic memory? Justify your answer.

**Answer:- (Page 197)**

In case our class involves dynamic memory allocation we had to write assignment operator code by our self as we had to write the user defined code for copy constructor.

**Question No: 37 ( Marks: 3 )**

Give the names of three ways to handle errors in a program.

**Answer:- (page 329)**

- Abnormal termination
- Graceful termination
- Return the illegal value

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